



Mobility in Germany – MiD 2023

Results report



a study by:



Performed by:

infas

in cooperation with:



infas 360



Results report for/a study of:

Federal Ministry of Transport

Department G 13 – Forecasts, Statistics and Special Surveys

Robert-Schuman-Platz 1

53175 Bonn

FE project number VB600001



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Project:

7555, Bonn, November 2025

Version 1.0

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The following citation styles are recommended:

Langform:

infas, DLR, IVT and infas 360 (2025): Mobility in Germany – MiD Final Report

Study by infas, DLR, IVT and infas 360 commissioned by the Federal Minister of Transport (FE-No. VB600001).

Bonn, Berlin. www.mobilitaet-in-deutschland.de

Short form:

infas, DLR, IVT and infas 360 (2025): Mobility in Germany (commissioned by the BMV)

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Mobility in Germany 2023 – the project

Being on the move and mobility shape our everyday lives. They are integral to social life to varying degrees, encompassing work and education, leisure activities, and daily needs. A precise understanding of these mobility needs and daily traffic patterns is essential for providing the necessary resources. While the past decades have seen a steady increase in car use, many efforts are now focused on giving greater attention to other modes of transport, such as public transport, cycling, and walking. This also includes new service models like sharing options and on-demand transport. Environmental aspects are demanding increasing consideration. It is evident, not only in Germany, that continuously growing car traffic, especially in urban areas, is reaching its limits.

Zen is encountering this. Nevertheless, it will remain a defining element of traffic and must be actively shaped.

But where do we stand on this path? How do citizens in Germany use public transport? What has changed in recent years? After a COVID-19-related dip in mobility, are we returning to previous levels, or will certain changes persist? What role do working from home and other digital activities play?

And last but not least: What about people with limited mobility? The Mobility in Germany (MiD) study comprehensively addresses these and other questions. Further documents beyond this report are available at www.mobilitaet-in-deutschland.de All

documents relating to previous MiD surveys from 2002, 2008, and 2017 are also available there. Further information and a list of all regional clients of the MiD 2023 can be found at the end of this report.

We hope you enjoy reading it!

Your MiD project team

Data basis

Following 2002, 2008 and 2017, the Federal Ministry of Transport (BMV) commissioned the infas Institute for Applied Social Sciences GmbH to conduct the Mobility in Germany (MiD) study for the fourth time in 2023.

On the client side, over 60 regional partners were again involved, each commissioning different regional analyses. As has been the case since 2008, the work was carried out jointly with the Institute of Transport Research at the German Aerospace Center (DLR).

In addition, the project team was expanded in 2017 to include IVT Research GmbH and infas 360 GmbH.

The fieldwork phase of the current survey, which covered a period of more than twelve months, took place between April 2023 and July 2024. Study participants could take part in a multi-stage process, either in writing, by telephone, or online. Within the total sample of 218,101 households collected in 2023/24, 39,847 households comprised the nationwide baseline sample and 178,254 comprised regionally focused samples.

We would like to extend our sincere thanks to all study participants!

A total of 420,979 people were surveyed, reporting on 1,087,393 journeys on their respective reference dates. For a further sub-segment, the journeys were broken down into 94,337 stages. In addition, data were collected on 67,231 day trips and 69,788 trips with overnight stays. A summary of the survey content is provided at the end of this report.

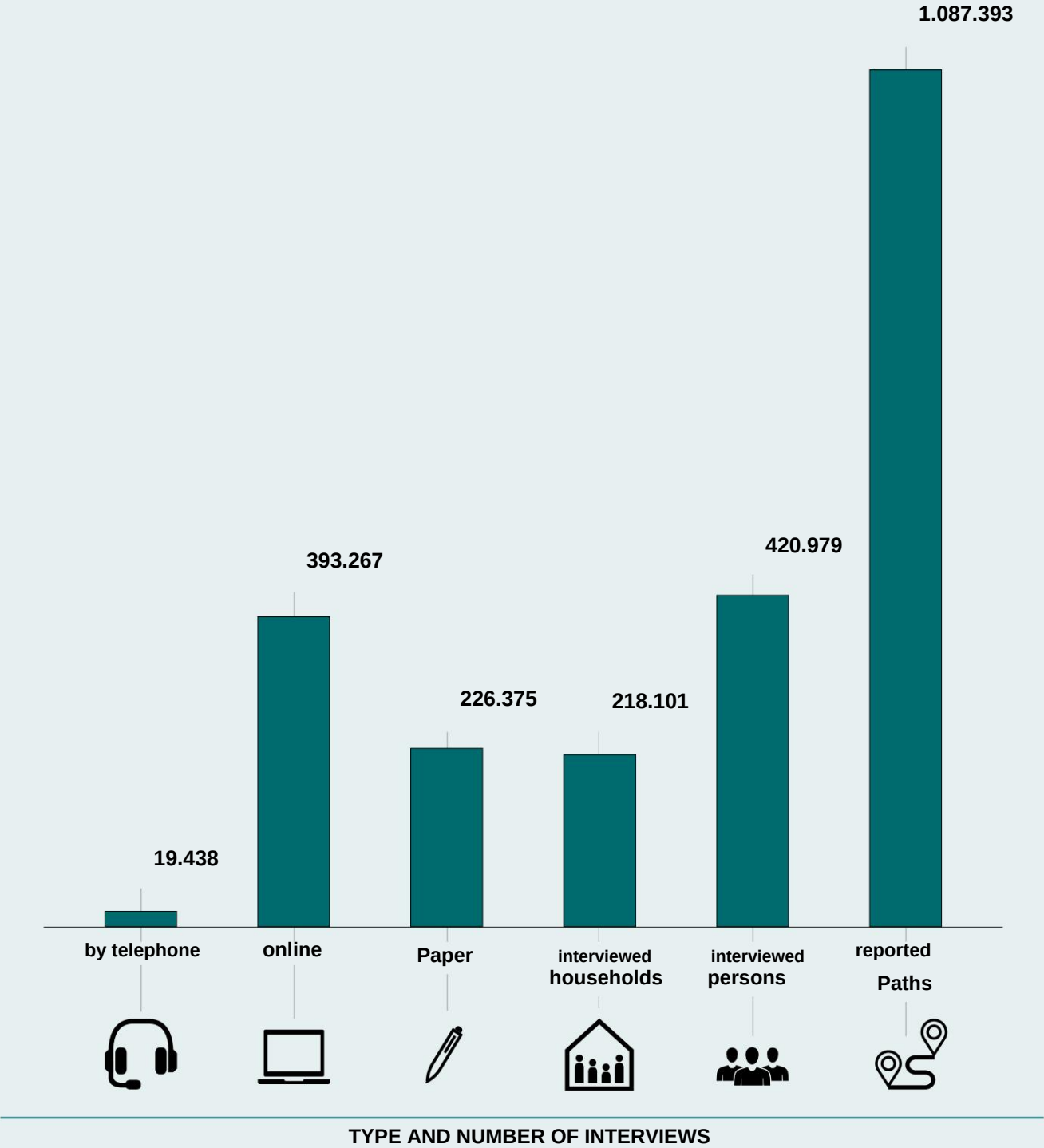
The evaluation of the nationwide base sample and the regional supplements will be integrated, as was the case in 2017. The extrapolation of the results will provide comprehensive indicators of the everyday mobility of the resident population in Germany for the twelve-month period of 2023/24.

It refers to all journeys made by the population within Germany.

We would like to point out that the results of sample surveys can always differ from the actual values in the general population. The possible range of this deviation is called the "confidence interval".

Thanks to the exceptionally large sample size, the interval for mobility indicators in the MiD study is very small, and the results are highly precise.

However, the interval increases as the subset being analyzed becomes smaller. If results show only small differences and relate to a subset (e.g., a population subgroup or a region with a small sample size), they must be interpreted with appropriate caution. This is particularly important to consider when comparing data over time: Small differences do not necessarily represent a genuine change in that direction. A detailed explanation of the statistical background is provided in the methodology report.



Results telegram

Traffic volume and traffic performance slightly reduced

Nationwide, traffic volume and performance have changed somewhat compared to the last MiD surveys. Despite a slight increase in population, traffic volume in 2023 rose from 257 million to approximately 250 million compared to 2017.

The number of journeys per day decreased slightly. Transport performance also declined slightly, from 3.2 billion to approximately 3.0 billion passenger-kilometers per day. This puts it roughly back in the same range as in 2008. Overall, these changes are only slightly above the random fluctuations caused by sampling.

It should be noted, of course, that the COVID-19 pandemic fell between the two most recent MiD surveys. This led to significant temporary declines, particularly in public transport. However, since MiD did not conduct any measurements during this period, this information will have to suffice. Other sources can be consulted if desired.

This overall trend masks regional differences. Particularly in large cities, traffic volume and traffic performance figures, after an increase in 2017, have fallen back to a similar absolute level as in 2008. In rural areas, however, there are hardly any changes in this regard.

The so-called "mobility rates" have continued to decline. While an average of 90 percent of German citizens were on the go on an average day in 2008 and 85 percent in 2017, this figure reached only 82 percent in 2023. This has led to a slightly reduced average daily number of just under three trips per person. A somewhat above-average decline is noticeable among people from households with a higher economic status.

Distances, daily mileage, and travel times change slightly, but not significantly. The average distance traveled decreases slightly from just over to just under 12 kilometers. The total mileage per person per day shows the greatest change, decreasing from almost 40 to around 35 kilometers. Despite the shorter distance, the daily travel time remains at approximately 85 minutes.

More walking and cycling at reduced prices Car shares

The structure of the reduced traffic volume, i.e., the distribution of journeys across different modes of transport (modal split), shows, both proportionally and extrapolated, a slight decrease in private motorized transport (PMT), increases in walking and cycling, and stability with a slight upward trend in public transport. Cycling is among the winners in some urban areas. The share of journeys made solely on foot, however, is rising consistently in both urban and rural areas. Conversely, this also applies to the proportional changes in PMT: car use has decreased.

The nationwide modal split for 2023 was 26 percent for journeys made exclusively on foot, 11 percent for cycling, 40 percent for journeys made by car drivers, and 13 percent for journeys made by car passengers. Public transport, including long-distance services, accounted for 11 percent of the total traffic volume. This analysis of the "main mode of transport" categorizes journeys made using various modes of transport according to a hierarchy and assigns them to one of the aforementioned modes.

If, instead of the route-related modal split, the usual mode of transport used by individuals is considered, a decline in car use is also evident. The proportion of people who spend the majority of their everyday mobility behind the wheel has decreased from 44 to 41 percent. Slight increases, however, are seen in the proportions of those who primarily cycle, public transport users, and those who regularly switch between modes of transport.

A plus for the combination of walking, cycling, bus and train

An analysis of transport performance, i.e., passenger-kilometers traveled, shows a significant increase for walking and the now often electrically assisted bicycle. Public transport remains roughly stable overall, with slightly fewer long-distance kilometers and a minor decrease in local transport, continuing its slow recovery from the COVID-19 slump. Private motorized transport performance is 10 percentage points below the 2017 MiD result.

In the longer time series since 2002, cycling has benefited most, increasing by almost half during this period. Walking has also seen a slightly less pronounced increase. Public transport as a whole has maintained its level, with slight proportional shifts in favor of local public transport. The Deutschlandticket (Germany Ticket) has also contributed to this, with almost one-sixth of adults opting for it more or less regularly in 2023/24.

For the car sector, a decline can be recorded for the first time after growth up to 2017. Overall, it can be said that the previously highest result for a Mobility in Transition (MiT) from 2017 has now shifted in 2023 primarily in favor of non-motorized mobility – apart from the many new e-bike motors, which now assist more than a third of bicycle kilometers.

These changes, however, do not correspond to increased satisfaction. On the contrary: Car traffic is rated largely unchanged, while the ratings for pedestrian and bicycle infrastructure have declined significantly. The reasons are likely manifold, but a more critical view of this infrastructure, coupled with its increasing use, is a strong factor.

Local mobility in the plus

Walking and cycling can be considered part of so-called "local mobility." They often cover relatively short distances. Nevertheless, their contribution should not be underestimated. If purely pedestrian journeys, sections walked to a bus stop, and cycling journeys are combined in terms of their mileage share, they accounted for almost one-tenth of all kilometers traveled in 2023.

Walking, in particular, deserves more attention. In 2023, it accounted for more than a quarter of all journeys. Analyses of pedestrians show that they often differ from the average in their choice of other modes of transport and contribute to more environmentally friendly mobility.

Cycling can also claim this benefit. Its mileage share is growing – thanks to electric propulsion. However, satisfaction with cycling is declining, and the MiT results suggest that improved infrastructure for both cycling and walking could lead to further growth for these two mobility options.

Mobility occasions – stable despite some changes

The structure of travel purposes has changed only slightly. This is not surprising, as this area has traditionally shown a high degree of stability. Nevertheless, a slight increase in both volume and traffic performance for the leisure sector can be observed in 2023.

This coincides with changes in work-related traffic due to working from home.

The proportion of traditional commuter traffic, involving journeys to and from work, is declining, while business trips during working hours are increasing, particularly for regular tasks in sectors such as skilled trades or service industries. The effects of working from home explain a significant portion of the overall slight decrease in mobility, accounting for two-thirds or more, depending on the assumptions made.

Approximately 50 million cars are in operation, compared to around 80 million bicycles.

Contrary to the somewhat reduced use of cars, the car fleet is developing rapidly. It has now grown to almost 50 million vehicles, most of which are also available in private households. This means that, statistically, there are now about 1.2 cars per household. Car ownership is increasing in most cities as well as in rural areas.

Outside of cities, more than 90 percent of households now own at least one car. In metropolitan areas, this figure is still 40 percent. Nationwide, 20 percent of households were car-free in 2023, compared to 22 percent in 2017. Conversely, the proportion of households with more than one car is rising. And the bicycle fleet is also growing, from around 75 million to 81 million bicycles, of which almost a sixth are now electric.

The increased car availability associated with the growth of the passenger car fleet encompasses all age groups – both household ownership and availability independent of personal ownership. Furthermore, a cohort analysis shows that, as in previous editions of the MiD survey, over 90 percent of respondents in the mid-30s to 60s age group had access to a car in 2023. Thus, contrary to often-expected expectations, from this stage of life onward, the younger generations do not differ from previous ones.

However, actual car use is declining almost across the board, especially in younger age groups, but hardly at all among those surveyed of retirement age. Looking at all cars, the average daily driving time per vehicle fell from almost 50 minutes in 2008 to just over 40 minutes in 2023.

Around 5 percent of cars available in private households are now fully electric. This share is still low. But the usage patterns of this new segment have largely adapted to those of the rest of the (newer) fleet. Unlike after 2017, electric cars no longer exhibit specific usage patterns, but are becoming a normal part of the fleet.

Within the passenger car fleet, the trend towards SUVs, off-road vehicles, and vans remains unbroken. Their share in the surveyed households is now even higher than the 20 percent threshold that applied in 2017.

The growing fleet of cars also leads to a greater use of public parking spaces.

While in 2017 around 8 million of the approximately 43 million cars available in private households were parked publicly at home, in 2023 this figure is about 10 million out of now around 46 million cars, an increase in space requirements of over a fifth.

The average distance-weighted car occupancy rate has remained almost stable compared to 2017. In 2023, it remained at approximately 1.4 people with only a slight decrease, although it has decreased somewhat compared to 2008.

Overall, the number of people holding a car driver's license has hardly changed. In 2023, a total of 88 percent of those aged 18 and over held a driver's license. As always, however, a more nuanced view is necessary. The percentage decreases slightly in the under-30 age group, remains largely stable in the middle age groups, but continues to rise, particularly among older women.

Children and seniors on the go

Among others, children and young people, as well as senior citizens, are groups with specific mobility needs. The MiD 2023 study will dedicate special analyses to both groups.

For children, it often represents the dominant view from the back seat of a car. Approximately two-thirds of children's mobility takes place in this form. While activity levels are declining for both children and teenagers, the distances traveled to and from daycare and school remain relatively constant. Among young people aged 14 and over, the mix of transportation modes is changing, with increasing independence, particularly towards public transport. Bicycles play a role in both subgroups, but are not present in all households.

Among senior citizens, a slight decline in mobility levels becomes apparent from the age of 65, and a more pronounced decline from 75. The number of trips per person per day decreases from around three to closer to two. Despite this, car availability and usage remain high. Furthermore, mobility limitations play an often underestimated role even among younger senior citizens. At the age of 65, the "handicap value" is approximately one-fifth.

Everyday mobility also depends on economic status – with some recognizable consequences.

Restrictions on participation due to "mobility poverty"

Observations on "mobility poverty" show that it is often more a matter of "activity poverty" and less a lack of mobility. The affected groups are often elderly, but sometimes also young adults due to economic reasons.

Monday.

The decisive factors are, relatively equally, either infrastructural or personal financial or health-related reasons. In particular, health deficits lead to restrictions in participation.

With the exception of financial constraints, this is often not directly related to the economic household situation. Nevertheless, different levels of mobility are evident along the dividing lines of such a classification. As was the case in 2002, 2008, and 2017, it remains true that well-off individuals realize larger and wider activity radii – and to a greater extent than less well-off individuals with a strong car orientation.

Multifaceted mobility among migrant women and migrants

The MiD 2023 survey examines the mobility of migrants in more detail than the 2017 survey. However, this is a complex picture, and a clear distinction must be made between different regions of origin, making simple conclusions impossible. On average, mobility needs and patterns in this group do not fundamentally differ from those of the rest of the population. However, a somewhat lower overall level of mobility, a lower car ownership rate, and a greater reliance on public transport can be observed. The analysis underscores the importance of public transport, especially for people without a car.

Car availability is up, Car use is declining

Young adults in larger cities are somewhat less car-oriented than their peers in previous years.

It should be noted that, at the latest from the age of 30 onwards, every age cohort can report a higher availability of passenger cars than between 2002 and 2017.

This is especially true for senior citizens. Their car ownership continues to grow, as it did between 2002 and 2017. This is primarily due to the increasing number of older women driving themselves and more frequently owning a car than previous generations of senior citizens.

However, the gap between availability and usage continues to widen even among those aged 70 and over, as typical car use decreases. Only above this age do they become more similar.

Carsharing organizations continue to find their members primarily in large cities. There, more than one in five households now has at least one membership, almost twice as many as in 2017. While in 2017 a total of 22 percent of carsharers almost never used the service despite being members, this figure had increased to a good quarter by 2023.

The share of the newly developing on-demand transport sector is different. Its slice of the modal split pie remains significantly less than one percent. Furthermore, similar to carsharing, a considerable portion of this usage is currently concentrated on just a few services in some major cities.

Mobility impairments affect one in seven people.

According to their own figures, 14 percent of the population is affected by a mobility impairment. This primarily includes walking or visual impairments. This affects all population groups and all age groups. However, the extent increases with age.

In the group of people aged 70 and over, this applies to every third person.

The survey also inquires about the use of assistive devices. These play a particularly important role in the daily mobility of older citizens. For example, among those aged 70 and over with a reported disability, approximately one-third rely on a rollator.

· wheelchair or even electric mobility scooter required.

The results show that the transport infrastructure is insufficiently geared towards the needs and requirements of people with mobility impairments. They receive inadequate support, and their health limitations are accompanied by a sometimes significant and involuntary reduction in their everyday mobility.

Day trips and overnight trips are also a focus.

The MiD 2023 not only records overnight trips undertaken in the last three months beyond the cut-off date, as in previous editions, but also day trips. Both are recorded across all occasions and destinations, including trips abroad.

Based on this, the projection leads to 1.7 billion day trips and 0.5 billion overnight trips per year for the resident population aged 14 and over.

This equates to an estimated average of 24 day activities or excursions and official business trips per quarter, beyond the usual daily routine, thus with a broad definition.

Even on these trips, the car dominates, provided they cover a considerable distance. For day trips, 96 percent of which are within Germany, the car accounts for 68 percent overall. For overnight trips within Germany, the private motorized transport share is around two-thirds (69 percent).

He achieves about half of these trips when traveling to other European countries.

Additionally, the category of day trips can also include journeys made by long-distance commuters who switch between different residences and frequently travel on weekends. This group comprises between 1 and 2 percent of the total population.

If a 100-kilometer distance threshold is used, these journeys are mostly made by car. However, long-distance rail travel also plays a role. The distances are often considerable. For two-fifths of long-distance commuters, the distances are between 100 and 250 kilometers, and for another two-fifths, between 250 and 500 kilometers. A further fifth travels even longer distances.

The bottom line?

Traffic is characterized by routines. These change slowly. The decrease in car traffic and the increase, primarily in pedestrian traffic and, more sporadically, in bicycle traffic, is therefore noteworthy and a signal of possible changes. However, a closer look shows that a trend reversal cannot yet be spoken of. The current traffic mix is similar to what existed in 2008 and only changed in 2017 with an increase in private motorized transport.

the.

From a health and environmental perspective, the positive trend towards non-motorized mobility is encouraging.

However, the results suggest that – with more care taken with their infrastructure – even more would be possible with footpaths and cycle paths.

The Germany ticket was able to stabilize public transport after its Corona-related slump and could even help to increase demand in the longer term.

Nevertheless, outside of major cities, the car remains by far the number one mode of transport, especially when considering mileage – even though the gap between availability and usage is widening.

Beneath the surface of the average figures for passenger transport activity, various and sometimes contradictory trends can be observed. These often reflect regional differences and efforts to improve cycling and public transport options.

1. MiD: Task and concept

"Mobility in Germany" (MiD) is the largest national survey on everyday travel behavior in Germany. Comparing MiD 2023 with its predecessor surveys from 2002, 2008, and 2017 allows for in-depth analyses of the development of mobility behavior and traffic in Germany over a period of two decades. Accordingly, MiD 2023, like its predecessors, represents a central component of German transport statistics and an important data basis for policymakers and businesses. Experts in data analysis, statistics, and modeling from Germany and abroad will use it for analyses and forecasts of transport demand.

The current MiD 2023 provides a comprehensive dataset for describing the everyday mobility of the German resident population. To account for recent societal developments such as the increasing popularity of working from home and online shopping, or the availability of new means of transport, the MiD 2023 has been expanded with new questions compared to previous surveys, while "old" questions have been adapted and updated.

The unique study design of MiD 2023 not only includes a national sample but also integrates data from in-depth surveys commissioned by over 60 regional partners who participated in the national survey, in order to provide a more detailed regional analysis of everyday mobility findings. Together with the data from the in-depth surveys, the national sample data form a comprehensive dataset on mobility behavior that is unparalleled in its breadth and depth in Germany.

As a result, the entire MiD sample includes data on more than one million journeys from over 420,000 respondents from nearly 220,000 households.

Survey design and study content: A combination of innovation and continuity

The primary aim of the MiD 2023 is to provide a comprehensive and up-to-date inventory of everyday mobility in Germany. However, great emphasis was placed on comparability with previous surveys during the creation of the MiD 2023.

Thus, the imputation procedures introduced at MiD 2017 for the core variables of the pathways-

The factors of distance, travel time, and primary mode of transport used were also applied in the MiD 2023. The regional statistical area type ("RegioStaR"), developed by the Federal Ministry of Transport (BMV) in cooperation with the Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR) and first introduced into the MiD in 2017, is also included in the MiD 2023. Finally, the weights for the MiD 2023 projection were developed using the same procedure as in 2017. They are based on the most up-to-date population development data from the 2022 census and its update for December 2023.

The period between the 2017 Mobility Index (MiD) and the current 2023 MiD was marked by several events that had a significant impact on mobility. Foremost among these was the COVID-19 pandemic from 2020 to 2022, which not only led to an overall reduction in traffic but also to a temporary shift away from public transport and a return to private cars. At the same time, the introduction of the €9 ticket and the subsequent nationwide ticket for public transport also falls within this period. Finally, there were significant changes in land use, particularly in urban areas, to promote cycling infrastructure, as well as technologically relevant developments in the field of pedelecs and e-bikes, where considerably more powerful models than ever before are now available.

Study participants could choose between the same data collection instruments as in 2017. They had the option of computer-assisted telephone interviews (CATI), paper questionnaires (PAPI) and computer-assisted online interviews (CAWI), although in 2023 significantly more CAWI and significantly less CATI was used compared to 2017.

New in-depth topics

The MiD 2023 addresses various socially relevant aspects that were either not covered at all or not in sufficient depth in previous versions. These include, for example, the topics of "mobility impairments" and mobility while traveling with and without overnight stays. The MiD 2023 contains specific questions about necessary mobility aids such as rollators, wheelchairs, or the need for accompaniment by other people. In addition, randomly selected

Participants were asked very detailed questions about day trips and overnight trips, as well as the means of transport used to reach and within the destination. This allows for more comprehensive analyses of travel mobility than previous MiD surveys.

In addition to the previously mentioned home office and online shopping, the topic of electromobility, including pedelecs, is covered in greater detail in the current MiD 2023 survey than in previous surveys. This allows for a more precise analysis of both the ownership and use of these new modes of transport.

All these innovations contribute to making the MiD 2023 a comprehensive empirical data basis for mapping current mobility in all its facets in Germany.

Mobility Inventory 2023

This report focuses on presenting everyday mobility in Germany in 2023 and compares it at key points with results from the previous MiD time series. Following this brief introduction, the first chapter explains the study design and the key concepts of MiD 2023 in more detail.

The results are then presented with regard to the key indicators of transport planning and research. These primarily include the modal split in terms of traffic volume and transport performance. The following chapters deal with the availability of different modes of transport to households in Germany, their use, and the underlying travel purposes. This also includes analyses of intermodality and multimodality, i.e., the use of different modes of transport on the same journey or the use of different modes of transport on different journeys throughout the day and week.

The following sections are dedicated to more detailed analyses of the use of individual modes of transport, such as private car ownership, electromobility, public transport, car sharing and on-demand services, cycling, and walking. These chapters-

These chapters primarily aim to illustrate which modes of transport are used, how frequently, by which groups of people, and for what reasons, as well as the significance of their use in terms of overall traffic volume and transport performance. Furthermore, these chapters examine specific topics such as electromobility and pedelecs in more detail, the societal importance of which has increased further since the MiD 2017.

The following chapters address specific topics that were either newly added to the MiD 2023 or were collected in significantly greater detail than in previous surveys. This includes examining the mobility of specific population groups. Specifically, it encompasses analyses of potential differences in the mobility of women and men, as well as evaluations of the specific mobility of children and people with health-related mobility limitations.

The report then examines work-related mobility in more detail, including long-distance commuters. Following this, the section on "Digital" investigates the significance of working from home and online shopping for traffic development. Both of these topics were explored in greater detail in the 2023 Mobility in Germany (MiD) survey than in the 2017 survey. Accordingly, one chapter provides an up-to-date overview of the significance of these two topics for mobility after the end of the pandemic. Another chapter focuses on mobility for day trips and trips involving at least one overnight stay. These two topics were also recorded in significantly greater detail in the 2023 MiD survey than in previous surveys.

Finally, the results of the MiD 2023 are compared with the findings from other data sources and their overall message is placed in context.

The results presented in this report thus aim, on the one hand, to provide a comprehensive and up-to-date overview of people's everyday mobility in Germany. At the same time, the analyses presented in this report only draw on a fraction of the wealth of data available.

MiD 2023. Accordingly, the report is also intended to serve as a basis and inspiration for further analyses and uses of MiD 2023.

This report uses both masculine and feminine forms in some places and not in others, depending on whether a generally accepted masculine or feminine form exists for the word in question. Regardless of the specific word choice, however, the analyses always refer to all people living in Germany, unless they explicitly concern evaluations of specific subgroups of the population.

2. Study design and key concepts

To better understand this report and to contextualize the results, the survey design and key terms used in the MiD 2023 are explained below. A detailed description of the survey design and fieldwork can be found in the MiD 2023 methodology report.

Study design structure: the two data collection phases of the MiD 2023

In the first stage of the data collection, a household survey was conducted, collecting information on household size, details about individual household members, the means of transport available in the household, and several other characteristics. This information was provided by one household member on behalf of all household members.

In a second stage of data collection, all household members were individually interviewed about personal characteristics, their travel patterns on a predetermined date, their day trips in the past four weeks, and their trips involving at least one overnight stay in the past three months. Where possible, each person was interviewed individually.

There was no age restriction for the survey. Up to the age of ten, parents were asked to answer on behalf of their children. Between the ages of ten and thirteen, either the children themselves or their parents could respond. From the age of fourteen, individuals were expected to participate in the survey themselves. If this was not possible, the interview could be conducted on their behalf by another household member.

The data collection phase for the MiD 2023 extended continuously over more than twelve months. After the initial sampling, a specific reference date was set for each household address to collect data on travel patterns. This date then determined the successive assignment of a contact for mailings or presentation at the telephone studio. Due to the division into the two phases described above and the resulting lead times for announcements and follow-ups, the data collection work spanned from April 2023 to July 2024.

The resulting dataset for the MiD sample comprises information from nearly 421,000 individuals in 218,000 households. Over one million journeys were recorded as part of the MiD survey. Of the MiD sample, 39,847 households are included in the federal sample commissioned by the Federal Ministry of Transport, and 178,254 households are included in regional in-depth samples. Unlike the 2008 survey, the 2023 MiD datasets, like the 2017 MiD datasets, also include the cases from the regional in-depth samples.

In the first MiD survey in 2002, a methodological study led to a shift from a purely postal-based method to a combined written and telephone survey. For the subsequent MiD in 2008, the postal-based method was omitted for route data collection. In the 2023 survey, as in the MiD in 2017, all three methods were used in both data collection phases (households and individuals/route data collection): telephone, written, and online surveys. This procedure, already specified in the call for proposals, was designed to offer respondents as many options as possible, closely aligned with their individual preferences, and to achieve good coverage of the target population.

Core survey and module questions

The content of the two interviews is shown in Figure 1. A total of seven levels of data analysis can be distinguished: households, individuals, routes, cars, day trips, trips with at least one overnight stay, and stages.

During the development of the questionnaire, attention was paid to the comparability of the instruments between the MiD 2023 and the MiD 2017, as well as earlier surveys. However, due to societal changes, the inclusion of new questions also became necessary.

To avoid overwhelming respondents with an overly extensive questionnaire, only questions relating to key characteristics of the core survey were posed to all households. The additional content was grouped into eleven survey modules, and each household was presented with only a subset of the questions. This approach was only feasible for the telephone and online surveys.

The following modules are distinguished:

- Additional personal mobility features
- Local mobility and cycling
- Infrastructure and digital supply
- Satisfaction
- Vehicle features –
- Overnight trips
- Homeoffice
- The stage
- Day trips
- Participation –
- Public transport

Definition of key terms

The study is based on a number of terms, the definition and precise delimitation of which are important for the interpretation of the results. Therefore, all terms relevant to the report are presented here.

Route: A route is defined as a journey undertaken by a person traveling from one place to another outside of their home, either on foot or using other means of transport. The outward and return journeys are counted as two journeys. If a transfer occurs on the way to a destination between-

Even when using different modes of transport, it remains a single route. Circular routes are also recorded.

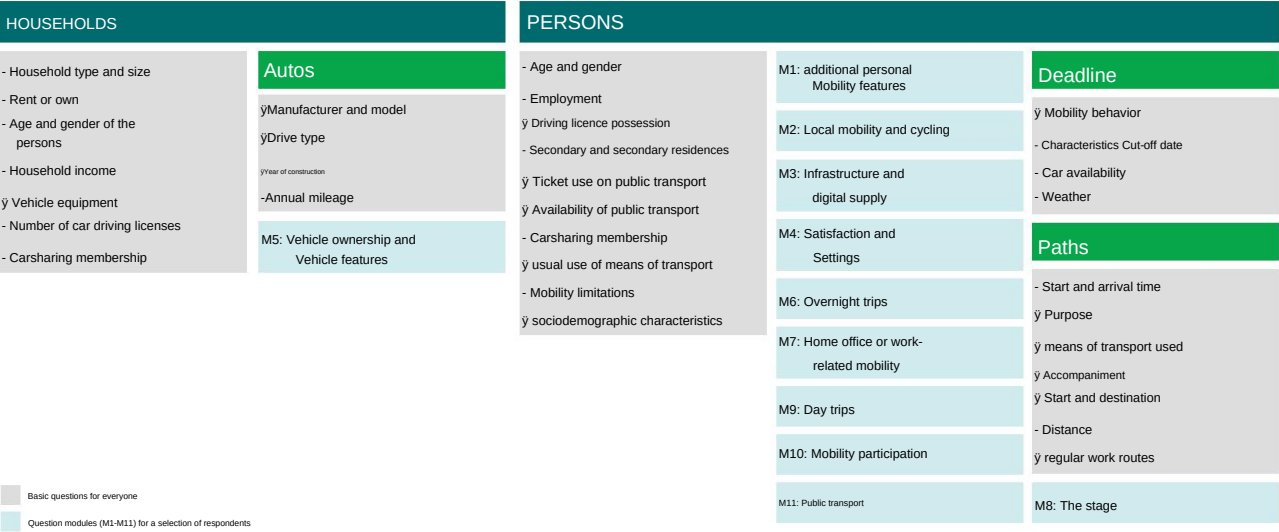
For each journey, all means of transport used are recorded, but not their time share and the distance share (with the exception of a small sample of the stage module) that is allocated to the different means of transport for each journey.

Business travel: Business travel encompasses all journeys undertaken in the course of one's work. Most business travel journeys identified in the MiD (Mobility Data Database) are recorded via the separate survey module on regular work-related journeys, which is aimed at specific occupational groups (bus drivers, sales representatives, etc.). However, even professions not inherently characterized by a large number of work-related journeys involve official travel. For example, if a business trip was undertaken on the reference date, this journey is recorded in the journey log under the "business purpose" category. The business travel represented by the MiD therefore comprises the separately recorded regular work-related journeys and the official journeys reported in the journey log.

Figure 1

Cap 2, Abb 1

Survey program in the main field: Core survey and additional modules (M)



Regular work-related travel: This includes all regular journeys undertaken by individuals during working hours as a result of their job. Typical occupational groups with regular work-related travel include tradespeople, postal workers, and delivery drivers. Commuting to work or occasional business trips are not considered regular work-related travel.

Main means of transport: For each journey, all means of transport used were recorded. If only one means of transport was used, this was considered the "main means of transport." If, however, a journey consisted of several stages for which different means of transport were used, the means of transport with which the longest leg of the journey was most likely traveled was declared the main means of transport. This was based on a hierarchy of transport options.

Main purpose of journeys: As part of the MiD (Mobile In-Depth Survey), a differentiated recording of journey purposes with more than 40 detailed purposes was carried out in online and telephone interviews. For the variable "main purpose of journeys," the number of purposes was reduced to the seven most important ones, which were also recorded in the written survey. In addition, the return journeys were

The primary purpose of each route is determined by the preceding route and the routes home, or, in the case of a chain of routes, by the purpose with the highest priority. Thus, the primary purpose of a route can be categorized as follows: work, official business, education, shopping, personal errands, leisure, and accompanying.

The purpose of a journey is "work," referring to the commute to the workplace, while "official journey" refers to journeys undertaken during working hours. These journeys might include a doctor's appointment or a visit to a government office. For "accompanied journeys," the reporting person has accompanied another person, for example, a mother taking her child to school. In this case, the journey for the child would be listed as "education," while the mother's journey would be categorized as "accompaniment."

Household type: The classification of household type is based on the age of the household members and whether children live in the household. The household type comprises the following four categories.

together:

- Young households: All persons living in the household are between 18 and 34 years old.

Figure 2

Cap 2, Abb 2

Regional Statistical Spatial Typology (RegioStaR)

Regional Statistical Region type RegioStaR 2	1 City region		2 Rural Region	
More nuanced regional statistical Region type RegioStaR 4	11 Subways urban region	12 Regiopolitane urban region	21 Rural regions near urban areas	22 Peripheral rural Region
Regional Statistical Room type RegioStaR 17	111 Metropole 112 Big City 113 Medium-sized town 114 Urban area 115 Small-town, rural area	121 Regiopole 123 medium-sized town 124 Urban area 125 Small-town, rural area	211 Central City 213 Medium-sized town 214 Urban area 215 Small-town, rural area	221 Central City 223 Medium-sized town 224 Urban area 225 Small-town, rural area
Summary regional statistical Room type RegioStaR 7	71 metropolitan areas (111) 72 regional centers and major cities (112, 121) 73 medium-sized cities, urban area (113, 114, 123, 124) 74 Small-town, rural area (115, 125)		75 Central Cities (211, 221) 76 medium-sized cities, urban area (213, 214, 223, 224) 77 Small-town, rural area (215, 225)	

- Family households: At least one person in the household is under 18 years old.
- Households with adults: All persons living in the household are 18 years of age or older. At least one person of middle age (between 35 and 64 years) lives in the household, or both young and older people live together in the household.

- Households with persons aged 65 and over: All persons living in the household are 65 years of age or older.

Economic status: A household's economic status is derived according to the principle of equivalent income. This, in turn, is derived from net household income and weighted household size and has become established in social and poverty research for analyzing income distribution. Without weighting household size, families would be disproportionately represented in the lower income brackets.

The weighted household size is determined from the number and age of the household members.

Children under 14 are weighted with a factor of 0.3. The first person aged 14 or older in the household receives a weighting factor of 1, and all subsequent persons aged 14 and older receive a factor of 0.5. For a family of four with two children under 14, this results in a weighted household size of $1 + 0.5 + 0.3 + 0.3 = 2.1$ instead of an unweighted household size of 4. Based on the matrix of net household income and weighted household size, each household is assigned to one of five categories, ranging from very low to very high.

Regional Statistical Spatial Type (RegioStaR): The spatial type represents a central analytical framework in this report. The "Regional Statistical Spatial Typology for Mobility and Transport Research (RegioStaR)," developed by the Federal Ministry of Transport with support from the Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR) (see www.bmv.de/regiostar), was first applied at the 2017 Mobility in Germany (MiD) survey. It serves as a tool for analyzing the interrelationships between transport and spatial structures and for visualizing spatially diverse mobility behaviors. The regional statistical spatial typology is also included in the catalog of spatial

The organizational structures of the federal and state statistical offices are stored there and can be used for special analyses of official statistics.

The typology is based on settlement structure characteristics at the level of municipal associations.

It is therefore significantly more detailed and spatially precise than established spatial typologies at the level of districts and independent cities. Furthermore, the results are more comparable over time

because changes in municipal associations have little impact on the typology results, whereas districts become increasingly larger and more heterogeneous due to territorial reforms, leading to larger breaks in the time series (e.g., the merger of the urban districts of Aachen or Hanover with their respective surrounding districts).

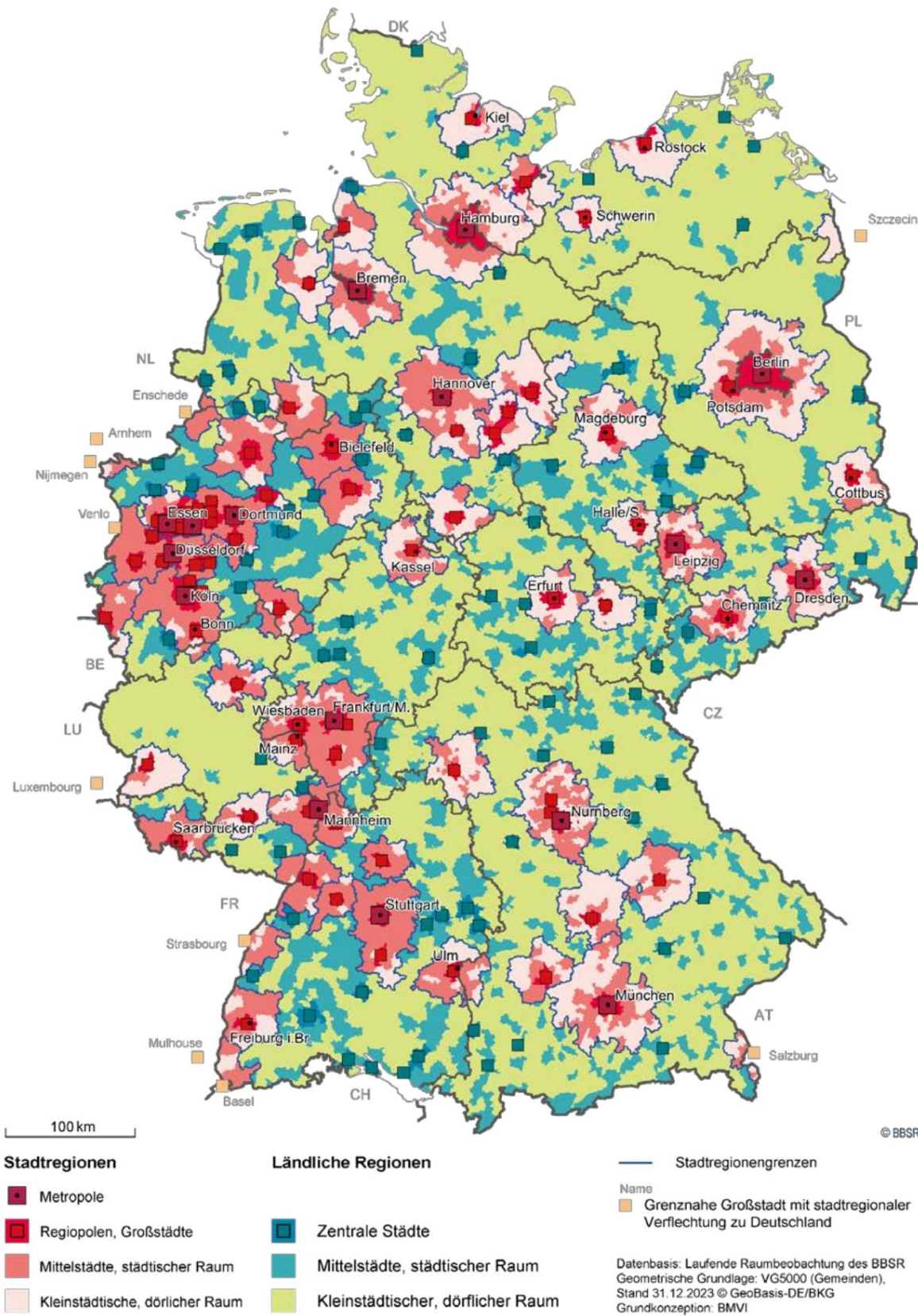
The regional statistical spatial typology is hierarchically structured. For the allocation and use of data at the level of municipal associations, the typology branches out from a broad region type, differentiated into urban and rural regions, through a region type further differentiated into four subtypes, and ultimately into 17 spatial types. These 17 spatial types (RegioStaR 17) form the core of the spatial typology and are the basis for various summaries into spatial and municipal types (Figure 2) and are included in datasets and other output products of the MiD.

In this report, the combined regional statistical spatial type (RegioStaR 7) is used as a compromise between clarity and differentiation; its spatial distribution in Germany can be seen on the map in Figure 3.

External data input: The MiD analysis uses not only the survey data from the added spatial categories, but also numerous locally sourced characteristics. These include, for example, the quality of public transport and local amenities at the place of residence, as well as other characteristics. An overview is provided in the data usage manual.

Figure 3
Cap 2, Abb. 3

Combined regional statistical area type (RegioStaR 7)



3. Everyday mobility: key figures at a glance

This chapter provides an overview of key indicators of everyday mobility in Germany.

In this process, diverse individual data points are combined to create meaningful indicators. The indicators selected here are those available in the MiD time series since 2002 and which were also central to previous analyses. These include the proportion of mobile persons, journeys and distances per person, characteristics of individual journeys such as length and duration, and finally, traffic volume and traffic performance. The selected indicators are presented both in aggregate and broken down by population group and spatial type.

Furthermore, specific subdivisions are made according to means of transport and purpose of travel.

Out-of-home

consumption: How many people are out and about every day?

The out-of-home travel rate, or mobility rate, indicates what percentage of the population is out and about on an average day, meaning they have made at least one trip outside the home. In the MiD 2023, this rate is 82 percent, which is lower than in 2017 (85 percent) and significantly lower.

significantly below the levels of the MiD in 2008 (90 percent) and 2002 (87 percent).

This trend continues the declines already observed before the Corona pandemic. It is similar for all population groups and regions.

As before, clear differences emerge over the course of the week: While an average of 84 percent of the population leaves their homes from Monday to Friday, this figure drops to 79 percent on Saturdays and only 71 percent on Sundays. During the week, work, school, or other fixed daily routines ensure a high level of basic mobility. This framework disappears on weekends. Those who are mobile often use Saturdays or Sundays for longer journeys or leisure activities – which is reflected more in the daily distance traveled or the length of journeys than in the mobility rate.

The mobility rate fluctuates only moderately throughout the year. In winter, at 79 percent, it is slightly below the levels of spring (83 percent) and summer and autumn (both 82 percent). The weather also has a certain influence: In rainy-

Some interpretive notes

The following presentations focus on the data in the MiD 2023. For selected content, supplementary data from the MiD time series are also presented. A frequent challenge lies in the insignificance of many changes. This places high demands on reliable measurement, requires meticulous methodological rigor and careful interpretation – and, not least, a neutral perspective.

Despite the extraordinarily large sample size, changes should therefore only be considered as actual changes if they amount to two percentage points or significant changes in mean and absolute values.

Equally crucial is the time period under consideration. The years 2020 to 2022 must be excluded here due to the effects of the coronavirus pandemic and the fact that no MiD measurements were taken during this period. The year 2023 represents the first extended period after this disruption. However, it is by no means certain that the situation in this year under consideration already provides a good starting point for comparison with the pre-coronavirus years from 2002 onwards or with the MiD in 2017. Establishing such a connection may take more time.

Changes should therefore be considered not only in the "long run", but also against the background of short-term effects.

Under clear or heavily clouded skies, the mobility rate is slightly below average (below 81 percent), while under sunny or partly cloudy weather it is slightly above average (over 83 percent).

Overall, the seasonal and weather-related differences are relatively small. The mobility rate varies more significantly depending on sociodemographic characteristics (Table 2). It is particularly high among 30- to 49-year-olds, reaching up to 87 percent. Most people in this age group have already...

People have a comparatively large number of professional and private obligations that require mobility. Children and young people are also mobile more often than average. However, the rate decreases with increasing age: people aged 70 and over leave the house significantly less often; for those over 80, the mobility rate is only 61 percent.

This means that the mobility rate of the oldest population group is 21 percentage points below the population average. Compared to the MiD 2017, this difference remains stable.

Table 1

Mobility rate and travel time by day of the week, season and area type			
	Mobility quota (except on the reporting date) (Drink)	Travel time	
		per person and day	per mobile person and day
all the person	%	h:min	h:min
IN TOTAL	82	1:24	1:43
WEEKDAY			
Monday	83	1:20	1:37
Tuesday	84	1:23	1:38
Wednesday	84	1:24	1:41
Thursday	84	1:24	1:40
Friday	85	1:31	1:47
Saturday	79	1:27	1:51
Sunday	71	1:13	1:46
SEASON			
Winter	79	1:15	1:36
spring	83	1:25	1:44
Summer	82	1:26	1:47
Autumn	82	1:24	1:43
ROOM TYPE			
urban region			
Metropole	84	1:32	1:51
Regional center and major city	82	1:25	1:45
medium-sized town, urban area	81	1:22	1:42
small-town, rural area	81	1:22	1:42
rural region			
central city	80	1:20	1:42
medium-sized town, urban area	81	1:18	1:37
small-town, rural area	80	1:19	1:40

Table 2

Mobility rate and travel time by personal characteristics and economic status

	Mobility quota (except on the reporting date Drink)	Travel time	
		per person and day	per mobile person and day
<i>all the person</i>	%	<i>h:min</i>	<i>h:min</i>
IN TOTAL	82	1:24	1:43
ALTER			
0 to 9 years	84	0:58	1:11
10 to 19 years	81	1:12	1:31
20 to 29 years	84	1:31	1:52
30 to 39 years	87	1:32	1:48
40 to 49 years	87	1:34	1:50
50 to 59 years	85	1:38	1:55
60 to 69 years	80	1:31	1:52
70 to 79 years	73	1:12	1:39
80 years and older	61	0:49	1:22
GENDER			
Men	83	1:28	1:48
Women	81	1:18	1:37
TASK			
Full-time employed (including trainees)	87	1:41	1:57
Part-time/marginally employed, employed	88	1:32	1:46
with other scope or without specification of scope	83	1:33	1:52
Kind	84	0:59	1:12
Pupil, including preschooler	82	1:05	1:20
Student/in	83	1:30	1:51
Housewife/Househusband	72	1:02	1:30
Retired person, pensioner, including early retirement	71	1:08	1:36
ECONOMIC STATUS			
very low low	74	1:16	1:44
	76	1:16	1:41
medium	80	1:21	1:42
high	84	1:26	1:44
very high	86	1:27	1:43

Economic status also influences mobility rates: While 86 percent of people with very high economic status are mobile on a daily basis, the figure is only 74 percent for those with very low status. The gap between these groups has widened slightly compared to the 2017 Mobility in Germany (MiD) survey (2017: 8 percentage points, 2023: 12 percentage points). This is due to a greater decrease in mobility among the lower-status groups and indicates inequality in participation outside the home or in access to and/or use of mobility options.

This report examines this trend in more detail in its later chapter on participation.

Employment status also influences mobility rates: Full-time and part-time workers, as well as school and university students, are particularly mobile. Mobility rates are lower for pensioners (71 percent) and homemakers (72 percent). These and other differences between various groups reflect the different

Table 3

Number of trips, trip length and daily distance by day of the week, season and room type					
	Paths		average distance	Daily distance	
	per person and day	pro mobile Person and day		per person and day	pro mobile Person and day
<i>all persons, all paths</i>	<i>Number of paths</i>	<i>Number of paths</i>	<i>km</i>	<i>km</i>	<i>km</i>
IN TOTAL	2,9	3,6	12	35	43
WEEKDAY					
Monday	3,1	3,7	11	34	41
Tuesday	3,2	3,8	11	34	40
Wednesday	3,1	3,8	11	34	41
Thursday	3,2	3,8	11	36	43
Friday	3,3	3,9	12	39	46
Saturday	2,6	3,4	14	37	48
Sunday	1,9	2,7	17	31	45
SEASON					
Winter	2,7	3,5	11	31	40
spring	3,0	3,6	12	35	43
Summer	3,0	3,7	13	37	46
Autumn	3,0	3,7	12	36	44
ROOM TYPE					
urban region					
Metropole	2,9	3,5	11	32	38
Regional center and major city	2,9	3,6	11	32	39
medium-sized town, urban area	2,9	3,6	12	35	44
small-town, rural area	2,9	3,6	14	41	51
rural region					
central city	2,9	3,7	11	30	39
medium-sized town, urban area	2,9	3,6	12	34	43
small-town, rural area	3,0	3,7	14	40	51

The daily routines and obligations of the respective groups are reflected and will be examined in more detail from different perspectives in the following chapters.

An average of 84 minutes of travel time per day – as shown by the travel time

The average daily travel time in the 2023 Mobility in Germany (MiD) survey was 84 minutes, slightly below the level of the previous MiD survey in 2017 (85 minutes). If only those individuals who were mobile on the day of the survey are considered, the travel time increases to 103 minutes. Despite minor fluctuations, the mobility time budget has thus remained relatively stable overall since the 2008 MiD survey.

A clear rhythm emerges over the course of the week: People travel the longest on Fridays (91 minutes) and the shortest on Sundays (73 minutes).

Among mobile individuals, the travel time on Sundays is at least 98 minutes – an indication of less frequent but longer travel on this day of the week (Table 1).

Seasonal and weather-related influences are also evident: In summer, travel time averages 86 minutes; in winter, 75 minutes. On sunny days, it rises to an average of 92 minutes; in rain or snowfall, it drops to below 79 minutes. The differences depending on the season and weather suggest that favorable conditions such as sunshine and warm temperatures lead people to travel longer distances or require more time – for example, by making additional detours or traveling at a slower pace. Conversely, unfavorable conditions such as cold, rain, or snow tend to result in shorter and fewer journeys overall.

The differences between population groups are quite pronounced. Males, including children, spend an average of about 10 minutes longer traveling than females (88 minutes vs. 78 minutes). This is primarily due to differences in adulthood. The longest travel times are recorded by people between 50 and 59 years old (98 minutes), the shortest by children under 10 (58 minutes) and the very elderly (80 years and older) (49 minutes). Full-time employees spend the most time traveling, at almost 101 minutes per day, followed by homemakers (62 minutes) and retirees.

In contrast, travel time is significantly less for pensioners (68 minutes). With increasing levels of education, travel time increases – from 62 minutes for those without a degree to 96 minutes for those with the highest educational qualifications. A similar pattern emerges with regard to economic status: Travel time is 87 minutes for those with very high status and 76 minutes for those with very low status.

These differences reflect both structural conditions and differing daily routines. Longer travel times are often associated with high employment rates, more complex daily schedules, or a larger radius of action. Shorter travel times, on the other hand, indicate more limited mobility needs or restricted opportunities for participation – for example, in old age, with low levels of education, or due to economic disadvantage.

Overall, travel time varies widely between population groups – for example, according to

Age, employment status or education – and yet it remains Remarkably stable for its average size.

Route after route – on average three routes daily

The average number of trips describes the number of trips a person makes on average in a day. If only the trips of people who are mobile at all on a given day are considered, this value – the average number of trips for mobile people – is by definition higher.

out of.

In the 2023 Mobility in Germany (MiD) survey, the number of trips per day, as shown in Table 3, is slightly below the 2017 level (3.1), with a value of 2.9 trips per day for all individuals. The number of trips made by mobile individuals in the 2023 MiD survey is also somewhat lower than in 2017 (3.7), at 3.6 trips per day. Whether this represents a temporary fluctuation or a permanent trend cannot yet be definitively determined based on the available snapshot for 2023/24.

As was the case before the pandemic, clear differences emerge throughout the week: From Monday to Friday, the number of trips is consistently above the weekly average; on Friday, it is highest at 3.3 trips per person. The level is significantly lower on weekends. On Saturdays, an average of 2.6 trips are made.

Table 4

Number of trips, trip length and daily distance according to personal characteristics and economic status

	Paths		average distance	Daily distance	
	per person and day	pro mobile Person and day		per person and day	pro mobile Person and day
<i>all persons, all paths</i>	<i>Number of paths</i>	<i>Number of paths</i>	<i>km</i>	<i>km</i>	<i>km</i>
IN TOTAL	2,9	3,6	12	35	43
ALTER					
0 to 9 years	2,6	3,1	8	21	25
10 to 19 years	2,6	3,2	11	27	35
20 to 29 years	3,1	3,8	14	44	55
30 to 39 years	3,4	4,0	13	44	51
40 to 49 years	3,4	4,0	13	44	51
50 to 59 years	3,4	4,0	14	47	54
60 to 69 years	3,0	3,7	12	36	43
70 to 79 years	2,2	3,0	10	22	31
80 years and older	1,6	2,7	7	12	20
GENDER					
Men	3,0	3,7	14	41	50
Women	2,8	3,5	10	29	37
TASK					
Full-time employed (including trainees)	3,5	4,1	15	53	61
Part-time/marginally employed, employed with	3,6	4,1	10	37	43
other scope or without specification of scope	3,3	4,0	11	36	43
Kind	2,5	3,1	9	22	26
Pupil, including preschooler	2,6	3,2	9	23	28
Student/in	2,7	3,3	13	36	45
Housewife/Househusband	2,2	3,1	9	19	27
Retired person, pensioner, including early retirement	2,1	3,0	9	20	28
ECONOMIC STATUS					
very low low	2,6	3,6	8	21	29
	2,7	3,5	10	26	34
medium	2,9	3,7	11	33	41
high	3,0	3,6	13	39	47
very high	3,0	3,6	14	44	51

On Sundays, only 1.9 trips per person are made. Among mobile individuals, the average is 2.7 trips on Sundays, which is roughly one trip less than during the week.

Compared to MiD 2017, the number of journeys on all weekdays has decreased slightly, although the basic pattern hardly changes over time: weekdays remain characterized by work, education and other regular obligations, while fewer journeys occur on weekends.

In the 2017 MiD (Measurement of Daily Travel), the average number of trips varied depending on the season, ranging from 3.0 in winter to 3.2 in other seasons. In the 2023 MiD, the winter value is 2.7, while the other seasons each show 3.0 trips. Despite the slightly increased interval, the underlying pattern – lower trip count in winter, slightly higher values in spring, summer, and autumn – remains. Seasonal and weather-related effects are therefore still present.

Average distance traveled: Differences between working days and weekend

The average distance traveled in the MiD 2023 is 12 kilometers, which, despite a slight decrease in the decimal place not shown here, is roughly the same as in the MiD 2017.

Over the course of the week, however, clear differences become apparent: Sundays see the longest journeys at 17 kilometers, followed by Saturdays at 14 kilometers. Those who are mobile on weekends generally travel longer distances – for example, for excursions, visits, or other leisure activities. During the week, the distance traveled is consistently below average, with a minimum on Tuesdays at just under 11 kilometers. This can be attributed primarily to the fact that, in addition to work-related travel, mobility on weekdays is more strongly characterized by short, regular trips within the immediate living environment – such as for shopping, errands, or childcare.

Seasonal influences are only minimally evident here: in summer the average distance traveled is 13 kilometers, in winter it is 11 kilometers.

During the warmer months, somewhat longer lengths were therefore worn.

Ways were taken, most likely due to increased leisure mobility.

Overall, the average trip length in the MiD 2023 remains almost stable compared to the MiD 2017, with a slight downward trend, but continues to vary significantly over the course of the week depending on daily routine and trip purpose.

35 kilometers per person per day – with a clear weekly rhythm

The average daily distance traveled per person in the 2023 Mobility in Germany (MiD) survey is 35 kilometers, significantly lower (around 10 percent) than in the last two surveys (2017: 39 kilometers; 2008: 38 kilometers). The average daily distance also decreased among mobile individuals – from 46 kilometers in the 2017 MiD survey to 43 kilometers. This indicator primarily reflects the trend toward lower overall mobility in everyday life, as evidenced by the decline in the average daily distance traveled and the associated lower mobility rate. This can be attributed to a variety of factors, such as the trend toward working from home and digital activities – and will be examined in more detail in the relevant chapters of the report. The barely perceptible decrease in individual trip lengths is considerably less significant.

A clear pattern emerges over the course of the week: longer daily distances are covered on weekdays than on weekends. The daily distance is highest on Friday, averaging 39 kilometers, and lowest on Sunday, at only 31 kilometers. However, when considering only mobile individuals, Sunday still has one of the highest daily distances at 45 kilometers – further evidence that on this day, journeys are made less frequently but for longer distances, for example, for leisure or visiting purposes.

Seasonal differences are also apparent. In winter, the daily distance is 31 kilometers, in summer 37 kilometers. Spring and autumn fall in between, with 35 and 36 kilometers respectively.

The seasonal pattern thus follows the well-known pattern of more active, more extensive mobility in the warmer months and reduced movement in winter. Compared to previous surveys, the difference between summer and

Winter in the MiD 2023 is somewhat more pronounced, as was already the case with the number of journeys per capita.

This is how much travel takes place in Germany: around 250 million journeys and 3 billion kilometers per day

The sum of all journeys made in a day (traffic volume) and the sum of the kilometers traveled (traffic performance) represent the key indicators for the analysis of the nationwide burden on the transport infrastructure and the climate and environmental impacts of passenger transport.

In the 2023 MiD (Mobility in Days) survey, approximately 250 million journeys are made in Germany on an average day – around 3 percent fewer than recorded in the 2017 MiD (257 million) and significantly below the 2008 MiD (275 million). Passenger transport performance also declines by around 6 percent. While approximately 3.2 billion passenger-kilometers per day were recorded in the 2017 MiD, this figure drops to just under 3.0 billion in the 2023 MiD.

While the overall trend remains relatively stable, it again shows slightly lower values than in the two previous surveys. Whether the observed decline in everyday mobility since the MiD (Mobility in Germany) is due to

Whether 2017 marked a break in the trend compared to the upward growth trajectory since the 1950s cannot yet be definitively assessed, as has been emphasized elsewhere. After all, 2023 was a year characterized by the recovery from the COVID-19 pandemic as well as other factors such as high inflation and economic pressures.

Throughout the week, traffic volume and performance vary, with daily averages of approximately 250 million trips and 3 billion kilometers differing. On Fridays, traffic volume, at around 275 million trips, is above the average for all days, while on Sundays, with just under 160 million trips, fewer trips are made. The daily routine, with commutes, commercial traffic, and school routes, results in a higher level of mobility during the week than on weekends. The traffic performance indicator also shows Friday as the day with the highest mobility (just under 3.3 billion trips).

Passenger kilometers). On Sundays, it is lower, at around 2.6 billion passenger kilometers.

Seasonal fluctuations are also noticeable. The highest transport volume is recorded in summer, with an average of 3.1 billion passenger-kilometers per day.

Table 5

Population, traffic volume and performance by spatial type						
all persons, all paths	Population		Traffic volume		Transport service	
	Number in millions	%	Number of journeys in millions per day	%	Person-kilometer in millions per day	%
IN TOTAL	83	100	249	100	2.983	100
ROOM TYPE						
urban region	53	63	158	63	1.846	62
Metropole	14	17	43	17	459	15
Regional center and major city	12	14	36	14	388	13
medium-sized town, urban area	22	26	64	26	780	26
small-town, rural area	5	6	16	6	219	7
rural region	5	37	92	37	1.137	38
central city	12	6	15	6	157	5
medium-sized town, urban area	13	15	37	15	434	15
small-town, rural area	31	16	40	16	546	18

The figure is reached. In winter, it is significantly lower at around 2.6 billion passenger-kilometers per day. Traffic volume also varies: in spring and autumn it is just over 250 million trips per day, while in winter it averages only about 225 million trips.

Spatial types and mobility shares of the city and Country: quite different

The previous analyses and data in the tables have also shown differences along the lines of the region types used. For example, there are different usage patterns between urban and rural areas.

These factors are structural in nature, but they also affect traffic patterns. Small towns and villages, in particular, regardless of whether they are classified as urban or rural regions, differ significantly from other regions. Here, the average daily distance traveled is 41 or 40 kilometers per person per day, exceeding the average of 35 km and significantly higher than the 32 km figure for metropolitan areas and regional centers.

In aggregate, this leads to proportions of the transport performance, that is, the total number of passenger journeys made on an average day.

in kilometers, which are only one or two percentage points above the population share. This seemingly minor difference, however, means that the transport performance in this type of area is 18 percent or 16 percent higher than would be expected based on the population share, as illustrated in Figure 4. In areas classified as central cities, the share of transport performance is 14 percent lower than the population share. Despite the small differences in the aggregate, different spatial structures thus lead to differences in transport performance, primarily due to greater distances to be covered and a more decentralized supply – and initially independent of the modes of transport used.

What the time series shows: fewer journeys and fewer kilometers than in 2017

The snapshot presented in the projection of traffic volume and performance for 2023/24, as well as the individual indicators on which it is based, is an important benchmark. However, its significance can only be fully grasped in the context of the MiD time series from 2002-2008.

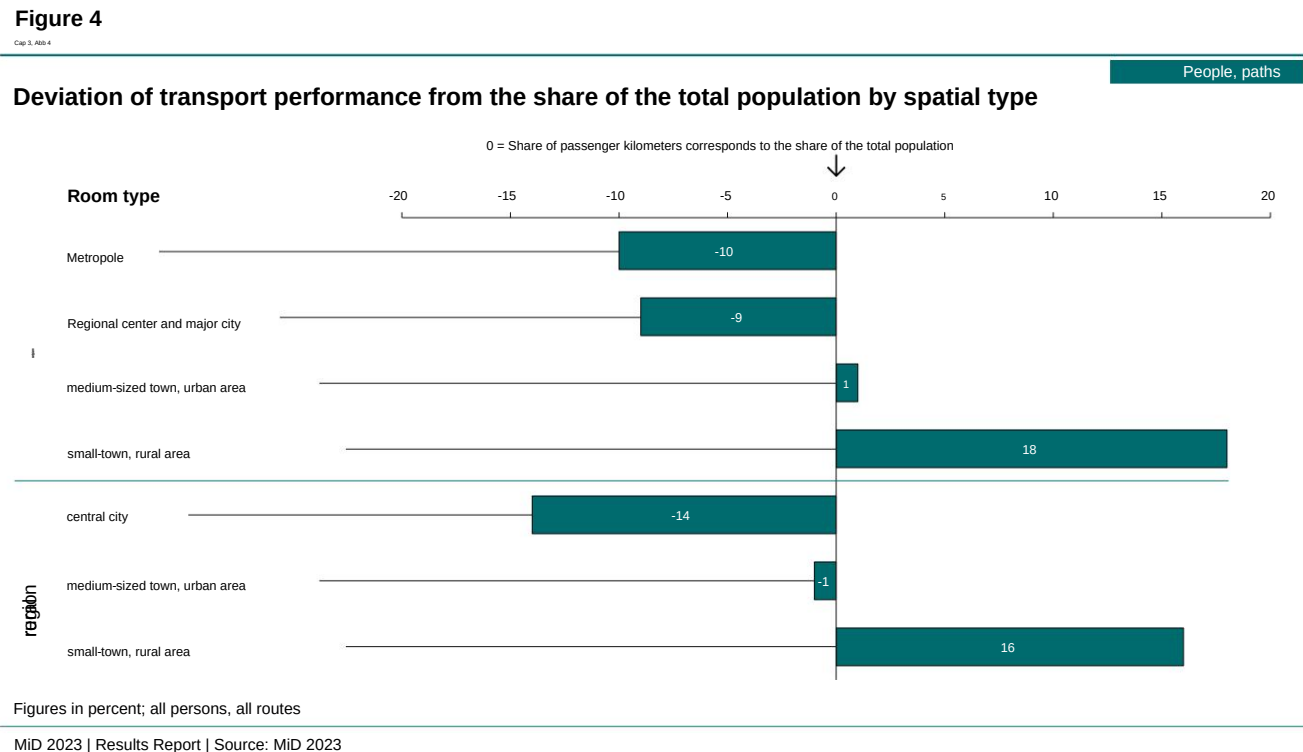


Table 6

General key figures, traffic volume and performance according to regional statistical area type (Regiostar7)

	2002	2008	2017	2023	
Population	81,6	80,7	82,2	83,5	Number in millions
Percentage of mobile people every day	87	90	85	82	%
Share of mobile persons on weekdays	90	92	88	84	%
Share of mobile people Saturday	83	87	82	79	%
Share of mobile people on Sunday	76	83	73	71	%
Routes per person and day	3,3	3,4	3,1	2.9	Number
Daily distance per person and day	33	38	39	35	km
Travel time per person	01:16	01:22	01:25	01:24	h:min
Traffic volume (trips) per day	270	275	257	250	Number in millions
ROOM TYPE					
urban region					
Metropole	38	45	48	43	Number in millions
Regional center and major city	46	37	39	36	Number in millions
medium-sized town, urban area	74	62	64	63	Number in millions
small-town, rural area	20	15	16	16	Number in millions
rural region					
central city	19	18	15	15	Number in millions
medium-sized town, urban area	37	46	37	37	Number in millions
small-town, rural area	36	52	40	40	Number in millions
Transport performance (passenger kilometers) per day	2.717	3.080	3.214	2,983	in Mio.
Urban region – metropolis	346	455	558	459	in Mio.
Urban region – regional center and major city	417	402	434	388	in Mio.
Urban region – medium-sized cities, urban area	769	683	820	780	in Mio.
urban region – small-town, rural area	242	189	222	219	in Mio.
Rural region – central city	164	182	174	157	in Mio.
Rural region – medium-sized towns, urban area	363	500	437	434	in Mio.
Rural region – small-town, village area	416	669	569	546	in Mio.

Total figures for 2023 regarding volume and performance are rounded within the margin of error.

The data for 2017 and 2023 are categorized – first in terms of overall levels, and then, in the following chapters, differentiated by mode of transport and purpose of travel. Table 6 shows the overall values for each of the four survey periods without this differentiation .

The trends in parameters such as journeys per day and person, and in relation to other person-related daily values, have already been discussed. Differences in these parameters also lead to changes in the figures for traffic volume and traffic performance.

The 2023 result for trip volume, at approximately 250 million per day, represents the lowest value in the MiD time series. Despite a larger population, the mileage of 3 billion per day is lower than in 2017 and 2008, but higher than the first value in the series from 2002. At the same time, 2023 is the year most likely to be a survey year marked by crises in the MiD – still influenced by the after-effects of the COVID-19 pandemic and additionally by the economic consequences of the war in Ukraine. While 2008 was also associated with an economic crisis, its effects, with the peak of the financial crisis occurring in autumn 2008, extended more into 2009. A certain dependence on the economic cycle is therefore evident, also reflected in the high level of mobility in 2017, particularly with regard to the total transport volume.

In addition to this anomaly, Table 6 allows

A comparison is also made within the unit of analysis based on the regional types according to RegioStaR 7, with its urban-rural differentiation. Here, it is noticeable that the decline in 2023 compared to 2017 is concentrated more in metropolitan areas and large cities than in small towns and rural regions. At the same time, within the data collection periods, both the volume and the service volume shift from rural to urban areas. Thus, various effects intersect: absolute population trends towards large cities and differing changes in mobility levels in both areas. This is accompanied by non-uniform changes in the age structure, as well as intervening effects such as the impact of the increasing prevalence of working from home. Beneath a seemingly stable "mobility surface," changes are therefore discernible that warrant further investigation. A logical approach is to examine modes of transport and reasons for travel, as presented in the following chapters. Worthwhile.

However, it also involves a closer look at different population groups and life situations, as the following chapters do with varying focuses.

4. Transport use, multimodality and user segments: a question of place of residence, preferences, life situation and age

So far, the key indicators have been presented without differentiating between modes of transport. However, different modes of transport are used to complete journeys.

The choice and suitability of transportation depend heavily on its specific characteristics, the available infrastructure, personal access to cars and bicycles, individual preferences, and the availability of public transportation. Walking and cycling are limited in range and exposed to weather conditions. Cars, on the other hand, offer weather protection and flexibility in overcoming even long distances, as well as the ability to transport passengers and goods. However, this requires access to a car, either from one's own household or through other private or commercial means. Furthermore, traffic jams and the search for parking increasingly hinder the predictability of car use in many regions. The availability and quality of public transportation, in turn, depend significantly on the place of residence, the intended destinations, the timetable, and the level of comfort offered.

A decrease in car traffic, a gain for pedestrian traffic. Bicycles are generally stable, offering little benefit to public transport.

One of the most important indicators of the MiD (Mobility in Germany) is the first variant, the modal split. It expresses the percentage share of each mode of transport in the total volume of traffic and thus in all journeys or passenger-kilometers traveled. It can also be extrapolated to include absolute values per day or year, again differentiated by mode of transport. The MiD follows the principle of the primary mode of transport. Journeys made solely on foot constitute the walking share, while journeys made exclusively by bicycle constitute cycling (regular bicycles and e-bikes combined). If public transport was also used on the journey, it falls into the category of public transport – specifically, local transport (all regional public services including taxis and on-demand services) together with long-distance transport (public services on longer routes by bus, train, and plane). For certain questions, such as kilometer-based analysis, these two "public" categories can be evaluated separately. And finally, car traffic (MIV – motorized individual transport with cars including motorized two-wheelers, trucks and other motor vehicles), where a distinction is made between drivers (MIV journey) and passengers (MIV ride-along).

MiD observes the choice of transport mode on two different levels.

These will be examined in this chapter:

- A key statistical parameter is the distribution of means of transport across the routes traveled.

The MiD survey uses a point-in-time approach to capture this data. All individuals report on their journeys and the means of transport used on a randomly selected date. Because the survey covers all days of a year, this results in a reliable annual result – across all seasons, regions, and population groups.

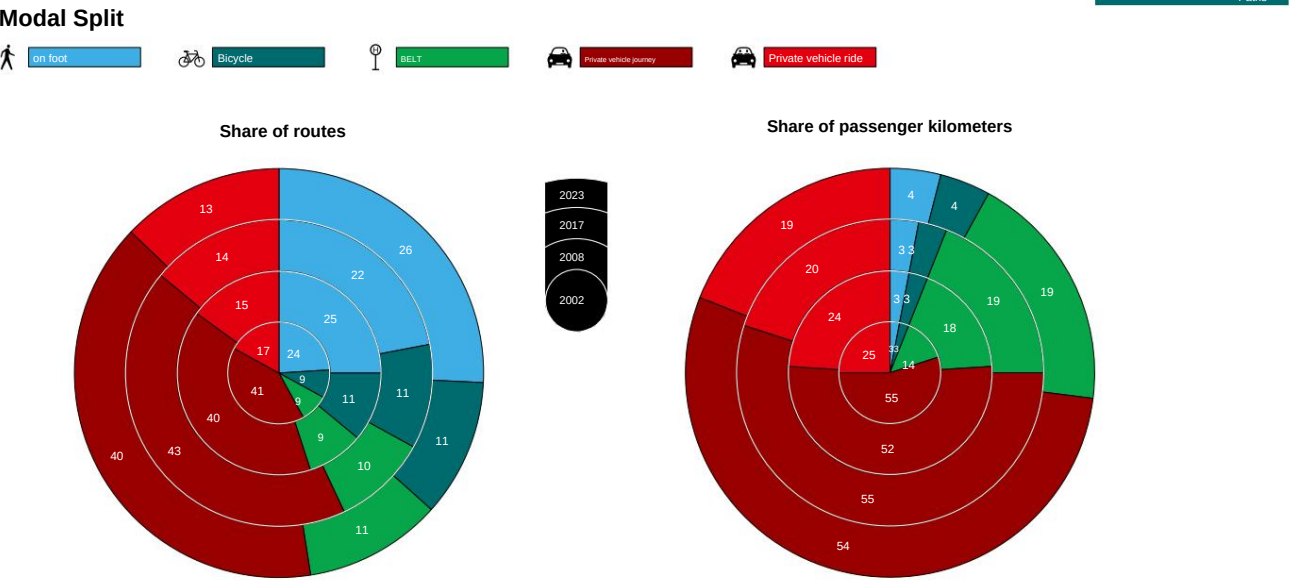
- In contrast, a more personal, classifying characteristic independent of a single reporting day is the usual use of means of transport.

This is collected separately for each mode of transport. The combination of the individual usage frequencies can be used in the analysis to delineate mobility segments and, for example, to describe their socio-demographic composition.

The modal split has remained relatively stable over the previous MiD (Mobility in Germany) time series, with only minor changes in the nationwide overall figure. This is changing somewhat in the current survey, with private motorized transport (PMT) losing ground in terms of trip share. Nevertheless, quantitatively speaking, it remains by far the most important mode of transport and continues to exceed the 50 percent mark. Nationwide, PMT accounted for only 53 percent of trips in 2023, including both drivers and passengers. While the decrease compared to 2017 is only four percentage points and compared to 2008 only two, this is still the lowest PMT share in the previous MiD time series. As explained, this figure includes car use, but also mopeds, motorcycles, and commercial vehicles. Since the 2017 survey, car sharing has also been included. In 2023, PMT trips accounted for 40 percent and passenger rides for 13 percent.

Figure 5

Cap 4, Abb 5



Figures in percent; all routes

MiD 2023 | Results Report | Source: MiD 2023

In this analysis at the journey level, cycling remains stable at 11 percent nationwide, with pedelecs now accounting for a good 2 percentage points of that. Public transport slightly improves its share from 10 to 11 percent – at the journey level, this almost exclusively refers to journeys using public transport – but can also be considered virtually stable when looking at the time periods covered by the MiD survey.

We will see. What is not apparent in the MiD is the collapse of public transport during the Corona pandemic and the process of winning back passengers.

In contrast, pedestrian traffic is a winner in the MiD 2023, both proportionally and in absolute terms. The share of journeys made exclusively on foot grew significantly from 22 percent in 2017 to 26 percent six years later and two to three years after the Corona phase.

Table 7

Traffic volume and traffic performance 2002 to 2023

	Trips in millions per day				Passenger kilometers in million per day				Route length in km 2023	
	2002	2008	2017	2023	2002	2008	2017	2023	Mean	Median*
IN TOTAL	270	275	257	249	2.717	3.080	3.214	2.983	11,9	3,8
on foot	64	69	56	65	88	98	93	113	1,7	1,0
Bicycle	25	29	28	27	82	96	112	117	4,3	2,1
BELT	25	26	26	27	387	554	605	561	21,1	7,2
Private vehicle journey	111	110	111	99	1.496	1.589	1.754	1.611	18,5	6,7
Private vehicle ride	45	41	36	31	665	735	650	581	16,3	6,7

*This value represents the midpoint of the given distribution and helps in interpreting the mean, which is often influenced by high individual values. Example: Trip length for walking routes: The median is 1.0 km. Therefore, 50 percent of the reported walking routes are up to 1.0 km long, and 50 percent exceed this value.

However, it should be noted that while some of the proportional changes in the distribution of journeys may be compared to the last MiD 2017, they closely resemble the results for pedestrian, bicycle, and car traffic from 2008. In this comparison over time, public transport is the primary beneficiary, increasing from 9 to 11 percent. It is therefore advisable to also consider the longer time series.

The mileage balance: an almost even more important figure.

Modal split shares can be considered not only in terms of distances traveled, but also in terms of transport performance or passenger-kilometers. Since the average distances between the different modes of transport vary significantly, this perspective reveals a different distribution. For example, in 2023, private motor vehicle (PMV) drivers accounted for 54 percent of journeys, compared to 55 percent in 2017, while PMV passengers accounted for 19 percent, down from 20 percent.

Together with private motorized transport, approximately three-quarters of all passenger kilometers will still be covered, if

Model-based small-area estimation for counties and independent cities

Despite the substantial regional expansion in the MiD 2023, the realized sample size only allows for analysis where corresponding additions beyond the nationwide base sample have been made. The minimum sample size required for this in the MiD 2023 is 1,000 surveyed households. Therefore, even at the level of the more than 400 districts and independent cities, analyses are far from possible everywhere. However, since the need for such regional data is high, at least for the modal split indicators, a small-area estimation method was used for the first time in the MiD 2017. This method combines known MiD results with external structural data of the regional units under consideration and determines the relationships in a statistical model. Provided a sufficient number of satisfactory estimators are obtained and the necessary structural data are available, a modal split result can thus be estimated even for sub-areas where few or no surveys were conducted.

Since the MiD (Model-in-Depth) method is calculated nationwide, only estimates available for all regional units can be reviewed and included. The method was refined in 2023 and extended beyond districts and independent cities to include municipalities with at least 10,000 inhabitants. The model-based estimates are provided in a separate report. This report explains the method used and points out the limitations of this application. These limitations primarily stem from the need for a uniform data basis across the entire country and the availability of various traffic data, as well as—with regard to interregional comparability—the highly heterogeneous situation concerning district and municipal sizes due to differing regional concepts in each federal state. Nevertheless, the results offer a starting point for regions that could not be adequately covered for such "small-scale" indicators in the 2023 MiD.

The results show that in order to achieve greater reliability in the estimation models, further data and comprehensive separate research are needed, such as that initiated in the research project "ReMo – Regional Indicators of Mobility" supported by the mFUND funding program of the BMV.

even with a slightly declining share. Public transport comes in second place with just under a fifth, of which about 60 percent is local public transport and 40 percent is private passenger transport.

Following pedestrians are bicycles and walking, each accounting for 4 percent of total kilometers traveled, both showing a significant increase of one percentage point compared to 2017. If walking segments, such as those used to reach public transport, are also included in these walking distances calculated according to the primary mode of transport concept, the share of walking increases by another percentage point, to the detriment of public transport. In this calculation, walking accounts for 5 percent and public transport for the entire journey for 18 percent.

Absolute values as a point of reference – less in private motorized transport, more on foot and electrically on bicycles.

The percentage analysis can be expanded by using extrapolated absolute values. Since the number of trips per person per day decreased slightly again in 2023, and average distances also declined slightly, this represents an important addition, particularly at the kilometer level. Of the nearly 3 billion passenger-kilometers measured daily in 2023, approximately 2.2 billion were attributable to private motorized transport, around 200 million fewer than in 2017, but approaching the levels seen in 2008.

If only vehicle movements are considered, the private motor vehicle (PMV) mileage decreases slightly by a good 130 million kilometers per day from 2017 to 2023, despite a growing car fleet and an increase in population.

The situation is different for pedestrians and cyclists. Both absolute mileage totals are increasing and are higher than in 2008 and 2017. For pedestrians, this also applies to the number of trips compared to 2017. The increase in kilometers is significant in both cases, resulting in the highest totals in the time series to date: 113 million kilometers daily on foot and 117 million by bicycle – with a small contribution from children carried on new types of bicycles. In cycling, the biggest driver is the e-bike. Of the 11 percent of cycling in the modal split, about a quarter is e-bikes. And of the 4 percent of bicycle kilometers, they account for almost a third.

A closer look at the mileage of public transport shows that it decreased slightly overall from just over 600 million kilometers per day in 2017 to around 565 million kilometers per day in 2023. If only local public transport is considered, it remained almost unchanged at approximately 300 million kilometers per day. This slight decrease in public transport mileage is primarily due to long-distance buses, which did not reach their pre-pandemic levels in 2023. Changes in domestic air traffic also played a role, but to a lesser extent, and here the MiD (Municipal Data Collection) reaches its limits of accuracy even with this comprehensive sample.

Further significant regional differences Modal Split

Nationwide, the changes are minor, despite the moderate trend away from cars and towards cycling and walking in terms of kilometers traveled. This makes the differentiation by federal state and the distinction between spatial types all the more important. The modal split results reveal significant differences between urban and rural regions. The results for the federal states and the spatial typology, both presented in [Table 8](#), continue to show familiar urban-rural disparities. These include, above all, the role of the car outside cities, the role of public transport in metropolitan areas, and the role of cycling in both large cities and many medium-sized towns. A common feature across all regions is the increase in walking – compared to the survey date of 2017. The comparison of the federal states, aside from the urban-rural differences also present there, shows similarities among the larger states, but also some notable features. These include, above all, states where the share of cycling appears to have potential for improvement. This applies first and foremost to the car-dominated Saarland, but also to Rhineland-Palatinate, Thuringia, Saxony, and Hesse. They all remain below the 10 percent mark in terms of their cycling modal share.

Distribution of journey lengths: high proportions in the immediate vicinity for all modes of transport.

For the modes of transport analyzed according to the main transport concept, not only the average trip lengths but also the respective distribution of these lengths across different categories are important. Figure 6 presents this for 2023 and 2017, differentiated by mode of transport.

A comparison over time reveals no significant changes, with one exception. Overall, six out of ten trips remain below the 5-kilometer threshold and can therefore be classified as local travel. The change concerns cycling routes, caused by the increasing use of e-bikes. While 82 percent of cycling trips fell below 5 kilometers in 2017, this figure dropped to slightly less than 80 percent in 2023.

Table 8

Main means of transport 2023 by federal state and regional statistical area type (RegioStaR7) by route					
	on foot	Bicycle	Public transport	Private motor vehicle	Journey Private motor vehicle
Line percentages	%	%	%	%	%
IN TOTAL	26	11	11	40	13
Schleswig-Holstein	22	15	8	42	13
Hamburg	31	16	24	21	8
Lower Saxony	24	15	8	41	12
Bremen	30	22	15	24	7
North Rhine-Westphalia	25	10	10	42	13
Hesse	27	9	12	39	13
Rheinland-Pfalz	27	7	9	43	14
Baden-Württemberg	27	11	9	40	13
Bayern	23	11	10	42	14
Saarland	21	3	10	51	15
Berlin	35	14	27	18	6
Brandenburg	28	12	9	38	13
Mecklenburg-Vorpommern	27	14	7	38	14
Saxony	24	8	11	44	13
Saxony-Anhalt	31	11	9	36	13
Thuringia	30	6	9	42	13
ROOM TYPE					
urban region					
Metropole	31	15	21	25	8
Regional center and major city	29	15	14	32	11
medium-sized town, urban area	25	11	9	42	14
small-town, rural area	21	9	8	47	15
rural region					
central city	28	13	8	38	12
medium-sized town, urban area	25	10	6	45	15
small-town, rural area	22	6	7	51	14
Rounding-related deviations from 100% adjusted					

The average trip length increases from 3.9 to 4.3 kilometers across all bicycle types between 2017 and 2023. This change is particularly noticeable in the e-bike segment, where the average trip length is 5.6 kilometers – compared to 3.9 kilometers for regular bicycles.

Another important indicator from this analysis is the distribution of private motorized transport (PMT) journeys. Around 40 percent of these journeys fall into the segment of less than 5 kilometers, i.e., the local area. However, not all of these short car journeys are suitable for walking or cycling. This is due to accompanying persons, items being carried, and also to journey chains where several longer and shorter car journeys follow one another before a person, for example, returns home. MiD cannot provide information on the various "necessities for carrying items" and other limitations. However, the journey chain analysis shows that about a third of the short car journeys are embedded in such a chain, meaning they are not readily suitable for walking or cycling on that section.

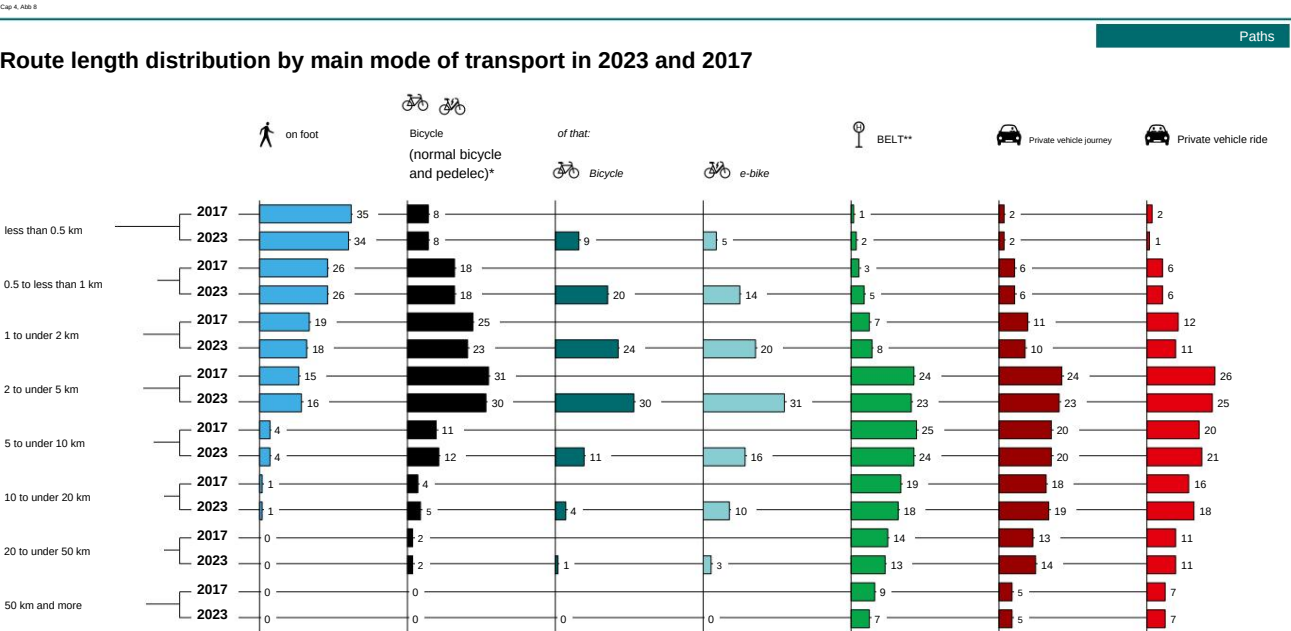
High-resolution modal split: how our traffic is distributed under a closer magnifying glass

The current presentation of transport mode usage for the main modes of transport helps to create a compact time series and to classify the significance of the categories established. However, the modes of transport used are recorded in greater detail within the framework of MiD (Mobility in Germany). This allows for the analysis of high-resolution modal split shares of traffic volume and transport performance. A distinction is made between different vehicle types in private motorized transport, individual public transport services, and bicycles with and without electric assistance.

Both traffic volume and traffic performance are dominated by private motorized transport. His shares, shown in red in Figure 7 and Figure 8 , total 53 percent for journeys and 73 percent for passenger kilometers.

Of this share of private motorized transport, 92 percent of the total traffic volume is attributable to the use of private cars as journeys or rides. In terms of transport performance, even 93 percent of passenger-kilometers traveled by private motorized transport are attributed to journeys made by private car.

Figure 6



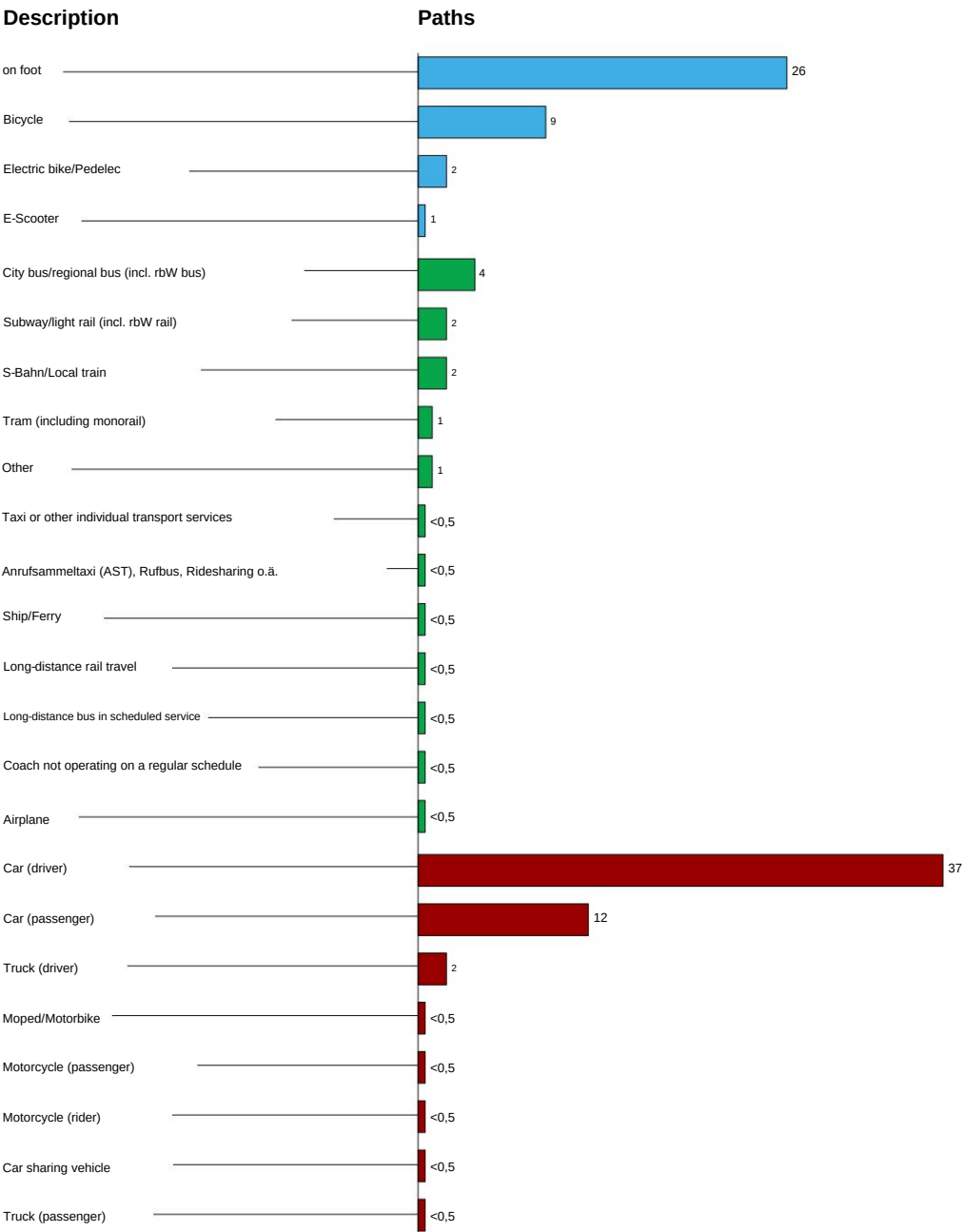
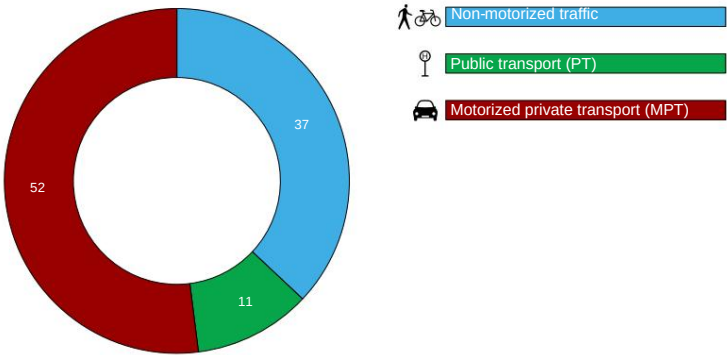
Figures in percent; all routes; 2017 not shown separately due to the still small share of e-bikes; ** Values for public transport and regional rail services can be found in Figure 43.

Figure 7

Cap 4, Abb 5

Paths

High-resolution modal split of the paths



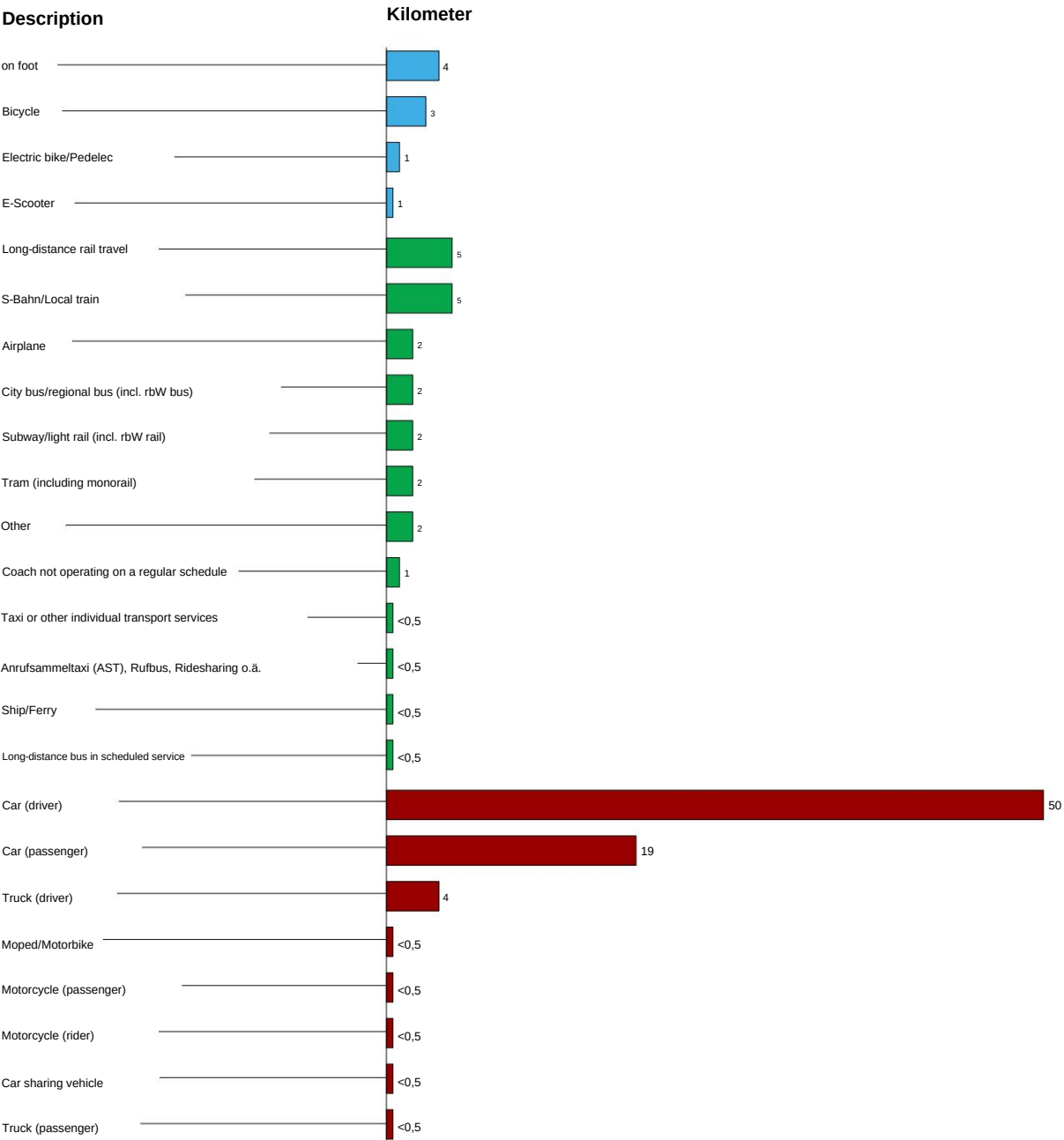
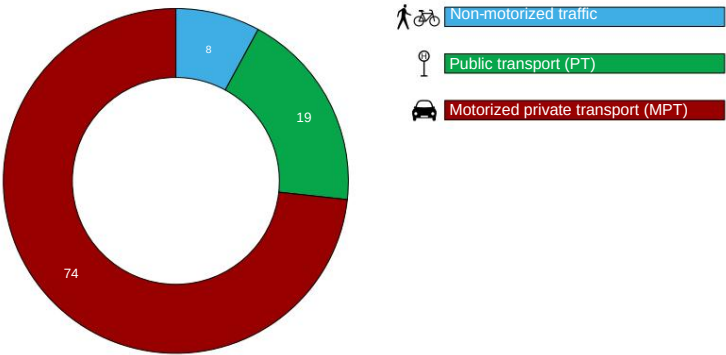
Figures in percent, all routes

Figure 8

Cap 4, Abb 7

High-resolution modal split by passenger kilometers

Paths



Figures in percent, all routes

Passenger cars account for the majority of all traffic. Only journeys involving trucks represent a significant share of total traffic volume, at 2 percent and total transport performance, at 4 percent. Other private motorized transport options, such as mopeds and motorcycles, account for very small shares of less than 0.5 percent.

In this analysis, public transport can be further differentiated. Public, primarily urban, rail-based transport options such as light rail, subway, and commuter rail have a comparatively lower volume of ridership in this representation. However, they provide a greater transport capacity, especially due to the commuter rail services. City buses, in turn, record higher passenger volumes than kilometers traveled. Other forms of local public transport, such as taxis or various on-demand services (AST, call bus, ridesharing – which will be discussed in more detail in a separate chapter), have very low volume and service shares.

In contrast, long-distance transport services, especially rail, have a small share in terms of volume but are very significant in terms of transport performance. Rail accounts for 5 percent of total long-distance mileage, representing a quarter of the public transport sector's total mileage. If other long-distance public transport options, such as long-distance buses and airplanes, are included in this mileage balance, the share within public transport rises to around 40 percent. Compared to 2017, the importance of scheduled long-distance buses has decreased even more than that of domestic air travel. However, even in the comprehensive MID sample, their shares are so small that an analysis must be omitted here. This analysis will be conducted within the framework of the MiD supplementary modules on day trips and overnight journeys in a later chapter of this report.

Interim conclusion: multiple causes for few changes

Viewed from a distance, the modal split shows relatively few changes – and some of these changes are reminiscent of the conditions measured in 2008. Nevertheless, an initial interim assessment calls for an investigation into the underlying causes. The MiD 2023 survey measures mobility one to two years after the end of the COVID-19 pandemic. The hygiene measures implemented during the pandemic – mandatory masks in public places including buses and trains, curfews, and social distancing rules – all had a significant impact on mobility between 2020 and 2022, but understandably, they have been almost forgotten. However, some trends have persisted or evolved from them. These include, first and foremost, working from home, increased online activity, and the Deutschlandticket (Germany-wide public transport ticket). Certain services remain significantly reduced, such as domestic air travel and long-distance bus services. Other trends have continued, such as the increase in walking.

In addition, there are further effects beyond everyday mobility, such as steady population growth, a constantly changing age structure, urbanization, densification, and urban sprawl. Last but not least, economic effects play a role that is difficult to quantify but should not be underestimated. It is also striking that key changes, such as the slight reduction in car traffic and the growing importance of walking, are occurring nationwide rather than regionally. This suggests broader societal shifts in mobility patterns rather than city- or region-specific effects, such as those resulting from local initiatives.

Given these events, it can be assumed that mobility behavior in 2023 differed from what it would have been had there been no COVID-19 pandemic and the period since 2023 not been marked by economic hardship. Simple causal relationships are therefore misleading. Moreover, mobility often occurs routinely, a fact frequently underestimated, and the supply and demand system was definitely not in a stable state. Premature interpretations based on the MiD 2023 data regarding stagnation or even a decline, which would be significant with respect to car mobility, are therefore misguided.

Excursion: Stage recording – a closer look at walking

Until 2008, the MiD (Mobility in Germany) survey recorded only journeys for the reporting days. This included routes from the starting point to the destination, specific to each trip. When multiple modes of transport were used on a single journey, these were recorded as multiple entries, but the individual segments or stages were not recorded in more detail. This changed with the 2017 survey. For a randomly selected subset of journeys involving multiple modes of transport or a walking segment, the stage characteristics were recorded in greater detail. A special report on local mobility is available for both 2017 and 2023.

This section provides a brief overview of the 2023 results. Data was collected on 94,337 stages. The proportion of journeys with stages is lower than often expected. Approximately four-fifths of the journeys use only one mode of transport. These are generally journeys made solely on foot, by car, or by bicycle. To limit the effort required from respondents, very short walking segments, estimated to be well under 50 meters, were not included in the data collection.

The results show that most stages included in the MiD route evaluation according to the concept of Main means of transport are not shown, foot stages or paths from a car park to

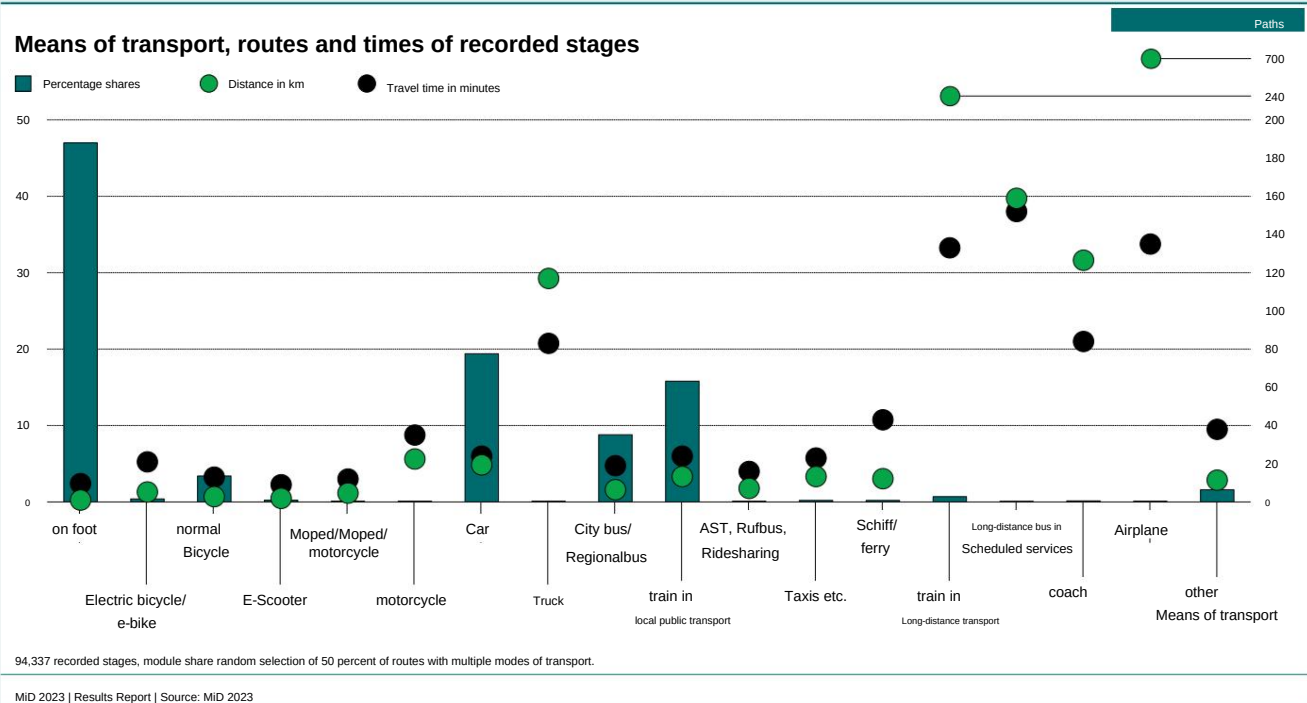
The destinations (or vice versa) are largely journeys to or from stops using public transport. Other combinations, such as cycling and public transport, play only a minor role.

This means that most stages only cover short distances.

The situation is somewhat different when changing trains within public transport, for example from a bus to the subway or a tram to a long-distance train. These stages are also shown, but not for short transfers within a product, such as changing from one subway to another on the same platform.

On average, routes with stages consist of two segments. This figure is higher due to transfers and, as a rule, at least two walking sections on public transport. Beyond these unsurprising findings, the key insight—beyond the differentiated use of this data in traffic models, for example—lies in the estimation of the transport performance of pedestrian traffic. This already amounts to 4 percent in the main transport concept. If the kilometers from the walking stages are added, this share rises to 5 percent. Thus, in absolute terms, pedestrian traffic is at a higher level of transport performance than cycling, which accounts for 4 percent—further evidence of the importance of this mode of transport.

Figure 9



Therefore, it is inappropriate to conclude at this time that this would represent a trend reversal. Nevertheless, the MiD (Mobility in Germany) is an important additional benchmark for structural and absolute developments in mobility over the past and coming years, and with the reversal of the trend in private motorized transport and pedestrian traffic, it at least underscores the need for close monitoring.

Groups with the greatest need for improvement in car use only play a larger role again in later life. Even then, the proportion of car use decreases, although it remains high "until the very end." Private motorized transport thus almost always plays the most important role, even for the youngest, who quite obviously often gain their first mobility experiences outside the home in the back seat of a car.

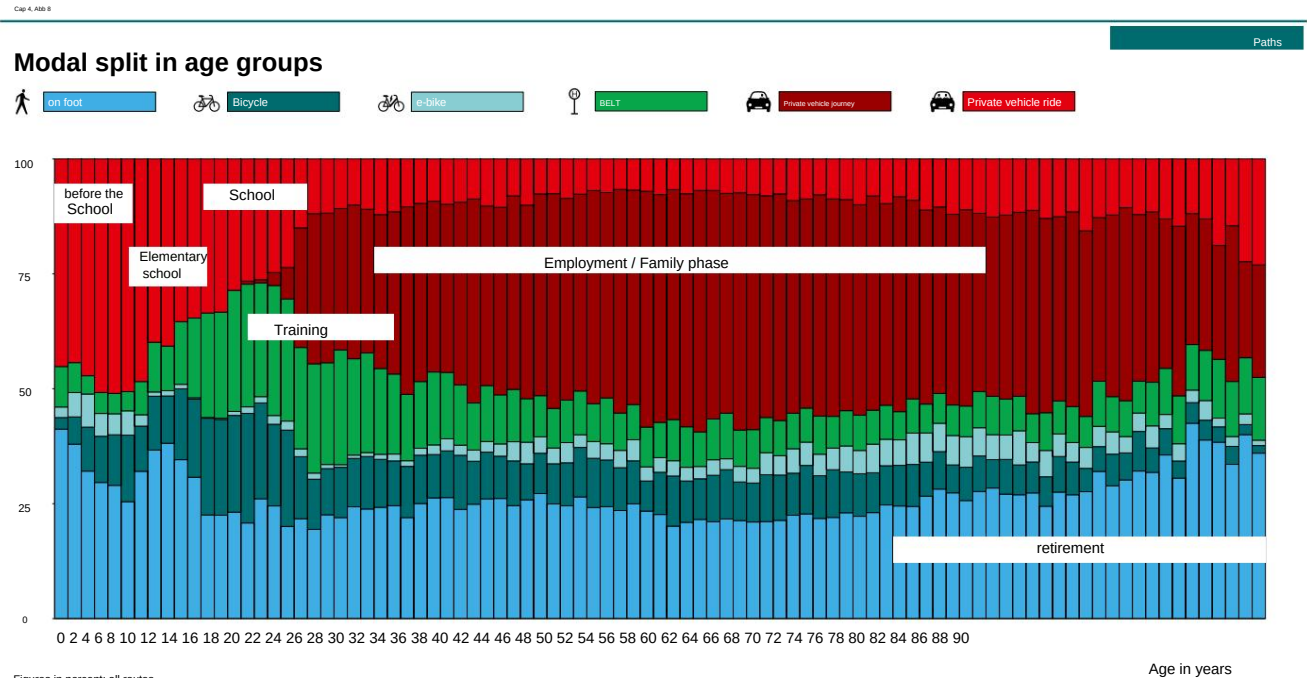
Regarding the people – and the modal split according to age groups

With over 400,000 respondents, the MiD offers analytical possibilities not found in other studies. One such possibility is the differentiation of results by age group. When this is done, as shown in Figure 10 , a modal split that "flexes" according to life phases and age becomes apparent, specifically in terms of travel shares. Walking shares are particularly striking, with high values in both the early and late stages of life. This is followed by cycling, with the highest shares among young people. The e-bike, also shown, is clearly visible among children being transported in trailers or cargo bikes, and then again only in the middle to older age groups. The share of public transport decreases significantly once a driver's license can be obtained, and in the middle age groups...

Typical transport use and mobility segments – here too a slight decrease for the Car orientation

This overview across age groups and modal split is intended only as a transition to a different perspective on transportation use. Beyond the results on traffic volume and performance, the MiD (Mobility in Germany) survey has, since 2002, allowed for an analysis of individually typical transportation use. For this purpose, all individuals aged 14 and over are surveyed about their usual use of various transportation options. This includes long-distance buses and walking. Surveys also inquire about car sharing and, since 2023, the use of cargo bikes, rental bikes, and rental scooters. Unlike the data on modes of transport used on a randomly selected day, the responses allow for a more nuanced analysis.

Figure 10



Means of transport allow conclusions to be drawn about general usage patterns. Combined, they enable insightful segmentation.

In a comparison over time, the information provided in 2023 shows a decline in typical car use compared to 2017 (Figure 11). The share of daily and weekly car use fell from 76 percent in 2017 to 72 percent in 2023. This is noteworthy given the otherwise stable data. The response distributions for all other modes of transport remained virtually unchanged. The proportions of regular walkers, cyclists, and public transport users also did not increase. The size of the groups of people who almost never cycle or use public transport also remained unchanged. Any changes in usage therefore occurred within the frequency categories surveyed, ranging from monthly to daily.

The "new" modes of transport also shown in the figure are still at a low level, but are becoming more "measurable". In 2023, the number of at least occasional carsharing users – not to be confused with carsharing members – rose to 7 percent of the respondents aged 14 and over included in this study, compared to 3 percent previously. This represents a significant increase, especially in large cities with such a high demand.

Regarding shared bicycles, there was no such increase. The 2023 result was the same as in 2017, with a usage rate of 5 percent. The percentage of people using cargo bikes at least occasionally also reached 5 percent in 2023, although, as with car sharing, this was almost exclusively in urban areas.

Rental scooters have a share of 6 percent predominantly occasional use, due to supply, especially in larger cities (each without illustration).

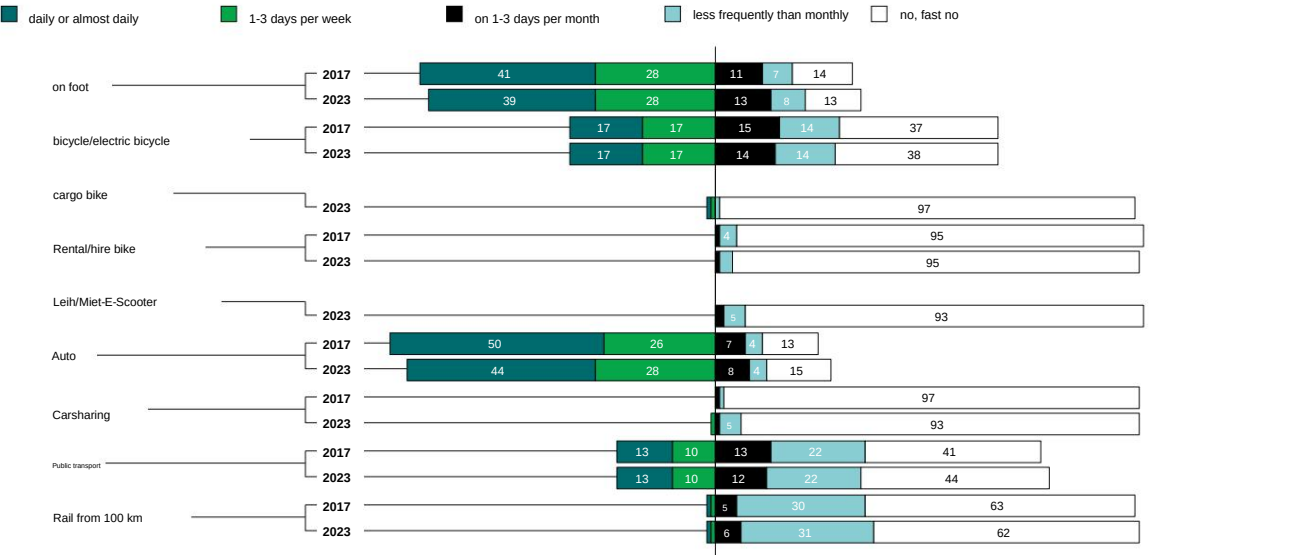
The benefits of walking alone, without combining it with other means of transport, should not be underestimated. Only 13 percent do not do this at all, while around 40 percent cover at least some of their journeys completely on their own two feet every day or almost every day. In contrast to the increases in the modal split, the usual participation in pedestrian traffic remains unchanged.

The two long-distance transport options considered in this survey result in different usage rates. For long-distance rail travel over distances of 100 kilometers, usage remains comparable to 2017. This still results in a non-use rate of approximately six out of ten respondents. Long-distance buses, on the other hand, have seen a decline. In 2023, 93 percent of respondents were...

Figure 11

Cap 4, Abb 9

Typical use of transport over time



Figures in percent, persons aged 14 and over, excluding representatives, cargo bike, rental/hired bike, rental/hired e-scooter, car sharing

Those who do not fly. In 2017, this figure was 90 percent. Typical aircraft usage, which is also not included in the figure, was surveyed in slightly different categories. Here, around 30 percent of respondents stated that they never fly.

Car orientation in 2023 on the decline

The data on the individual modes of transport can be linked together. Based on daily and occasional use, eight groups were formed. These range from "low-mobility users" who hardly travel during the week to those who travel almost exclusively by car. It is also possible to consider "regular multiple users" or multimodal users – defined here as those who typically use a car, public transport, and bicycle at least once during the week. Pedestrians are not included in this grouping, as the subgroup of those who walk almost exclusively during the typical week is very small and overlaps significantly with the "low-mobility users."

A review of the period from 2002 to 2008, 2017 and 2023 shows stable shares of 7 to 8 percent for the often-focused "multi-users". Anders

The largest segment, those who use cars daily, saw a decrease in their share in 2023, reaching only 41 percent from 44 percent previously. Conversely, the group of those who rarely use cars grew by two percentage points – a further indication of the declining car-centricity. The size of the public transport user segment, as well as the share of those who prefer cycling, increased by one percentage point. The share of the predominantly elderly and less mobile population also grew.

The categorization allows for further differentiation, as illustrated, for example, between urban and more rural residential environments. The table in Figure 12 shows the shares of the mobility segments in each region type for 2023.

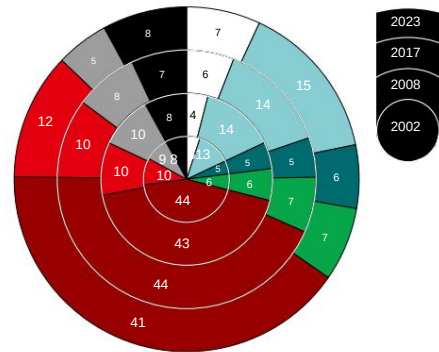
- Striking, but also plausible, are
- the higher proportions of people who use public transport in the metropolises and large cities,
 - likewise for the group of bicycle-oriented people, here also in the usually compact central areas
 - cities in rural regions,
 - the proportions of mixed and multiple users that are only marginally higher in urban areas compared to other regions

Figure 12

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Transportation user segments

persons



		urban region				rural region			
		Metropole	regional metropolis	suburban-	central	central	suburban-	central	suburban-
Line percentages									
	Low-mobility	7	8	7	6	...			
	Bicycle-oriented	20	20	13		8	19	13	8
	Public transport users without a car driving license	10	8	4	2	4	2		2
	Public transport users with a car driving license	18	9	5	3	4	2		2
	daily car-oriented	22	32	45	54	39	49	57	
	rare car-oriented	7	10	13	15	12	15	15	
	Mixed use	6	5	6	5	4	4	4	
	Regular multiple uses	10	9	8	7	8	7	6	

Rings: Data in percent per survey year; Table: Column percentages within the RegioStaR7 categories 2023
Definition based on typical daily/weekly usage; deviations from 100% are possible due to rounding.

– as well as the stark urban-rural differences in the Shares of car-oriented people.

A comparison with the situation in 2017 reveals a further decline in the number of people who use cars daily. The decrease is approximately 5 percentage points in all region types except for rural areas. A further change is observed in the number of people who use bicycles.

Their share is growing by around two percentage points in metropolitan areas and large cities, and even more significantly in central cities. Unlike the nationwide cycling modal split, this indicates a growing focus on cycling. These changes should be interpreted with caution, but can be seen as an indication of changing mobility patterns.

Assessments of cycling, public transport and walking routes in 2023 were significantly more critical than in 2017.

This change has numerous facets, and the subsequent chapters of the report will repeatedly address and further explore the topics of modal split and transport use. Before delving into these, however, another overarching finding should be presented. Following the 2017 study, the 2023 MiD (Mobility in Germany) survey again offers a comparative perspective on satisfaction with the most important modes of transport. This survey considered cars, public transport, cycling, and walking. No other study provides this directly comparable analysis, differentiated by population groups and spatial types, and presented over time. This results in fascinating insights into these subjective, yet often action-guiding, assessments.

These results indicate – despite or perhaps even because of the emerging changes in mobility practices – potential headwinds. The slight decrease in traffic volume and service performance, and the somewhat less frequent car use, are not due to greater satisfaction with alternative options.

On the contrary: cycling conditions, as well as the situation regarding public transport and walking, were viewed much more critically in 2023 than in 2017. Figure 13 presents the complete results.

2023. For comparison, it also includes the 2017 Top Box values, i.e., the added grade percentages for the ratings "very good" and "good".

It almost seems as if more attention and discussion surrounding these alternatives is leading to a closer look and growing demands.

Although slightly better ratings are maintained, especially in medium-sized cities, the grades awarded are nevertheless reduced in metropolitan areas, in other cities and in rural areas.

Even pedestrian traffic, which has previously performed quite well, has to accept significant losses in its rating.

The increased frequency of walking may have led to a more sensitive perception of existing deficits than before. men.

It is also striking that in all evaluations, those who rarely or never use the mode of transport being assessed are the most critical. This is especially true for public transport users compared to car users, and vice versa. The situation is somewhat different for those with limited mobility: they often rely on good local transport options but are relatively dissatisfied with the existing footpaths and cycle paths.

The bicycle, as an important link in this chain, now only achieves an average of around 40 percent very good or good ratings. In 2017, this figure was still at 50 percent. In larger cities, which generally place great hopes on cycling, it currently falls below the 40 percent mark. Around a third of those surveyed there even rate cycling as adequate or worse. While there are city-specific exceptions, which the regional MiD report also identifies, overall, significant shortcomings and a considerable need for improvement are evident.

The same applies to public transport.

While the Deutschlandticket, with its affordable price and ease of use, has significantly reduced or completely eliminated long-standing obstacles, existing service deficiencies elsewhere apparently weigh more heavily in the public's judgment. Even major cities are showing a decline in this regard. And in small towns and villages, satisfaction, measured by the sum of the two highest rating levels, remains low.

Figure 13

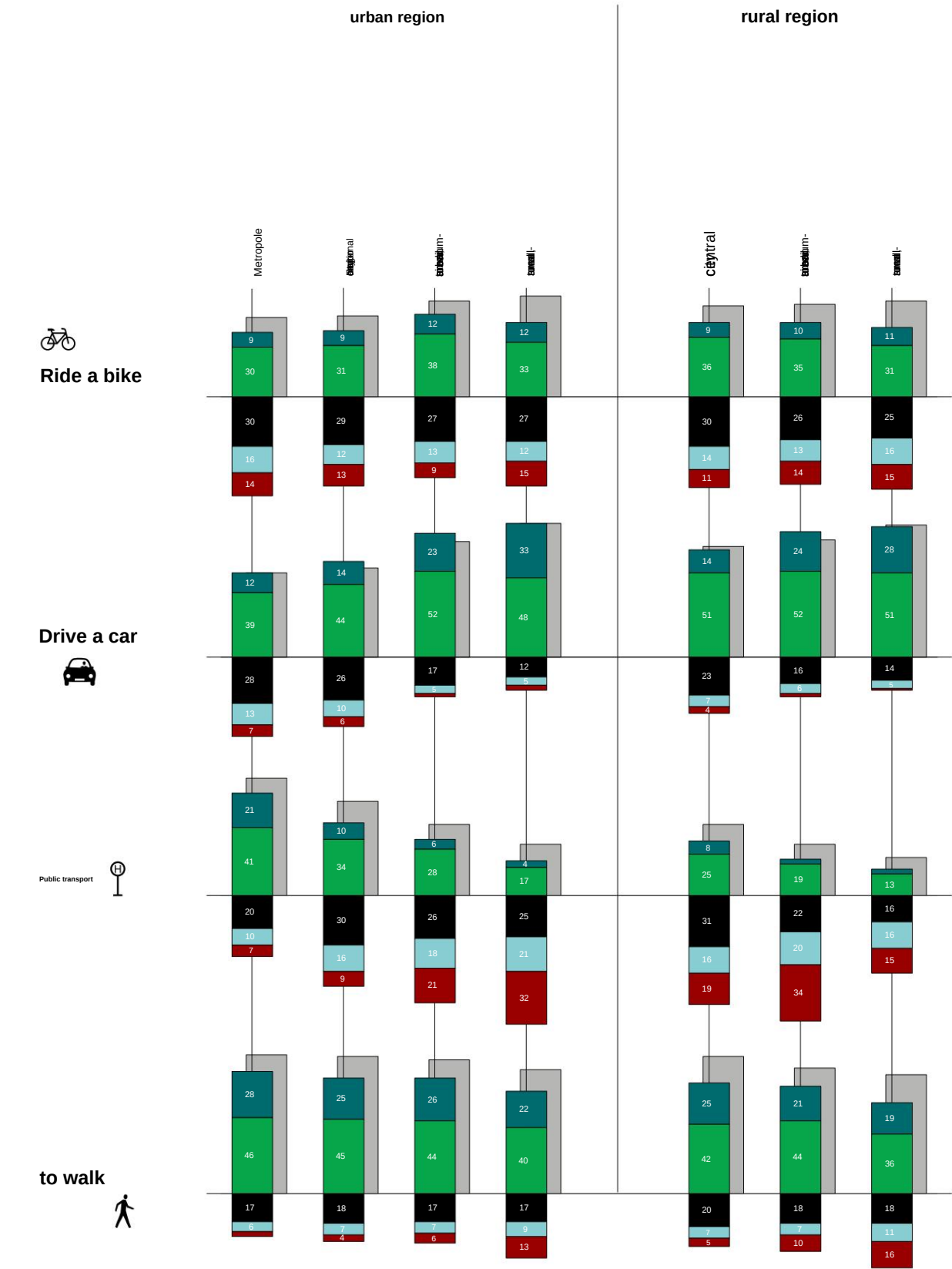
Chapter 6, AHS 13

Satisfaction with the traffic situation at the place of residence by means of transport

persons

very good good satisfactory sufficient deficient/inadequate

2017: very good/good



Figures in percent; only respondents in the "Satisfaction and Attitudes" module

The figures are now in the 20 percent range. The two worst rating levels are shared by half of the respondents. The need for improvement identified for cycling infrastructure therefore also applies to public transport.

The quality of footpaths is not immune to this. While it still receives the best ratings relatively speaking, the growing criticism of this infrastructure is unmistakable. Nationally, two-thirds of respondents see no problems in this regard, but the more critical view of the remaining third should give us pause.

In large cities, this ratio is somewhat more favorable. However, even there, more than a quarter of citizens are dissatisfied with the opportunities for getting around on foot.

Car traffic claims the satisfaction levels

The car does not follow the more critical perspectives on cycling, public transport, and walking. It largely maintains its high ratings and even more frequently takes the top spot in 2023. In rural areas, roughly four out of five respondents are satisfied with car traffic. In large cities, this figure is still one in two. This result is better than the result for cycling – and in some cities with fewer than a million inhabitants, it is also more positive than the result for public transport.

The trend observed in the MiD 2023 survey towards a decline in very frequent car use – not yet in ownership and availability, as will be shown – is, according to these results, unlikely to be attributable to perceived deterioration of car traffic conditions. Other factors must therefore be at play. Investigating these factors will be the task of the following chapters.

Use of small-scale feeds: Locations of households and destinations

The MiD (Municipal Data Analysis) can use not only direct survey results for evaluation, but also provided characteristics such as the location of households, starting points, and destinations of journeys. A simple characteristic in this series is, for example, the Regi-oStaR classification, which is used extensively in the report. It provides an assignment at the municipal level. Below this level, however, further characteristics are available. This section will present and use some classifications from a comprehensive catalog of provided characteristics as examples (a detailed description of these and all other indicators can be found in the separate MiD Data Usage Handbook; the characteristics originate from the BBSR (Federal Institute for Research on Building, Urban Affairs and Spatial Development) and infas 360 databases):

- **Locations within the town or city for household residences,** destinations and starting points of journeys, insofar as sufficiently precise information is provided by the respondents (for the related evaluations at the destination level, see a further digression in the later chapter on the Route purposes, also with a special feature regarding the location of destinations on the outskirts of a municipality in commercial or shopping centers, in the broadest sense (Sense also on the “greenfield site”) – here, various location categories are used, from the outskirts of a town to the central town center of a city or municipalities differed
- a **local supply index** for the residential address of a A household, which shows in five stages how well typical infrastructures are used in terms of shopping, Shopping, health and education can be achieved

- an index classification of household residential areas according to the **degree to which the criteria for a "15-minute city"**

The underlying question behind the inclusion of these characteristics is to what extent trip lengths, daily distances traveled, travel times, and mode choice differ depending on these environmental factors. This overview cannot present all the detailed results, but aims to illustrate the findings and encourage further analysis using the MiD dataset.

Daily values under scrutiny

The first step involves presenting the values at the individual level. This means that the daily travel distances per person are evaluated according to the various assigned characteristics. The result is shown in the green area of Table 9. To differentiate the overarching categories of local location, local amenities, and proximity to a 15-minute city by regional type, a subdivision into urban and rural regions according to the RegioStaR classification is performed in addition to the overall values. However, no clear picture emerges for any of the characteristics. All selected mobility indicators are close to each other. At most, the daily distances are sometimes slightly longer in peripheral and less compact municipalities or districts, depending on the location, local amenities, and proximity to a 15-minute city. Similarly, the average number of trips per person tends to be slightly higher in the more compact areas.

Table 9

Mobility indicators at the daily and route level according to local location, local amenities and Classification as a 15-minute city

LOCATION OF THE RESIDENCE ADDRESS	Daily values (per person and day)		Travel time (min)	Path values	
	Paths	distance		Travel time (min)	distance km
IN TOTAL	Number of km				
total	2,9	35	82	28	12
central town center further	3,0	32	83	28	11
town center central town	2,9	36	82	28	12
location decentralized	2,8	37	83	29	13
town location	2,8	34	82	29	12
outskirts of town	2,8	37	82	26	12
CITY REGION					
City region as a whole,	2,9	35	85	29	12
central town center,	3,1	33	88	29	11
further town center,	2,9	35	85	29	12
central location,	2,8	35	84	30	12
decentralized location	2,8	33	83	30	12
outskirts of town	2,8	37	89	28	12
RURAL REGION					
rural region, central town center,	2,9	36	78	27	12
further town center,	3,0	31	77	26	11
central town location,	2,9	36	77	27	13
decentralized town	2,9	40	81	28	14
location	3,0	41	78	25	12
outskirts of town	2,8	38	70	23	11
LOCAL SUPPLY INDEX					
IN TOTAL					
overall	2,9	35	82	28	12
very low low	2,8	39	83	29	14
medium	2,9	35	81	28	12
high	2,9	33	80	28	11
very	2,9	34	85	29	12
high	3,0	32	84	28	11
CITY REGION					
City region as a whole:	2,9	35	85	29	12
very low, low,	2,8	38	84	30	13
medium,	2,9	35	85	29	12
high,	2,9	34	83	29	12
very	2,9	35	88	30	12
high	3,0	32	87	29	11
RURAL REGION					
rural region overall very low low	2,9	36	78	27	12
medium high	2,9	41	81	28	14
very high	2,8	36	76	27	13
	2,9	33	73	25	11
	3,0	32	80	27	11
	3,0	32	76	26	11
15-MINUTE CITY INDEX					
IN TOTAL					
overall	2,9	35	83	28	12
very low low	2,9	40	81	28	14
medium	2,9	38	81	28	13
high	2,9	34	80	28	12
very	2,9	32	84	29	11
high	3,0	32	91	30	10
CITY REGION					
City region as a whole:	2,9	34	85	29	12
very low, low,	2,9	41	84	29	14
medium,	2,9	38	82	29	13
high,	2,9	34	83	29	12
very	2,9	32	85	29	11
high	3,0	32	92	31	11
RURAL REGION					
rural region overall very low low	2,9	36	79	27	12
medium high	2,9	40	79	27	14
very high	3,0	38	79	27	13
	2,8	34	76	27	12
	3,0	32	80	27	11
	3,2	29	80	25	9

Path values under scrutiny

But what picture emerges when we move beyond the daily timescale and consider individual journeys? The table for journey lengths and travel times on the right-hand blue page shows this result. However, even here, no clear correlations exist. Only minor effects on journey lengths, analogous to the daily mileage totals, can be observed. The result must therefore be interpreted similarly to the barely perceptible differences in the daily values.

A supplementary correlation analysis shows no clear relationship for either level of analysis – day and routes. No significant correlations were observed for any metric.

This can be interpreted as an indication that both the total daily values per person and the distances a person travels throughout the day are hardly influenced by their immediate place of residence and the availability of services. The activity radii of individuals throughout the day blur any potential differences, and everyday mobility apparently mostly takes place not in confined spaces, but across municipal or district boundaries – thus proving more complex than often assumed. Furthermore, the relatively stable "guideline" of an average travel time of approximately 90 minutes is confirmed once again, proving relatively robust across various conditions. This suggests, among other things, that time saved due to short distances and good local amenities is sometimes used for longer journeys or other daily activities.

The slightly increased number of journeys per person per day in the well-served areas, though not significantly so, underscores this conclusion.

Therefore, when planning such projects, it must be taken into account that short distances offer certain advantages, but do not necessarily reduce the overall level of mobility – a very specific rebound effect.

And what about the choice of transportation?

In the third step, the question remains, since distances and durations do not show any clear dependencies, whether the choice of transport mode yields a different picture.

For this purpose, the modal split can be differentiated according to the characteristics presented, and further differentiated between urban and rural regions.

Unlike the mobility indicators considered previously, a clear correlation can be observed here. The more compact and better-served an activity environment is, the higher the proportions of walking and cycling. For example, in urban areas with very good local amenities, these figures are 35 and 13 percent, respectively, while in areas with very poor local amenities, they are only 20 and 9 percent. The other characteristics used and their differentiation show a similar pattern, as can be seen in the corresponding figure. Public transport also benefits from nearby amenities, at least in urban areas, although it is less dependent on the location of residential areas within the town or city. Conversely, the proportion of private motorized transport is lower in well-served areas than in less well-equipped urban or municipal categories.

A classification

Here, too, classification and interpretation are required. Unlike the ambiguous, barely existing, or leveled-out differences in daily and route values for time and distance, the choice of transport mode can apparently be influenced by appropriate spatial concepts. According to the presented results, good local amenities reduce car traffic and offer advantages for walking and cycling, thus contributing to easing traffic demands in this regard – although not necessarily to a lower traffic volume, as explained in the first part of this digression.

Nevertheless, a frequently discussed, important area of action is confirmed when it comes to strengthening, above all, short-distance options such as walking or cycling.

Table 10

Modal split according to location within the city, local amenities and classification as a 15-minute city

	on foot %	Bicycle %	BELT	MitV trip %	Private vehicle ride
LOCATION OF THE RESIDENCE ADDRESS			%	km	
IN TOTAL					
total central	26	11	11	40	13
town center further town	33	13	11	33	10
center central town location	25	11	10	40	13
decentralized town location	23	10	11	43	14
	21	9	14	42	13
outskirts of town	19	8	11	50	13
CITY REGION					
City region as a whole, central	27	12	13	36	12
town center, further town	35	16	14	27	9
center, central location,	27	13	13	35	12
decentralized location	24	11	13	39	13
	23	10	15	39	13
outskirts of town	19	9	14	45	13
RURAL REGION					
rural region, central town center,	24	9	7	47	14
further town center, central	32	10	7	39	12
town location, decentralized	22	9	6	48	15
town location	21	6	6	52	15
	14	6	12	54	15
outskirts of town	18	6	5	58	13
LOCAL SUPPLY INDEX					
IN TOTAL					
overall	26	11	11	40	13
very low low	20	8	9	49	15
medium	23	10	10	43	14
high very	26	12	11	38	13
high	28	13	12	35	12
	35	12	14	30	10
CITY REGION					
City region as a whole: very	27	12	13	36	12
low, low, medium,	20	9	11	45	15
high, very	24	11	13	40	12
high	28	13	13	34	12
	28	15	14	32	12
	35	13	17	27	9
RURAL REGION					
rural region overall very low low	24	9	6	47	14
medium high very	20	6	6	53	14
high	21	9	6	48	16
	23	11	6	46	14
	30	10	7	41	12
	35	10	6	37	12
15-MINUTE CITY INDEX					
IN TOTAL					
overall	26	11	11	40	13
very low low	19	6	8	52	15
medium	21	9	7	49	15
high very	24	11	9	42	14
high	30	13	12	34	11
	37	17	20	20	7
CITY REGION					
City region as a whole: very	27	12	13	36	12
low, low, medium,	18	6	8	52	16
high, very	21	9	8	46	15
high	24	12	11	40	14
	29	13	14	33	11
	36	17	21	19	7
RURAL REGION					
rural region overall very low low	24	9	7	47	14
medium high very	20	6	7	53	15
high	20	9	6	52	14
	24	10	6	45	15
	32	11	6	38	13
	42	13	7	29	10

5. Journey purposes – diverse reasons for mobility

Mobility is generally a means to an end. This means that a wide range of different reasons underlie the volume of traffic. The reasons for leaving home vary considerably depending on life circumstances and individual circumstances. These reasons and motivations form the basis for a fundamental understanding of mobility requirements and thus of trends in mobility behavior. In contrast to all passive forms of mobility data collection, such as mobile phone data or GPS tracking, a survey like the MiD provides important and comprehensive insights into the purposes of mobility. For the reported journeys, the survey includes both the respondents' classifications of broader purpose categories (e.g., work, shopping, leisure) and more detailed journey purposes (e.g., travel to school, shopping, errands).

B. shopping for daily necessities, sports). This chapter presents analyses of the purpose structure of mobility and highlights relevant changes since the last MiD in 2017.

Changes in work-related traffic – more working from home, but also more service trips during work activities

Regarding travel purposes, a block of work-related journeys (including business trips made regularly or occasionally during work) is contrasted with trips for personal needs and leisure. In 2023, the total number of all work-related journeys remained smaller than the total of all other purposes combined. However, actual commuter traffic, i.e., journeys to and from work, showed a slight decrease compared to previous measurements since 2002 – an effect of the home office trend, which this report addresses in a separate chapter. Journeys made during working hours for official or business purposes continued to increase.

In summary, these two work-related purposes accounted for 32 percent in 2023, compared to 27 percent in 2017. This is due to a continued higher employment level, with approximately 46 million people employed in 2023 compared to 44 million in 2017.

Distinctiveness between commutes, occasional business trips and regular professional trips

In the MiD (Mobile Inventory of Departures), these categories have been recorded separately since 2002. The specific recording of regular work-related journeys is compiled for each day and is intended to facilitate participation for individuals with such activities and the numerous journeys associated with them. This daily data is divided into equal individual journeys and added to the journey data during the evaluation, with the kilometers being distributed equally among the specified number per day. Information on the mode of transport is also recorded. Information on start and arrival times is not available.

Travel times are estimated based on the means of transport and the distance traveled.

Despite numerous prompts in the questionnaire and considering the many hybrid situations (such as a tradesperson traveling directly to their first work location, or the lack of a central work location), the distinction between regular work-related journeys and those undertaken for work purposes is not always clear. Extensive data validation and corrections were performed, but a small portion of regular work-related journeys, unlike occasional business trips, could also be classified as normal commuting routes.

However, a detailed analysis reveals that the boundaries are becoming increasingly blurred, and some respondents find it difficult to distinguish between them. This is also related to the increased proportion of journeys made on foot in this sector. Therefore, some analyses, such as the MiD short report, decided not to separate the three categories mentioned and to group them together as work-related journeys. In the present long report, however, they are documented separately for reasons of transparency.

Fewer supply routes, increased share for leisure traffic

Changes are also evident in the proportions of trips made for purposes other than work-related travel. Shopping and errands are down, while leisure trips are slightly higher than in 2017, both in terms of proportion and distance traveled. These shifts reflect increased online activity, not only in professional contexts but also for personal needs. (See also Chapter 17)

The trip lengths for each purpose in 2023 are shown in Table 11. The results are close to those determined in 2017. Trip lengths for supply, leisure, and accompanying purposes remained very stable. Commuting to work has decreased slightly on average by 0.7 kilometers. This is even more true for business trips, which in 2023 were on average several kilometers shorter than in 2017. On average, the length of work-related trips is around 14 kilometers.

Nevertheless, the time required by respondents is changing. More time is needed per trip for all purposes. This is also reflected in the increasing overall time per trip across all purposes, which now averages half an hour; in 2017 it was almost three minutes less.

Compared to the average trip length of 12 km, shopping, education, errands, and accompanying others are relatively short. The average distance of work-related trips is slightly above average, meaning these trips contribute somewhat more to the overall traffic volume than they do to the total volume, but even here they still account for less than half. In contrast, education trips, at 6 km, are comparatively short and make up only 3 percent of the total traffic volume. The share of shopping and errand trips in passenger-kilometers decreases from 24 percent to 16 percent compared to the total volume of trips, primarily due to the shorter shopping trips.

Since 2017, there have been different developments in traffic volume and performance for various road purposes.

Traffic volume and performance have decreased overall since the 2017 MiD (Mobility in Germany) survey. However, the differences vary depending on the purpose of the trip. The volume and performance of shopping and errand trips have declined particularly significantly. For educational journeys, the number of passenger-kilometers has decreased considerably more than the volume of trips, meaning that educational journeys have become shorter.

In contrast to these routes with declining key figures, recreational routes are at a similar level to 2017. With a high increase.

Table 11

Absolute values per day for travel purposes 2002 to 2023													
	Paths in millions per day				Passenger kilometers in millions per day					Travel time in Min. 2023		distance in km 2023	
	2002	2008	2017	2023	2002	2008	2017	2023	Central			value	Median* Mean value Median*
IN TOTAL	270	275	257	249	2.717	3.080	3.214	2.983			28,4	15,0	11,9 3,8
Profession	57	58	70	79	920	1.039	1.213	1.134			29,1	20,0	14,3 4,9
Training	17	19	18	16	111	147	131	97			22,5	15,0	6,0 2,5
Shopping	56	55	41	34	268	264	217	169			19,4	10,0	5,0 2,0
Completion	34	35	37	28	256	268	376	288			28,0	15,0	10,1 3,8
Leisure time	83	84	71	72	1.005	1.186	1.098	1.114			39,5	20,0	15,5 3,9
Accompanying	23	24	21	20	158	186	179	181			20,7	15,0	9,1 2,9

information *This value represents the midpoint of the given distribution and helps in classifying the mean, which is often influenced by high individual values. Example: Trip length for walking routes: The median is 1.0 km. Therefore, 50 percent of the reported walking routes are up to 1.0 km long, and 50 percent exceed this value.

Official travel is a clear outlier. However, the total kilometers traveled by individuals for work purposes have only increased slightly, meaning that these distances have also become significantly shorter. Traditional commutes, on the other hand, have decreased.

When professional paths are combined, the most noticeable difference is a decrease in mileage (see the delimitation in the information box on page 65).

Increases in the environmentally friendly transport network across all sectors

Road purposes

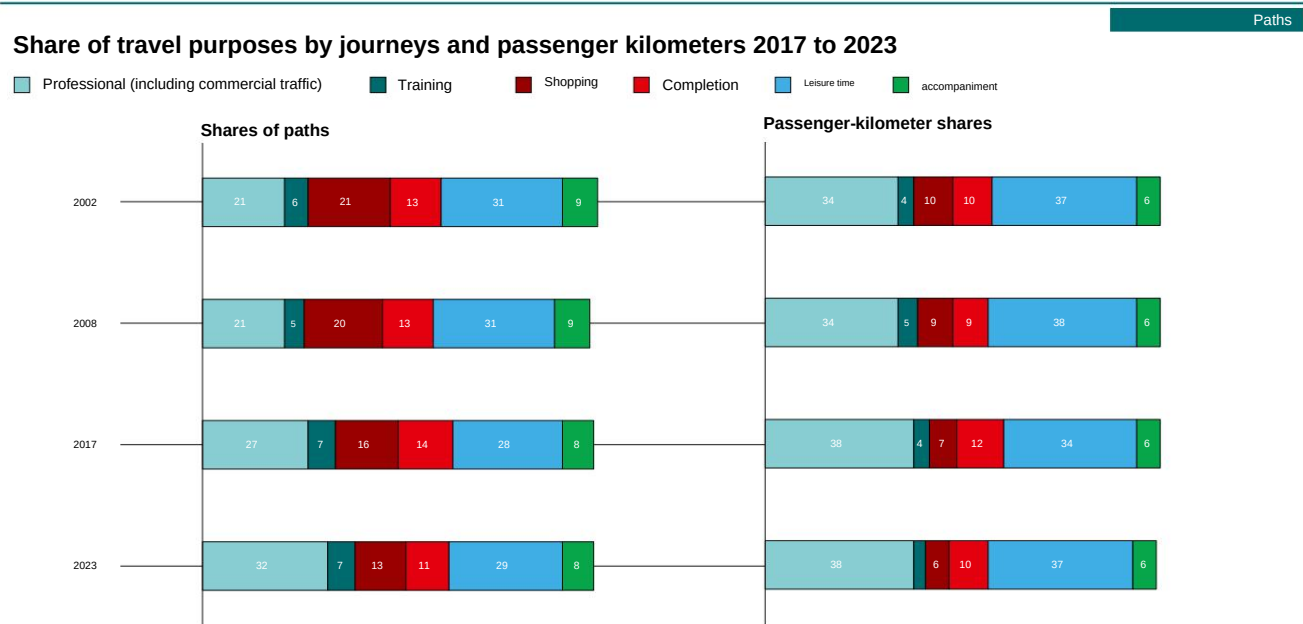
For each of the seven main trip purposes considered, the share of walking in the total trip volume has increased since 2017 (Figure 15). The increase in the proportion of walking is particularly significant for work-related trips. For these trips, the share of walking rose from 7 percent to 26 percent in the MiD 2023. This is almost entirely due to changes in regular work-related trips, i.e., trips that are routine and recurring during work or are part of daily activities (tradespeople, couriers, postal workers, caregivers, and other similar fields). Conversely, the share of private motorized transport (PMT) for work-related trips has declined significantly. For the other trip purposes, the share of walking has increased in the low single-digit percentage range. In isolated cases

Further slight increases in environmentally friendly modes of transport can be seen in the modal split of journey purposes, such as gains for public transport in educational journeys or cycling in companion journeys. Overall, it can be stated that the share of private motorized transport has stagnated or declined across the board.

Strong increase in regular professional Paths

The increase in business trips compared to 2017 is attributable to a significantly higher number of regular work-related journeys. In the 2017 MiD (Minimum Daily Travel Time), three-quarters of business trips were regular work-related journeys. In the 2023 MiD, this share rose to 86 percent. The total number of regular work-related journeys in Germany increased from just under 30 million per day in 2017 to just over 40 million per day in 2023. At the same time, the number of other business trips decreased from 7 million to around 6 million. Excluding regular work-related journeys, the share of business trips among all journeys remained unchanged at 3 percent in both the 2017 and 2023 MiDs. The average trip length also remained virtually unchanged at 32 km compared to 31 km in the 2017 MiD. The reduced average mileage per trip of 13 km compared to the previous 19 km for all business trips

Figure 14



Figures in percent

Table 12

Overview of route length and duration, traffic volume and performance shares by route purpose 2023

	medium distance	medium Travel time	Portion Paths	Portion Passenger kilometers
	km	Min.	%	%
IN TOTAL	12	28	100	100
WAY PURPOSES				
Work	15	29	15	19
official	13	22	17	19
Training	6	22	6	3
Shopping	5	19	13	6
Completion	10	28	11	10
Leisure time	15	40	29	37
accompaniment	9	21	8	6

All paths; deviation from 100%: rounding differences

MiD 2023 | Results Report | Source: MiD 2023

The shorter routes are therefore due to the shorter length of regular work-related journeys.

In 2023, individuals who made at least one regular work-related journey averaged approximately 7 journeys and 70 km per day for regular work purposes. These individuals also made an additional 1.2 business trips, covering a daily distance of 37 km. Individuals who made one business trip but not a regular work-related journey on the reference date averaged 2.3 journeys and 74 km per day for this type of business trip (Table 14).

More than half of all work-related journeys are now official journeys – especially on business trips. weekend

Compared to 2017, the number of commutes to and from work decreased on all weekdays in 2023. On an average Friday in the 2023 survey period, there were almost 10 million fewer commutes than on a Friday in 2017. This occurred despite increased employment and is an effect of the work-from-home trend (see also the thematically specific Chapter 17 for further details). Among the

Table 13

Change in traffic volume and performance by trip purpose compared to 2017

	Paths	Person-kilometer
	%	%
IN TOTAL	-4	-8
WAY PURPOSES		
professionally overall	12	-7
Work	-13	-16
official	50	5
Training	-11	-26
Shopping	-18	-22
Completion	-23	-24
Leisure time	1	1
accompaniment	-6	0

all ways; Mid2023 & Mid2017

MiD 2023 | Results Report | Source: MiD 2023

Table 14

Trips per person per day, daily distance traveled by persons with at least one business trip

	in total		regular professional Paths		other official routes	
	Trips per day	Daily distance	Trips per day	Daily distance	Trips per day	Daily distance
	Number	km	Number	km	Number	km
PERSONS WITH OFFICIAL responsibilities	6	84	5	47	2	37
PATHS (TOTAL)						
People without a regular career path	2	74	0	0	2	74
People with regular work commutes	8	107	7	70	1	37

only persons with official business on the cut-off date

MiD 2023 | Results Report | Source: MiD 2023

On normal working days, Monday to Friday, the decline in commuting distances is smallest on Wednesday, but at 6 million, or minus one tenth, it is still considerable.

Due to the increase in work-related travel, the total volume of journeys for work and business purposes, taken together, is higher than in 2017 on all weekdays. On Mondays and Tuesdays in 2023, approximately half of all work-related journeys were traditional commutes. This proportion decreases towards the end of the week.

On Fridays, this proportion drops to only 45 percent. The proportion of business trips among work-related trips is particularly high on weekends, reaching 68 percent on Saturdays (Figure 17).

The importance of the days of the week for the number of shopping trips decreases slightly.

Saturday is the most popular day for shopping in Germany. With almost 54 million trips, the number of shopping trips on a Saturday is significantly higher than on Mondays through Thursdays. Friday, with around 40 million trips, is the second most popular shopping day.

From Monday to Thursday, a similar number of shopping trips take place daily. Given that only a few shops are open on Sundays, the number of shopping trips is significantly lower on Sundays (Figure 17).

Compared to 2017, the total number of shopping trips is almost one-fifth lower. This decline in shopping trips is evident on all days of the week.

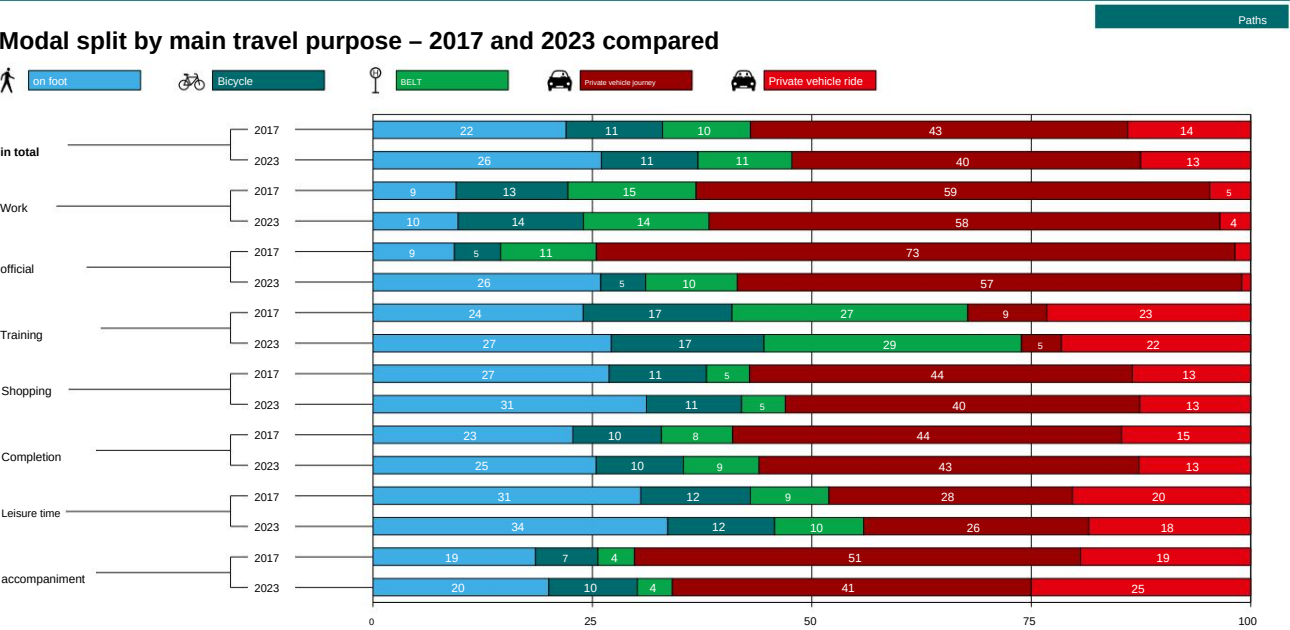
The decline is most pronounced on Saturday, followed by Wednesday. Sunday, with its already low overall level of shopping, sees the smallest decrease. Changes in shopping behavior do little to alter the ranking of weekdays for shopping purposes: despite slightly larger declines on Saturday, it remains the day with the most shopping trips, just as it was in 2017, ahead of Friday.

Shopping times vary depending on the Location of residence

Shopping trips undertaken from Monday to Friday mostly take place in the morning or afternoon. Looking at the time when shopping trips end, i.e., the presumed start of actual shopping activity, the hour between 10 and 11 a.m. emerges as the peak demand period. The afternoon peak accounts for a smaller proportion of the total number of trips and is significantly less pronounced than the morning peak. In large cities, the afternoon peak is reached approximately one hour later than in rural communities and small and medium-sized towns. Furthermore, a proportionally higher number of shopping trips occur in the evening hours in large cities.

The curve in Figure 18 differentiates between different locations of the residential sites of the

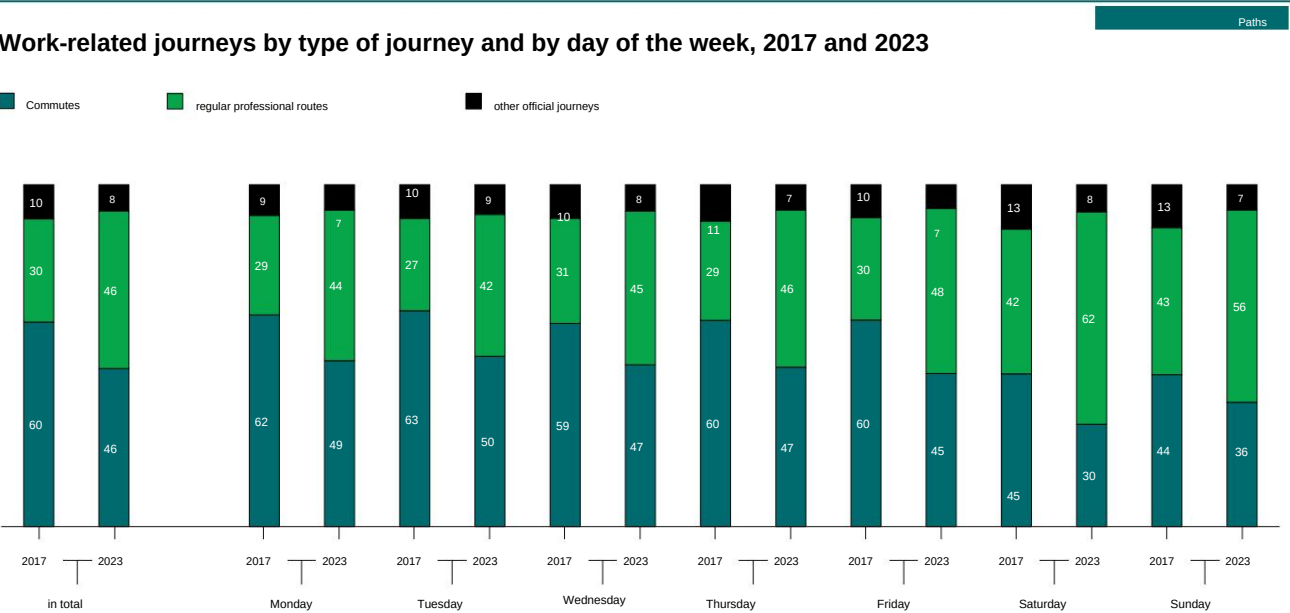
Figure 15



Figures in percent, MID2023 & MID 20217, all routes; deviation from 100%: rounding differences.

MID 2023 | Results Report | Source: MID 2023

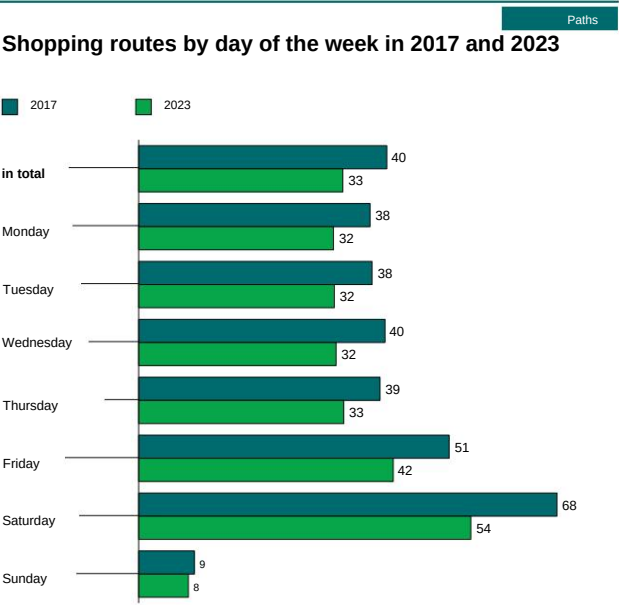
Figure 16



Figures in percent, MID 2023 and MID 20217, journeys for work and official purposes; deviation from 100 percent: rounding differences

MID 2023 | Results Report | Source: MID 2023

Figure 17



Figures in millions of trips, MID2023 and MID20217; shopping trips only, main trip purpose

MiD 2023 | Results Report | Source: MiD 2023

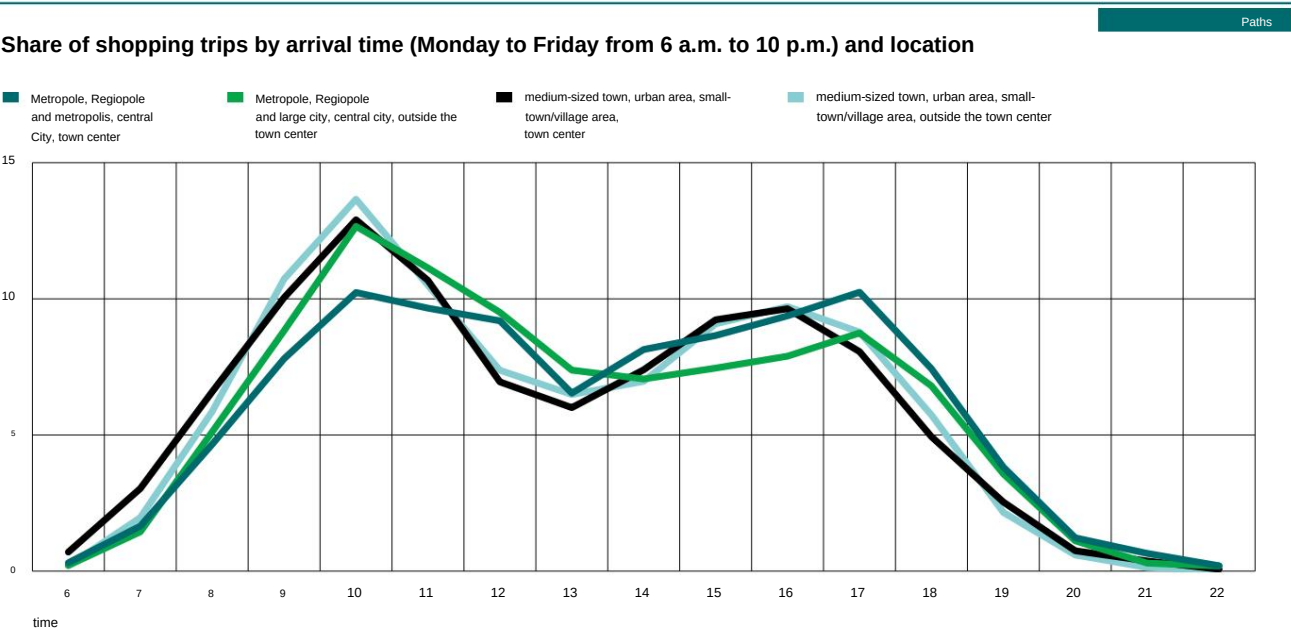
The survey revealed that in medium-sized towns, urban areas, and small-town/village areas, there are no significant differences in shopping times based on location. This is different in the categories of metropolis, regional center, large city, and central city: The shopping behavior of the population in large city centers is characterized by later shopping times and less pronounced peak demand. The pattern of shopping demand among the population of large cities outside the city centers is a mixture of urban and rural shopping behavior in terms of timing.

Shopping, errands and leisure – Route purposes in detail

Analyses of the motivations behind mobility usually utilize the previously presented summary categories of main trip purposes. However, behind these very broad categories – especially for shopping, errands, and leisure – lies a wide range of different reasons for mobility. For example, the category "leisure" could encompass activities as diverse as singing in a choir, visiting a cemetery, or taking a stroll downtown. But is a cemetery visit leisure or an errand?

And isn't a stroll through the city center perhaps also shopping? To better understand the reasons for mobility-

Figure 18



Figures in percent

MiD 2023 | Results Report | Source: MiD 2023

In addition to the standard questionnaire, the MiD also asks about the detailed purposes of each trip in a supplementary module. Respondents first indicate the overarching purpose of each trip, e.g., "Leisure" or "Errand," and are then asked to provide further details. This provides information on the detailed purpose of each trip as well as a self-assessment-based classification of the trip according to the overarching purpose.

The following analyses of detailed travel purposes, comparing the MiDs of 2017 and 2023, refer to individuals aged 14 and over and focus on the outbound journeys to the respective activities. Return journeys and connecting journeys are not included, unlike previous analyses of primary travel purposes.

Fewer trips required when shopping for non-daily necessities

Examining the detailed purposes of the trips provides an accurate insight into the significance of certain purchases for the volume of shopping trips and its change since 2017 (Table 15).

The category of "everyday shopping" is particularly significant. This includes trips to supermarkets, bakeries, and other grocery stores. These trips accounted for four-fifths of detailed shopping trips in 2023. Relatively speaking, this decline is considerably less pronounced than for other types of shopping trips, meaning that everyday shopping trips dominated shopping mobility even more strongly in 2023 than before.

In 2017, the number of shopping trips categorized as "shopping excursions" plummeted by more than half, while "purchases of other goods" declined by about a third. Only minor changes were observed among visits to services categorized as "shopping." Although altered online and ordering behavior seems like a likely explanation for the decline in shopping trips, overlaps with demographic and economic factors are also likely to play a role. However, a shift in perception cannot be ruled out, leading to certain shopping activities now being interpreted more as leisure than as shopping, as the following analyses demonstrate.

In 2023, city strolls were viewed less often as a shopping activity and more often as leisure.

The detailed trip purpose "general city/shopping trip" exists in the categories of shopping trips as well as leisure trips and errands. While, as just explained, shopping trips in the sense of a shopping activity have declined sharply, they have gained importance in the sense of leisure or errands: In 2017, less than one percent of leisure trips were declared as shopping trips, while in 2023 this figure rose to three percent. The share of shopping trips among errands increased even more significantly, from four percent to ten percent. In other words, shopping trips still take place – but they are apparently associated more with leisure or errands than with traditional shopping. This may also reflect the crisis in the retail sector in many city centers, which is characterized primarily by the fact that less money is being spent while still spending time in the city.

Table 15

Traffic volume according to detailed Shopping trip purposes – 2017 and 2023		
	2017	2023
Paths	%	%
Shopping for daily needs, shopping for other goods,	73	79
general city/shopping trip, services	17	14
(hairstylist, shoemaker, etc.), other	8	4
	1	1
	2	3
MiD2023 & MiD 2017; shopping trips only, persons aged 14 and over		

Proportional importance of different Leisure activities are shifting

Besides the increasing importance of shopping trips as a leisure activity, there are further changes in the composition of leisure journeys. "Active sports" is replacing meeting friends, relatives, and acquaintances as the most frequent leisure activity. Nevertheless, meeting relatives, friends, and acquaintances remains an important purpose of leisure journeys. While in 2017

While 22 percent of leisure trips were attributed to this purpose in 2020, this figure dropped to 17 percent in 2023. A similar trend is seen for visits to restaurants and cafes. In this case, the share decreased from 10 percent to 8 percent. However, both meetings and restaurant visits are reported not only as leisure trips but also as errands, so their overall significance is greater than the percentage of leisure trips would suggest.

A wide range of reasons for leisure trips leads to different distances and durations.

Depending on the leisure activity, the average trip length and duration of leisure trips vary considerably (Table 16). While the categories "holiday" and "day trip, short trip" account for only 6 percent of the trips, they represent 33 percent of the traffic volume of the leisure trips recorded in detail.

In both cases, the routes are average

The journeys are particularly long, around 150 km and 50 km respectively, and take several hours on average. Especially for less common leisure activities, such as "visiting a cultural institution" or an "event," like a concert, the average distance is relatively long at 17 km and 20 km respectively.

A large proportion of leisure trips are relatively short across all trip purposes. However, there are also a small number of significantly longer trips. This leads to considerable differences between the arithmetic mean and the median. A median of 3 km for restaurant visits, for example, means that half of the trips are shorter and the other half are longer than this value. Particularly for trips that are expected to take place close to home, such as accompanying children to the playground or going for walks, the median trip length is often within a range that can be covered on foot in under an hour.

Table 16

Share of detailed recreational route purposes in traffic volume and performance				
	2017		2023	
	Paths	km	Paths	km
	%	%	%	%
general city/shopping trip, private errand for another person	1	0	3	2
	0	0	0	0
Care for family members, acquaintances	0	0	1	0
Visiting/meeting friends, relatives, acquaintances	22	29	17	21
Visit to cultural institution	4	1	3	3
Attending an event	5	2	5	6
Actively participating in sports	16	15	18	11
Restaurant, pub, lunch, bar, club	10	4	8	4
Allotment garden, weekend house	2	0	1	1
Day trip, short trip	2	11	3	10
Holiday (minimum 4 nights)	2	17	3	23
walk, drive	14	10	14	6
dog walking	8	1	9	2
Kirche, Friedhof	3	0	1	0
Accompanying children	1	0	3	1
miscellaneous	4	1	4	4
other meetings	7	3	6	4
CAWI/CATI; 14 years and older; purposes not grouped by main route purpose; deviation from 100%: rounding differences				

The journey as a leisure activity

In the categories "walk, drive" and "dog walking," as well as in parts of "active exercise" and "general city/shopping stroll," the journey itself is often the activity. These journeys are characterized by long travel times over frequently short distances and a resulting low average speed. However, this does not mean that these journeys are not also partially undertaken using private or public transport.

With a share of 18 percent of the detailed recreational route purposes, "active participation in sports" is the most significant active purpose. This category includes not only routes taken for personal enjoyment, such as jogging, but also routes to football practice or the gym. "Walking, cycling" accounts for 14 percent and "walking the dog" for 9 percent of recreational routes.

The latter is also the category with the shortest leisure routes, averaging 4 km. "Walks, strolls" are also rather short at 7 km compared to other leisure routes, but still seem long for a route purpose that likely mainly includes footpaths and cycle paths.

Utilizing small-scale feeds: Examples of the location of shopping destinations and other routes

The previous chapter already presented some results from this area in a digression on data input in addition to the directly queried characteristics. This will be continued here with regard to travel destinations and selected purposes. If respondents were able or willing to specify their travel destinations precisely, as part of a voluntary response—such as participation in the survey—this information can also be included in the analysis.

The destinations are assigned cross-regional categories. The evaluation is only performed in aggregate form for these categories and, for data protection reasons, not with precise localization for specific areas. However, the categorization is carried out as accurately as possible before the address-specific data is aggregated and subsequently deleted.¹

The previously described assignment to inner-city locations (a feature of infas 360) can thus be applied not only to residential addresses but also to some route addresses. This requires precise address information. Accurate matching based on this information was possible for roughly half of the starting and ending points. For the remaining...

More specific questions can be further specified by adding additional characteristics via the target specification. One example that is interesting for examining shopping traffic is the allocation to the location of shopping centers, preferably on the outskirts of cities, i.e., often in what is termed a "greenfield site." In the current MiD (Municipal Survey of Germany), approximately 8,000 reported shopping trips fall into this category. Due to the aforementioned limitations, this does not constitute a reliably representative sample, but it does provide insight into this type of shopping trip in relation to all other trips for this purpose.

The most revealing aspect is once again the modal split, i.e., the mode of transport chosen for these journeys. The graphic shows the results differentiated between urban and rural regions. It is clearly evident that shopping trips to the aforementioned shopping centers differ from other shopping trips.

Their share of trips made on foot is only 15 percent in the urban area, compared to 37 percent for other shopping trips. In rural areas, the figures are 11 and 24 percent, respectively. Conversely, the situation is different for the private motorized transport (PMT) share, drivers and passengers combined: If the route leads

¹ This is done using the KLASER grid cell system from Destatis (small-scale analysis grid for the census), but is aggregated into larger areas in a so-called "data protection-compliant mixed grid" from which it is no longer possible to trace back to individual persons. Further details can be found in the MiD Data Handbook.

Active leisure activities with high demands
Traffic performance of motorized vehicles
connected means of transport

One reason for the high average length of leisure trips lies in the definition of a trip in combination with the question about the means of transport used in the MiD (Mobile Inventory of Leisure Trips). For example, a portion of a leisure trip might be covered by car, and trips by motorcycle or car can also be considered leisure trips. An examination of the aforementioned active trip purposes in relation to the primary means of transport used reveals a more nuanced picture. With a share of over three

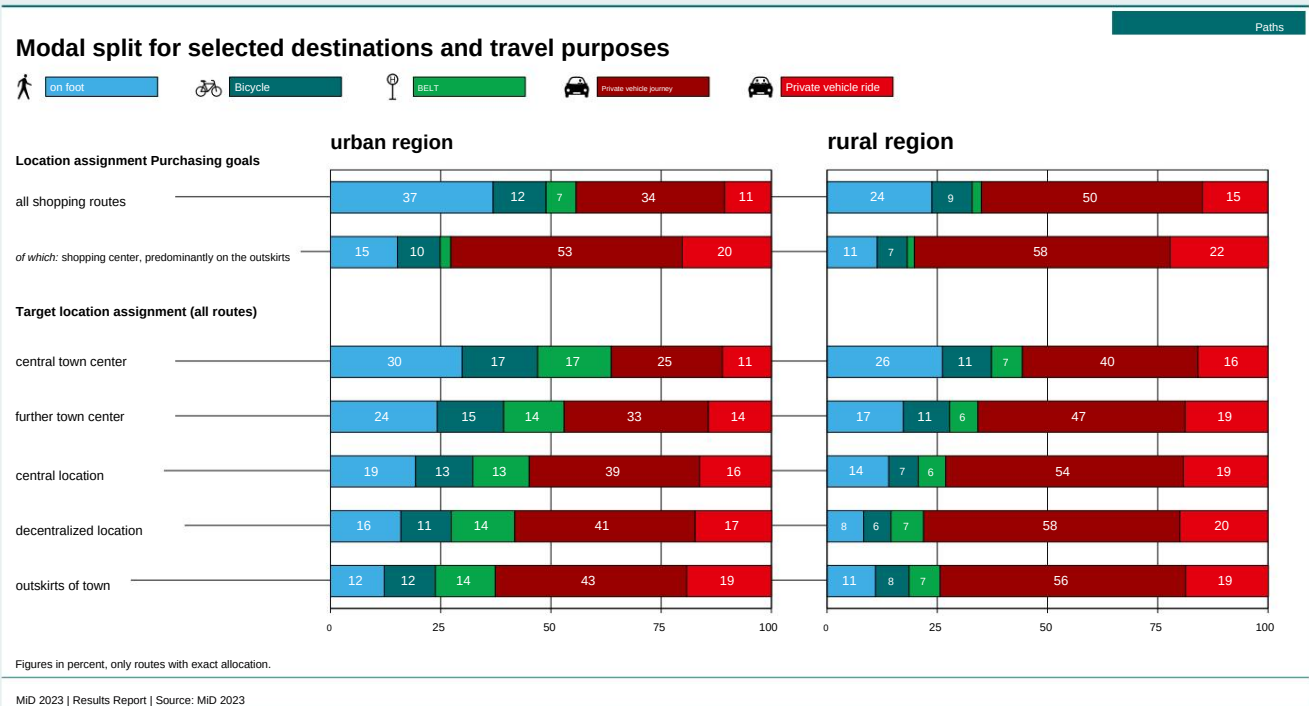
In a quarter of cases, most walks are indeed made on foot. However, over half of the kilometers traveled are accounted for by other means of transport.

The average walking distance is four kilometers. The average duration is just over an hour. Journeys by car (drive) are significantly longer at 20 kilometers, but take less time at 49 minutes.

For shopping trips to "greenfield" locations, private motorized transport accounts for three-quarters of trips in urban areas and even four-fifths in rural regions. For other shopping trips, the figures are just under half and two-thirds, respectively. Therefore, the location planning of shopping facilities significantly influences shoppers' subsequent choice of transportation.

Beyond this specific purpose-related perspective, modal split analyses can also be performed for all destinations within the city. These analyses also show a clear location dependency: the more central the destination, the higher the proportions of walking and cycling. In urban areas, this also applies to the proportions of public transport.

Figure 19



More restaurant visits in metropolitan areas, more walks in the countryside – leisure activities by type of space

The relative importance of different leisure activities in leisure traffic volume varies depending on the type of area and also appears to reflect the local amenities: In particular, in "metropolises," the proportion of trips to visit a restaurant or bar is significantly higher at 12 percent than in other types of areas. On the other hand, walks or walking the dog are proportionally more important in rural regions. "Visiting friends, acquaintances, and relatives" is of similar proportional importance in "metropolises" and in "small towns and villages in rural regions." This leisure purpose seems to be more prevalent in the

to be of lesser importance to "medium-sized cities" (Figure 20).

Doctor visits have decreased significantly and Men visit the doctor less often

Compared to 2017, the number of trips undertaken to access medical services has decreased significantly, both proportionally and in absolute terms. This may reflect developments such as the introduction of sick notes issued over the phone or video consultations, which allow people to access these services without leaving their homes. Similarly, the number of visits to government offices, banks, or post offices has also decreased considerably, possibly due to the increasing digitalization in these areas.

Table 17

Route length and duration according to detailed leisure purpose – mean and median in comparison				
detailed recreational trail purpose	mean		Median	
	Route length	, route duration	, route length	, route duration
	km	Min.	km	Min.
IN TOTAL	17	50	5	30
general city/shopping trip, private errand for another person, care of family members, acquaintances, visiting/meeting friends, relatives, acquaintances, visiting a cultural institution (e.g. cinema, theatre, museum)	14	72	5	40
Attending an event (e.g., football match, market, concert)	11	34	4	15
Actively participating in sports (training, jogging, gym, etc.)	11	34	5	20
Restaurant, pub, lunch, bar, club	21	43	6	20
Allotment garden, weekend house	17	49	7	30
Day trip, short trip	20	46	6	25
Holiday (minimum 4 nights)	10	39	5	20
walk, drive	7	30	3	15
dog walking	10	34	3	15
Kirche, Friedhof	48	119	29	75
Accompanying children (e.g., playground, etc.) other meetings (e.g., courses, hobbies, clubs, other groups)	149	161	67	125
	7	64	4	55
	4	45	3	40
	7	27	3	15
	7	39	2	20
	15	44	5	20
	11	30	4	15
CAWI/CATI; 14 years and older; purposes not grouped by main route purpose				

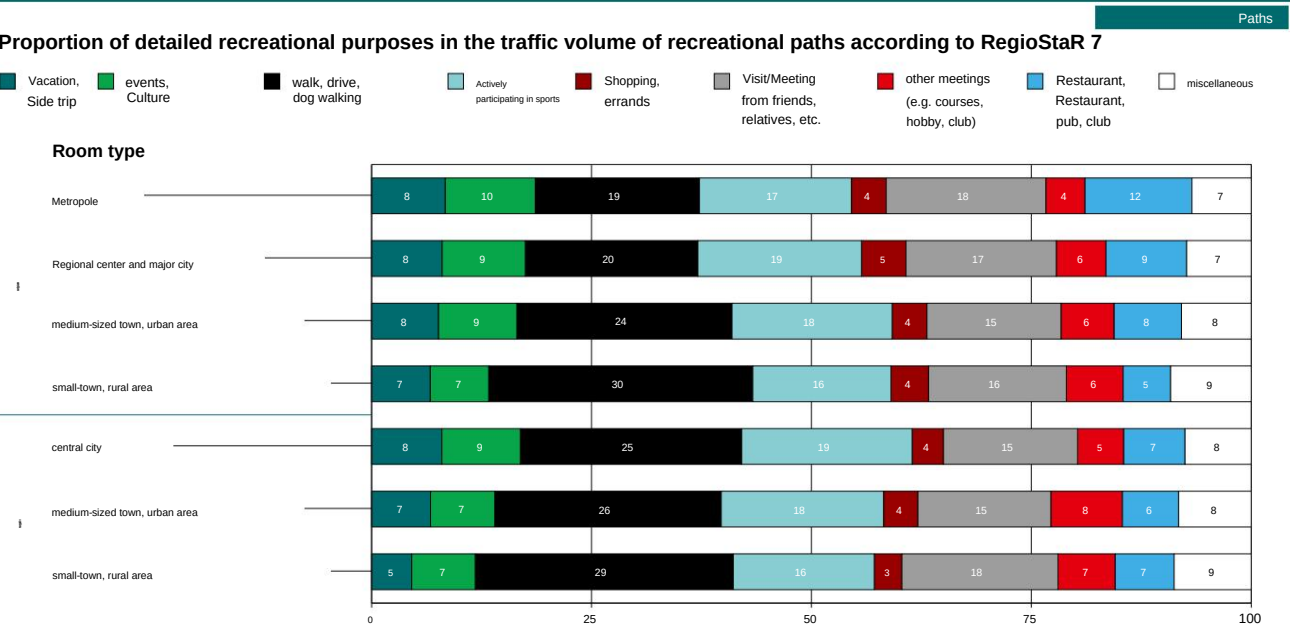
Table 18

Route length, journey duration and proportions of journey volume and passenger-kilometers of main transport modes for selected leisure purposes

detailed recreational trail purpose	distance	Travel time	Paths	Passenger kilometers
	km	Min.	%	%
DETAILED RECREATIONAL TRAILS TOTAL				
on foot	3	52	39	8
Bicycle	9	48	13	7
BELT	33	68	10	21
Private vehicle journey	27	43	27	43
Private vehicle ride	31	47	11	21
general city/shopping stroll	14	72	100	100
on foot	4	80	42	13
Bicycle	8	93	8	5
BELT	20	65	20	28
Private vehicle journey	24	65	19	33
Private vehicle ride	29	54	11	22
Actively participating in sports	10	39	100	100
on foot	5	49	27	12
Bicycle	11	46	24	27
BELT	12	48	6	7
Private vehicle journey	12	26	34	40
Private vehicle ride	17	32	8	14
walk, drive	7	64	100	100
on foot	4	66	78	48
Bicycle	13	78	8	15
BELT	21	61	2	7
Private vehicle journey	20	49	7	21
Private vehicle ride	14	46	5	9
dog walking	4	45	100	100
on foot	3	46	87	69
Bicycle	5	39	2	2
BELT	7	67	0	1
Private vehicle journey	10	35	9	23
Private vehicle ride	12	29	2	5

CAWI/CATI; 14 years and older; deviation from 100%: rounding differences; purposes not grouped by main route purpose

Figure 20



Figures in percent; 14 years and older; purposes not grouped by main route purpose; only recreational routes (categories grouped together)

MiD 2023 | Results Report | Source: MiD 2023

Table 19

Traffic volume by detailed trip purpose (completion) 2017 and 2023

	2017	2023
	%	%
General city/shopping stroll; services (hairdresser, shoemaker, etc.)	4	10
	7	8
Doctor's visit, other medical services, government office, bank, post office, ATM	23	15
	9	7
private errands for other people, other errands, care for family members, acquaintances, visiting/	6	8
meeting friends, relatives, acquaintances, restaurant, pub, lunch, etc.	17	17
	2	4
	15	16
	4	4
dog walking	3	4
Kirche, Friedhof	5	3
Accompanying children (e.g., playground, etc.) other meetings (e.g., courses, hobbies, clubs, other groups)	1	1
	3	3

MiD2023 and MiD 20217; CAWI/CATI; 14 years and older; purposes not grouped by main route purpose; only service routes

MiD 2023 | Results Report | Source: MiD 2023

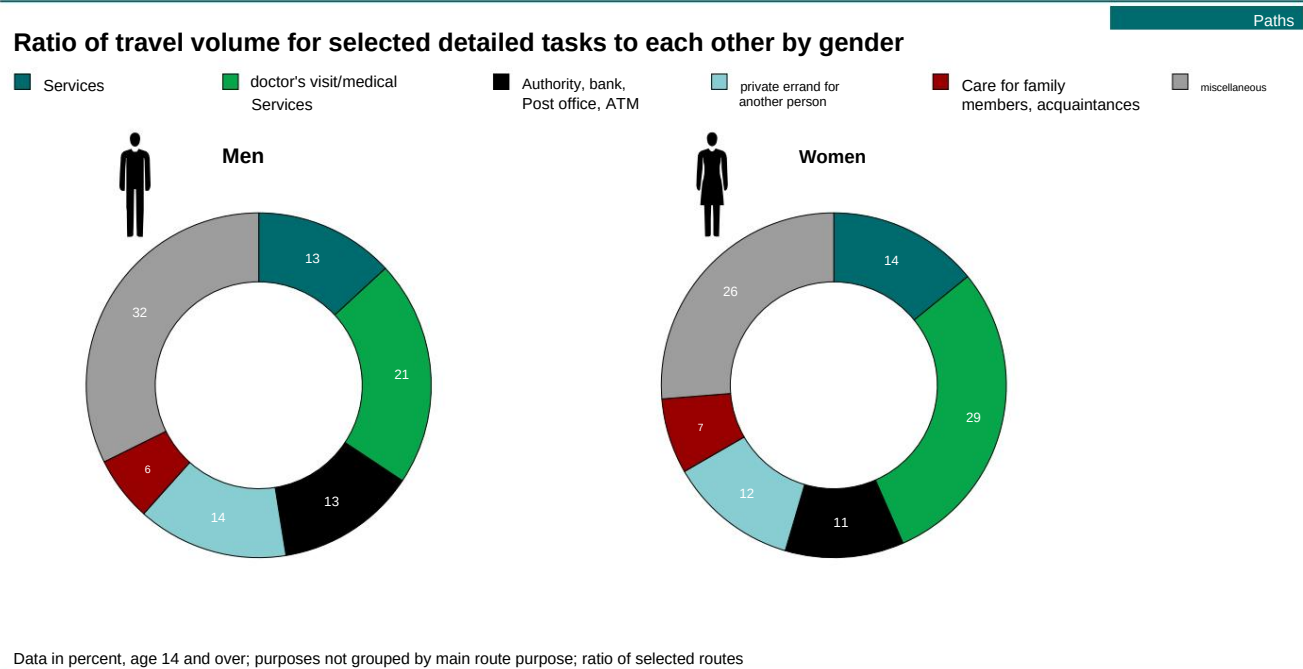
Women make more trips for errands than men (Figure 21). At the same time, the relevance of the primary purpose of these trips differs. For example, there are significant gender differences in the importance of doctor's appointments. A higher relative proportion among more errands means that women make trips to medical services considerably more often than men. Relatively speaking, however, men visit government offices, banks, and the post office more frequently and run errands for others.

Although it may have changed only slightly in proportion, structural changes are evident beneath its surface. Here, too, working from home plays a role, as do shifts in the proportions of different employment sectors and various other changes. One finding that has been valid in the MiD (Mobility in Germany) since 2002, and of course before, remains: the purpose structure is diverse, and the travel patterns are correspondingly below average. It should also be noted that the choice of transport mode is less purpose-specific than one might expect, even if the proportions vary. All modes of transport play a role in all purposes, from walking and cycling to public transport and private motorized transport.

Travel purposes accounted for – more variable than expected?

A detailed look not only at the main purposes of travel, but also at the specific data collected in the areas of shopping, errands, and leisure reveals interesting sub-findings beneath the surface of the broader purposes. The proportions of these overarching categories change only slightly, yet still suggest digital influences – from working from home and online errands, potentially resulting in more leisure time. However, this could also lead to newly utilized free time and activities, or different routines and patterns, as the specific purposes sometimes indicate. And although the aggregated work-related traffic

Figure 21



6. Automobility: growing number of vehicles, growing availability, but declining usage.

For decades, cars have been the dominant means of everyday transport in Germany. They account for approximately three-quarters of all passenger transport in daily traffic. As a result of this high usage, combined with the large amount of land required by car traffic, the transport infrastructure is significantly shaped by the demands of private motorized transport. Furthermore, private household cars represent one of the key challenges with regard to the decarbonization of transport.

As this chapter shows, the dominance of car traffic will continue in 2023, although developments with significant consequences are emerging. On the one hand, long-standing trends impacting car use, such as the aging of society, will persist. On the other hand, changes in car usage compared to 2017 will be evident in 2023, affecting, for example, the utilization of infrastructure for moving and parked traffic. Thus, the transformation of car use in recent years has implications that naturally affect transport planning and policy, but also extend to broader societal and industrial policy issues.

Fewer households without any room types Cars and more households with multiple cars

After decades of strong growth in motorization since the economic miracle era, the proportion of households without a car in Germany remained relatively stable at around one-fifth for a long time, at least since the 1990s. The first three MiD surveys, conducted in 2002, 2008, and 2017, also recorded a car-free household share of approximately 23 percent each year.

During these years, there were roughly the same number of households without a car as households with several cars in Germany. This ratio changed significantly for the first time in 2023

(Figure 22): the proportion of households without a car fell to just under one-fifth. Households with several cars, on the other hand, now make up almost 30 percent of all households. While, as expected, a significantly larger proportion of households without a car still lives in metropolitan areas and larger cities than in small towns and rural regions, the trend toward fewer households without a car and more households with several cars is evident in all types of areas.

Motorization and vehicle numbers have increased significantly – especially in medium-sized countries. cities

One key indicator that summarizes the development of carless and multi-motorized households is motorization, i.e., the number of cars per 1,000 people (Table 20). The MiD (Mobility in Germany) records cars available in private households regardless of registration type. This includes both privately registered vehicles (2023: 91 percent of cars in private households) and commercially registered vehicles used as company cars in private households (2023: 9 percent of cars in private households).

The number of cars per 1,000 people has increased significantly in all spatial types since 2008, by an average of 12 percent across Germany. Combined with the varying population trends in these spatial types, this results in the average change in the respective number of cars in cities and municipalities within these spatial categories. Since 2008, the number of cars has increased in all spatial types except for the most rural regions, where the population has declined. Even in metropolitan areas, where a different trend is often assumed, the number of cars has grown by 12 percent since 2008. The increase is particularly pronounced in medium-sized cities, where the average number of cars in 2023 was between a quarter and a third larger than it had been 15 years earlier.

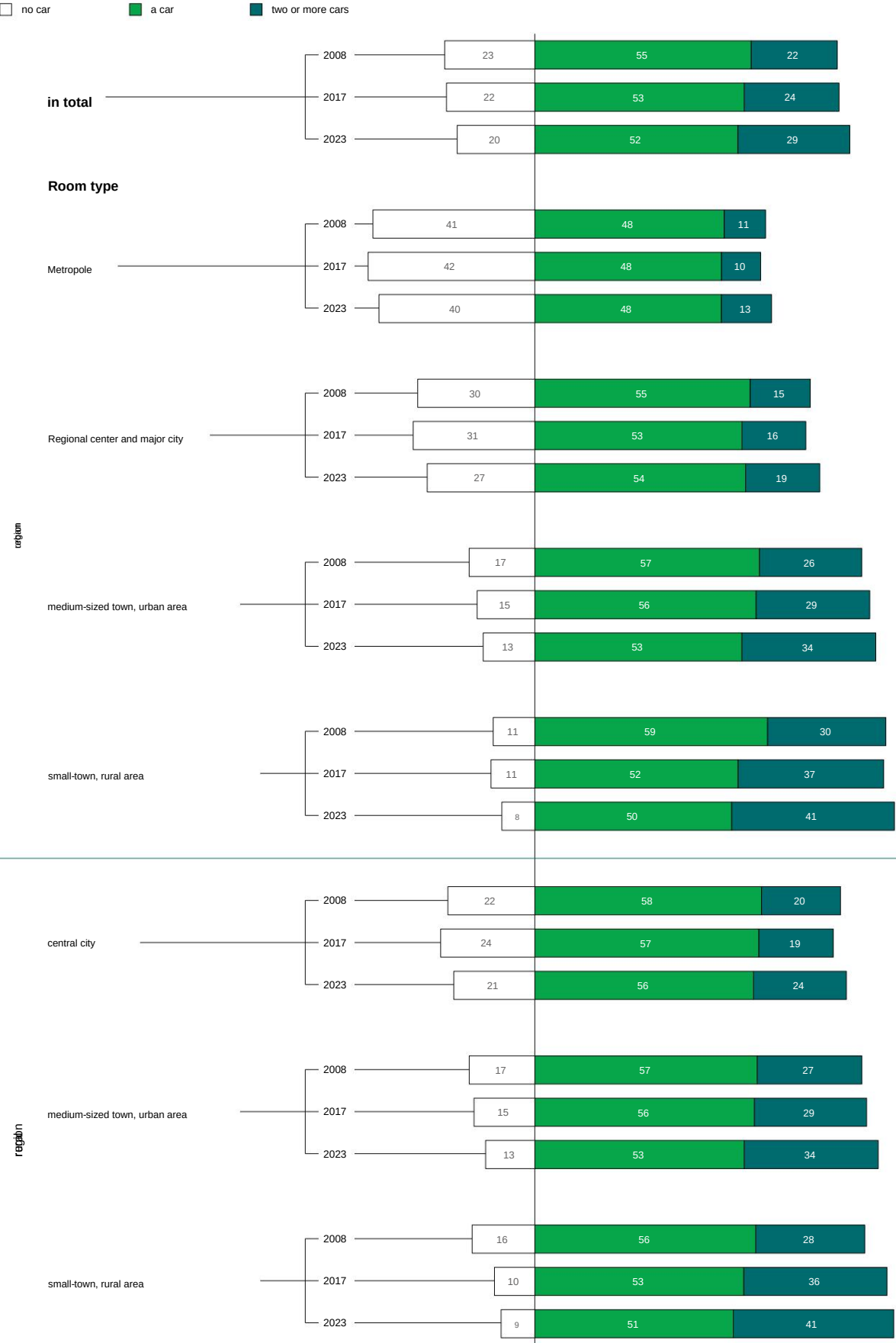
The proportion of people living in households without a car has changed little.

Despite this dynamic development in motorization, the proportion of people living in households with cars has remained relatively stable at around 85 percent since the first MiD in 2002. Demographic trends in terms of age structure and household size overlap with motorization trends across age groups. Figure 23 shows the proportion of people living in households with cars by age. According to this figure, over 90 percent of young children live in households with cars. In the age group where young people typically leave their parents' home, the proportion of those living in a household with a car decreases. According to MiD 2023, about a quarter of young adults in their early twenties live in a household without a car.

Figure 22

Car availability in households by room type 2008 to 2023

households



Figures in percent

While the proportion of young people without a car has noticeably increased over the last 20 years, this has not yet led to a change in the proportion of people in households with a car from their mid-30s onward (see also the chapter on Millennials). From the age of 35 to about 65, 9 out of 10 people in Germany again live in a household with a car. Only at an older age does this proportion of people with a car decrease. It is noteworthy that the cohort effects on car ownership, clearly evident for the years 2002 to 2017, are not reflected in the data.

The trend of senior citizens not continuing in 2023: In 2008 and 2017, the proportion of people aged 60 and over with a car was significantly higher than in the preceding MiD (Middle Ages). This was due to the fact that subsequent cohorts who had lived with cars gradually replaced earlier cohorts, among whom many women in particular had never driven a car. The fact that the proportion of senior citizens in households with a car hardly differed from 2017 in 2023 indicates that these cohort effects are fading. Instead,

Table 20

Car stock and motorization (cars per 1,000 people) 2008 to 2023 by room type in private households

	2008	2017	2023	change 2023 vs. 2008
NUMBER OF CARS	My.	My.	My.	%
IN TOTAL	40,0	43,3	46,4	16
ROOM TYPE				
urban region				
Metropole	5,1	5,5	5,7	12
Regional center and major city	4,8	5,5	6,0	25
medium-sized town, urban area	9,6	12,0	12,9	35
small-town, rural area	2,6	3,3	3,4	31
rural region				
central city	2,5	2,5	2,8	11
medium-sized town, urban area	7,1	6,5	7,3	3
small-town, rural area	8,4	8,0	8,3	0
	2008	2017	2023	change 2023 vs. 2008
CARS PER 1,000 PEOPLE	Number	Number	Number	%
IN TOTAL	494	527	556	12
ROOM TYPE				
urban region				
Metropole	380	372	400	5
Regional center and major city	442	455	500	13
medium-sized town, urban area	528	584	598	13
small-town, rural area	564	649	648	15
rural region				
central city	500	527	547	9
medium-sized town, urban area	528	556	597	13
small-town, rural area	537	607	631	18

Increasingly, health reasons are becoming the dominant cause for the decline in car ownership among the elderly (see also the chapter section on reasons for not having a car in the household).

Percentage of adults with their own car in all Age groups increased

Figure 23 also shows, broken down by age, the percentage of people living in households with at least as many cars as adults. For adults, this percentage can be interpreted as the proportion of people with their own car. This proportion continued to increase noticeably for all age groups in 2023 compared to 2017, especially for people aged 50 and over. This trend, in which more and more adults own a car that they hardly ever have to share with other household members, has continued for many age groups since 2002 and corresponds to the increase in the proportion of households with multiple cars shown above.

Less private motor vehicle travel per person per day – with the exception of senior citizens

Unlike the availability of cars, the driving performance per person per day in 2023 is lower than in 2017 (Figure 24).

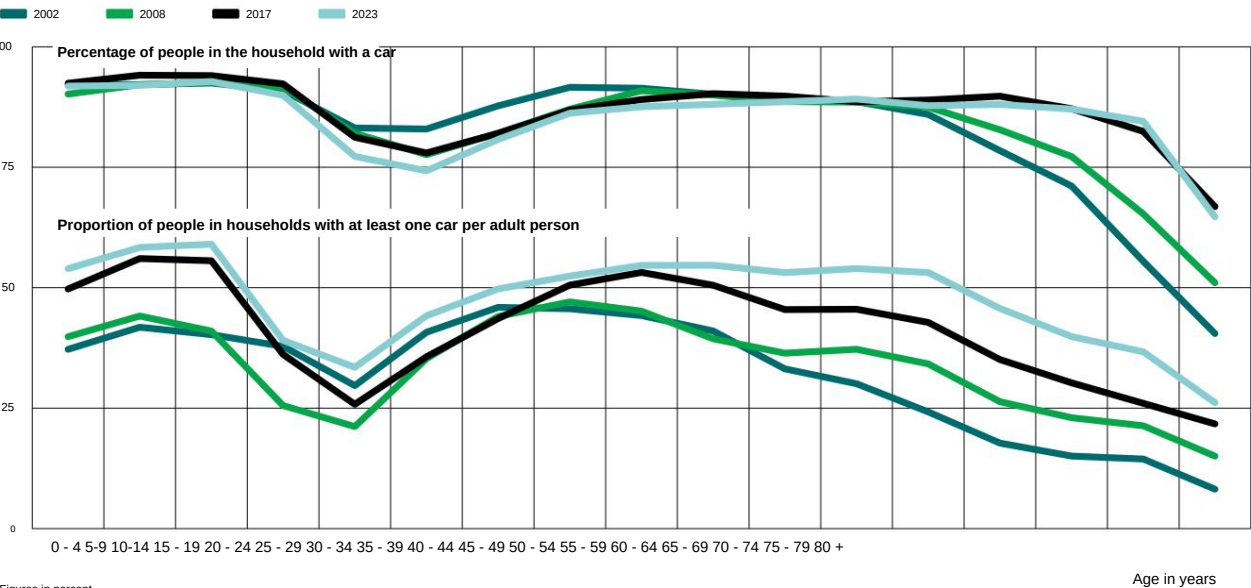
This applies to the adult population as a whole as well as to most age groups. Factors such as the end of the COVID-19 pandemic with still high rates of working from home, or economic influences, are likely to play an important role. However, given the continued increase in motorization, which essentially creates the structural prerequisites for everyday car use, the lower per capita mileage in 2023 should be interpreted with caution. It does not yet indicate a reversal of long-term trends.

An important long-term trend with regard to the However, the trend in per capita mileage continued into 2023: For people aged 60 and over, the mileage per capita increased. Unlike other age groups, the difference in driving performance between older people and

adult middle age consisted, further back-gone.

Figure 23

Percentage of people in households with different car equipment by age, 2002 to 2023



Private motor vehicle driving performance is primarily driven by men and increasingly by people over 50.

The combination of an aging population, the increasing number of cars owned by senior citizens, and their comparatively higher mileage compared to other age groups is increasingly shaping the kilometers traveled by car in Germany. Figure 25 illustrates this development: The left side of the figure shows, for orientation, the familiar distribution of the population by age and gender in 2002 and 2023. The right side of the figure shows how private motorized vehicle mileage in Germany was distributed among individuals by age and gender in these years.

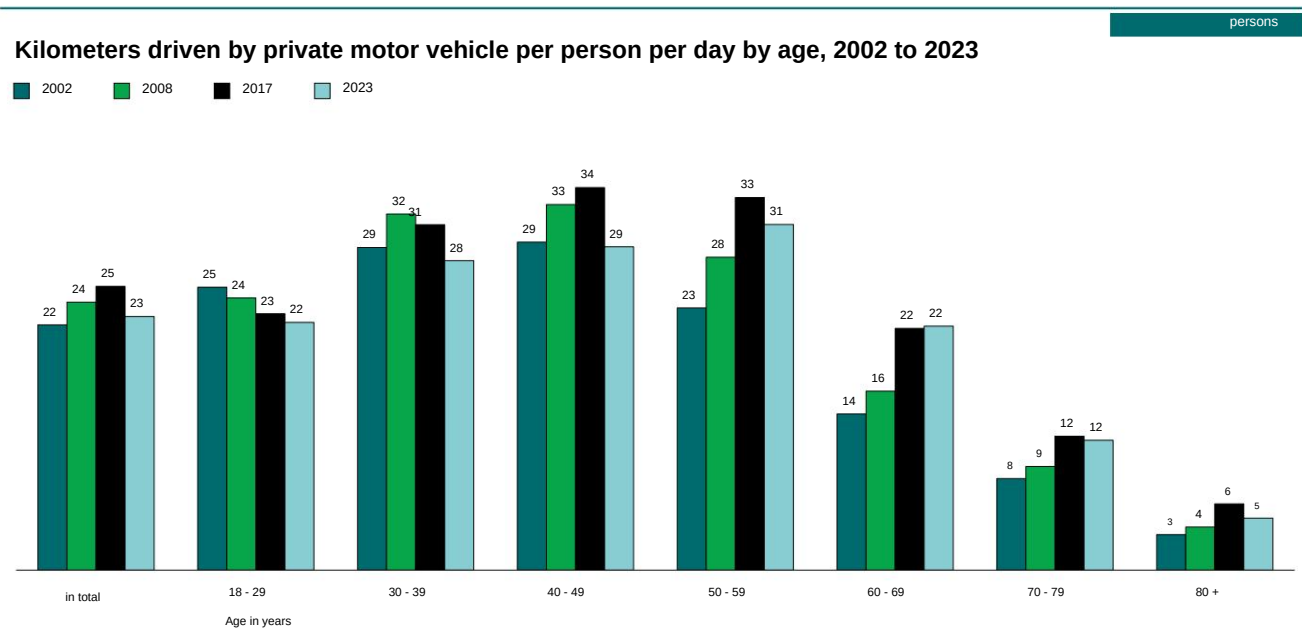
The initial finding is the significant male preponderance: men were behind the wheel for approximately two-thirds of private motorized transport (PMT) journeys in both 2002 and 2023, with no significant change between these years. More pronounced changes are evident with regard to age: in 2002, individuals aged 50 and over accounted for just under a third of PMT journeys, while in 2023 their share had risen to almost half. The increasing importance of people over 70 in terms of driving performance is even more striking: their share doubled between 2002 and 2023, rising from 4 percent to 8 percent.

Millennials: From their mid-30s onwards, their car features are not significantly different from those of previous generations. with

In recent years, the development of car ownership has been a particular focus on the so-called "Millennials." This term refers to the generation born roughly between 1980 and the mid-1990s, meaning those who came of age between the late 1990s and the early 2010s. This generation exhibited lower car ownership rates in their early adulthood than previous generations, as evidenced in Germany, for example, by the registration figures by age published by the Federal Motor Transport Authority, which declined significantly for young adults in the 2000s. This sparked a discussion about a possible reversal of the long-term trend toward car ownership among young adults and the potential reasons for this shift.

With the MiD 2023 survey, data is now available that allows for the analysis of car ownership among millennials aged approximately 30 to their mid-40s. The following section examines the car ownership (specifically, the number of cars per adult) of this group at different age levels. Car ownership data from 2023 is a more reliable indicator for this analysis.

Figure 24



Figures in percent; persons aged 18 and over; without regular work-related commutes

Since car use (e.g., private vehicle kilometers per person per day) has decreased for all age groups, possibly – see above – due to specific influencing factors that were effective in 2023. When evaluating car ownership, it should be noted that this characteristic is available at the household level in the MiD (Mental Income Data) and therefore cannot initially be assigned to specific age groups in mixed-age households. In order to nevertheless calculate the motorization of adults by age, vehicles within households were distributed proportionally among the adults.

For example, if a person in their mid-fifties lives with an adult child in their early twenties and they share a car, the car was attributed to each of them equally. In this way, using data from studies conducted between 2002 and 2023, the motorization rates of adults were calculated for different age groups, from before 1930 up to the 1990s.

In addition to the expected increase in motorization from generation to generation for older people as a result of the cohort effects described above, it shows

Figure 26 shows one thing above all: For the age of mid-20s, the motorization rate for those in the 1980s and

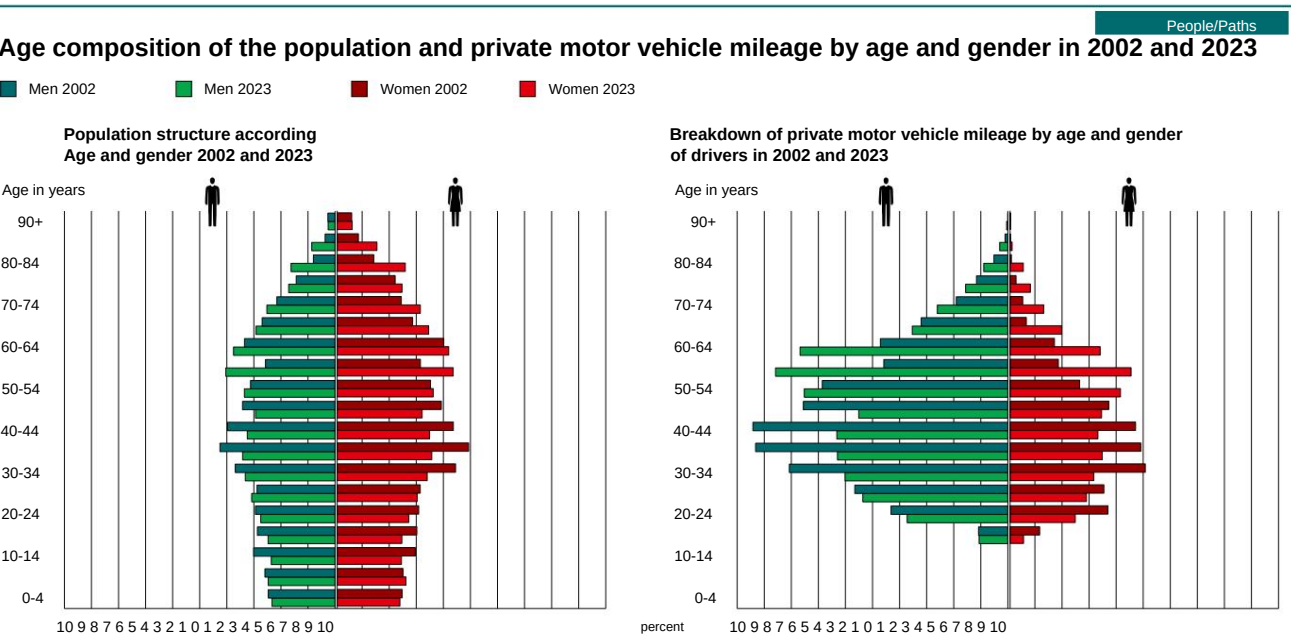
Those born in the 1990s were indeed below those born in the 1970s; however, with increasing age, the motorization of millennials then increased more rapidly, so that from about the mid-30s onwards there are no longer any significant differences between millennials and earlier generations.

Many people without a car in their own household can share cars.

Given the high level of car ownership in society as a whole, the question arises as to what extent people from households without their own car can share other vehicles. Figure 27 addresses this question, illustrating car availability and use for people with and without a driver's license, as well as for those with and without a car in their own household. Availability and use refer to both driving and riding along and are based on the respondents' self-assessment.

As expected, virtually all individuals with a driver's license from households with a car have at least occasional access to a car and use these vehicles at least monthly. For individuals without a driver's license from households with a car, car availability and use apparently refer to...

Figure 25



Figures in percent

Carpooling. It turns out that the mobility options of this group are strongly influenced by car availability, presumably mostly within their own household: Three-quarters can share a car at least occasionally, and over half do so at least monthly. However, the widespread availability of cars in society as a whole is particularly evident among those with a driver's license but no car in their household: Two-thirds of them have occasional access to a car, and almost a third use a car at least monthly. Differences between urban and rural areas or between men and women are relatively small; only between age groups are there larger discrepancies. Young people without their own car can travel by car much more frequently than older people, often likely due to sharing their parents' car. Even over two-fifths of people who have neither a driver's license nor a car in their household can share a car at least occasionally. Overall, these findings demonstrate how strongly the widespread availability of cars in society extends to those who, formally speaking, do not initially have access to one. Their mobility is also significantly influenced, at least by the option of using a car.

Affordability and health in certain situations

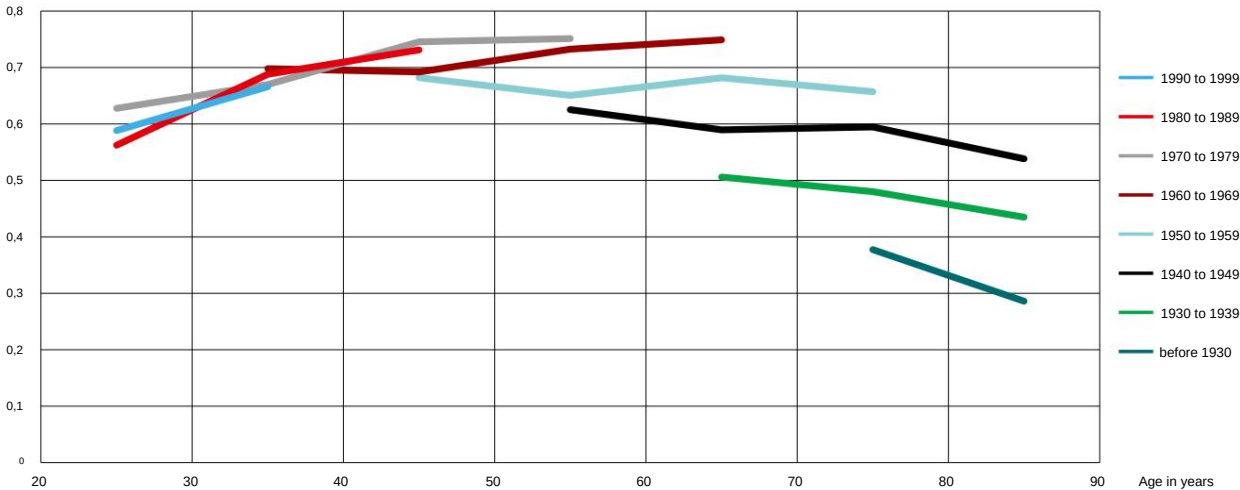
Groups cite important reasons for foregoing a car

Given the high overall car ownership rate in society, the question arises as to why households without a car choose to forgo one. The reasons for this were surveyed at the household level, allowing for multiple responses. Therefore, for Figure 28, the reasons for a household not owning a car were assigned to each individual within that household. This allows for a meaningful breakdown of the reasons for not owning a car by age.

The results show that roughly half of the people in households without a car have no major constraints preventing them from owning one. The most frequently cited reasons for not owning a car are "not needed" and "conscious decision to forgo one or cannot afford one" among all individuals in car-free households. Major constraints such as financial affordability or health reasons play a less significant role overall. However, financial affordability is the dominant factor in rural areas and for households with low or very low economic status.

Figure 26

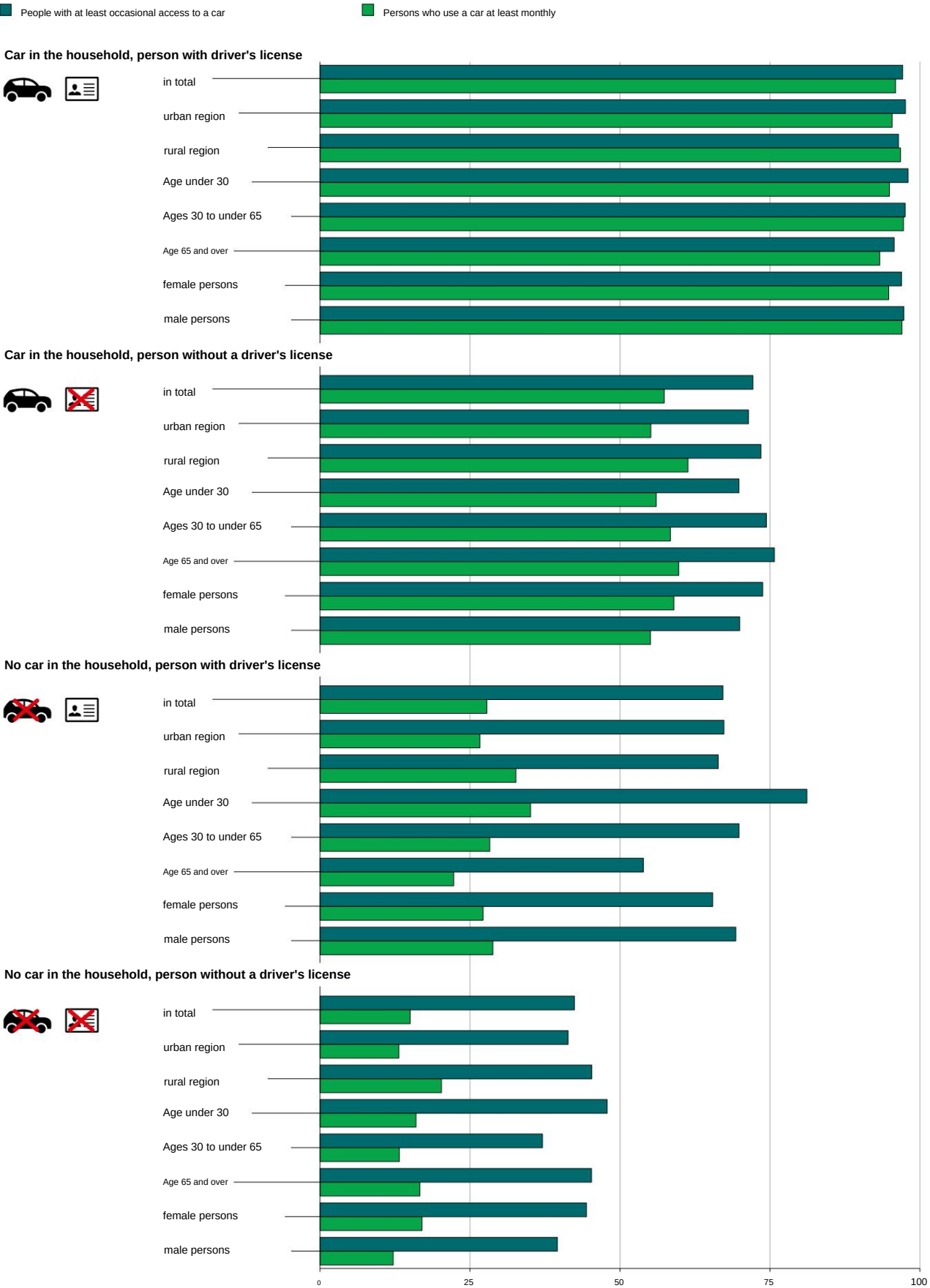
Cars per adult by age and birth year cohort



Figures in percent; persons aged 20 and over;
The different birth cohorts can be recorded across an age range of more than 21 years, lying between the MIDs 2002 and 2023, as illustrated by the following example: those born in 1970 were recorded at ages 32 to 51; those born in 1979 were recorded at ages 23 to 44. Thus, the birth cohort from 1970 to 1979 was recorded from the early 20s to the early 50s.

Figure 27

Population by car availability and use, for self-driving and ride-sharing respectively, 2023



Figures in percent, persons aged 14 and over

However, few reasons for foregoing a car shape the landscape in a particular group as much as health reasons for people aged 65 and over: For half of the people in car-free households in this age group, health plays a role. Here, it is worth recalling the significant decline in the proportion of elderly people in households with a car, which was shown in [Figure 23](#) and is more pronounced than would be expected based on cohort effects of the catching-up motorization in this age group. This suggests that these cohort effects are becoming less significant for the decline in car use with increasing age, while health-related reasons for abandoning the car are becoming increasingly important.

On average, cars are being used less and less each day.

In the MiD system, the detailed information at the route level allows for the allocation of personal journeys to the individual vehicles of the household, provided that the journeys were made with cars from the household. This provides a car dataset with information on the intensity of vehicle use in private households in Germany. The evaluation of relevant key indicators since 2008 shows that long-term trends continued in 2023 ([Table 21](#)): Almost half of the cars are not driven on an average day; in 2008, only a third of the cars remained parked during the day. The average number of trips per car per day has decreased significantly, as has the average mileage, which was approximately 26 km in 2023. (Extrapolating this daily value results in an annual mileage of approximately 10,000 km, which is less than what is estimated from other sources, such as "Traffic in Figures".)

The average annual mileage for passenger cars is approximately 12,500 km, which is known to be lower. This discrepancy is due to the underrepresentation of long-distance trips in the data collection, which primarily reflects everyday traffic. The average daily driving time in 2023, at 42 minutes, is also significantly lower than the figures for 2008 or 2017.

Given the backdrop of increasing motorization over many years, only moderate growth in per capita mileage up to 2017, and even a decline in per capita mileage in 2023, the decrease in these key figures is a logical consequence. The trend described above towards private vehicle ownership for very large proportions of the population leads to less vehicle sharing and reduced usage intensity.

This individualization of car ownership is also reflected in the continuing decline in occupancy rates.

Significantly fewer cars were on the road during peak hours in 2023 than in 2017.

The changing usage patterns of cars are evident not only in the key daily usage indicators, but also in a different utilization of the road network over time. [Figure 29](#)

Firstly, it shows what proportion of cars from private households are on the roads during the different 168 hours of a week.

However, since the number of vehicles has increased, the figure also illustrates the absolute number of vehicles on the road at these times. Due to the decreased usage intensity of cars, this demand curve is...

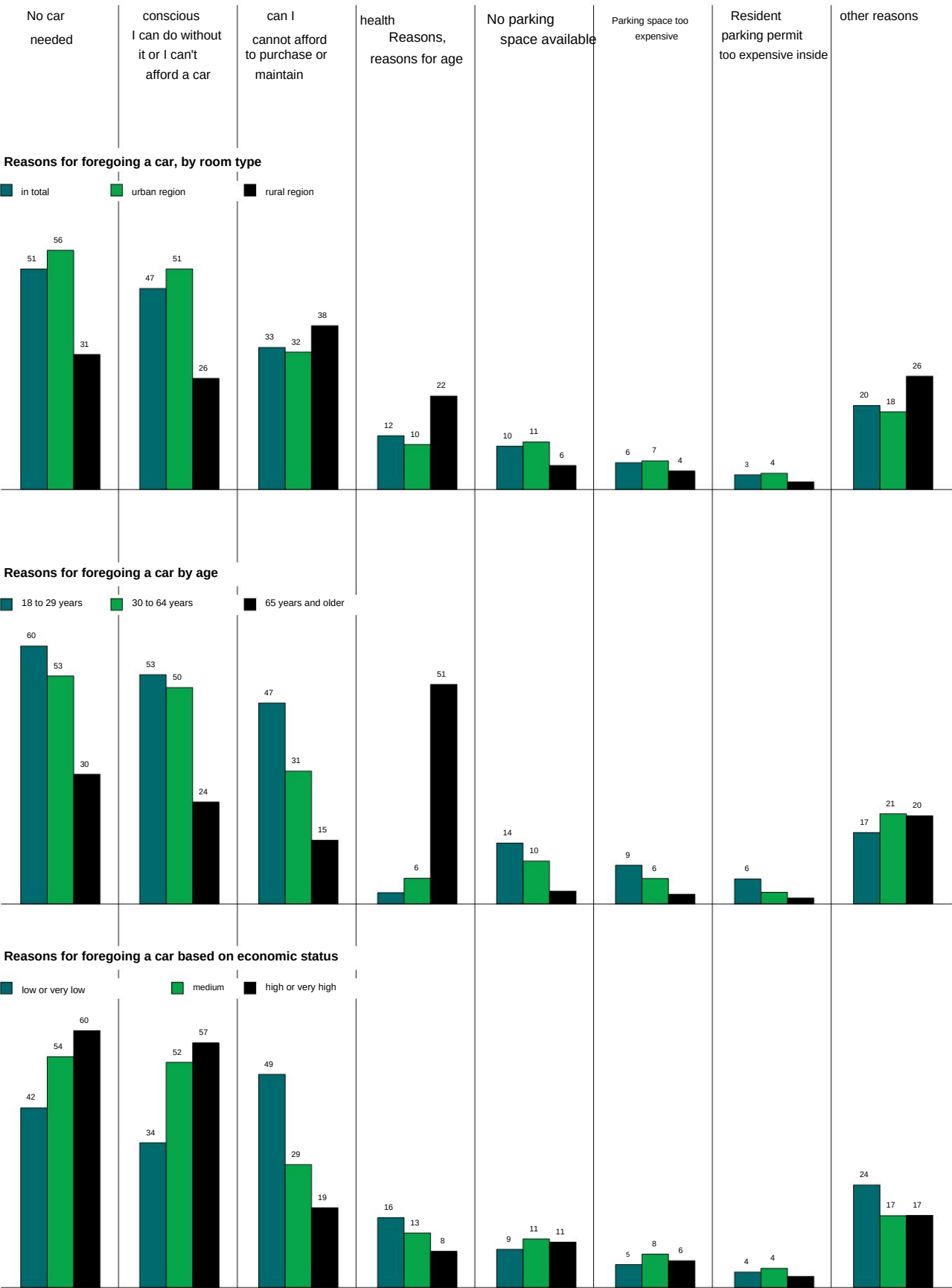
Table 21

Key figures for car usage per day				
	2008	2017	2023	
Percentage of cars with journeys on the reference date	67	59	54 %	
Average number of trips per car per day	2,3	1,9	1.6	number.
Average mileage per car per day	31,3	30,2	26,2	km
Average travel time per car per day	49	46	42	Number
Distance-weighted car occupancy rate (excluding regular work commutes)	1,54	1,47	1,46	-

Figure 28

Proportion of people from households without a car, by reasons for the household's car-free status

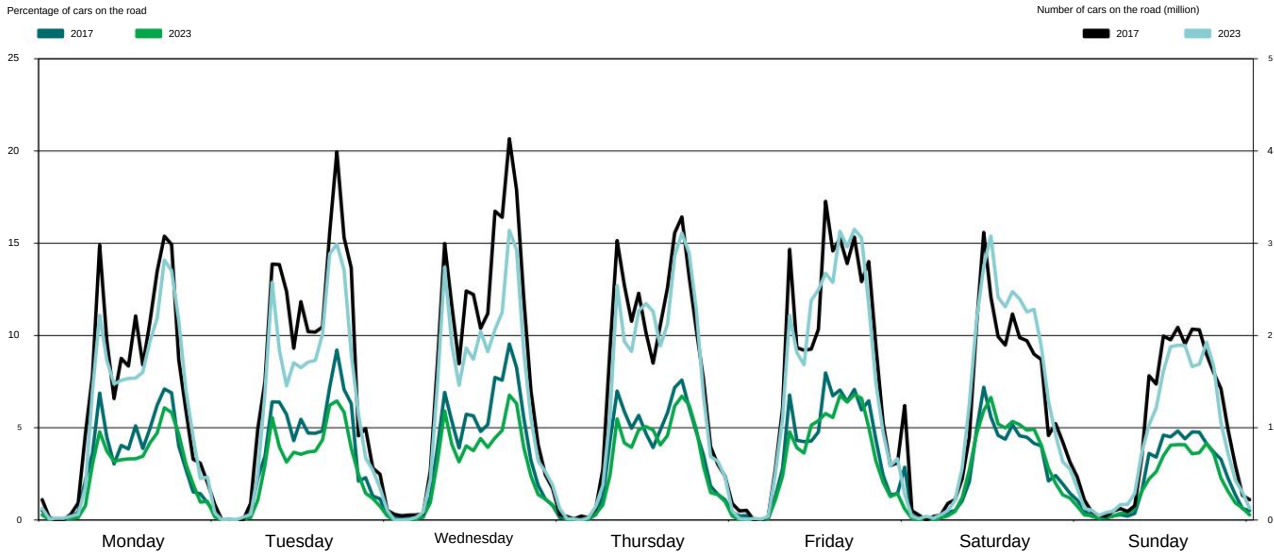
persons



Figures in percent, share of people from households without a car, multiple responses possible

Figure 29

Cars on the road during the week in 2017 and 2023



MiD 2023 | Results Report | Source: MiD 2023

2023 was mostly below the trend line of 2017. This is particularly evident during peak hours of car traffic demand on Tuesday and Wednesday afternoons. At these times, the total number of cars on the road is almost a quarter lower than in 2017.

A plausible reason for these declines is less commuting, which significantly impacts peak hours and is likely to have decreased compared to 2017, perhaps as a result of working from home. In contrast to the relief during peak hours, especially on Tuesdays and Wednesdays, demand on Friday afternoons in 2023 was slightly higher than in 2017. For the overall network, this suggests a somewhat more even and therefore more efficient utilization of the road infrastructure. However, this can vary in individual situations, particularly because commercial and freight traffic still overshadows the demand from privately owned cars shown here.

Table 22

Times per car per day, traveling and parked at different locations

	2017	2023
hours per day	h:min	h:min
IN TOTAL	24:00	24:00
Drive	00:46	00:42
Parking at work	01:42	01:13
Parking while shopping	00:27	00:20
Parking at home	20:15	21:06
Parking at other locations	00:50	00:39

MiD 2023 | Results Report | Source: MiD 2023

In 2023, cars were parked at home for longer than before.

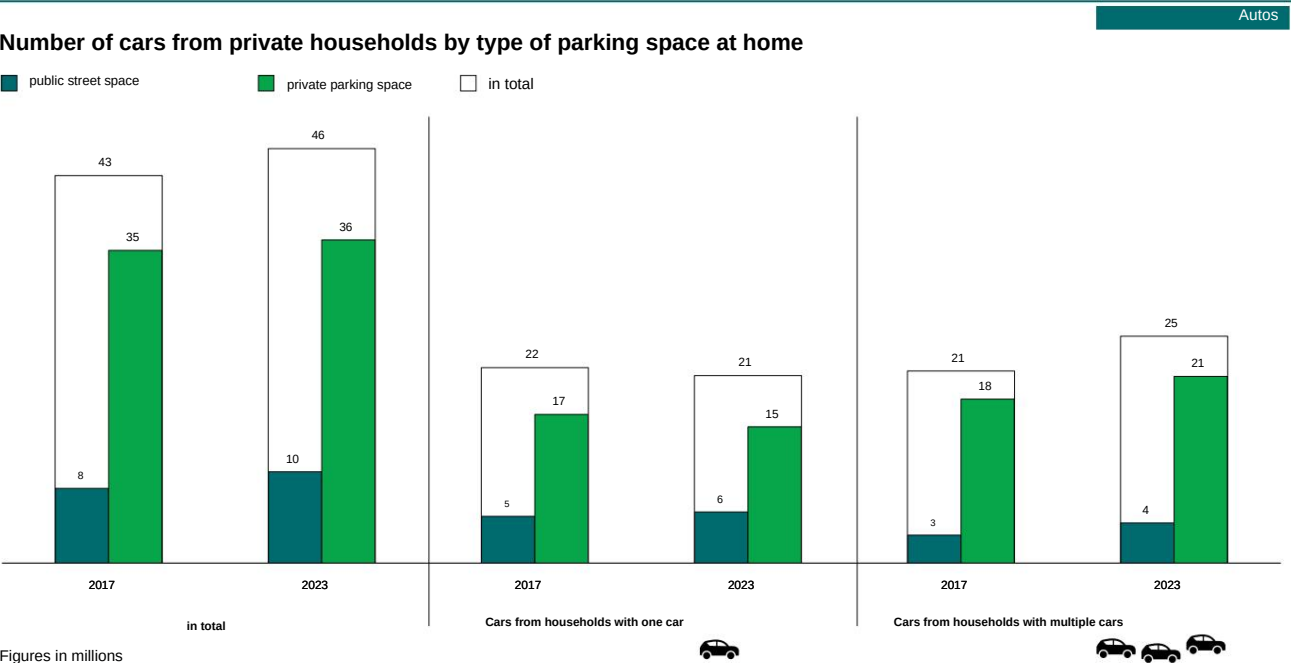
Shorter travel times per car per day also mean that cars are parked for longer periods. More importantly, fewer trips per day mean that in 2023 cars were parked at home for longer and less frequently elsewhere than in 2017 (Table 22): Parking times at home increased by approximately 5 percent to about 21 hours. In contrast, parking times per car per day at work decreased significantly, presumably again as a result of the increased prevalence of working from home. Since parking is also linked to considerable infrastructure requirements for stationary traffic, such shifts in parking space needs are of important importance for traffic planning, for example, for parking space regulations.

While 8.4 million out of 43.3 million cars (19 percent) were parked at home in public street space in 2023, this figure rose to 10.2 million out of 46.3 million (23 percent) – an increase in absolute numbers of over one-fifth. The number of vehicles parked in public street space belonging to households with multiple cars has increased particularly sharply. Thus, while working from home and decreasing usage intensity have provided some relief to the traffic infrastructure for through traffic, the MiD results regarding parking suggest an increase in conflicts over space and traffic planning challenges.

In 2023, a good fifth more cars were parked at home in public street space.

Between 2017 and 2023, however, it wasn't just the parking durations of privately owned cars that changed. There was also a shift in the type of parking space recorded by the MiD for home parking (Figure 30): the proportion and absolute number of vehicles parked in public street space increased. During 2017

Figure 30



7. Electromobility – electric cars

Electrifying the vehicle fleet is a key prerequisite for decarbonizing passenger transport. Furthermore, electromobility is a highly relevant industrial policy issue for Germany. This chapter deals with the development of electromobility in relation to cars in German private households (electromobility in the form of pedelecs as part of cycling is covered in Chapter 10). For the successful market ramp-up of electromobility, user acceptance and the integration of electric cars into people's everyday lives and into the existing structures of the transport system, e.g., with regard to parking spaces and charging facilities, are fundamental prerequisites.

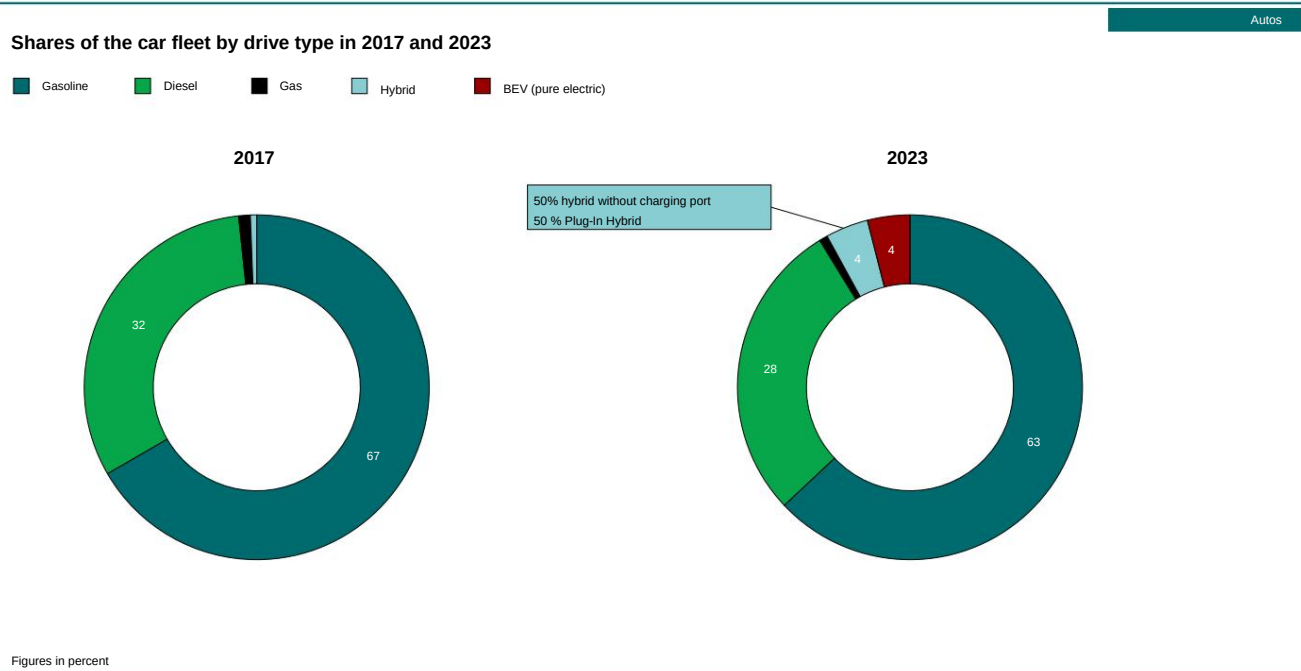
As with the MiD 2017, the MiD 2023 also collected information on the drive systems of the vehicles used. This allows for analyses of electric vehicle usage, their integration into everyday life, and the framework conditions, such as the household context or the spatial structures in which electric vehicles are used. In 2017, the share of electric vehicles was still extremely low, users were early adopters, the usage patterns of electric vehicles differed significantly from those of conventional cars, and the everyday suitability of electromobility could hardly be assessed due to these circumstances. This chapter

The study therefore focuses primarily on the question of how closely electric cars have become similar to the rest of the car fleet in terms of usage and important contextual factors. As it turns out, this is the case with regard to many important indicators, even if differences still exist.

The number of electric and hybrid cars has increased more than tenfold since 2017.

The number of electric and hybrid cars in private households, as determined by MiD in 2017, was approximately... 300,000 vehicles. By 2023, this number of vehicles in these categories had grown to approximately 3.8 million. The 2017 survey did not differentiate between hybrid vehicles without an onboard charging port and hybrid vehicles with a charging port for external charging, so-called plug-in hybrids. The MiD 2023 (Mobility Index 2023) differentiates between these vehicles, so that Figure 31 shows the vehicle shares both as compared to 2017 and according to the differentiated data collection in 2023. While gasoline and diesel vehicles made up virtually the entire vehicle fleet in 2017, these drive types still accounted for a good 90 percent of cars in the survey period of the MiD 2023. Just over 8 percent of the cars in the projected MiD 2023 were purely electric vehicles, i.e., battery-electric vehicles.

Figure 31



Electric vehicles (BEVs) or hybrid vehicles.

The following analyses focus on comparisons with 2017, which is why many of the presentations are based on the 2017 MiD powertrain classification, where the various groups of hybrid vehicles are summarized again. The interpretations of these comparisons primarily refer to pure electric vehicles (BEVs), because the hybrid vehicle group includes both those that can be charged externally and those that do not have a charging port, making hybrid vehicles a very heterogeneous group.

90 percent of electric cars in private ownership
Households only took to the streets in 2020 or later.

In Germany, passenger cars typically remain in the vehicle fleet for between 15 and 20 years before they are, for example, replaced. For example, through scrapping, vehicles disappear from the fleet. This results in a large number of relatively old vehicles in the car fleet. The situation is quite different for electric vehicles. Since a significant market ramp-up of electromobility only began a few years ago, 90 percent of the electric vehicles in the MiD 2023 were manufactured in 2020 or later (Figure 32). It should also be noted that the survey period for the MiD extends to

the period lasted into 2024, so vehicles first registered in 2024 are also included in the sample. For the sake of simplicity, the results of MiD 2023 are referred to here as being for the year 2023.

Typically, car usage patterns are heavily dependent on the vehicle's age; for example, the average annual mileage of newer vehicles is significantly higher than that of older vehicles. A comparison of the predominantly very new electric vehicles with the overall vehicle fleet, which is on average considerably older, would therefore be misleading. For this reason, the following comparisons include the category "petrol or diesel, young" alongside the different drive types. This vehicle group comprises petrol or diesel-powered vehicles with a first registration no earlier than three years before the respective Minimum Indication (MiD) date (i.e., vehicles first registered from 2014 onwards for the MiD 2017, and from 2020 onwards for the MiD 2023).

These young vehicles with conventional drive systems are significantly more comparable to electric vehicles in terms of age structure. A similarity in usage patterns and contextual factors between electric vehicles and this vehicle group suggests that electric cars are now being used similarly to the previously dominant combustion engine vehicles.

Figure 32



Proportionally, the same number of new private cars were among young BEVs as among young combustion engine vehicles.

Overall, slightly less than 10 percent of vehicles used by private households are not privately owned but rather company cars registered for commercial use (Figure 33). In 2017, the proportion of company cars among purely electric vehicles was significantly above average at over 50 percent. At that time, the pioneers of electric cars essentially found their way into everyday use in private households via their initial use as company cars. This changed considerably by 2023. Currently, less than a third of electric cars in private households are company cars, meaning that electromobility has increased significantly among privately owned vehicles. There is no longer any significant difference in the company car rate between electric cars and newer cars with conventional drive systems.

While the average annual mileage (estimated by the respondents) was still noticeably below the benchmark for the existing fleet and even lower than that of the newer vehicles in the comparison, their mileage has since increased significantly and become more similar to that of the comparison vehicles. In addition to total mileage, the distance of individual trips is of interest, given the still clearly limited range of electric cars. In 2017, the differences between electric and other cars were particularly pronounced with regard to longer trips of 30 km or more. These were undertaken less frequently with electric cars – an indication that range anxiety was prevalent. In 2023, this proportion of trips over 30 km for electric cars, at 13 percent, is even higher than the benchmark for the entire fleet (12 percent) and only slightly below the figure for the comparison group of newer vehicles (15 percent).

Significant convergence in the use of electric cars and other cars since 2017

In addition to the harmonization of registration types as private or company cars, electric cars and the rest of the vehicle fleet have also become significantly more similar with regard to key usage indicators (Table 23). While electric cars in 2017 were in terms of the

Electric cars are still mostly just one of several cars in the household

Another indicator of how much drivers trust electric vehicles for all purposes is how many other vehicles are present in the household. If there are other vehicles in the household besides the electric car, this may indicate a higher risk of electric vehicle failure.

Figure 33

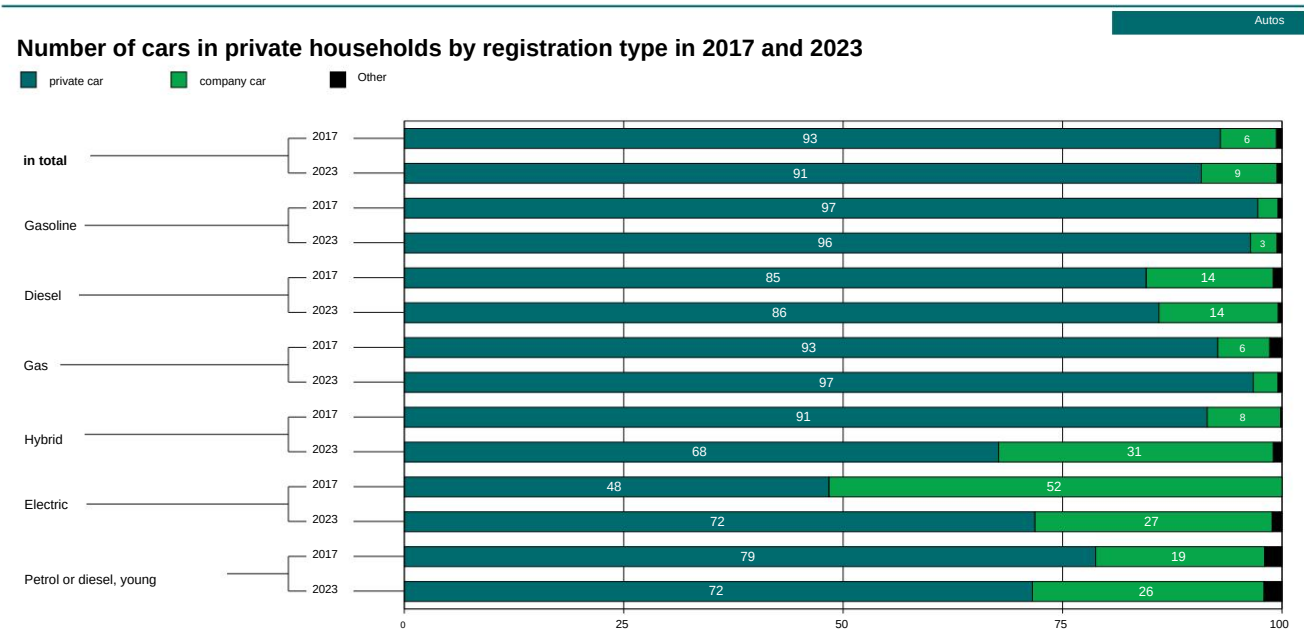


Table 23

Key indicators of car usage with different drive systems in 2017 and 2023

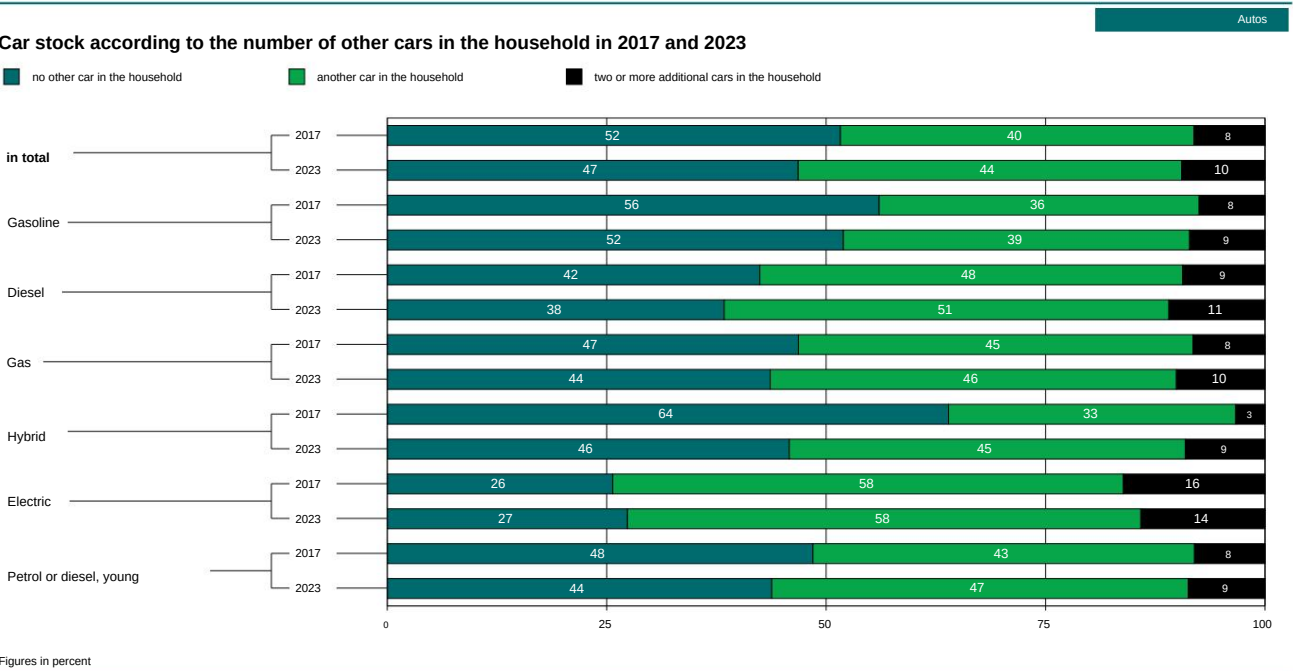
	Average distance of individual journeys		Percentage of individual journeys over 30 km		Average estimated Annual mileage	
	2017	2023	2017	2023	2017	2023
	km	km	%	%	km	km
IN TOTAL	15	17	11	12	14.700	13.700
Gasoline	13	14	8	9	11.800	11.400
Diesel	20	21	15	16	20.600	18.500
Gas	19	20	19	17	19.000	15.600
Hybrid	11	20	9	15	14.500	15.900
Electric	12	17	3	13	13.100	14.600
Petrol or diesel, young	18	21	12	15	17.800	16.800

MiD 2023 | Results Report | Source: MiD 2023

For long journeys, such as on vacation, other vehicles may be used. In this respect, only a moderate convergence took place between 2017 and 2023 (Figure 34): In 2017, approximately half of the cars in the total vehicle fleet and among the newer vehicles used for comparison were the only vehicle in the household. These proportions reflect the increasing prevalence of multiple vehicles in households.

Overall, sales have declined slightly. However, the trend for electric cars has been somewhat different. However, only about a quarter of electric cars are currently the sole vehicle in a household. Electromobility therefore remains primarily a phenomenon of households with multiple cars.

Figure 34



There are hardly any differences left in the distribution of Cars on space types between electric and other drives

In the early stages of the electric mobility rollout, electric cars were better suited to the needs of drivers outside of major cities: the high fixed costs but lower operating costs were particularly advantageous given the high mileage driven in these areas. Urban populations use cars less for everyday errands but more for long-distance journeys, so range limitations were more of a hurdle for drivers in metropolitan areas. Finally, large cities are more prone to street parking – people who park their cars on the street overnight – for whom charging infrastructure can pose a greater problem than for households with their own parking space. Accordingly, in 2017, electric cars were proportionally less common in metropolitan urban areas than other drive types (Figure 35). This has changed, and by 2023, there are hardly any differences between the different spatial types with regard to the distribution of electric cars. These are indicators that some factors that hindered the market ramp-up of electromobility in metropolitan areas a few years ago, such as range and access to charging infrastructure, are at least now less of an issue.

are less relevant. The following analyses provide further insights regarding parking spaces and charging infrastructure.

Convergence of electric vehicles with the rest of the vehicle fleet in terms of parking space type at home

Having a private parking space at home often provides the (need to be created) opportunity to charge regularly. This leads to the understandable expectation that electromobility is significantly more attractive for households with a private parking space than for those without. Accordingly, in 2017 only 8 percent of electric vehicles were parked at home in public street space – about half as many as in the comparison group of newer combustion engine vehicles. Overall, the proportion of cars parked at home in public street space has increased slightly, but particularly significantly for electric vehicles, of which 12 percent were parked in public street space in 2023. This does not yet match the 19 percent of electric vehicles parked on the street in the comparison group of newer vehicles, but it represents a significant convergence.

It shows that electromobility is increasingly becoming an accepted option even for people without their own parking space.

Figure 35

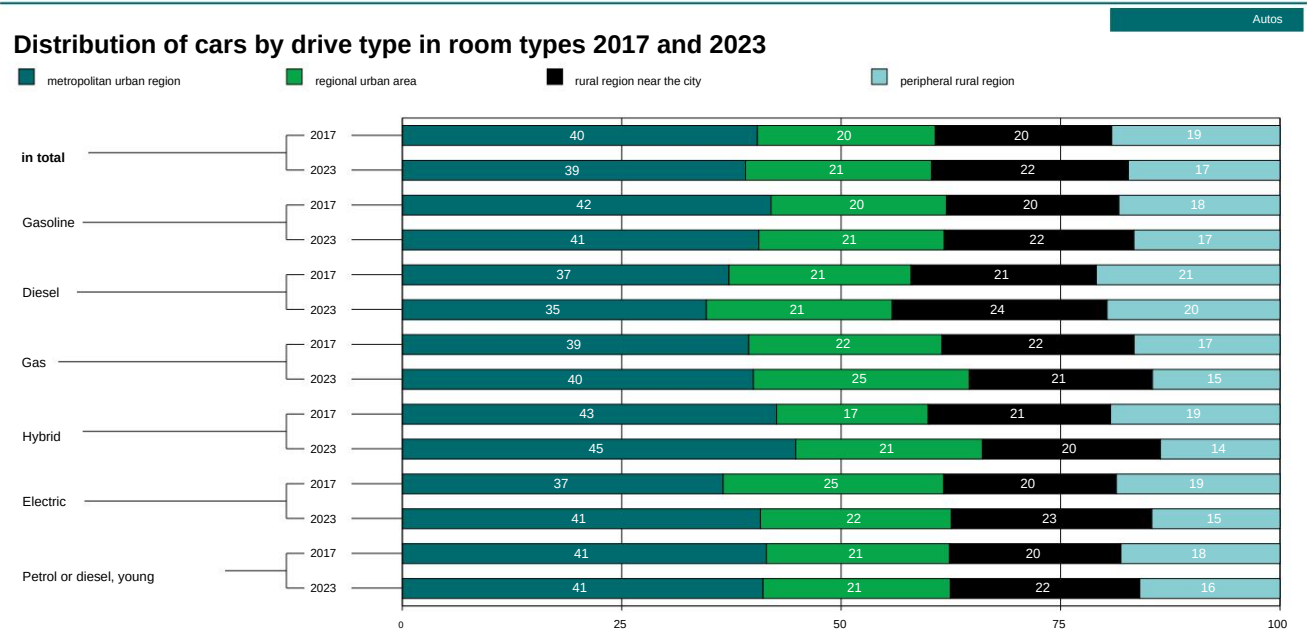
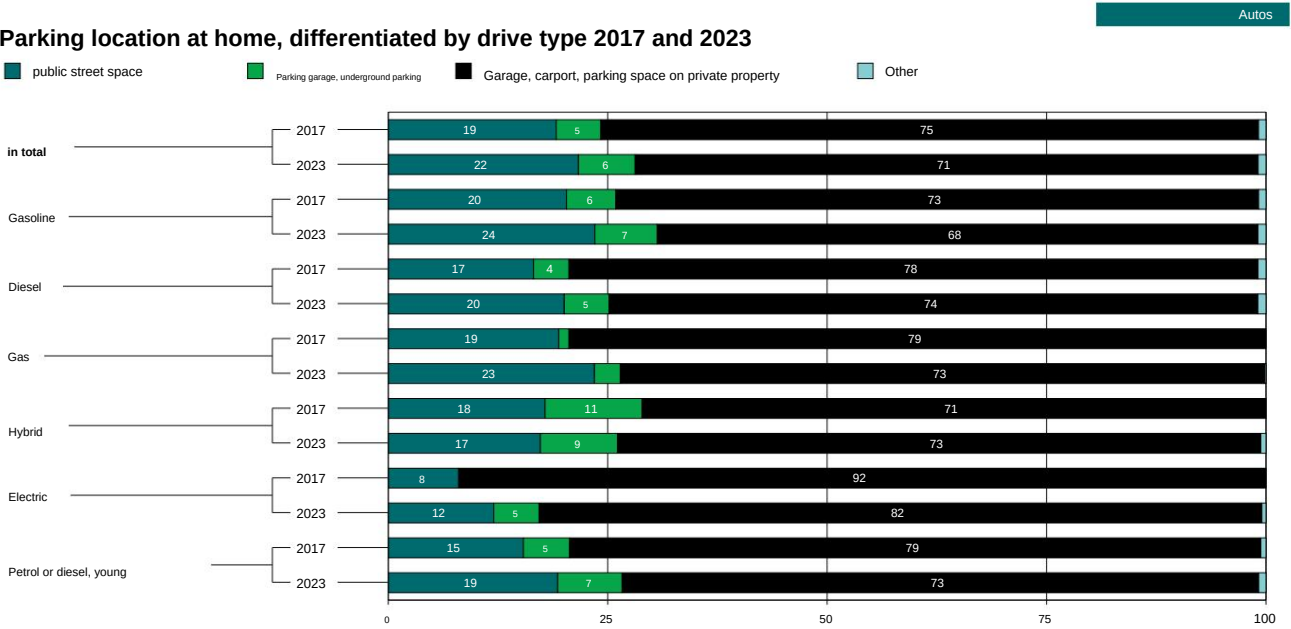
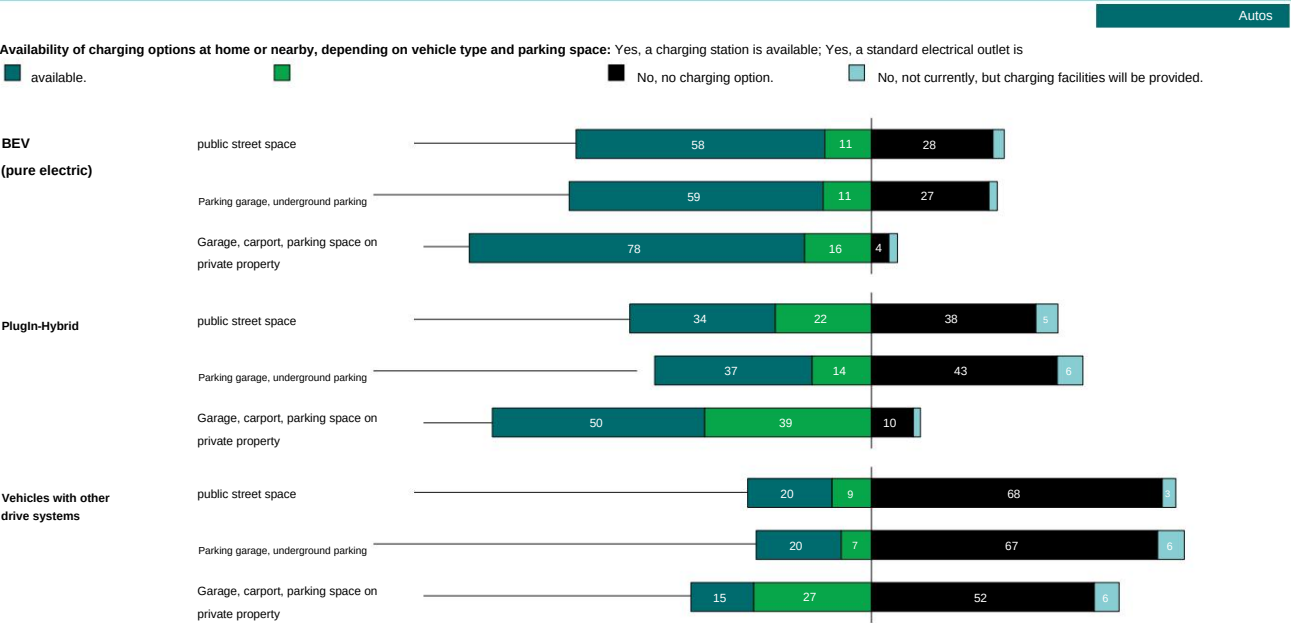


Figure 36



Figures in percent
MiD 2023 | Results Report | Source: MiD 2023

Figure 37



Figures in percent
MiD 2023 | Results Report | Source: MiD 2023

Charging facilities are available in underground car parks and parking garages are just as scarce as public streets.

To understand possible obstacles to the adoption of electromobility by private households, the respondents in the MiD survey were also asked about charging options at or near their place of residence.

For the illustration in Figure 37, these charging infrastructure availabilities were evaluated at the level of individual household cars, so that the graphic shows the charging infrastructure situation for vehicles according to drive type and parking space type. As expected, there is a charging option available for almost all electric vehicles parked at home on private property. Charging options in public streets are noticeably less common, although a charging option is also available for about two-thirds of the BEVs parked there. Considering standard electrical outlets, households estimate that a charging option would even be available at almost one-third of the locations in public streets where combustion engine vehicles are parked. Perhaps unexpectedly, parking garages and underground car parks do not (yet) fare any better than public streets with regard to charging infrastructure.

The lack of charging infrastructure is a major issue, particularly in parking garages and underground car parks, which often have favorable structural conditions for charging infrastructure.

Fazit E-Autos:

Significant normalization of electromobility – but differences and obstacles still exist.

The analyses of electromobility in 2023 compared to 2017 reveal significant changes over these six years. The usage patterns of electric cars no longer reflect the specific driving profiles of early adopters. Instead, the patterns and contextual factors of electric vehicles are becoming increasingly similar to those of other vehicles. This finding is important because there had been speculation as to whether electromobility would change mobility patterns overall, whether mobility would, in a sense, adapt to the requirements of the new vehicles. The MiD results suggest the opposite, namely that electric vehicles are increasingly being used for established everyday patterns that were previously handled by other vehicles. One obstacle to this is the still partially existing

8. Public transport – multifaceted mobility requirements in transition

Public transport is a backbone of mobility in Germany, especially in metropolitan areas, and indispensable for climate-friendly and sustainable urban transport. The MiD data allows for the derivation of the status quo and trends in public transport demand and the drawing of important conclusions for its further development. The following analyses focus on typical ground-based public transport modes and concentrate on local public transport. Airplanes, taxis, long-distance buses, and ride-hailing services, which are also commonly considered part of public transport, are not included in this chapter due to their high degree of heterogeneity and small sample size in the MiD. However, they are addressed in the separate sections on sharing services and demand for overnight and day trips.

Public road passenger transport (ÖSPV; mainly bus and tram) accounts for 7 percent of the total passenger transport volume and 5 percent of the passenger transport performance (Table 24).

On average, journeys using public transport are 9 km long and take about half an hour. Journeys using regional rail services (primarily S-Bahn and regional trains) are less frequent, accounting for only 2 percent of total journeys; however, due to their significantly longer average journey length of just over 26 km, they represent 5 percent of passenger transport services. Regional rail journeys take more than an hour on average. Long-distance rail travel accounts for less than one percent of journeys. However, the very long distances, averaging more than 200 km, result in a 5 percent share of passenger kilometers. On average, passengers spend almost three hours on long-distance journeys.

Traveling with a long-distance component. All travel times for public transport include access times to the stop or station and departure times to the destination.

Public transport availability at the place of residence: a key factor for the use of Bus and train

Depending on where you live, public transport services vary considerably. High-quality services in urban areas contrast sharply with the often sparse public transport connections in rural areas, which sometimes only serve a basic need. This disparity in public transport services leads to a very uneven distribution of its use across the population.

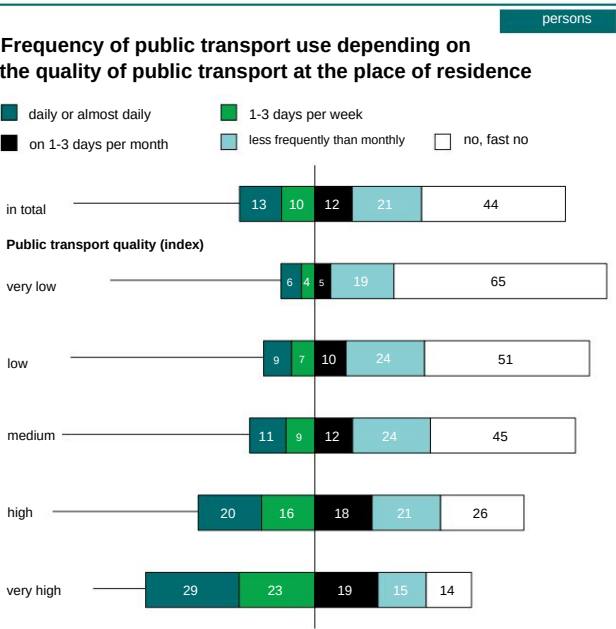
Figure 38 shows the frequency of public transport use depending on the quality of public transport services at the respondents' place of residence. The benchmark for the quality of public transport services is an objective variable that is not requested but rather provided to the respondents. This variable comprises the distance from the respondents' residence to the nearest bus or tram stop or train station, as well as the frequency and type of services offered there, and is used to create five quality levels (infas 360, see also the MiD handbook on data usage).

Nearly half of the population has relatively poor public transport options at their place of residence. 28 percent of the population have access to a medium level of public transport, only 16 percent to an objectively good level, and 12 percent to a very good level. This last group typically lives in large cities and metropolitan areas or near heavily used public transport routes. Figure 38 also shows that public transport usage is strongly influenced by the availability of public transport.

Table 24

Key figures for demand in public transport, regional rail and long-distance rail services				
Means of transport	share of all Because of	Percentage of passenger- kilometers in all journeys	medium distance	medium Travel time
	%	%	km	min.
Public road passenger transport (ÖSPV)	7	5	9	34
Local rail passenger transport (SPNV)	2	5	26	66
Long-distance rail passenger transport (SPFV)	< 1	5	211	179

Figure 38



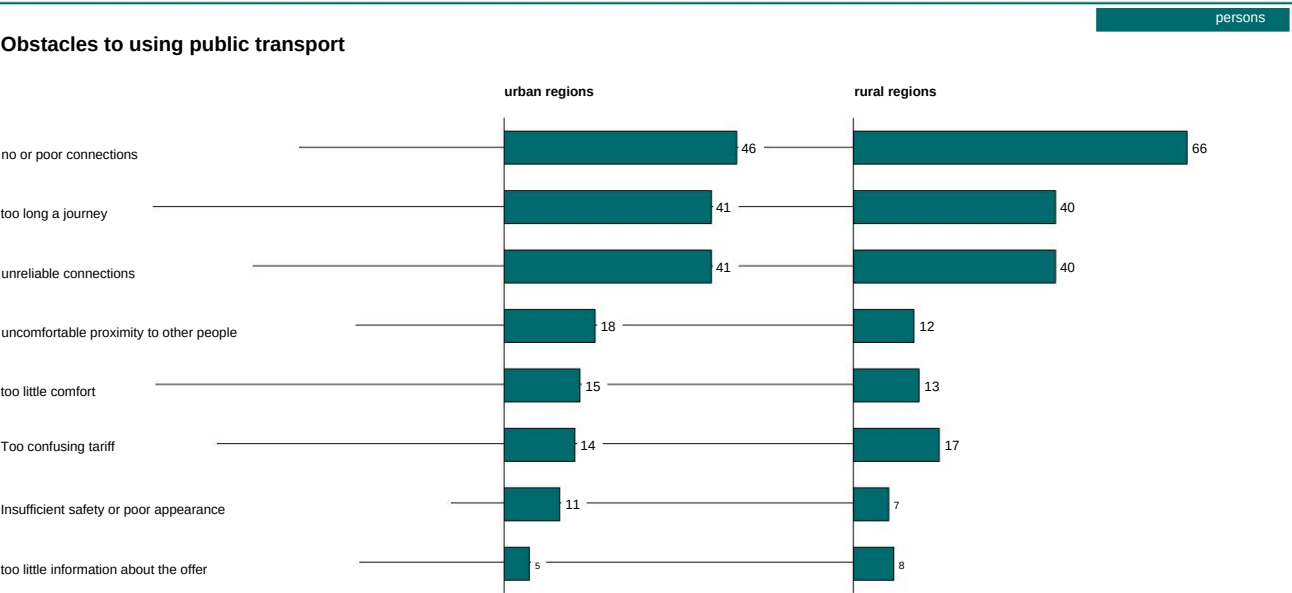
Figures in percent
MiD 2023 | Results Report | Source: MiD 2023

which depends on the availability of public transport. While only about 35 percent of people in locations with very poor public transport use it for their mobility, about 85 percent of people in locations with very good public transport use it. 29 percent of people in these locations even use public transport daily or almost daily.

Multifaceted obstacles to the use of public transport

The survey on obstacles to public transport use in the MiD (Municipal Survey of Germany) also confirms that inadequate local services and connection quality problems are important reasons why many people do not use public transport: For more than half of the population, no or poor connections are an obstacle to using public transport; for 40 percent, it is excessively long travel times, as well as unreliability. The confusing fare system, apart from the Deutschlandticket (Germany Ticket), prevents around 15 percent of the population from using public transport. Further reasons for not using public transport include comfort and safety concerns, such as uncomfortable proximity to other people, a generally low level of comfort or safety, and the poor public image of public transport. Overcoming some of the aforementioned obstacles to using public transport is relatively challenging. However, 16 percent of the respondents also indicated that...

Figure 39



Figures in percent; multiple responses possible, breakdown according to RegioStaR 2
MiD 2023 | Results Report | Source: MiD 2023

Public transport brand channel: a different representation of public transport perception

As explained in this chapter on public transport, the MiD 2023 also included several questions about the reasons for not using public transport. The individual results are shown in Figure 39. However, these can also be combined and analyzed in stages.

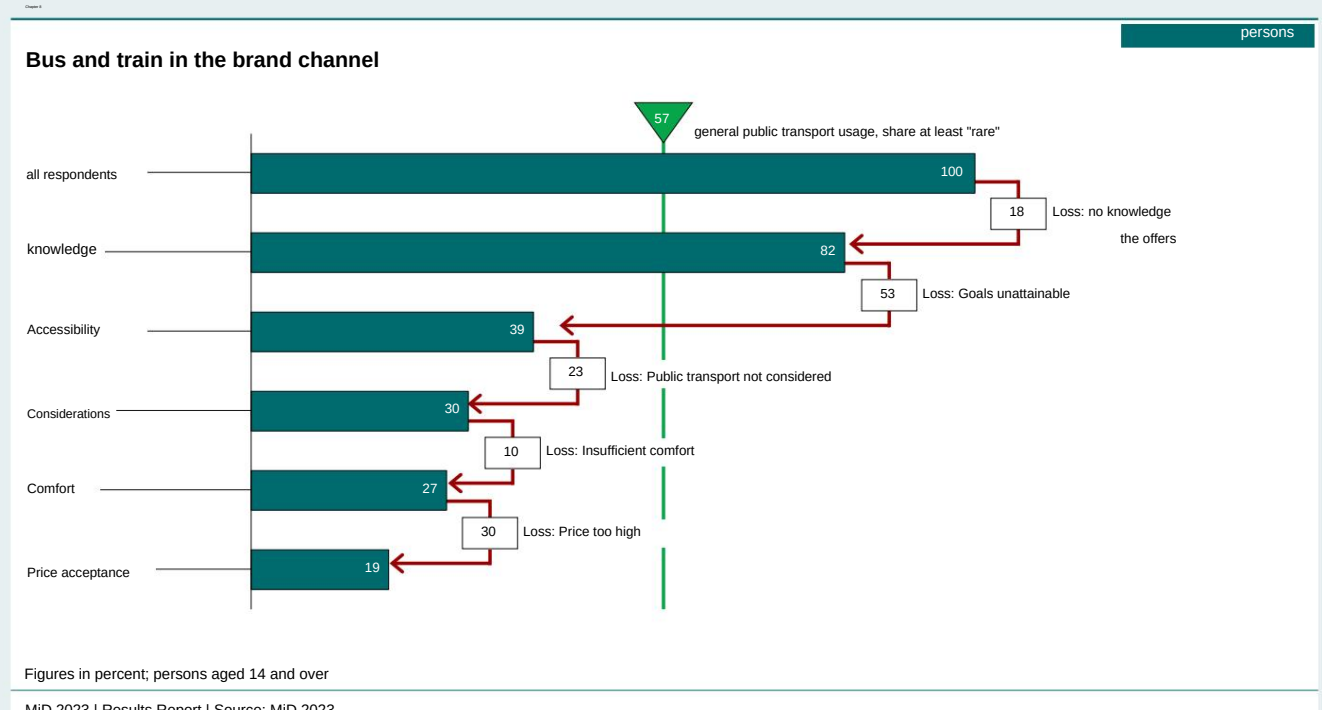
Based on the representation of purchasing decisions, a "public transport brand channel" emerges. It illustrates the gradual reduction of customer potential and documents the "losses" from stage to stage. The order is selectable.

The public transport-related analysis begins with all individuals aged 14 and over. The first stage plots the lack of knowledge, i.e., the percentage of respondents who state they are unfamiliar with public transport. At 18 percent, this is just under a fifth. The stages then continue with further characteristics. The greatest losses occur with regard to the accessibility of usual destinations by bus or train. A full half of the respondents who demonstrate some knowledge of public transport deny this. The decreases become smaller in the subsequent stages. However, the consideration of

Public transport plays a role in personal voting decisions, as do comfort requirements and, finally, price acceptance. Price acceptance was deliberately placed last, given the availability of the Deutschlandticket (Germany Ticket). In the end, just under a fifth of respondents answered in favor of public transport in all the listed aspects. This roughly corresponds to the proportion of regular customers, but is naturally not entirely identical.

Overall, this analysis offers a different perspective on public transport than usual. Buses and trains are losing potential customers primarily due to network quality, but also due to perceived comfort deficiencies – and even with the Deutschlandticket (Germany-wide ticket), still due to price perceptions. The challenges are therefore multifaceted from this perspective as well. Simply expanding services and offering low prices will not be enough to reach a significantly larger customer base. The quality of the passenger experience and good communication are also crucial.

Figure 40



A significant number of respondents indicated that they had too little information about public transport services, which appears to be a solvable obstacle to accessing public transport.

Declining and generally low satisfaction with public transport

The MiD survey also asks about satisfaction with different modes of transport. The results were already presented in the transport comparison in Chapter 4. Car use remains stable, while cycling and walking show declining ratings. This applies across all region types.

The top-box shares of public transport, meaning the two best rating categories combined, are declining by a few percentage points. In 2023, they will be 62 percent in metropolitan areas and 14 percent in rural, village-like areas, with the shares falling in between for other region types.

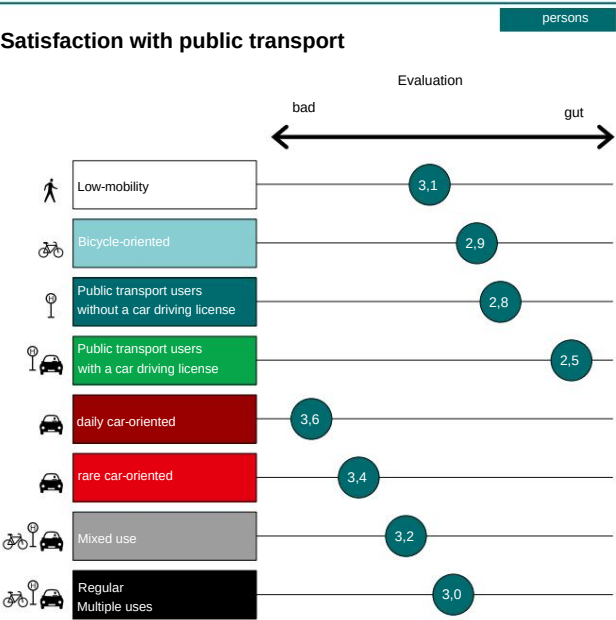
Will the results for 2023 – always referring to persons aged 14 and over – be calculated according to the criteria in Chapter 4 ? The weaknesses become apparent from a different perspective when considering the mobility segments also presented (Figure 41). While current users of public transport vote somewhat more favorably, though not truly satisfied, non-users almost consistently give average ratings of around 3 or lower.

Those who favor cycling give public transport a rating of 2.9, while those who prefer cars give it a 3.6, and the overall average rating of 3.3 also indicates significant perceived quality deficiencies. The fact that these deficiencies are particularly noticeable among non-passengers suggests that their needs are too often not being met. The preceding sidebar on a public transport brand channel also illustrates where these shortcomings lie.

The Germany Ticket achieved widespread use shortly after its introduction.

The aforementioned obstacles also include an overly complex fare structure. Shortly before the fieldwork for the MiD 2023, a significantly simplified fare system was introduced with the Deutschlandticket (Germany Ticket), making the MiD a suitable indicator of the acceptance of this new offer in the early stages after its introduction. As expected, the distribution of the Deutschlandticket depends on the frequency of public transport use, since the ticket, at its then-current price of €49, only becomes cost-effective after a certain number of journeys per month. During the survey period, an average of 16 percent of people possessed a Deutschlandticket (Figure 42). Among people who use public transport daily or almost daily, the Deutschlandticket is the most frequently used ticket type. 58 percent of this group were already using the Deutschlandticket in 2023 and 2024. The remaining individuals in this group largely used other season tickets. The less frequently public transport is used, the higher the proportion of people using cash fares. For example, half of those who use public transport one to three days a month buy single tickets or day passes. However, even in this group, 18 percent report using the Deutschlandticket (Germany Ticket). Since the fieldwork phase of the MiD study took place at the very beginning of the Deutschlandticket's introduction, it can be assumed that the proportion of Deutschlandticket users in these groups has increased further since then.

Figure 41

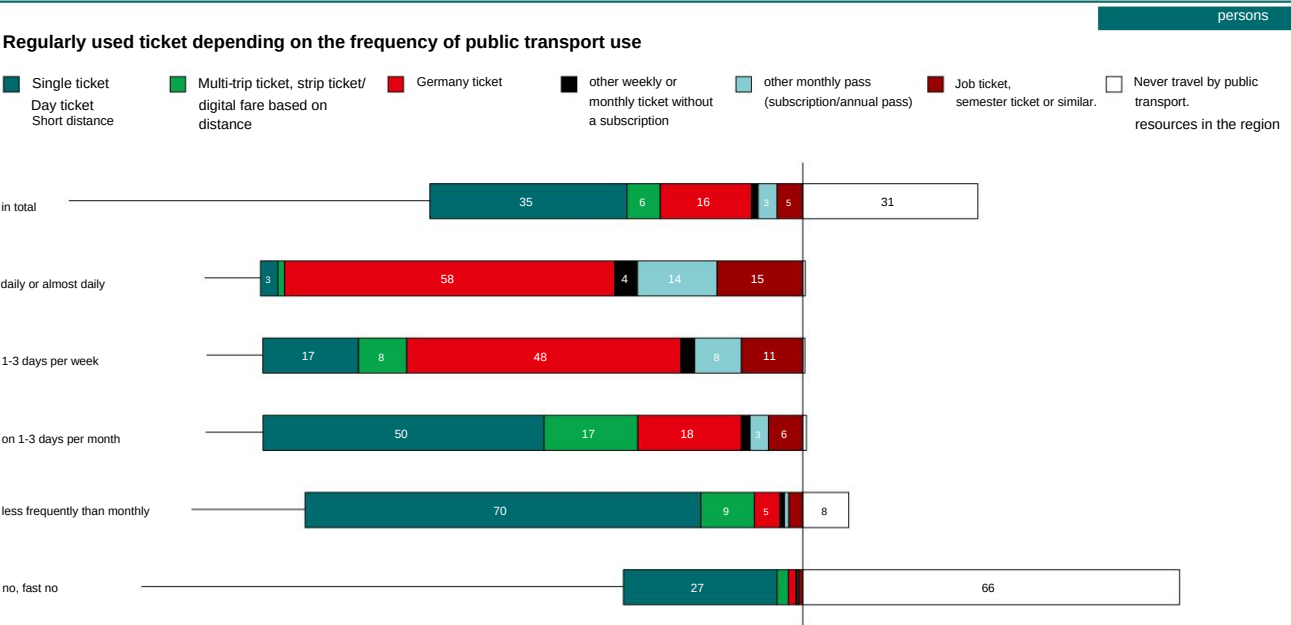


Average values, rating scale 1 to 6; persons aged 14 and over

The distance of most public transport journeys is between 5 km and 50 km.

Three-quarters of public transport journeys are purely local public transport journeys. In terms of journey volume, urban, road-based public transport is therefore significantly more relevant than regional rail transport. Figure 43 also shows, however, that the average distances of journeys in local public transport are

Figure 42



Figures in percent; to 100% Missing: other/no information

MiD 2023 | Results Report | Source: MiD 2023

are significantly shorter. The average distance of these journeys is 9 km. Almost half of the public transport journeys are shorter than 5 km, and 90 percent are shorter than 20 km. In contrast, regional rail journeys, averaging 26 km, are significantly longer than public transport journeys.

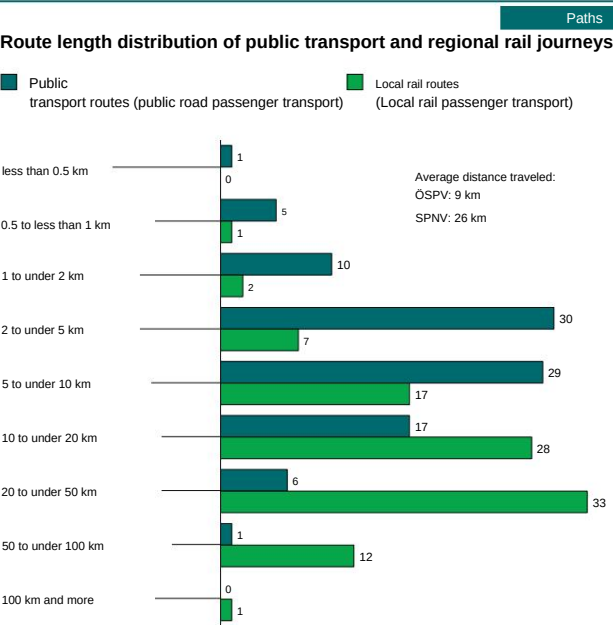
Approximately three-quarters of local rail journeys are between 5 km and 50 km long. Although local rail services can be used for longer distances in practice, and the Deutschlandticket (Germany Ticket) facilitates this use without any problems in terms of fare structure, only 13 percent of local rail journeys are longer than 50 km. This aligns with the fact that local public transport, by definition – including rail – is primarily intended for journeys up to 50 km.

Especially noticeable in rural areas

Differences in the use of buses and trains due to People with and without access to a car

The lengths of journeys made using public transport and their differences are influenced, among other things, by who uses public transport and regional rail services and for what purpose. Figure 47 illustrates, for different spatial types (RegioStaR), the proportion of journeys made using public transport and regional rail services by different groups of people, differentiated by age of majority, car availability, and driver's license possession. With the exception of metropolitan areas, there are significant differences between public transport and regional rail services.

Figure 43



Figures in percent

MiD 2023 | Results Report | Source: MiD 2023

The Germany Ticket – significant support for the resurgence of public transport “after Corona”

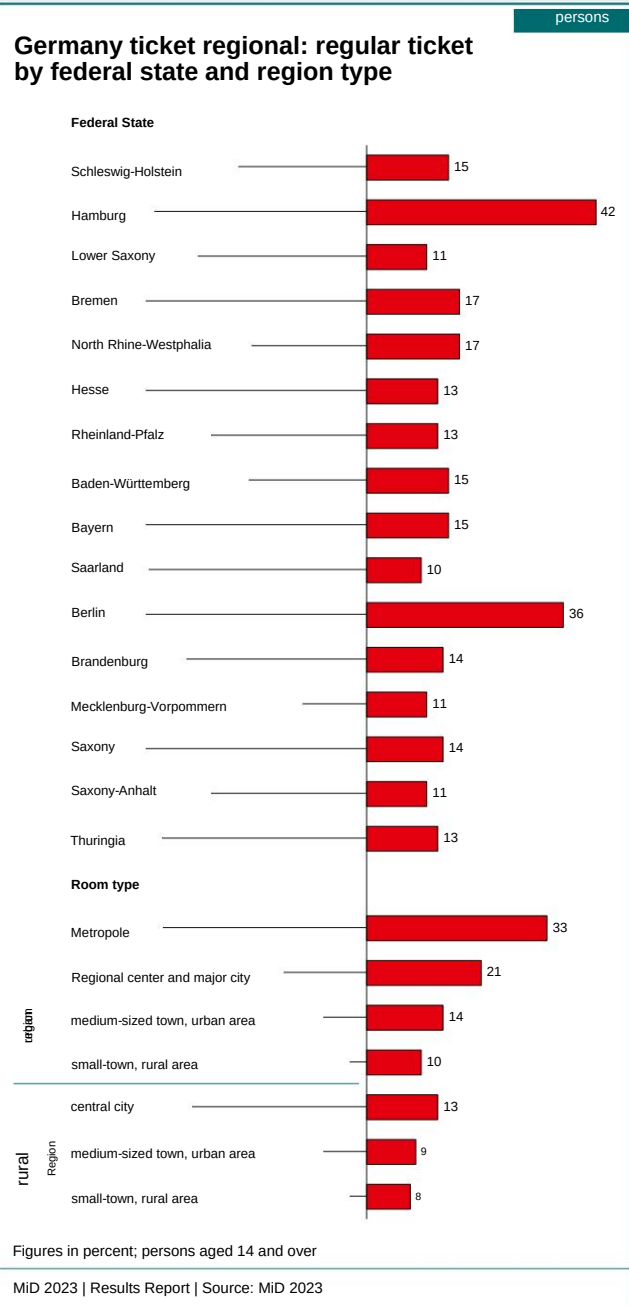
The 2023 MiD (Municipal Impact Survey) skips the COVID-19 slump and compares the situation in 2023/24 to the survey from 2017. However, the intervening period was marked by the profound disruption of the pandemic. Public transport, in particular, suffered during this time due to mandatory mask-wearing and other restrictions, unlike walking or cycling. By 2023, however, it had partially recovered. Important milestones in this recovery included the temporary €9 ticket, which also involved deliberate financial relief measures.

for the citizens. But more relevant is the Germany Ticket, available since spring 2023, with its uncomplicated nationwide usage offer for almost all local public transport.

A look at typical ticket usage has already shown that, at the time of the MiD 2023 survey, this new ticket represents the “usual” ticket for 16 percent of the people surveyed aged 14 and over, meaning it is generally chosen when using public transport.

This average value varies depending on the type of region and therefore also on the federal state.

Figure 44



The major cities, and Hamburg among them, are at the top for understandable reasons. Hamburg's particularly high score is due to an early and consistent restructuring of its fare system, a change that did not occur in this form elsewhere.

In 2023/24, the western German states were largely on par, with a few others in both the west and east lagging slightly behind. It's worth noting that the ticket is not the often-anticipated flop in rural areas. Nearly a tenth of the population there also opted for this new offer.

Looking at other aspects is also revealing. As early as the survey period of MiD 2023, it was found that exactly half of all public transport journeys were already being made using the Deutschlandticket (Germany Ticket). In terms of kilometers traveled, the share is even a few percentage points higher. The ticket is by no means solely a commuter ticket. Looking at the purposes of Deutschlandticket journeys reveals the broad spectrum of overall mobility. Work-related journeys account for around 30 percent, as do leisure journeys. Another almost third are for errands and shopping, and a further 9 percent are for accompanying journeys, such as picking up and dropping off or caring for other people. The average usage frequency per ticket is around 30 journeys per month, which is lower than most previous season tickets.

Finally, a brief overview of the buyer demographics is in order. The ticket is more commonly found in the pockets of younger people, even at a time when the transition from student and semester tickets to the Germany-wide ticket was often still in its early stages. Among older people, however, there is considerable catching up to do, largely due to their greater reliance on cars. Also noteworthy is the relatively balanced distribution of buyer shares across different economic statuses.

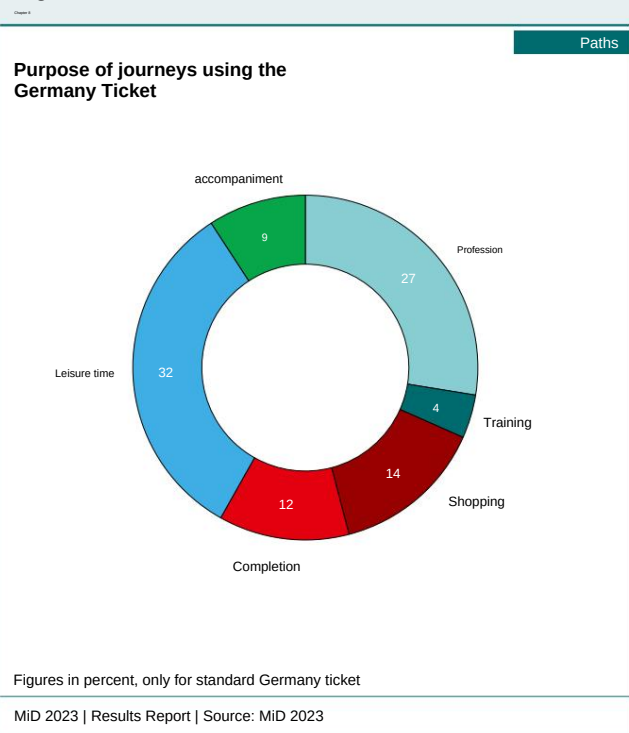
Although this rate is indeed highest in the lowest group, the difference between the other segments in this regard is not very large.

Even in the most affluent group, 17 percent are regular customers.

Chapter 15 examines in more detail the estimated benefits of participation that result from this .

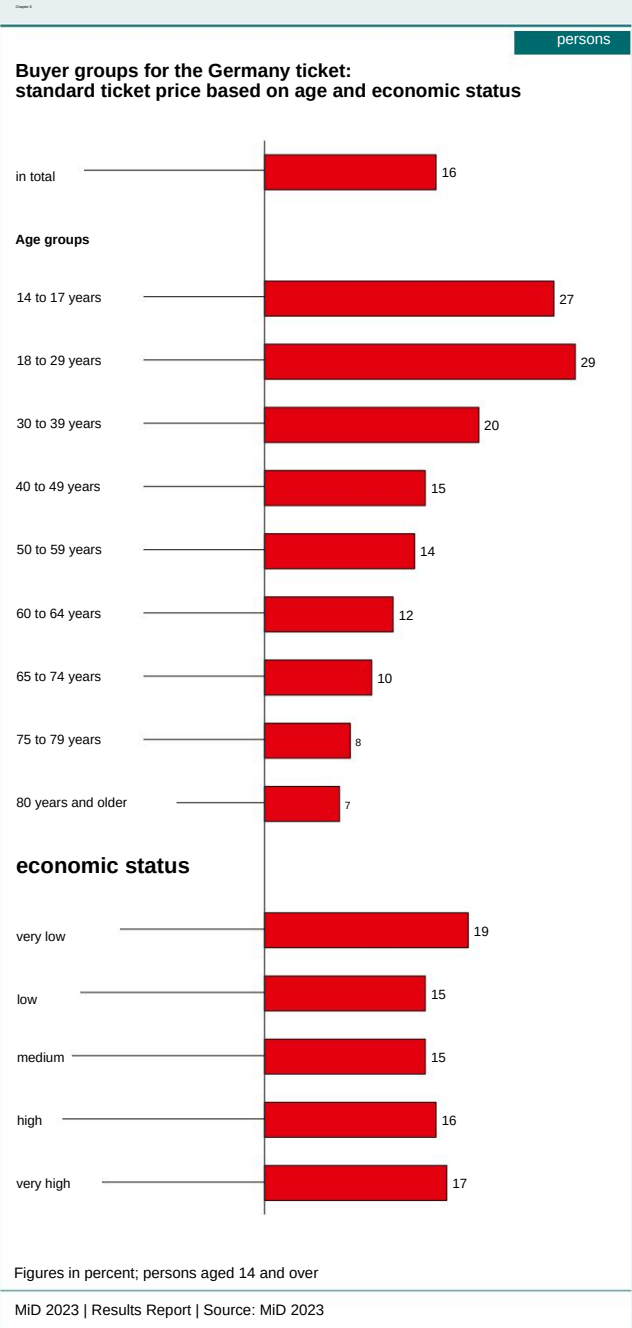
MiD cannot answer to what extent the ticket attracts new public transport customers or generates added value, to what extent the ticket generates more traffic or reduces the number of car journeys. However, the ongoing evaluation of the ticket, commissioned by the federal government, shows that the proportion of new customers is in the range of 5 to 10 percent, and even more people have now switched from cash-fare options to the new season ticket. The proportion of newly acquired

Figure 45



The number of journeys (induced journeys or additional traffic) is low, at around 5 percent. The ticket is also used predominantly regionally. In contrast, the rate of switching from cars – and to a lesser extent bicycles – is higher, reaching up to one-fifth of all journeys made with the Deutschlandticket.

Figure 46



It is evident that children and young people make up a larger proportion of passengers using public transport in all other types of areas than in regional rail services. The differences are particularly pronounced in rural, village-like areas, both in urban and rural regions: Here, public transport journeys are clearly dominated by people without a driver's license, and especially by children and young people, which underscores the great importance of school transport for public transport in rural areas. People with access to a car, i.e., an alternative means of transport, are clearly in the minority on buses.

In contrast, even in these types of areas, people with a driver's license and car access make more than half of their journeys on local public transport. This clearly demonstrates that for people with multiple travel options, rail is a significantly more competitive mode of transport than the bus, for example, in terms of travel speed and comfort.

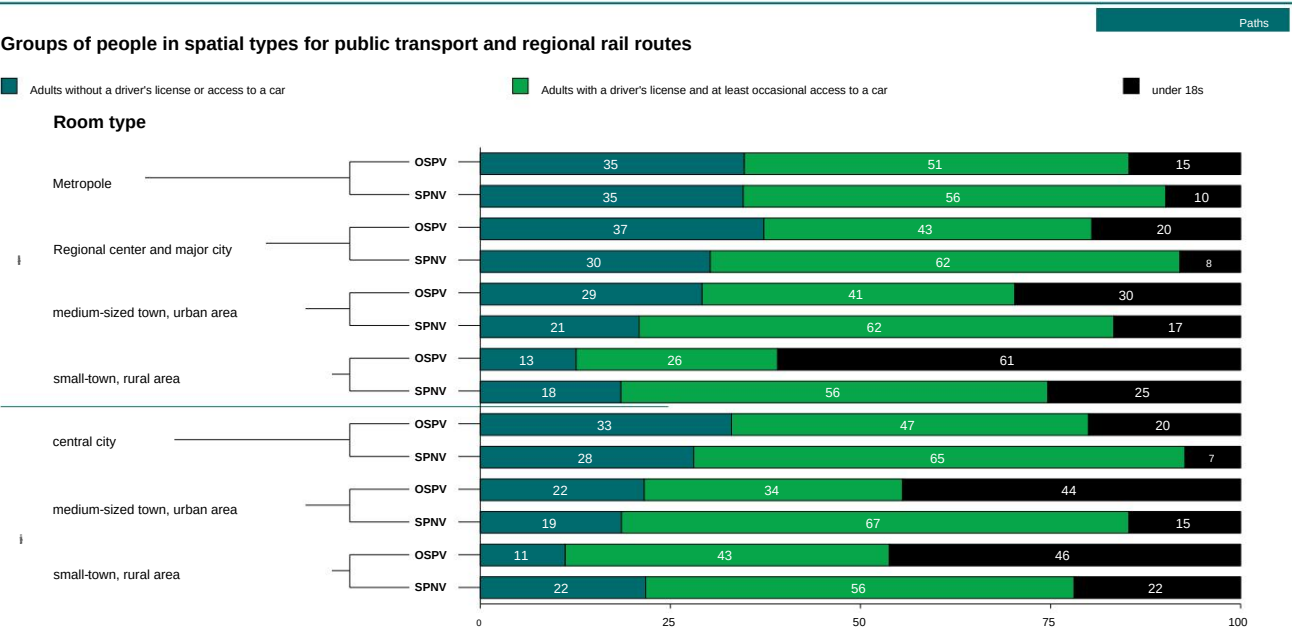
The purposes for using public transport vary significantly depending on the group of people

Leisure journeys account for approximately one quarter of all journeys made using public transport, making them the trip purpose with the highest share (Figure 48). Roughly one-fifth of all journeys on public transport are for work and education; business trips are for work-related travel.

Private routes account for 15 percent. 10 percent respectively. Eight percent of public transport journeys are for errands or shopping. Overall, public transport is used for a wide range of purposes and is by no means dominated by commuter traffic.

Regarding the employment status of individuals, the proportions of travel purposes differ significantly. For example, more than two-thirds of public transport journeys made by school pupils are to and from school. Only one-fifth of their public transport journeys are for leisure. The somewhat older group of university students uses public transport considerably less often for their commute, at 29 percent. In contrast, their use for leisure activities is higher at 37 percent, and their use of public transport for shopping, errands, and commuting to work is also proportionally higher than that of school pupils. The effect of the semester ticket, widely used in Germany based on the principle of solidarity, may also be relevant here. With this system, all students at a university purchase a semester ticket regardless of individual usage, for example, for their commute to university. Even students who live very close to the university and can walk or cycle have the ticket but do not use it for their commute.

Figure 47



School. Pupils, on the other hand, usually only receive a subsidized ticket for public transport if they are dependent on using it due to the distance to the school.

Employees use public transport for work-related or business purposes 62 percent of the time. Less than a quarter of their journeys are for leisure. This highlights the significant role of public transport for commuting and work-related travel. In contrast, pensioners primarily use public transport for leisure (43 percent), errands (30 percent), or shopping (22 percent). These four different groups therefore have very different expectations of what they consider a good public transport system: students primarily want to get to school quickly and reliably, employees want good public transport for commuting, and pensioners want easy access to leisure destinations and shopping facilities.

The diversity in public transport usage is also reflected in the different average distances of public transport routes for these purposes.

They range from shopping routes, which, with an average length of 7 km, represent the shortest routes in public transport,

...including commutes with an average distance of 17 km. The challenge for public transport is to provide a diverse range of services to be attractive to different groups of people and their specific travel needs.

Long-distance trains – longer distances within Germany

Public long-distance transport, with a focus here on long-distance trains, complements the public transport services over longer distances, usually from 50 km. Although the proportion of journeys made by long-distance train, regardless of distance, is less than one percent of all journeys, the share of kilometers traveled is 6 percent due to the significantly greater average distances. If only journeys of 50 km or more are considered, 14 percent are under 100 km and around two-thirds are between 100 and 500 km. Only one in seven of these train journeys covers more than 500 km, and the MiD (Minimum Distance Travel) only considers journeys within Germany.

The purposes of travel by rail in long-distance transport also differ from those in local transport. Leisure journeys dominate at 50 percent, while commutes account for 23 percent. Eleven percent of journeys...

Figure 48

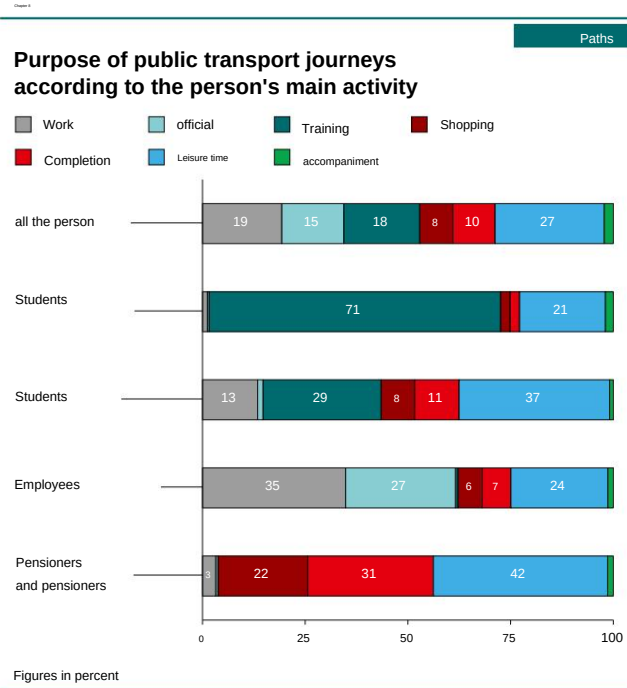


Figure 49



Long-distance rail travel is primarily used for business trips. Trips for training or errands each account for 6 percent of these journeys. Trips for accompanying persons and shopping play a negligible role, at 2 percent and 1 percent respectively. Long-distance rail travel is therefore mainly used for longer leisure trips, commuting, and business travel.

Typical everyday activities such as shopping and errands take place in the immediate living environment and

Therefore, they are practically never used here as a means of transportation.

More public transport: Young adults and the urban environment
The surrounding areas are leading the way

Public transport, as part of the integrated transport system, is a climate-friendly and city-compatible means of transport, the use of which is also supported by the single-family home.

Table 25

Development of the public transport modal split share of journeys depending on various socio-demographic factors				
Features	2002	2008	2017	2023
	%	%	%	%
IN TOTAL	8	8	9	9
ROOM TYPE				
urban region				
Metropole	19	18	19	20
Regional center and major city	11	11	11	11
medium-sized town, urban area	6	7	7	7
small-town, rural area	5	6	6	7
rural region				
central city	6	5	6	7
medium-sized town, urban area	4	4	5	4
small-town, rural area	5	4	4	5
AGE OF PERSON				
0 to 8 years	3	3	4	3
6 to 9 years	8	7	7	7
10 to 13 years	17	17	20	21
14 to 17 years	22	25	24	27
18 to 24 years	13	16	18	19
25 to 44 years	6	7	9	8
45 to 59 years	6	6	7	7
60 to 64 years	6	5	7	7
65 years and older	8	7	7	7
ECONOMIC STATUS OF THE PERSON				
very low low	12	13	15	15
	11	11	10	11
medium	7	7	9	9
high	7	7	7	8
very high	7	7	8	8

The implementation of the Germany Ticket is to be promoted. The time series of the MiDs since 2002 allows for analyses of trends in public transport use over the past 20 years. Overall, this shows only a very slight increase in the share of public transport in total traffic volume. From 2002 to 2023, this share rose by one percentage point, from 8 percent to 9 percent. This is a slight increase, but far from meeting political targets for public transport.

However, more pronounced changes are evident among certain population groups and in specific types of regions. These include, for example, medium-sized cities and urban areas, as well as small-town and rural areas within metropolitan regions—that is, the less densely populated hinterland of urbanized areas. Here, the share of public transport in the modal split increased by almost a third between 2002 and 2023. Finally, the Deutschlandticket (Germany Ticket) may also have contributed to this—at least in stabilizing the share of public transport back to the 2017 level after the COVID-19 pandemic.

Significant differences emerge when broken down by age group. While only minor changes are seen in the increasingly larger group of people aged 45 and over, young adults between 18 and 18 are particularly active.

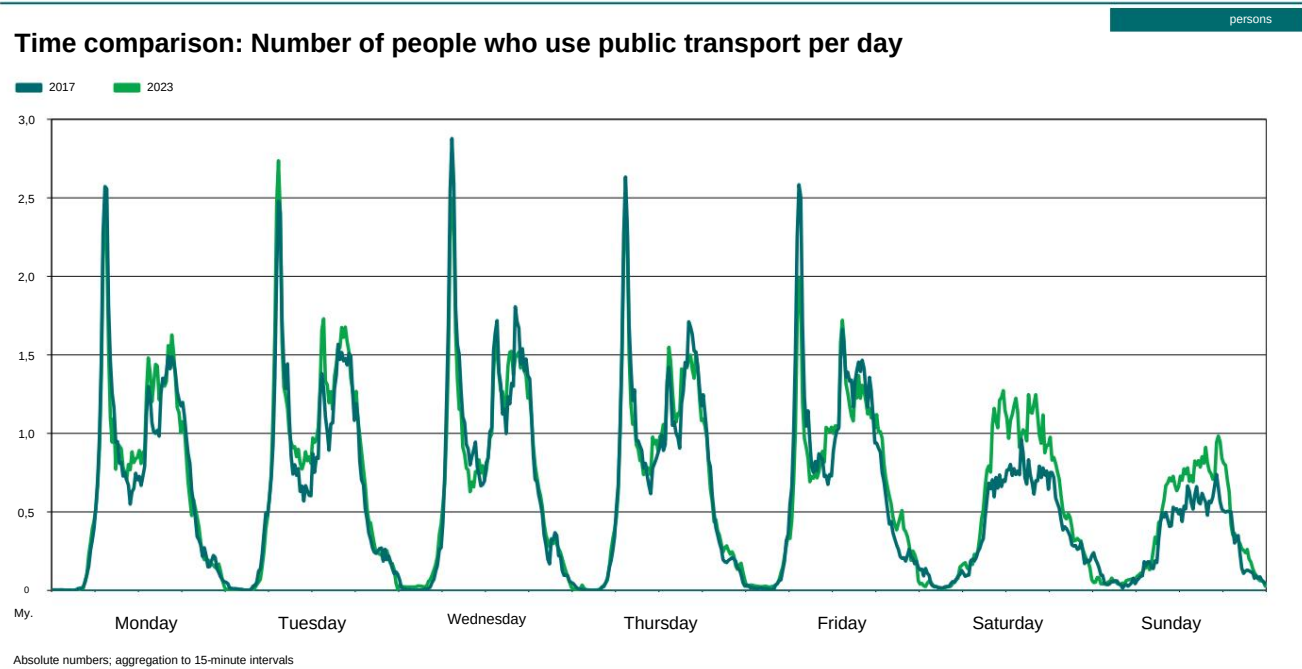
In 2023, a significantly higher proportion of journeys were made by public transport among 44-year-olds than in 2002. However, no further increase has been recorded for 25- to 44-year-olds since 2017.

There are smaller differences in development between different economic household status groups. Here, the relative increase of around 20 percent since 2002 is similar for all groups except the group with a low economic household status.

Changes in public transport demand curves:
Reduced peak hours, more traffic on weekends

A comparison of demand curves, i.e., the number of people using public transport at different times throughout the week (Figure 50), is revealing regarding changes in public transport usage patterns since 2017. This analysis considers the total journey time, i.e., the time from the start to the end of the journey. Data was aggregated into 15-minute intervals. At both points in time, a very pronounced morning peak in public transport usage is evident on regular working days, occurring between 7:00 and 8:00 a.m. and heavily influenced by school traffic. At midday and in the afternoon, there are two peaks Monday to Friday.

Figure 50



Further peak demand occurs due to the end of the school day and the end of the working hours for commuters. On Saturdays and Sundays, the number of people using public transport is more evenly distributed throughout the day. Overall, however, significantly fewer people use public transport on a weekend day than on a Monday to Friday day.

The aim is to improve local public transport services, as well as to implement further quality improvements in network, frequency, comfort, quality of stay and more, but also to increase public transport orientation among citizens. Public transport still has considerable potential to actually become the often-invoked backbone of changing mobility habits.

A comparison of 2017 and 2023 shows that the number of passengers using public transport during the morning rush hour between 7:00 and 8:00 a.m. has decreased. This is particularly noticeable on Friday mornings, which is related to a higher proportion of people working from home on that day (see [Chapter 17](#)). In contrast, the absolute number of passengers using public transport on Saturdays and Sundays was higher in 2023 than in 2017. A possible explanation for this is the Deutschlandticket (Germany Ticket), which, due to its widespread validity, can be used more easily for leisure travel than existing subscription products, which were often limited to specific routes or areas and therefore had limited use for leisure journeys.

Public transport situation in 2023 – further challenges

Taken together, the results regarding public transport show that it continues to be of great importance, but despite the introduction of the nationwide ticket and the gradual recovery to 2017 levels following the COVID-19 slump, it has not yet been able to meet the growth expectations placed upon it. Its share of total traffic volume has stagnated almost completely, and changes in transport performance, i.e., the total kilometers traveled, have not yet represented a breakthrough to a different level of demand. Furthermore, the subjective satisfaction scores collected in the MiD (Municipal Survey of Germany) are declining. In addition, it was shown that public transport usage still varies significantly according to a person's economic status; people with higher status still use public transport less than those with lower financial means, and the public transport customer base does not yet represent a cross-section of all respondents. Thus, numerous challenges remain. These include adequate financing, with the ongoing debate about the extent to which public transport should be funded by subsidies and to what extent by user contributions.

9. Sharing mobility: Car

sharing, rental bikes, e-scooters and on-demand services

Sharing services have become an integral part of urban mobility over the past two decades. Instead of permanently owning vehicles, they offer users flexible, time-limited access to cars, bicycles, or e-scooters. The first car-sharing services emerged in the 1990s and have since evolved from small local initiatives to professionally operated platforms. With the increasing prevalence of digital technologies and mobile applications, rental bicycles and e-scooters have also experienced significant growth since the 2010s. Today, a wide variety of models exist, differing in availability, pricing, and terms of use, thus catering to diverse needs in urban areas.

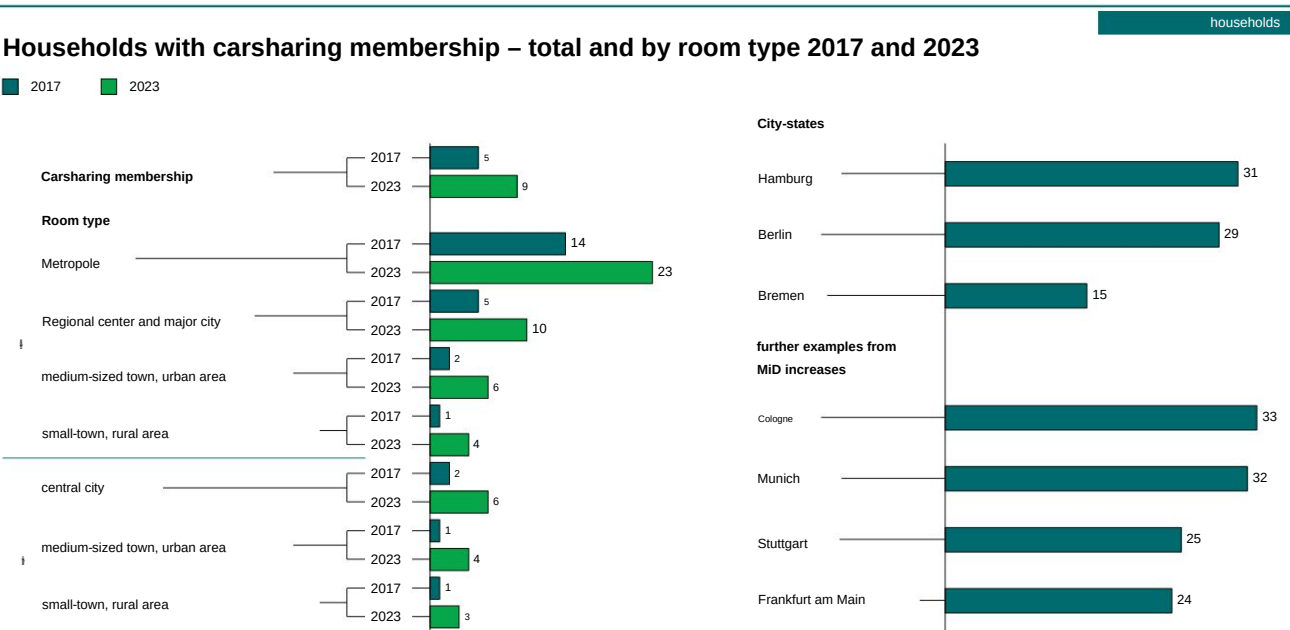
Despite the large sample size of MiD (Mental In-Depth Survey), the analysis is subject to strict limitations. However, users, particularly in the carsharing sector, exhibit specific characteristics and patterns of everyday mobility, which will be analyzed in more detail below. The focus is therefore on carsharing; ultimately, users of rental bicycles and rental e-scooters, as well as the demand in the on-demand sector, will be considered.

Carsharing memberships are predominantly found in urban areas

The proportion of households in which at least one person has a carsharing membership rose from 5 to 9 percent between 2017 and 2023. This represents significant growth, albeit from a not-so-low base. A more detailed analysis by geographical area reveals a clear concentration in urban areas. Almost a quarter of all households in metropolitan areas have access to carsharing. In regional centers and large cities, the figure is around 10 percent. In other geographical areas, however, the figures generally remain well below 10 percent. Since 2017, however, growth has been achieved in all geographical areas – particularly in small towns and more rural regions.

The recording of carsharing membership and usage is part of the MiD questionnaire. New services such as rental bikes and, most recently, e-scooters have been added, as well as commercial on-demand services within and outside of current public transport systems. Their share of overall mobility is very small – not least due to the often geographically limited availability. Less than one percent of all journeys reported in the MiD are attributable to these services – of which around half are carsharing, so that

Figure 51



A look at the city-states illustrates the potential that exists for carsharing in large cities. In Hamburg, the proportion of households with carsharing memberships exceeds 30 percent, while Berlin is just below that mark. The city-state of Bremen, comprising Bremen and Bremerhaven, has a significantly smaller population. However, a comparatively active carsharing scene there already results in 15 percent of all households having carsharing memberships.

MiD 2023 differentiates between carsharing with and without fixed stations.

At the individual or membership level, MiD enables, for the first time, a differentiation based on the type of carsharing. Two main forms are distinguished: station-based and station-independent (free-floating) carsharing. In the station-based model, vehicles are picked up from fixed locations and returned there after use.

In contrast, with free-floating carsharing, vehicles can be used and parked within a defined urban area. Both forms of carsharing serve different purposes.

Station-based carsharing vehicles can be booked in advance. They are particularly suitable for

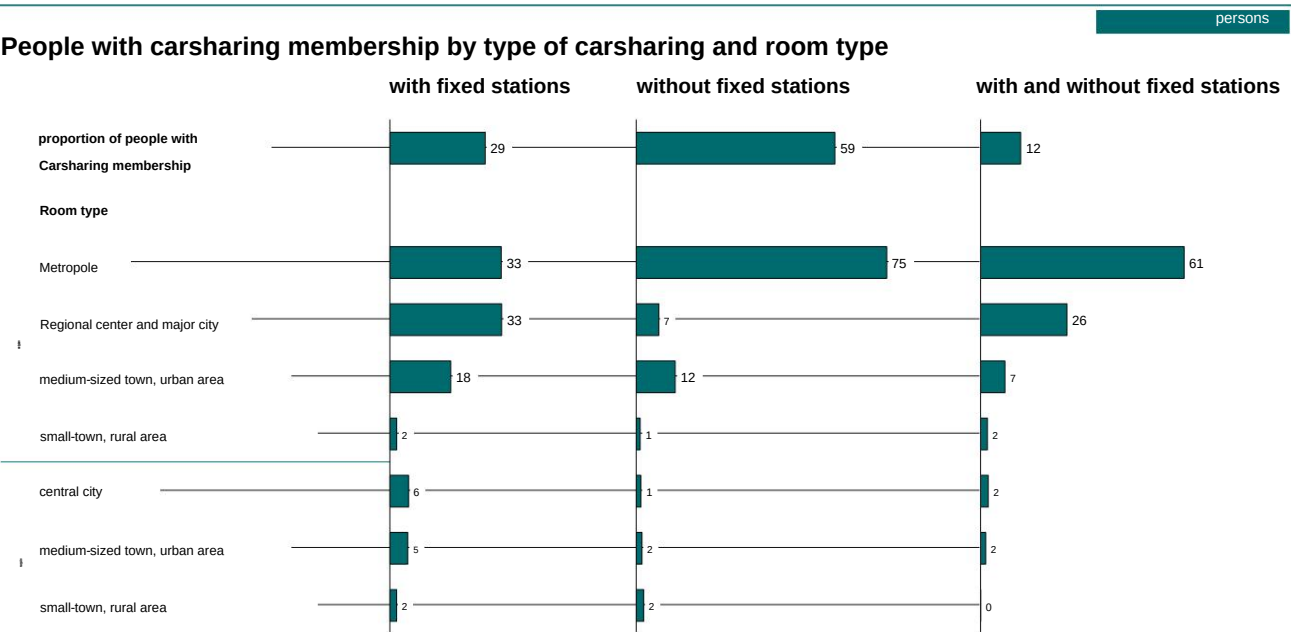
Planned trips and longer usage periods are preferred, as they are usually more cost-effective. Free-floating, on the other hand, allows for spontaneous use and is particularly attractive for short, flexible trips within the city.

Free-floating in metropolitan areas, station-based carsharing more prevalent in rural areas

At 59 percent, the majority of carsharing members have a membership for station-free carsharing. 29 percent are registered for station-based carsharing, and 12 percent have memberships in both forms. Depending on the type of carsharing, there is a significantly different geographical distribution. Users of free-floating services are primarily found in metropolitan areas. 75 percent of all respondents with such a membership live there. This is not surprising, given that there are only a few large providers offering free-floating carsharing in a limited number of major cities and, in some cases, their surrounding communities.

Station-based carsharing also has a distinctly urban character. Around a third of station-based carsharing members live in metropolitan areas, and another third in regional centers or

Figure 52



Figures in percent, persons aged 18 and over, excluding deputies

large cities. However, the service extends much further into rural areas. For example, 5 percent of respondents with station-based memberships live in medium-sized towns in rural areas.

Individuals with memberships in organizations offering both types of carsharing are also predominantly based in metropolitan areas, accounting for 61 percent. These may be respondents who hold memberships with both station-based and free-floating carsharing providers. However, they may also be registered with a provider offering combined carsharing, thus providing both pre-bookable and flexibly usable vehicles. Since the geographical focus, household and personal characteristics, and mobility patterns of this group show strong parallels to members of free-floating carsharing, the following analysis focuses on the two main forms of station-based and stationless carsharing. The first step involves examining the membership structure. This does not necessarily imply active use, particularly with free-floating providers. This aspect will be analyzed in the second step.

The classic profile of "young and male" is particularly prevalent in free-floating carsharing.

Table 26 describes the household and personal characteristics of the carsharing members of both services. To contextualize the results, the values of a comparison group are also presented. This group comprises individuals aged 18 to under 65 residing in the regional statistical areas classified as "metropolis" and "regional metropolis and large city," as carsharing is primarily concentrated in urban areas and older population groups are only minimally represented.

The results show clear differences between carsharing members and the comparison group. Men are overrepresented in both carsharing models, more so in free-floating carsharing at 63 percent (13 percentage points above the comparison value) than in station-based carsharing at 57 percent (7 percentage points above the comparison value). While station-based carsharing was also characterized by a high proportion of men in its early years, the gender ratio has changed.

The average age of users has largely been equalized here. The average age also indicates a stronger anchoring of station-based carsharing in the middle of society. At 42 years, it is only one year above the average of the comparison group, although that group excludes people over 65. Carsharing membership remains rare among older people aged 65 and over.

Highly educated and socially privileged – a characteristic of both car-sharing variants

In both carsharing models, the proportion of employed individuals is significantly higher, at 80 to 84 percent, compared to 72 percent in the control group. Furthermore, carsharing members tend to have a considerably higher socioeconomic status and above-average levels of education. While only 40 percent of the control group holds a university degree, the corresponding figure is around 65 percent in both carsharing models. Differences are also evident in household structure. Free-floating carsharing is disproportionately used by young households where all members are under 35 years old. In contrast, station-based carsharing is more common in family households.

Carsharing members more frequently use public transport Subscription and less frequently with a car in the household

Differences are also evident in the mobility indicators (Table 27). Users of both carsharing variants are more likely than average to have a public transport subscription and also more likely to live in households without a private car. This is particularly pronounced with station-based carsharing. In 52 percent of cases, no car is available in the household, while this figure is 35 percent for free-floating carsharing and only 29 percent in the comparison group.

Differences can also be observed in daily mobility behavior. Station-based carsharing users are generally somewhat more mobile; they leave their homes more frequently and travel more routes per day. Their average daily distance and travel time largely correspond to the values of the comparison group. The group of free-floating users is characterized by a very high

Daily distance. At an average of 47 kilometers, this is significantly higher than that of users of station-based carsharing (37 km) and the comparison group (38 km). This result reflects the socio-demographic composition of this group, with a high level of mobility and corresponding needs, which are disproportionately high.

is dominated by young men.

Everyday mobility of car sharing users
characterized by bicycle and public transport

The immediate use of carsharing, regardless of membership, plays a role in everyday mobility.

Table 26

Socioeconomic characteristics of carsharing members				
	Car sharing with fixed stations	Car sharing without fixed stations	Comparison group*	
GENDER				
Men	57	63	50 %	
Women	43	37	50 %	
ALTER				
Average	42	38	43 in years	
14 to 19 years	1	1	3 %	
20 to 29 years	18	24	20 %	
30 to 39 years	29	36	22 %	
40 to 49 years	22	21	18 %	
50 to 59 years	18	13	20 %	
60 years and older or < 70 years	12	5	17 %	
TASK				
employed	80	84	72 %	
Pupil, student, trainee	6	8	9 %	
Retired person, pensioner	5	2	8 %	
Miscellaneous	8	6	11 %	
SOCIO-ECONOMIC STATUS				
very low low	6	4	9 %	
	7	4	10 %	
medium	20	19	25 %	
high	42	42	39 %	
very high	25	31	17 %	
university degree (University of Applied Sciences/University)	65	64	40 %	
HOUSEHOLD TYPE				
young households (under 35 years old)	20	30	19 %	
Family households	39	35	31 %	
Households with adults	36	32	44 %	
Households with persons aged 65 and over	4	2	6 %	
*People aged 18 to under 65 from metropolitan areas or regional centers and large cities				

Both groups play only a minor role. As mentioned at the beginning, less than half a percent of all journeys reported in the MiD (Mobility in Germany) are made using this mode of transport. The overall frequency of use also shows that carsharing is used only rarely. Station-based carsharing shows slightly higher figures (Figure 53). Five percent of users use it at least once a week, compared to four percent for free-floating carsharing. Together with those who use carsharing at least monthly, this results in a share of around one quarter of all carsharing users.

Roughly the same number of members don't use carsharing at all or only very sporadically, with slightly more using free-floating carsharing. The carsharing option is therefore not, in most cases, an "everyday solution," which aligns with the service's philosophy and purpose and is thus within expectations. A significant number of members maintain their membership for various reasons without actually using it. Membership can also be useful as a "car for all occasions." However, both of these factors mean that actual usage on the road remains low compared to the significantly increased membership rates.

The user groups already analyzed several times in the report, which are based on typical mobility behavior

based, provide information about how carsharing users travel in everyday life (Figure 54). Users of station-based carsharing, in particular, show a strong affinity for bicycles and public transport. Seventy percent of users belong to groups that travel exclusively by bicycle or public transport, or combine both modes of transport as part of multimodal behavior. Only 27 percent use a car at all during a typical week, and then usually in combination with a bicycle and public transport. Bicycles and public transport also play a significantly larger role for free-floating carsharing users than in the comparison group, although to a lesser extent than in station-based carsharing.

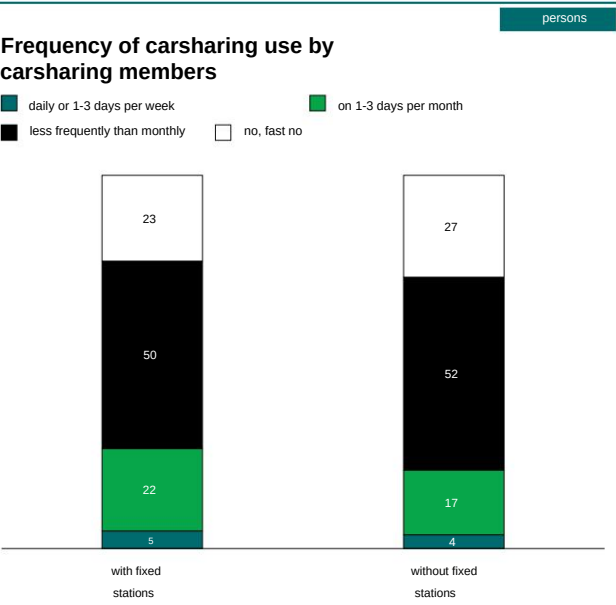
Rental bikes and rental e-scooters – so far only a fringe phenomenon

Another form of shared mobility is provided by rental bikes and rental e-scooters. Even though their presence in the public spaces of large cities sometimes suggests otherwise, they play only a marginal role in everyday mobility overall. The vast majority of respondents do not use these services at all; for rental bikes, the figure is 95 percent, and for rental e-scooters, it is 93 percent. Among the few people who report using them at all, the category "less than once a month" is dominant.

Table 27

Mobility indicators of carsharing users				
	Car sharing with fixed stations	Car sharing without fixed stations	Comparison group*	
Mobility rate (outside the office on the reporting date)	91	89	86 %	
Average number of trips (per person per day)	3,6	3,2	3.2	Number
Daily distance	37	47	38	km
Average travel time	1:41	1:42	1:38	h:min
No car in the household	52	35	29 %	
a car in the household	37	46	47 %	
two or more cars	11	19	24 %	
ÖV Subscription **	54	52	43 %	
Individuals with carsharing membership, without a representative; * People aged 18 to under 65 from metropolitan areas or regional centers and Major cities ** Germany ticket, monthly subscription ticket, annual ticket, regional/local job ticket, company subscription, semester ticket or comparable offers				

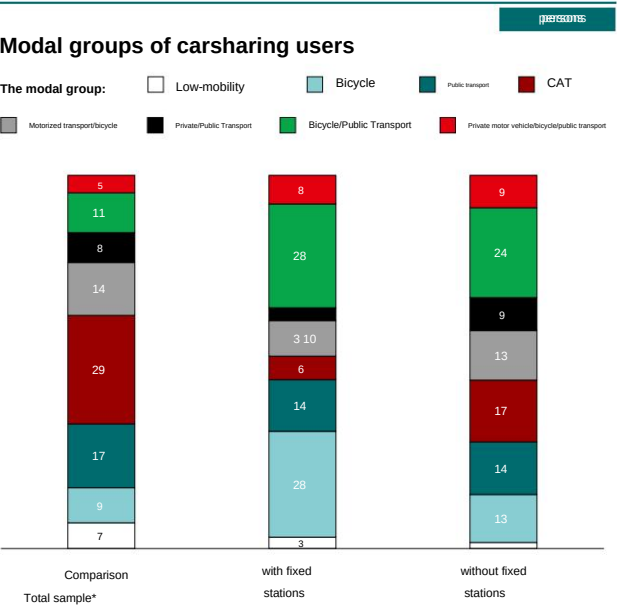
Figure 53



Figures in percent, persons aged 18 and over, excluding deputies, members only.

MiD 2023 | Results Report | Source: MiD 2023

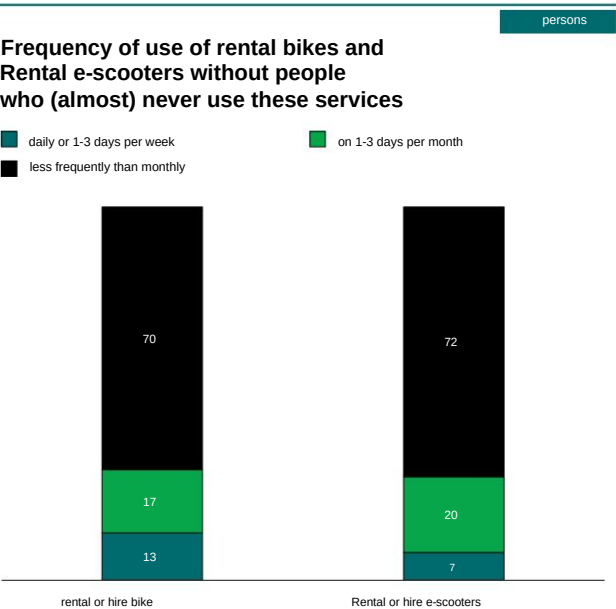
Figure 54



Figures in percent, persons aged 18 and over, excluding deputies;
*Persons aged 18 and over from metropolitan areas or regional centers and large cities

MiD 2023 | Results Report | Source: MiD 2023

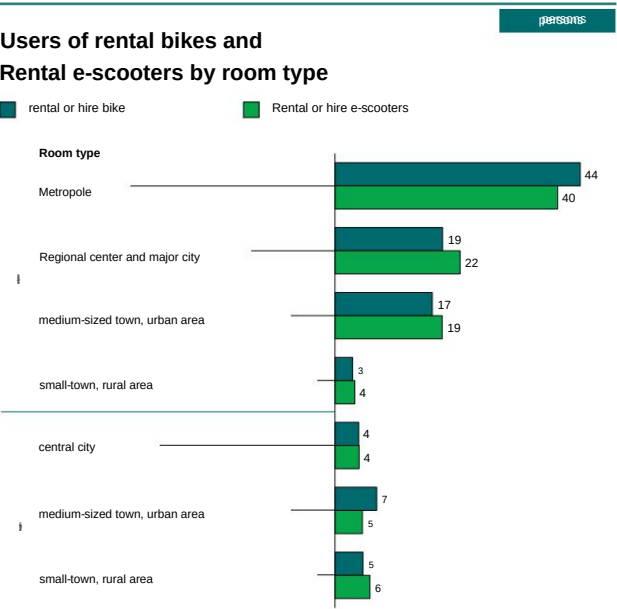
Figure 55



Figures in percent, persons aged 18 and over, excluding deputies

MiD 2023 | Results Report | Source: MiD 2023

Figure 56



Figures in percent, persons aged 14 and over who use the service at least "less than monthly", without deputies

MiD 2023 | Results Report | Source: MiD 2023

Table 28

Socioeconomic characteristics of users of rental bikes and rental e-scooters

	Rental and hire bikes available at least 1-3 days per month	Rental and hire e-scooters available at least 1-3 days per month	Comparison group*	
GENDER				
Men	69	72	50 %	
Women	31	28	50 %	
ALTER				
Average	35	31	40 in years	
14 to 19 years	4	12	9 %	
20 to 29 years	38	42	18 %	
30 to 39 years	27	25	21 %	
40 to 49 years	19	12	19 %	
50 to 59 years	8	7	22 %	
60 years and older or < 65 years	5	2	11 %	
TASK				
employed	72	67	73 %	
Pupil, student, trainee	20	25	13 %	
Retired person, pensioner	2	1	2 %	
Miscellaneous	5	7	11 %	
SOCIO-ECONOMIC STATUS				
very low	8	9	8 %	
low	11	8	8 %	
medium	26	20	24 %	
high	37	37	41 %	
very high	19	25	18 %	
university degree (University of Applied Sciences/University)	53	44	34 %	
HOUSEHOLD TYPE				
young households (under 35 years old)	42	38	16 %	
Family households	28	30	40 %	
Households with adults	29	31	44 %	
Households with persons aged 65 and over	1	1	* %	

Persons aged 14 and over, without a representative,
People aged 14 to under 65 from the urban area excluding the small-town and rural areas

(Figure 55). Only about one-tenth of this already small group uses the respective service at least once a week. For most, it is therefore occasional use.

Shared micromobility concentrated in urban areas

Rental bicycles and rental e-scooters also show a pronounced urban distribution (Figure 56). The majority of users live in metropolitan areas. Within urban areas and even in medium-sized cities, respondents report using such services, with the proportion being just under one-fifth. In rural regions, however, very few people indicate using these services. The service does not necessarily have to be available at the place of residence; it can also refer to use during a stay in larger cities.

Micromobility users with a clear profile: young, male, and educated.

Users of rental bicycles and rental e-scooters have a very specific socioeconomic profile (Table 28). A comparison group was also formed here to contextualize the values. In this

In this case, it consists of people aged 14 to under 65 from urban regions, excluding the small-town/village area within the urban region.

Both groups are predominantly young and male, disproportionately likely to be employed or in education, and possess a high economic status and a high level of education. They are particularly prevalent in households where all members are under 35 years old, while other household types are underrepresented. These characteristics are even more pronounced among users of rental e-scooters. On average, they are even younger and even more likely to be in education or training.

Significant differences are also evident in the mobility indicators (Table 29). Users of rental bicycles and rental e-scooters are significantly more likely to live in households without a car and are more likely than average to own a public transport subscription, with this effect being even more pronounced among rental bicycle users than among rental e-scooter users.

Both groups are also more mobile overall than the comparison group. Regarding the rental bike users-

Table 29

Mobility indicators of users of rental bikes and rental e-scooters				
	Rental and hire bikes available at least 1-3 days per month	Rental and hire e-scooters available at least 1-3 days per month	Comparison group*	
Mobility rate (outside the office on the reporting date)	91	86	86	%
Average number of trips (per person per day)	3,6	2,9	3.2	Number
Daily distance, average travel time	43	50	41	km
	1:45	1:43	1:35	h:min
No car in the household	41	32	20	%
a car in the household	38	41	43	%
two or more cars	20	27	36	%
ÖV Subscription **	63	59	36	%
Persons aged 14 and over, without a representative; * People aged 14 to under 65 from the urban area excluding the small-town/village area Space, ** Germany ticket, monthly subscription ticket, annual ticket, regional/local job ticket, company subscription, semester ticket or comparable offers				

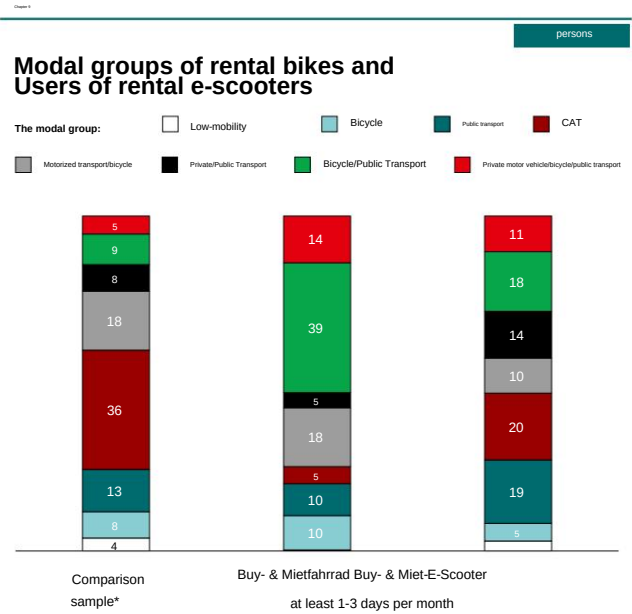
This is reflected in a high mobility rate and an above-average number of trips made. Among users of rental e-scooters, the daily distance traveled is particularly disproportionately high.

Multimodality shapes the everyday mobility of users of rental bicycles and rental e-scooters.

The differences between the groups are also reflected in very different mobility behaviors. The everyday mobility of bike-sharing users is characterized by a particularly high degree of multimodality (Figure 57). Three-quarters of the group combine several of the three modes of transport—car, bicycle, and public transport—in their daily lives. The focus of this multimodality is on the use of bicycles and public transport. Bike-sharing is used less frequently. The car plays a very minor role in this group. While 36 percent of people in the comparison group use only a car in their daily lives, this figure is only 5 percent for bike-sharing users.

Users of rental e-scooters are also significantly more multimodal in their travel habits than the comparison group and less reliant on cars. They thus clearly stand out from the comparison group.

Figure 57



Figures in percent, figures in percent, persons aged 14 and over, without representatives;
*Persons aged 14 to under 65 from the urban area excluding small-town and rural areas

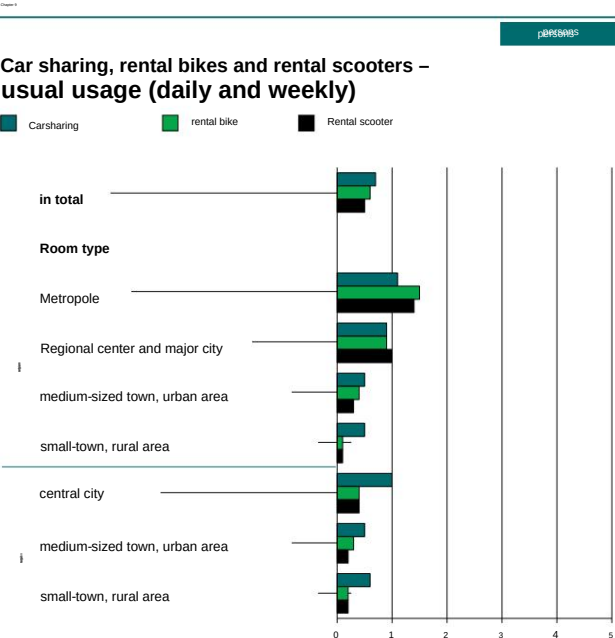
However, this behavior is less pronounced among users of rental e-scooters than among users of rental bicycles.

Broadened perspective: Sharing and on-demand Offers

Before assessing the sharing services, the perspective is broadened. The services are considered in relation to the total population, supplemented with on-demand services, and the resulting overall sector is subjected to a mileage analysis. Up to now, the frequency of use of carsharing, rental bikes, and rental e-scooters has only been reported for individuals with carsharing memberships or for those who use rental bikes or rental e-scooters at least occasionally. When considering the total population aged 14 and over, the proportions of daily or at least weekly use for all three services are below one percent – with the previously described differences between metropolitan areas and other types of urban areas (Figure 58).

For on-demand services such as call buses, no information is available on general usage frequency. In this case, the route level provides a useful basis for analysis. When reporting the modes of transport used on the reference date, the category "call-a-bus" was selected.

Figure 58

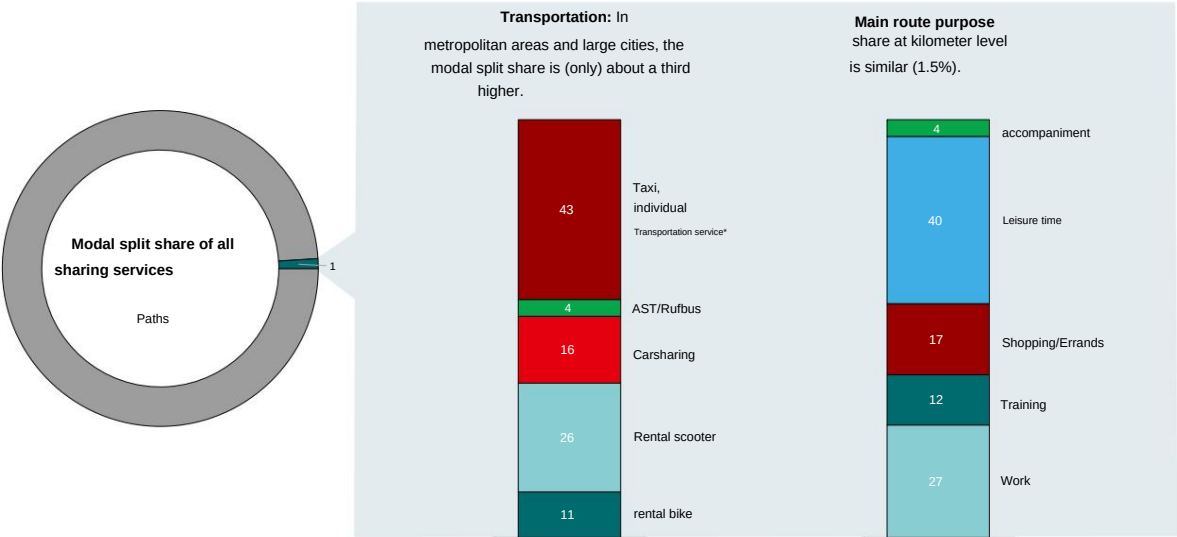


Figures in percent, persons aged 14 and over, no usual usage data collected for ride services

Figure 59

Car sharing, rental bikes, rental scooters, on-demand transport and ride services – in the modal split

Paths



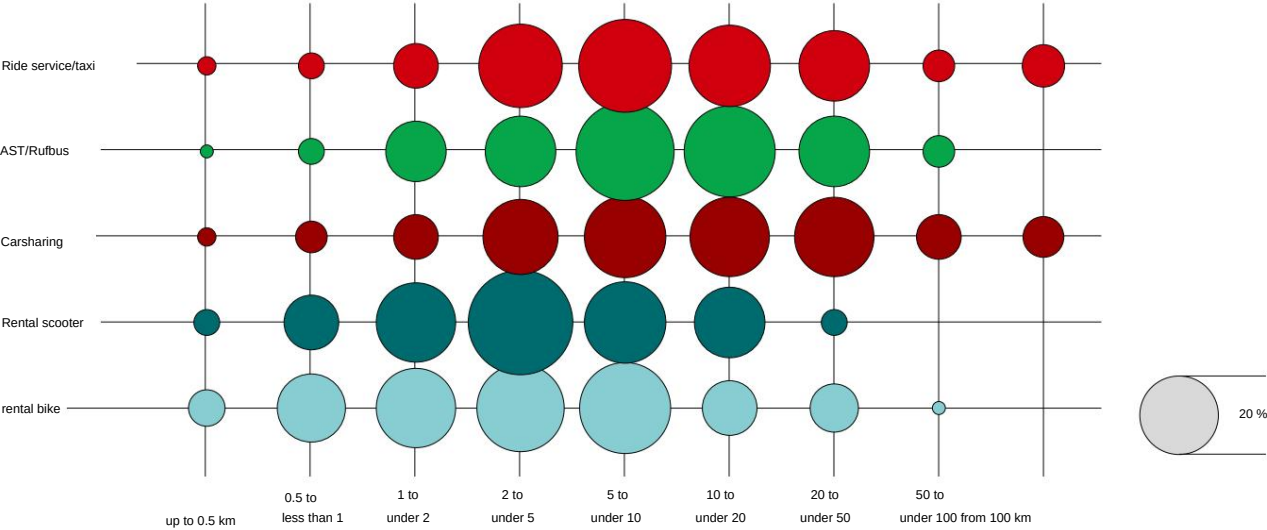
Figures in percent; * not further differentiated in the MiD

MiD 2023 | Results Report | Source: MiD 2023

Figure 60

Specific route lengths – shares in the various sharing services

Paths



Circle size shows line percentage per shared transport vehicle (extreme values for on-demand bus, rental bike and scooter have been adjusted for plausibility)

MiD 2023 | Results Report | Source: MiD 2023

Shared taxis (AST), on-demand buses, ridesharing, etc., are also available. These services are also rarely used overall, but they represent a significant share within the niche of sharing options. If the classic offerings in this sector – such as on-demand shared taxis – are considered together with carsharing and e-scooters, they account for roughly half of all sharing options. Carsharing shares the other half, with e-scooters making up the lion's share.

When the entire sharing sector is considered in terms of its combined mileage, it accounts for approximately one and a half percent. Usage is by no means limited to specific travel purposes, but rather encompasses all major travel objectives and closely reflects the general purpose structure. These services are even used for pick-up and drop-off. However, different distance patterns emerge. A graphical representation of the trip length distributions for these services shows longer trips in the on-demand sector, with a similarly balanced distribution of distances as carsharing trips. Understandably, e-scooter and rental bike trips are on average shorter, with a predominance of trips significantly under 5 kilometers.

These short trip lengths and the low usage rates presented explain why the use of sharing services is negligible at the trip level, even in large cities. Of the over one million trips recorded in the 2023 MiD (Mobility in Germany) survey, roughly 1,000 each were for rental bikes and car sharing. E-scooters accounted for slightly more, at almost 2,000 trips – with an average distance of just over two kilometers, slightly above the average walking distance and significantly below the corresponding figures for cycling. This corresponds to between 0.1 and 0.2 percent of all trips. On-demand transport services, such as call taxis, also accounted for no more than around 300 of the million trips recorded in 2023, more than half of which were in large cities and not in rural areas.

Small sharing balance sheet

The diagram illustrates a very specific niche. While carsharing in urban areas now boasts a considerable number of memberships, rental bikes and e-scooters are still only very limited in their use. All sharing services are used infrequently and therefore play a negligible role in everyday mobility. Nevertheless, users exhibit a distinctly different mobility behavior. They more frequently combine different modes of transport, increasingly use bicycles and public transport, and are less dependent on cars. In this way, these services promote multimodal mobility and contribute to more sustainable transport behavior.

They represent an important option, often for filling gaps in the existing transport network, and therefore do not necessarily aim to achieve high shares of traffic volume and transport performance. This is sometimes insufficiently considered in assessments and expectations. They possess "complementary power" and the potential to stabilize sustainable mobility behavior, although they do not shape mobility patterns as such. Since 2017, despite increasing membership numbers in carsharing, the actual use of these services has developed only slightly.

Rental bikes and e-scooters remain limited to urban areas and can only offer additional appeal in tourist locations. Overall, these services are not the primary cause of sustainable behavior, but they do contribute to it and remain a niche market.

10. Bicycles and pedelecs: Where electromobility is booming

Cycling in Germany is gaining in importance – with ambitious goals for its promotion and increase. For example, cycling infrastructure has been expanded for years. This includes the opening of new sections of express cycle routes, but also smaller-scale measures to promote cycling, such as the creation of pop-up cycle lanes, especially during the Covid-19 pandemic.

These are temporary cycle paths that can be set up quickly and cost-effectively, e.g. by repurposing existing motor vehicle lanes.

At the same time, the road safety of cyclists remains a challenge. The number of cycling accidents involving e-bikes and older people is a particularly significant issue.

The MiD 2023 provides information on the extent to which measures to promote cycling are reflected in mobility behavior and forms a basis for measuring cycling development.

In addition, it provides a statistical basis for classifying accident figures against the background of the cycling performance of different groups of cyclists with conventional bicycles and with pedelecs.

This chapter provides key figures on this topic and shows relevant developments regarding the stock and use of conventional bicycles and pedelecs since 2017.

There are now around 80 million (electric) bicycles in households.

MiD determines the number of available bicycles. Since 2017, a distinction has been made between "conventional" bicycles and pedelecs. In 2023, roughly one in five households did not have a "functional" bicycle ready for use. Overall, 76 percent of households owned at least one rideable bicycle.

Twenty-one percent of households now own at least one e-bike. In 2017, this figure was only 8 percent. Similar to 2017, almost three out of four respondents stated that they own at least one bicycle. Based on similar ownership figures, this translates to a projected fleet of approximately 81 million bicycles by 2023.

Bicycles, of which about 12 million are electric – a total of about five million more than in 2017. This means that the number of multiple bicycles owned per person is increasing, not the percentage of bicycle owners.

Pedelecs: Supplement and partial replacement for conventional bicycles

On average, the German population travels around 81 million km per day on conventional bicycles without pedal assistance. This figure has decreased by just over 12 percent compared to 2017. The total cycling distance, including...

However, pedelecs saw a 14 percent increase in the same period, reflecting a shift towards pedelecs. Their daily mileage has almost quadrupled, from just under 9 million km in the last survey to over 34 million km now. This has led to a situation where, in certain age groups, pederlec mileage is now almost on par with that of conventional bicycles, as illustrated in Table 30 .

Against this backdrop, the number of e-bikes in private households has naturally increased significantly since the last MiD (Mobility in Germany) survey. While in 2017 there were an average of around 50 e-bikes per 1,000 people in German households, this number had almost tripled by 2023, reaching just over 140. In both reference years, the number of e-bikes per capita was significantly lower in metropolitan areas than in all other regional types. However, this effect was less pronounced in 2023 than it was six years earlier.

With 842 bicycles per 1,000 people, conventional bicycles are still significantly more widespread than those with electric assist.

However, the decline in bicycle ownership since 2017 (880 conventional bicycles per 1,000 people) suggests that some cyclists have completely abandoned their conventional bicycles and replaced them with e-bikes. In contrast to the situation with e-bikes, metropolitan areas have the highest number of bicycles per capita of all spatial types. Furthermore, no decline in the number of conventional bicycles can be observed in metropolitan areas; rather, the figure has increased from 907 to 940 per 1,000 people since 2017. In contrast, in small towns and villages in rural regions, the number of bicycles decreased significantly from 848 to 757 per 1,000 people during the same period. Overall, the number of conventional bicycles in 2023 was less homogeneously distributed across the different spatial types.

than in 2017. The accompanying Table 31 shows the bicycle stock by regional type, household type and economic status.

The high number of conventional bicycles in households does not allow for direct conclusions about the availability of transportation from an individual perspective. Although, statistically, there is almost one conventional bicycle for every person in metropolitan areas, only 71 percent of people living there report having access to a functioning bicycle.

Therefore, people who own two or more bicycles are not a rare phenomenon. At the same time, across all room types, almost 30 percent of households do not have a conventional bicycle.

Such a disparity between the number of bicycles and the availability of transportation does not exist for pedelecs. Here, the proportion of those who have access to a pedelec exceeds the per capita number.

This suggests that some of these two-wheelers are used jointly by several people.

More and more e-bikes are being used by people with low usage intensity.

People who use a two-wheeler with electric assistance cover more kilometers on average. The average daily mileage for people using bicycles or e-bikes on the reference day is 9 kilometers for conventional bicycles and 14 kilometers for e-bikes. While this mileage for conventional bicycles has remained almost constant since 2017, the daily mileage of e-bike users has decreased by 2 kilometers. This suggests that, compared to the past, people with lower usage intensity are now also using e-bikes, whereas it is less common for people who never cycled before to do so. This results in an overall increase in the number of e-bikes in German households and a lower average daily mileage. Table 32 shows the frequency of use and the distances traveled for bicycles and e-bikes by age group, gender, and location.

Table 30
Transportation performance with conventional bicycles and pedelecs on an average day by age group

	2023			2017		
	conventional bicycles	Pedelecs total		conventional bicycles	Pedelecs total	
daily mileage in million passenger-kilometers	My. km	My. km	My. km	My. km	My. km	My. km
IN TOTAL	81	34	115	93	9	101
AGE GROUPS						
0 to 9 years	4	2*	6	4	0	4
10 to 19 years	10	1	11	12	0	12
20 to 29 years	10	2	12	12	0	12
30 to 39 years	12	3	15	14	0	14
40 to 49 years	11	5	16	15	1	16
50 to 59 years	16	7	23	17	2	19
60 to 69 years	11	8	19	11	2	13
70 to 79 years	5	5	10	7	2	10
80 years and older	2	1	3	1	0	2

Calculation without imputed values; total deviations due to rounding; * including passengers on e-bikes

No generally higher frequency of use due to Pedelecs

The fourfold increase in the transport performance of e-bikes since 2017 raises the question of whether this has led to an overall increase in bicycle use. Table 33 The study compares typical bicycle usage between 2017 and 2023, broken down by age group, region, economic status of respondents, and highest educational attainment. In the overall population, the proportion of those who regularly cycle decreased by almost 2 percent. While there was no decline in the cycling share of the modal split during the same period, the mobility-

Overall, the e-bike usage rate has decreased. While the proportion of regular e-bike riders has increased in all socio-demographic groups, these individuals were already regular cyclists before purchasing an e-bike. Therefore, the e-bike boom has not led to the development of new user groups for cycling.

Differences in usage frequency are also evident between the federal states. The three city-states of Bremen, Berlin, and Hamburg have the highest proportions of people who cycle daily. In Saarland, Thuringia, and Rhineland-Palatinate, on the other hand, 62 percent use a bicycle.

Table 31

Bicycle and pedelec stock by room type, household type and economic status

	Pedelecs per 1,000 inhabitants	Bicycles per 1,000 inhabitants	In total
	Number	Number	Number
IN TOTAL	142	842	984
ROOM TYPE			
urban region			
Metropole	92	940	1.031
Regional center and major city	133	852	985
medium-sized town, urban area	152	845	998
small-town, rural area	158	835	993
rural region			
central city	149	845	994
medium-sized town, urban area	165	802	967
small-town, rural area	159	757	916
HOUSEHOLD TYPE			
young HH (under 35 years old)	75	955	1.030
Family-HH	101	892	993
HH with adults	188	905	1093
HH with people aged 65 and over	182	593	775
ECONOMIC STATUS			
very low low	71	618	688
	130	634	764
medium	143	726	869
high	153	907	1.060
very high	152	1.086	1.238

Approximately 50 percent of the respondents aged 14 and over never or almost never use a bicycle. Figure 61 shows the breakdown of usage frequency by federal state . Similarly, Figure 62 graphically illustrates the breakdown by age group.

Satisfaction with cycling conditions is insufficient

One possible reason why cycling is not
The increase in users could be due to declining satisfaction:
Since 2017, the
Assessment of the general traffic situation for bicycles was
also reduced – also in relation to

Table 32

	Percentage of people using bicycles and pedelecs on the reference date		Daily distance traveled by bicycle and pedelec by persons using these means of transport on the reference date	
	conventional Bicycle	e-bike	conventional Bicycle	e-bike
	%	%	km	km
IN TOTAL	10	3	9	14
AGE GROUPS				
0 to 9 years	13	4	4	7
10 to 19 years	17	1	7	12
20 to 29 years	10	2	11	13
30 to 39 years	11	3	11	13
40 to 49 years	11	4	11	15
50 to 59 years	11	4	12	16
60 to 69 years	9	5	11	17
70 to 79 years	6	4	10	18
80 years and older	4	2	7	12
GENDER				
masculine	11	3	11	16
female	9	3	8	13
ROOM TYPE				
urban region				
Metropole	15	3	10	15
Regional center and major city	14	4	9	13
medium-sized town, urban area	10	3	9	15
small-town, rural area	8	3	9	15
rural region				
central city	12	4	9	13
medium-sized town, urban area	9	3	9	13
small-town, rural area	6	2	8	16

Table 33

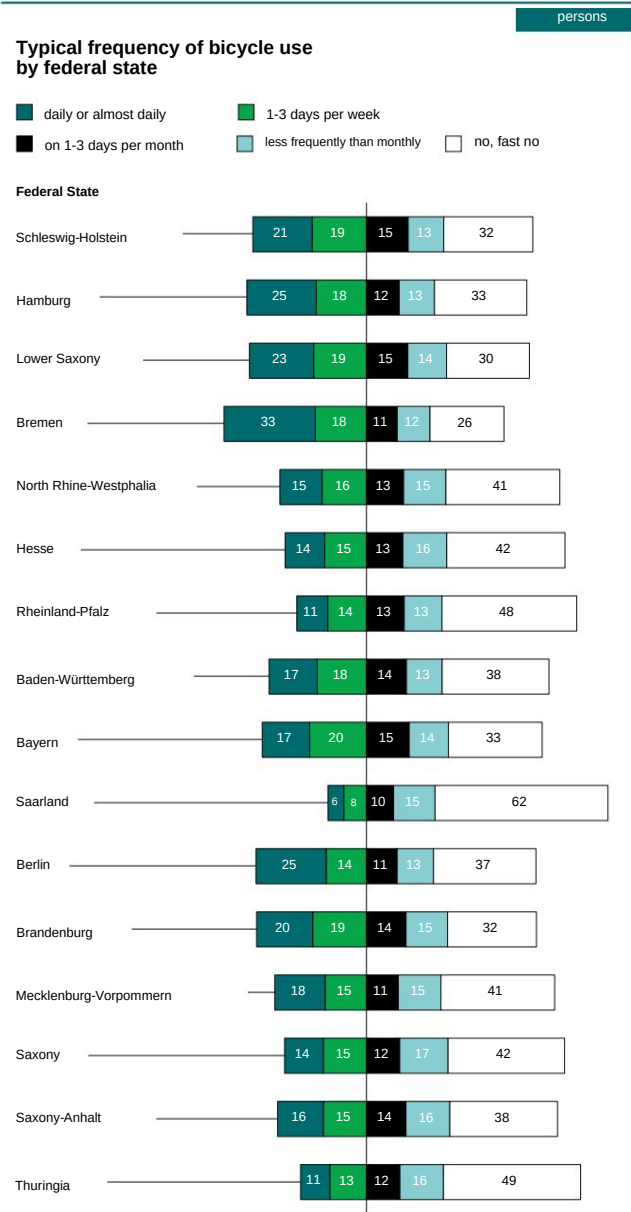
Percentage of people who use a bicycle or pedeleg at least once a month, by age group, type of location, economic status and educational attainment – 2017 vs. 2023

	2017			2023			People who do not regularly use bicycles: Change in 2023 compared to 2017
	People who use a bicycle at least once a month		People without regular bicycle use	People who use a bicycle at least once a month		People without regular bicycle use	
	no e-bike available	e-bike available		no e-bike available	e-bike available		
	%	%	%	%	%	%	Trend
IN TOTAL	45	5	50	33	15	52	↗
AGE GROUPS							
14 to 19 years	61	1	38	55	4	41	↗
20 to 29 years	48	1	51	40	6	54	↗
30 to 39 years	50	2	48	39	13	48	↔
40 to 49 years	53	3	44	39	16	45	↗
50 to 59 years	47	6	47	34	19	47	↔
60 to 69 years	41	10	49	28	21	50	↗
70 to 79 years	32	9	59	21	18	61	↗
80 years and older	18	4	78	13	8	79	↗
ROOM TYPE							
urban region							
Metropole	49	2	49	43	10	47	↗
Regional center and major city	44	4	52	35	13	51	↗
medium-sized city, urban area, small-town,	45	6	49	33	16	51	↗
rural area	45	6	49	30	16	54	↗
rural region							
central city	47	5	48	33	15	52	↗
Medium-sized town, urban area, small-town,	41	6	53	30	17	54	↗
rural area	42	5	52	26	17	58	↗
ECONOMIC STATUS							
very low low	42	3	55	30	7	63	↗
medium	41	4	56	27	13	60	↗
	40	6	54	30	14	56	↗
high	52	4	43	35	16	49	↗
very high	54	5	41	44	17	38	↗
EDUCATIONAL QUALIFICATION							
(still) without a secondary school	59	1	40	51	4	45	↗
leaving certificate, 8th grade at a secondary school	33	7	59	20	17	64	↗
Secondary school leaving certificate, intermediate school	43	5	53	29	16	55	↗
leaving certificate, 10th grade at the POS (Polytechnic Secondary School), university							
entrance qualification, Abitur (General Qualification for University Entrance)	51	3	46	38	13	49	↗
university degree	58	4	39	46	15	40	↗
(University of Applied Sciences/University)							
other degree	37	5	59	26	9	66	↗

People aged 14 and over. This is initially surprising, as in recent years the focus of transport planning has increasingly been on expanding and improving cycling infrastructure. One possible reason could be increased expectations for cycling infrastructure due to very positive examples from certain municipalities.

These high expectations could lead to disappointment due to slower progress in one's own community. Generally, satisfaction with the traffic situation for cyclists is decreasing.

Figure 61



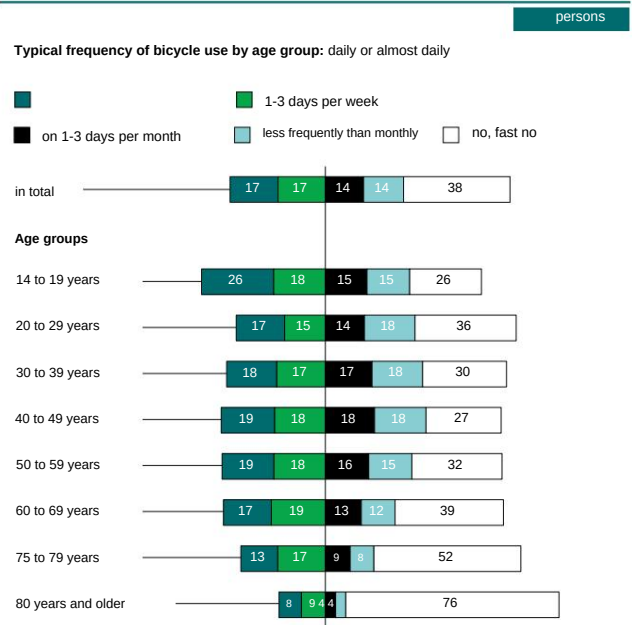
Figures in percent; persons aged 14 and over

Satisfaction with cycling decreases slightly with increasing age. Figure 63 shows a comparison of average cycling satisfaction between 2017 and 2023 by age group. School grades between 1 and 6 are used as the unit of measurement. Figure 64 relates satisfaction to regional types. It shows that locations with a higher share of cycling do not automatically equate to high satisfaction. Metropolitan areas have the highest share of cycling among the regional types, yet satisfaction with the cycling situation is lowest there. It should also be noted that both assessments such as the modal split of cycling are in the lower quadrant of expectations. The shares are consistently below the 20 percent threshold for each regional type, and the average grades are significantly below 2, sometimes even below 3.

Teenagers are the top cyclists.

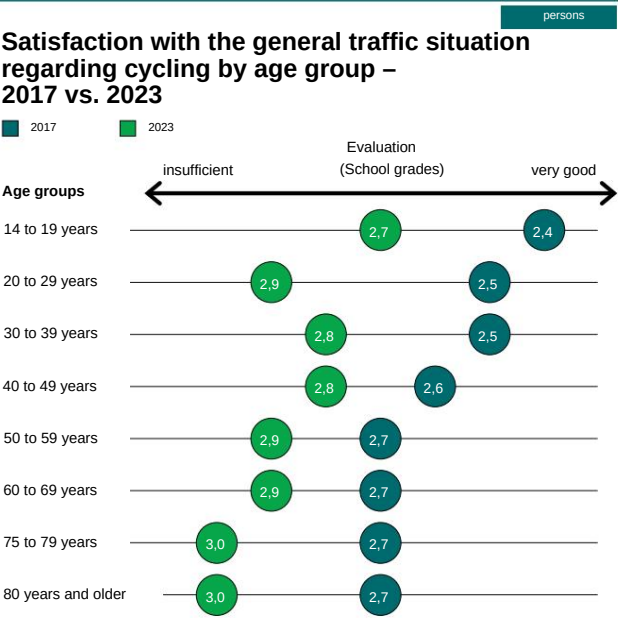
More than one in four people between the ages of 10 and 20 use a bicycle daily or almost daily. This makes the under-20s the age group with the highest rate of frequent bicycle use. Figure 62 shows the frequency of bicycle use by age group. Cities have the highest rate of bicycle use by type of area.

Figure 62



Figures in percent; persons aged 14 and over

Figure 63



Figures in percent; persons aged 14 and over
MiD 2023 | Results Report | Source: MiD 2023

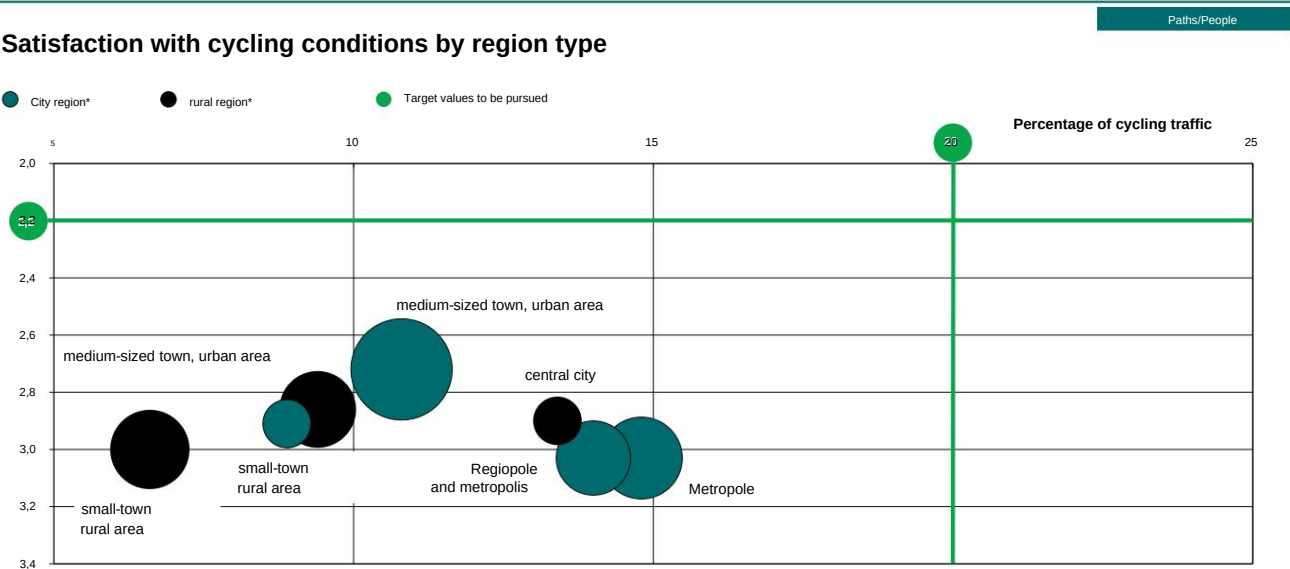
Those who use a bicycle at least once a week. In metropolitan areas, this figure is 41 percent, compared to 27 percent in small towns and villages. The percentage of people who never use a bicycle differs only slightly between the different types of regions and is consistently between 35 and 41 percent.

Pedelecs are now also trending among younger people.

Older people in particular are often considered the main target group for pedelecs: With the electric assistance, they can conserve their strength and still participate actively and nimbly in traffic.

This is true insofar as one in twenty people aged 60 to 69 uses an e-bike on an average weekday, the highest proportion among all age groups. Since 2017, however, the number of e-bike users has multiplied in all age segments, and the younger age groups in particular have caught up. While the proportion of daily e-bike users among those over 60 has increased two to threefold, the proportion among those under 40 has increased tenfold since 2017. This underscores that e-bikes have now become a widely accepted mobility option.

Figure 64



Note on cycling conditions
*Circle size proportional to population share
Share of cycling routes: figures in percent; Satisfaction: persons aged 14 and over, average rating on a scale of 1-6
MiD 2023 | Results Report | Source: MiD 2023

and can also be used by people who are fully physically capable.

Cycling to work is often electrically assisted

Another aspect that puts the classification of pedelecs as a means of transport primarily for retirees into perspective is their use for commuting: almost a quarter of all commutes made by bicycle are with electric assistance. A comparable proportion is seen for shopping trips, errands, leisure activities, and accompanying others. For educational journeys, which are predominantly made by very young people, the proportion is lower, so the pedelec plays only a minor role in this context. Figure 65 illustrates the proportions of journeys made with conventional bicycles and pedelecs in the total traffic volume according to trip purpose. When considering the modal split by travel performance, the proportion of pedelecs is higher than that of conventional bicycles across all trip purposes. This is due to the longer average trip length of pedelecs.

Cargo bikes remain a niche product so far.

Cargo bikes are considered an alternative, environmentally friendly means of transport for somewhat more complex everyday transport tasks, although they have a shorter range and lower payload than cars. Depending on their design, they are suitable for transporting both passengers and goods. The MiD 2023 survey therefore also recorded the typical frequency of cargo bike use, again focusing on the subgroup aged 14 and over, and thus not including potential younger passengers in the "cargo section."

Although cargo bikes can be a solution for everyday transport tasks, they are a rare sight on the roads: Over 97 percent of people in Germany never or almost never use this mode of transport. However, differences in usage frequency are evident between the various types of regions, as Figure 67 illustrates.

The primary use case for cargo bikes is urban mobility in metropolitan areas. There, twice as many people use such a vehicle at least occasionally compared to the national average. Even with a more precise differentiation of the

Figure 65

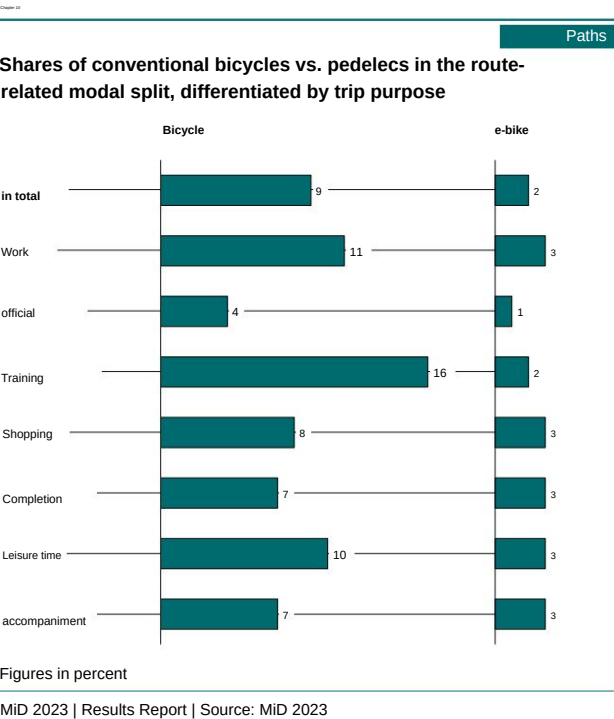


Figure 66

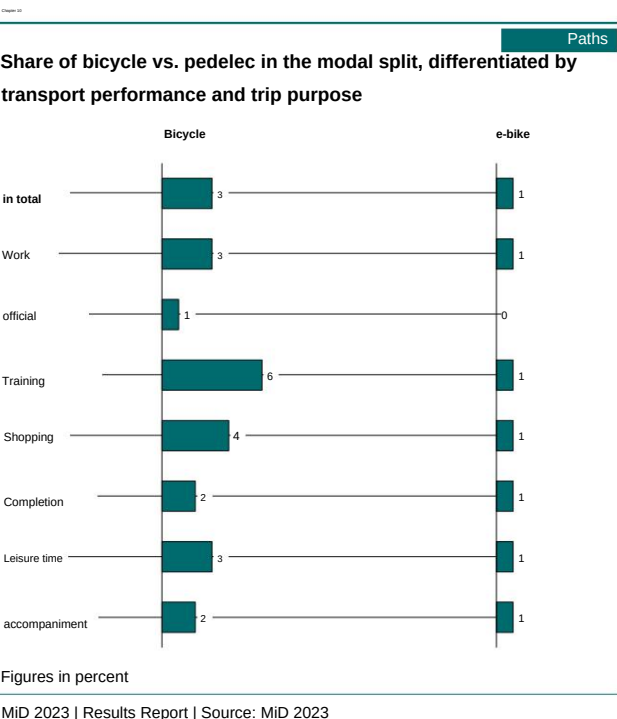
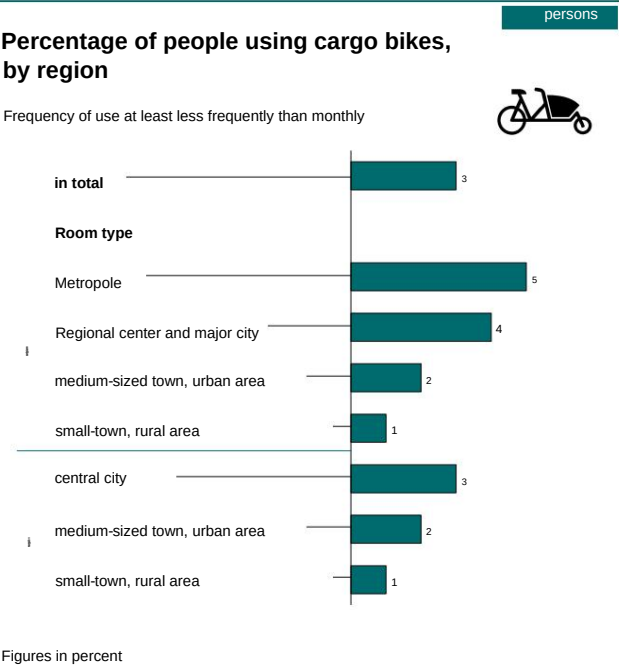


Figure 67



MiD 2023 | Results Report | Source: MiD 2023

Differences are apparent depending on the location. Within metropolitan areas, cargo bike use is concentrated in the city center. In these densely populated inner-city areas, over 2 percent of people use a cargo bike daily or almost daily. This percentage decreases, as with all other types of regions, towards the outskirts of a municipality, so that in the outskirts of even metropolitan areas, the proportion of people using cargo bikes daily is negligible. Thus far, the cargo bike is primarily a feature of some inner-city residential areas, and hardly beyond.

The higher the availability of cars on the reporting day, the lower the frequency of cargo bike use. This suggests that some people do indeed use cargo bikes as a car replacement. At the same time, there is no evidence that this use is primarily driven by economic considerations among low-income households. On the contrary: almost 4 percent of people from households with a very high economic status use cargo bikes, while in households with a very low economic status, only slightly more than 1 percent of people use them. Another group with a disproportionately high

Members of car-sharing organizations have a high usage rate. More than one in ten of these people use a cargo bike at least occasionally.

Another aspect of typical use:
Multimodality is the norm among cyclists.

In public discourse, the prevailing view is often that cyclists and motorists are two separate groups competing for space in traffic. It is therefore interesting to consider what proportion of kilometers cycled are attributable to individuals who also regularly use a car or public transport. Figure 68 illustrates this for different region types, showing the proportion of cycling activity attributable to individuals who also regularly travel by car and/or public transport. Regular use is defined as using the respective mode of transport one to three days per month or more frequently. The figures show that the group of cyclists who do not regularly use a car or public transport is small across all region types, reaching a maximum of 13 percent in regional centers and large cities. The highest proportion of cyclists who do not regularly use a car is found in metropolitan areas, at just over 40 percent. Overall, three-quarters of the kilometers cycled are attributable to individuals who also drive a car.

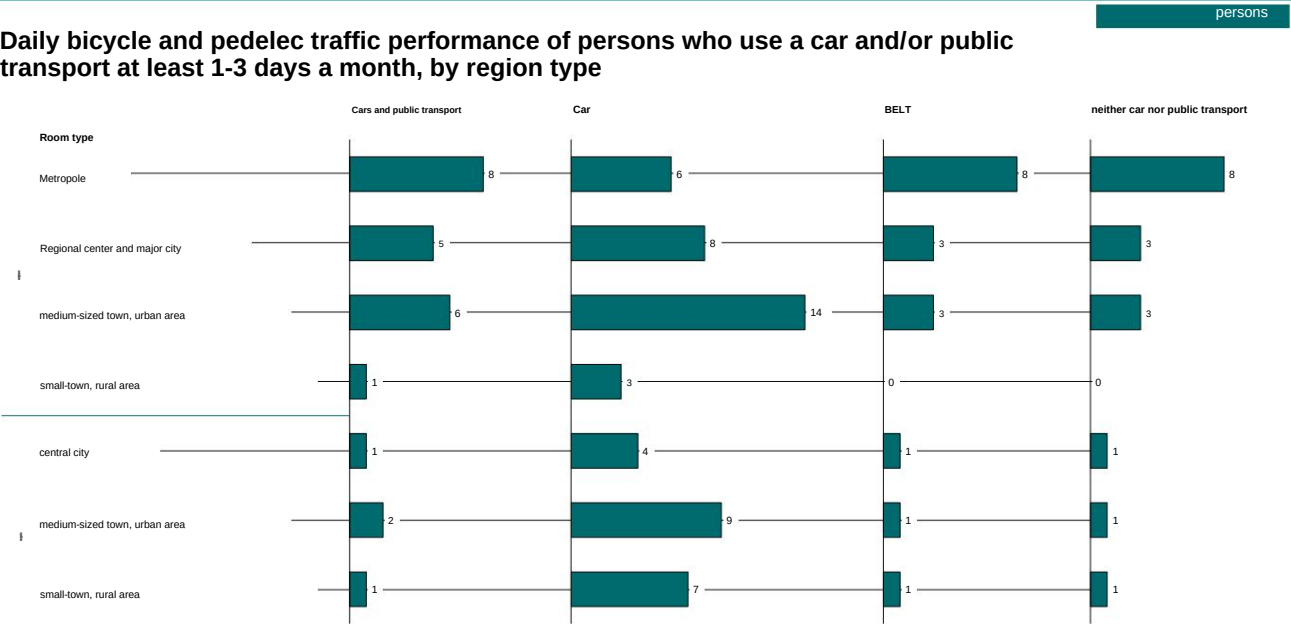
This indicates that portraying cyclists and motorists as clearly separate and competing groups inadequately reflects the actual overlaps.

Cycling and leisure traffic – only
there or also on other occasions?

One usage scenario that can lead to particularly high mileage on an individual level is the use of the bicycle as a sports device. A related use is leisurely rides. The question arises whether people who rarely cycle use their bikes disproportionately often as recreational or sporting equipment, rather than as a means of transport for everyday travel on other occasions.

To investigate this question, bicycle trips were divided into three categories: commuting, trips for sport or leisure, and all other bicycle trips. The comparison of kilometers traveled is shown in Table 34 .

Figure 68



Figures in million passenger-kilometers
MiD 2023 | Results Report | Source: MiD 2023

As expected, people who cycle daily show the highest proportion of commutes relative to their total mileage. However, for people who cycle weekly or monthly, the proportion of trips for exercise or leisure is only slightly higher than for daily users. This result should be interpreted with a caveat: The category "other cycle paths" may include additional trips for exercise or leisure.

The data may include trips whose purpose was only recorded in broad categories. These trips therefore cannot be more precisely categorized within the leisure trip category. However, the finding remains that the sole use of a bicycle for leisure – whether conventional or electric – is rather the exception. Those who are experienced cyclists generally not only continue to use their bikes for leisure but also rely on them for other occasions.

Table 34

Share of cycling mileage by trip purpose and frequency of cycling

typical frequency of use bicycle or e-bike	Wegezweck			Total daily mileage*
	Commuting routes	Sports or leisurely drive	Other routes by bike	
	%	%	%	My. km
daily or almost daily, 1-3 days per	25	11	64	59
week, 1-3 days per month	13	17	70	19
	7	15	78	3
less frequently than monthly	7	4	89	1
no, fast no	11	5	84	2

* Missing total mileage due to individuals with unknown usage frequency

A brief cycling report – more is more demanding.

High expectations are placed on the bicycle.

It still cannot fully meet these expectations. While the number of bicycles is changing, and many cyclists have modernized or even expanded their fleets in recent years, and e-bike assistance has led to more kilometers cycled, this still primarily reflects a consolidation of this mode of transport choice among those who already relied on bicycles before the advent of e-bikes and the ongoing debate about cycling infrastructure. Those who don't cycle at all have not been reached to the hoped-for extent. One reason for this is the low level of satisfaction with cycling conditions, which are often rated poorly. The underlying factors are likely—the MiD survey doesn't differentiate here—the condition of cycle paths and the sense of safety and well-being in traffic for those who have chosen to cycle. This requires less of the isolated improvements that are often observed, but rather the broader task of a consistently improved cycling infrastructure. This remains the Achilles' heel of cycling. Only when cycling infrastructure goes hand in hand with the comprehensive technological leaps in bicycle technology can significant increases in cycling be expected. This also includes a more bicycle-oriented mobility culture. The MiD 2023 study shows little success in this regard across Germany as a whole, but some successes have been observed in individual local areas. However, these are often cities that have been working on their cycling infrastructure for some time and often need many years to reap the rewards. But these examples exist, and they should encourage other municipalities to follow a similar path.

11. Walking – the basis of mobility

One of the most noticeable developments in everyday mobility in Germany since 2017 is the increase in walking – both in terms of the number of trips and the total kilometers traveled. Despite this trend, the importance of walking is often underestimated. It is the most primal form of human locomotion. It is the first form of movement learned and retains a central function even in old age. Although the distances covered on foot are usually short, walking still represents a significant component of everyday mobility for many people.

In urban and rural areas, the percentage increased by approximately 3 percentage points, while in the central cities of rural regions it increased by 5 percentage points. Only in very sparsely populated regions is there no clear picture; here the percentage remains largely stable.

Interpreting these figures is difficult. The coronavirus pandemic has given local mobility a temporary boost. Whether this will lead to a

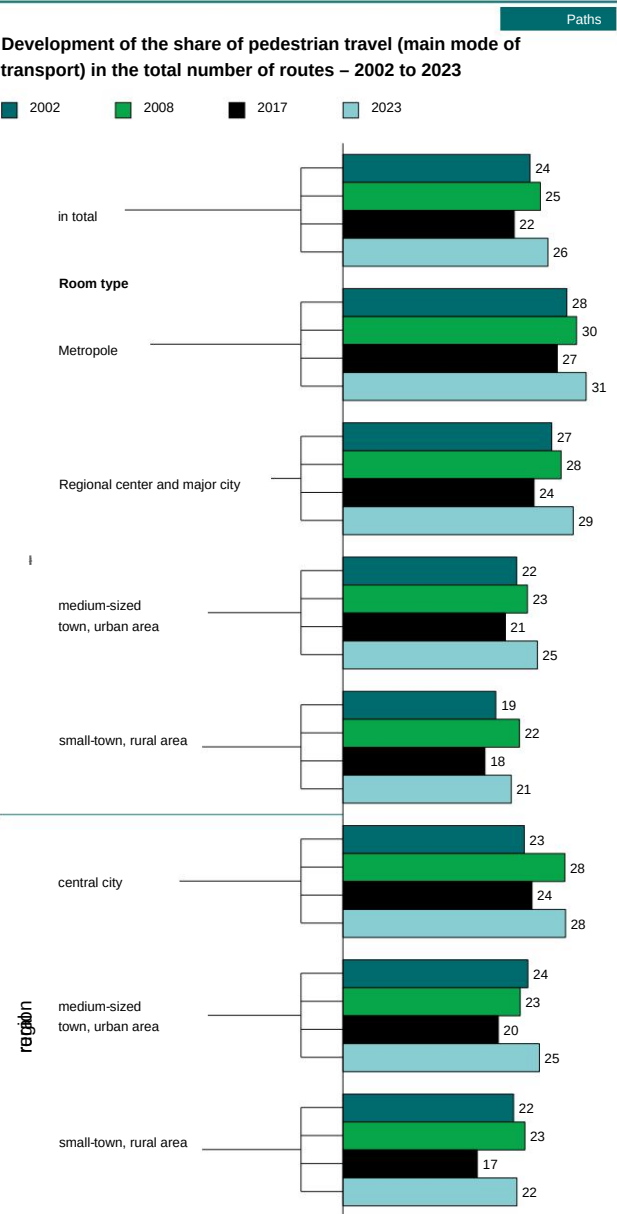
The MiD (Mobility in Germany) offers three approaches to examining pedestrian traffic. A key approach involves the data on mobility on the reference date. Following the established principle of the primary mode of transport in transport science, "walking" here refers to journeys made solely on foot. As soon as another mode of transport is indicated in addition to "walking," the entire journey is attributed to that mode of transport. This is based on the assumption that it accounts for the larger share of the journey. Furthermore, there is a question regarding the general frequency of walking, which allows for the formation of different user groups. Finally, for a subsample, journeys are broken down by stage, making the proportion of walking in intermodal journeys—that is, journeys involving multiple modes of transport—particularly visible.

This chapter applies all three evaluation methods. First, the time series is examined to contextualize the development of pedestrian traffic. Following this, the study investigates who travels on foot, their level of satisfaction with walking, and the role of walking in intermodal journeys.

Pedestrian traffic – higher share in urban areas and steady growth since 2002

Between 2017 and 2023, the proportion of journeys made on foot increased particularly sharply. However, looking at the entire time series from 2002 to 2023, the proportion of journeys made on foot was particularly low in 2017. Overall, the figure shows slight but steady growth from 2002 to 2023. This also applies to most spatial categories. For example, the proportion of pedestrian journeys in urban areas, with the exception of small-scale urban areas, has increased.

Figure 69



Figures in percent, all routes; MiD 2002, 2008, 2017, 2023

The extent to which the trend that may have existed around 2017 was interrupted, and the degree to which the year 2023 will still be affected by pandemic-related effects, will only become clear with longer-term developments.

gen.

In general, all MiD surveys show a clear correlation between population density and the proportion of journeys made on foot. The higher the population density, the higher the proportion of journeys made on foot. In 2023, the highest value of 31 percent was recorded in metropolitan areas, while the lowest, at 21 to 22 percent, was found in small towns and villages – regardless of whether these areas are located within an urban or rural region.

In pedestrian traffic, there is a significant increase in distances traveled.

Not only the proportion of journeys made on foot, but also their length has increased over time, from an average of 1.3 kilometers in 2002 to 1.7 kilometers in 2023 (Table 35). At this level, this corresponds to an increase of around a quarter. Walking, like other modes of transport, has thus shown a significant increase in distances traveled over the years. This also applies to the daily distance. If only mobile individuals who make at least one journey on foot are considered, the average daily distance traveled has increased from around 3 kilometers in 2002 to 4 kilometers in 2023.

Looking at the long-term data series, the total number of purely pedestrian journeys in Germany has remained more or less constant, even though it has increased compared to 2017 due to the lower figure. This applies both to the average number of these journeys made by mobile individuals with at least one such journey and to the extrapolation to the total population.

Looking at the proportion of people who made at least one trip on foot on the reference date, the time series shows a slight decrease. This is primarily due to the lower mobility rate; in other words, fewer people left their homes in 2023. However, when considering only mobile individuals, 40 percent of respondents reported making at least one trip on foot – the same percentage as in 2002. This doesn't mean that more people are walking, but rather that a comparable number are walking longer distances. In summary, the significant increase in pedestrian-kilometers is mainly due to the increased distances traveled on foot.

Walking is an integral part of everyday mobility for two-thirds of those surveyed.

Whether a person walks on a given day is partly a matter of chance. Behavior observed on a single day only ever represents a snapshot of a person's overall mobility. Therefore, asking about the general frequency of use is more helpful.

Table 35

Pedestrian traffic: Average and absolute values – 2002 to 2023					
	2002	2008	2017	2023	
average distance	1,3	1,4	1,5	1,7	km
Daily walking route only for mobile persons with a footpath	3,1	3,2	3,6	4,0	km
Number of pedestrian journeys by mobile persons with a pedestrian path	2,3	2,2	2,1	2.3	Number
Walking routes	64	69	56	65	million per day
Passenger kilometers on foot	88	98	93	113	million per day
Percentage of people walking on the reference date	34	38	32	32	%
Percentage of mobile persons traveling on foot on the reference date	40	43	37	40	%

the modes of transport as well as walking. In this way, a more comprehensive picture of the respondents' everyday walking behavior can be obtained, regardless of the reporting date.

39 percent of respondents aged 14 and over stated that they travel (almost) daily on foot, and another 29 percent do so one to three times per week (Figure 70). For two-thirds of respondents, walking is therefore an integral part of their everyday mobility. As expected, the data also shows that the more frequently people walk, the more likely they are to report walking on the survey day (Figure 71). Approximately half of those who walk (almost) daily report walking on the survey day. This proportion decreases significantly with decreasing frequency of walking. The following section compares individuals with and without walking on the survey day to provide a more detailed characterization of the group of pedestrians.

Women and people with the option to work from home were more frequently out on foot on the reference date.

Table 36 shows the socioeconomic characteristics of people who walked and those who did not on foot on the reference date. Overall, the differences between the two groups are small; however, some

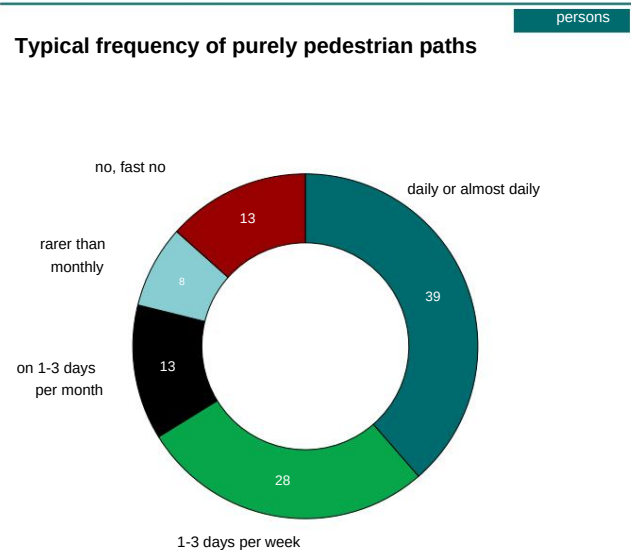
Significant differences are apparent. Similarities exist with regard to age, socioeconomic status, and household type. While the group of pedestrians includes slightly more children and older people, the proportion of "pedestrian households" in which all persons are 65 years and older is correspondingly slightly higher, and the socioeconomic status somewhat lower. However, the differences remain moderate.

However, differences emerge regarding gender and employment status: people who walked are disproportionately often women, while employed people were less likely to have walked on the survey date. At the same time, those who have the option of working from home are more likely to be pedestrians.

Pedestrians exhibit clearly different mobility patterns.

Even clearer differences between the groups emerge when looking at the mobility indicators (Table 37). People who walked on the reference date more often live in households without a car and are more likely to own a public transport season ticket. This clearly affects their everyday mobility behavior: While 48 percent of people who did not walk belong to the car-only group,

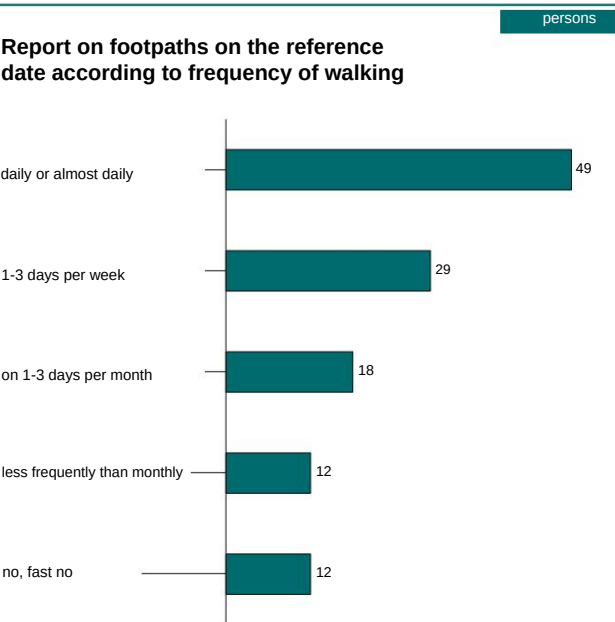
Figure 70



Figures in percent, persons aged 14 and over, excluding deputies

MiD 2023 | Results Report | Source: MiD 2023

Figure 71



Figures in percent, persons aged 14 and over, excluding deputies

MiD 2023 | Results Report | Source: MiD 2023

Table 36

Socioeconomic characteristics of mobile persons with and without walking on the reporting day

	Footpath on the day of the report		
	and	no	
GENDER			
Men	46	53 %	
Women	54	47 %	
ALTER			
Average	44	43 <i>in years</i>	
0 to 9 years	12	9 %	
10 to 19 years	9	10 %	
20 to 29 years	10	12 %	
30 to 39 years	13	13 %	
40 to 49 years	11	13 %	
50 to 59 years	14	17 %	
60 to 69 years	15	14 %	
70 years and older	16	13 %	
TASK			
employed	46	56 %	
Pupil, student, trainee	16	15 %	
Retired person, pensioner	23	18 %	
Miscellaneous	15	11 %	
HOME OFFICE OPTION EXISTS	49	39 %	
SOCIO-ECONOMIC STATUS			
very low low	9	6 %	
	11	10 %	
medium	29	29 %	
high	37	40 %	
very high	15	16 %	
UNIVERSITY DEGREE (University of Applied Sciences/University)	25	22 %	
HOUSEHOLD SIZE	2,6	2.8 <i>Number</i>	
HOUSEHOLD TYPE			
young households (under 35 years old)	9	9 %	
Family households	39	41 %	
Households with adults	31	35 %	
Households with persons aged 65 and over	20	16 %	

While only 37 percent of drivers use cars, pedestrians are more likely to travel by bicycle, public transport, or multimodal means. Eight percent belong to the group of people with low mobility who do not use any of the three modes of transport —car, bicycle, or public transport—in their normal weekday routine and therefore primarily travel on foot.

Furthermore, on the reference day, people who walk cover more distances overall and spend more time in traffic. Their average daily distance, however, is only around 30 kilometers, while people who do not walk cover an average of slightly more than 50 kilometers on the reporting day. Whether or not someone walks on this day is therefore a clearly differentiating factor and not merely a small difference compared to otherwise similar levels of mobility.

On foot: few commutes and many leisure routes

All journeys undertaken on foot represent a broad range of purposes (Figure 72). With two exceptions, the distribution of journeys on foot largely mirrors that of all journeys. Fifteen percent of all journeys are for work. Since commutes are disproportionately long, their share of journeys on foot is predictably lower, at only 5 percent. In contrast, journeys on foot are undertaken for leisure purposes far more frequently. Thirty-seven percent of all journeys on foot are for leisure, compared to 29 percent for all journeys.

A breakdown by spatial type reveals a striking high proportion of shopping trips made on foot in metropolitan areas. There, almost a quarter of all trips are for this purpose, while in small towns and villages it is only a tenth. In all spatial-

Table 37

Mobility indicators of mobile persons with and without walking on the reporting day

	Footpath on the day of the report		
	and	no	
AVERAGE NUMBER OF TRIPS (per person per day)	4,1	3.3	Number
DAILY ROUTE	31	52	km
AVERAGE TRAVEL TIME	1:52	1:37	h:min
CAR EQUIPMENT			
No car in the household	19	10	%
a car in the household	47	44	%
two or more cars	34	46	%
EXERCISES SUBSCRIPTION *	28	23	%
THE MOBILE GROUP			
Driver	37	48	%
Public transport user	10	7	%
cyclist	8	6	%
Car, Radio	20	22	%
Car, Belt	6	6	%
Row, OV	7	5	%
Car, Bicycle, Public Transport	4	4	%
person with limited mobility	8	2	%

* Germany ticket, monthly subscription ticket, annual ticket, regional/local job ticket, company subscription, semester ticket or similar Offers

Leisure trails make up the largest share of footpaths, particularly in rural areas. Leisure walks are generally quite long, averaging 2.5 kilometers. Shopping trips and walks to education, for example, average about one kilometer. Those who walk to work cover an average of 1.5 kilometers.

High satisfaction when walking with large Differentiated by location and frequency

When asked about their satisfaction with the traffic situation at their place of residence, respondents – as in 2017 – showed a high level of satisfaction with walking (Figure 73). On a scale of 1 ("very good") to 6 ("unsatisfactory"), walking received a grade of 2.3.

Only the car receives a similarly high rating. Bicycles achieve an average score of 2.9, and public transport 3.3. Compared to 2017, however, this represents a decline for walking. Back then, the overall rating was 2.0 (see also Chapter 4).

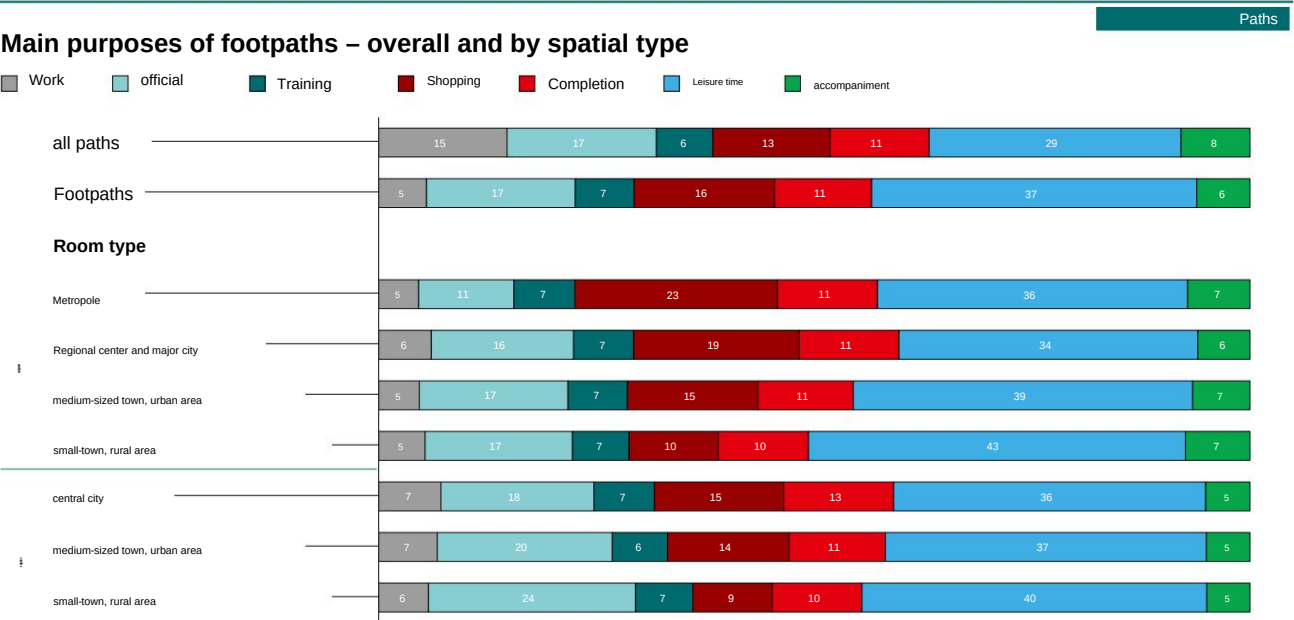
The more frequent use of the paths is one possible reason for the apparently growing demand – or perhaps a deterioration of the pedestrian infrastructure, especially the condition of the paths themselves. Against such a simple explanation...

However, Menhang points out that this assessment, like all other modes of transport, shows that the more frequently people walk, the better they rate the traffic situation in their place of residence. People who walk almost daily give an average grade of 2.2, while those who never walk only give it 3.1.

Significant differences also emerge between spatial types. In metropolitan areas, the rating for walking is 2.1, while in small-town and village areas of rural regions it is 2.8.

The greatest range is seen when both the type of space and the frequency of use are considered. People from metropolitan areas who walk almost daily give an average rating of 2.0, while people who never walk and live in the most sparsely populated areas are the least satisfied, with a rating of 3.8.

Figure 72



Intermodal routes almost always include a walking stage.

So far, only journeys completed entirely on foot have been considered. Walking also plays an important role on parts of journeys, particularly when accessing or egressing to other modes of transport. Since all modes of transport used for each journey were recorded in the survey, the proportion of journeys including a walking segment can be determined.

The mode of transport is listed as "on foot" for 38 percent of journeys. Since 26 percent of all journeys are purely on foot, 12 percentage points are attributable to intermodal journeys with a walking component. In fact, there are hardly any intermodal journeys without a walking segment. Even with the highest level of differentiation between modes of transport – the MiD (Mobility in Germany) distinguishes a total of 20 modes – only 13 percent of all journeys list more than one of these modes of transport, which is barely more than the number of intermodal journeys with a walking segment. Intermodal journeys can therefore essentially be equated with journeys that include at least one walking segment.

its.

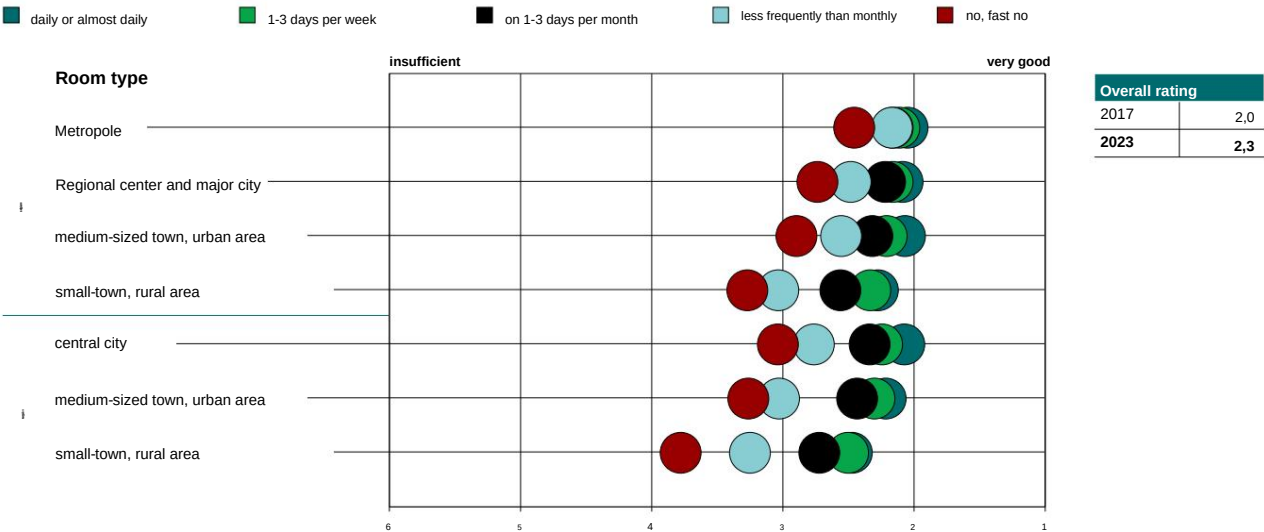
A breakdown by main mode of transport reveals:
Seven percent of all journeys by bicycle involve a

The survey reports walking stages for 11 percent of car journeys and 52 percent of journeys using public transport. Since stops are generally not located directly in front of people's homes, a significant underreporting of walking stages is likely for public transport journeys. The Swiss microcensus, in which all individual stages of a journey are recorded through personal interviews, shows considerably higher figures. The MiD survey reveals that, despite instructions in the questionnaire to include all modes of transport used on the journey, respondents often fail to notice walking stages or consider them worth reporting.

In the MiD (Mobile In-Depth Survey), the recording of stages for a randomly drawn subsample is also carried out in a module following the route recording, in order to ensure comparability of the survey with and without the additional question (see the information box in Chapter 4 on the stage module). Only respondents who had already indicated more than one mode of transport during route recording were included in the module. They were asked to report the mode of transport, duration, and distance for each stage of the journey. On average, 1.3 walking stages per journey were reported. 90 percent of the walking stages are shorter than 2 kilometers, with an average of one kilometer covered.

Figure 73

Satisfaction with the pedestrian traffic situation at the place of residence, differentiated by frequency of walking routes and spatial types.



Average of a six-point scale from 1=very good to 6=unsatisfactory, persons aged 14 and over, without representatives

The walking stages on intermodal routes are therefore not much shorter than pure walking routes of 1.3 kilometers. With an average of 10 minutes, the duration of walking stages also plays an important role. Given an average total duration of 28 minutes per trip, walking stages are thus of considerable significance.

High importance of pedestrian traffic

This and the other results presented in this chapter underscore the importance of a well-developed pedestrian infrastructure and a high quality of public spaces. When walking, structural features, safety aspects, route layouts, and street design are perceived particularly directly, as movement occurs at a slower pace. High-quality pedestrian infrastructure can therefore make a significant contribution to promoting walking.

12. Mobility and gender in the context of life phase and role models

Gender-specific analyses are standard practice in mobility evaluations. Many key indicators in this report are also differentiated accordingly.

Unlike previous MiD reports, gender-specific mobility behavior is addressed in a separate chapter this time. This encompasses far more than just the mobility patterns of men and women. Considering life phases and, in particular, integration into professional and family life, differences emerge that cannot be attributed to biological sex, but primarily to societal roles.

Numerous studies show that women perform a significantly larger share of care work. Time-use studies, for example, can describe very precisely which tasks related to housework, childcare, and eldercare are performed at which times of day. This reveals specific mobility needs and habits. This chapter aims to demonstrate the contribution that mobility surveys can make in this context. The chapter begins with "classic" gender-specific analyses and then transitions to a life-phase- and role-specific analysis.

Older women are about 20 years behind in terms of driver's license possession.

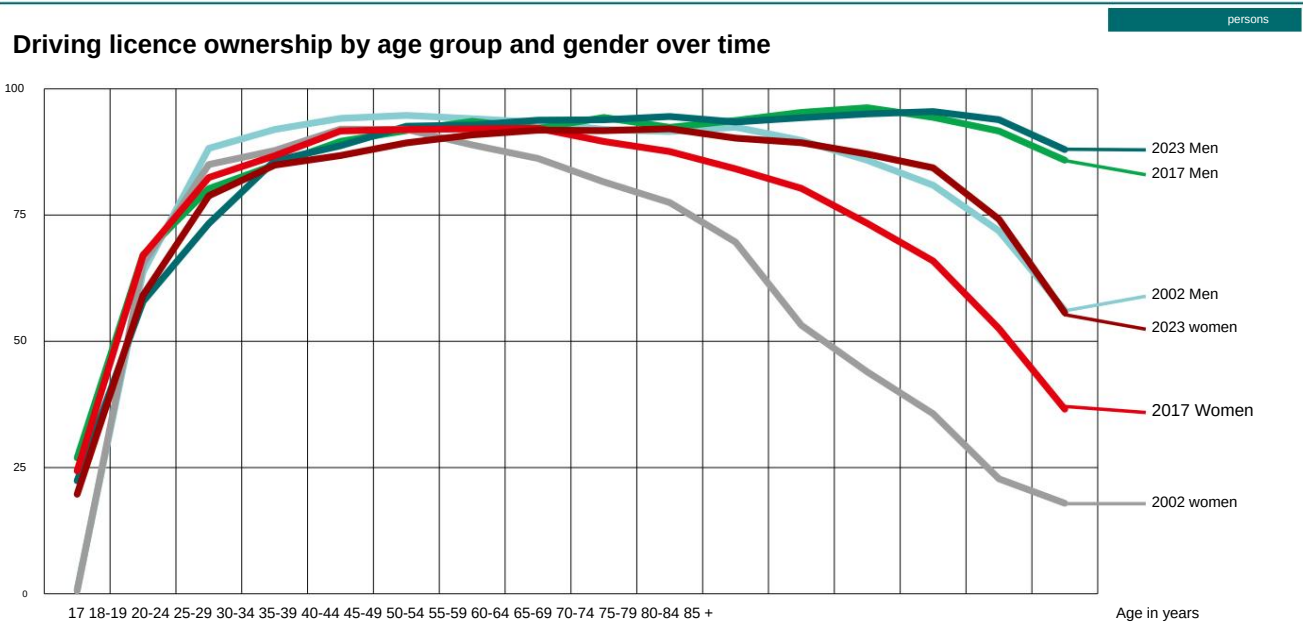
Having a driver's license is a key requirement for independently using a car.

Obtaining a driver's license has been considered part of growing up for many generations. Since 2008, it has been possible to obtain a driver's license at the age of 17.

Approximately one-fifth to one-quarter of a given age cohort of young women and men take advantage of this option. This involves bringing forward the date of obtaining a driver's license. In the various years of the MiD survey, the number of 18- to 19-year-olds holding a driver's license has remained consistent. Only the most recent MiD survey in 2023 shows a slightly lower rate of driver's license ownership – particularly among young men. However, this difference should not be overemphasized. Compared to 20 years ago, more young people are now studying, they are in education for longer, and they often start a family later in life.

Students in particular usually live in urban areas, are less dependent on owning a car and do not necessarily have the financial resources to obtain a driver's license.

Figure 74



Figures in percent, persons aged 17 and over, excluding deputies

At the upper end of the age scale, a distinctly different trend emerges. For men, the driver's license ownership rate remains consistently very high at well over 90 percent from middle age into old age – only declining slightly among those aged 85 and older. For women, however, the rate drops significantly in old age. From around age 60, the proportions of male and female driver's license holders begin to diverge.

In 2023, the trend in driver's license ownership among women closely mirrored that of men in 2002. For example, the rate for women aged 85 and over was 56 percent – exactly the same as for men in this age group in 2002. In the older age groups, women thus lagged behind men by approximately 20 years. It is foreseeable that this gap will gradually close in the coming years.

From adulthood onwards, women and men with different mobility profiles

In early childhood, there are hardly any differences in mobility indicators – percentage of time spent away from home, number of trips, daily distance traveled, and travel time. Children and

Young people, regardless of gender, participate in traffic in very similar ways. Up to the age of 10, both the travel time and the daily distances traveled by girls and boys are significantly below the national average.

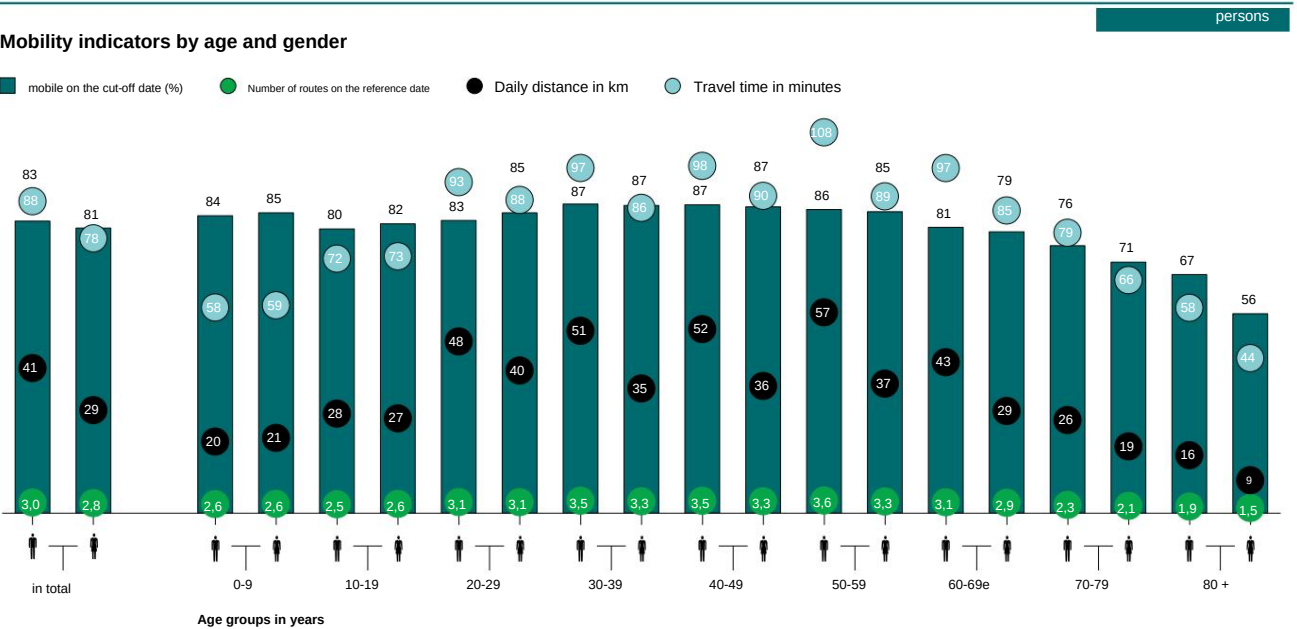
As they reach adolescence, their radius of action expands noticeably: they now need more time for their journeys and cover greater distances.

Gender-specific differences first become visible between the ages of 20 and 30. Despite a comparable proportion of time spent outside the home and the same number of trips in this age group, women's daily distance is now 8 kilometers lower. These gender-specific differences persist to varying degrees from this age into old age.

In the age range of 30 to 60 years, the proportion of time spent outside the home remains largely constant, both overall and between the sexes. However, women travel shorter distances and require less time to do so. The differences are most evident in the daily distance traveled.

While women of this age cover an average of 35 to 37 kilometers per day, men cover 51 to 57 kilometers. At their maximum, men cover a daily distance of...

Figure 75



10 kilometers further back, which corresponds to a weekly difference of 140 kilometers.

From the age of 70, the proportion of time spent outside the home decreases significantly. Now, for the first time, clear gender-specific differences are also emerging in this indicator. While 67 percent of men aged 80 and over still leave their homes, the figure is only 56 percent for women. The distance traveled decreases to 16 kilometers for men and 9 kilometers for women. With the exception of children and adolescents, gender-specific differences in mobility indicators are thus evident across all age cohorts, with daily distances being significantly higher for men.

Men, from the age they reach driving age, consistently spend more time in cars than women.

A similar picture emerges when looking at the modal split of journeys by age and gender. Up to about age 20, the differences are small. Boys and girls are driven by car for almost half of their journeys up to the age of 10, and walk for a third. In adolescence, public transport and other modes of transport that can be used independently and without adult supervision become increasingly important.

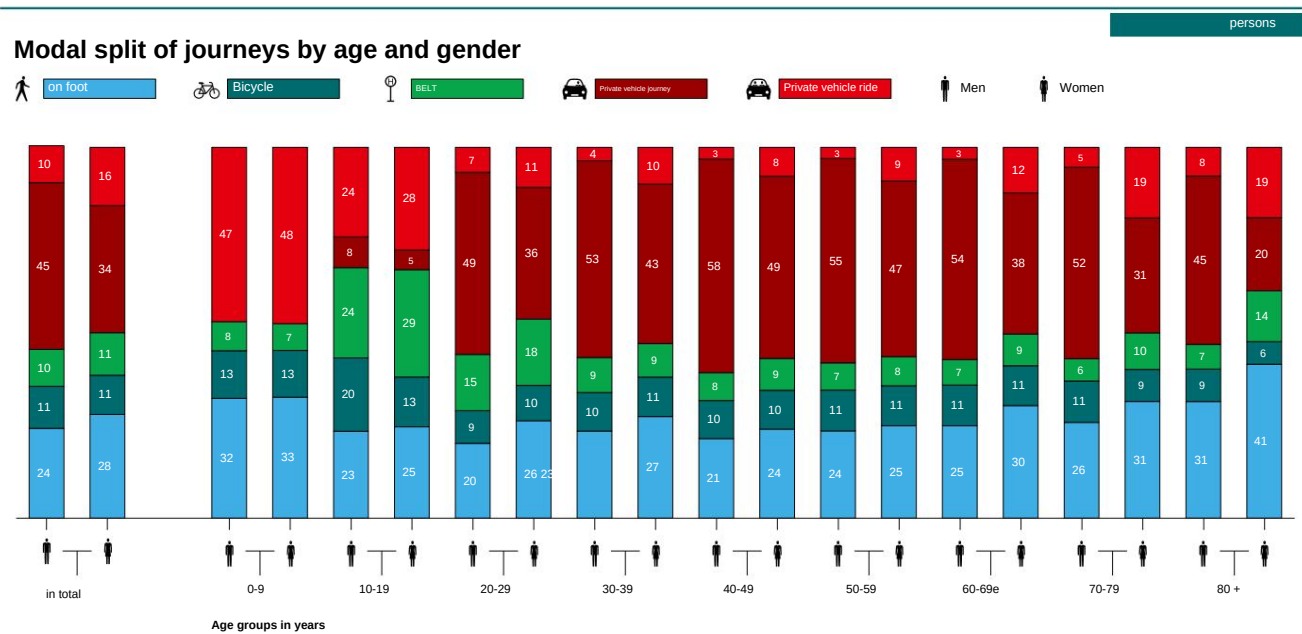
Cycling is gaining in importance, with boys cycling more and girls using public transport more often.

Upon obtaining a driver's license, the proportion of journeys made by car increases dramatically for both young women and young men. From this point on, with the exception of very elderly women, both groups are most often behind the wheel themselves and less frequently as passengers. Despite an overall low proportion of passengers, a clear gender difference exists across all age groups. The proportion of passengers in private vehicles is two to four times higher for women.

In the 20-60 age group, men have a significantly higher proportion of journeys made by private motorized transport – 8 to 14 percentage points higher than women. Women, on the other hand, use public transport more frequently and walk more often. Cycling plays a similar role in everyday transport for both genders, accounting for 9 to 11 percent of all journeys. Only beyond the age of 70 does the proportion of cycling become higher for men than for women.

Women.

Figure 76



Figures in percent, all persons

In general, gender-specific differences become significantly more pronounced from around the age of 70. For very elderly women, walking is the dominant mode of transportation, and the use of public transport is twice as high (14 percent) as for men. Men, on the other hand, still predominantly use cars at this age, accounting for 53 percent of their journeys.

Mobility behavior in young people is strongly dependent on their occupation.

The following section examines the differences in the mobility behavior of men and women, which become increasingly apparent from the age at which they obtain a driver's license. The initial focus is on the causes and underlying factors in young people. This is followed by an analysis of mobility in middle and older age.

This study examines the mobility of young women and men aged 18 to under 29. While children and adolescents experience relatively similar conditions due to schooling, different lifestyles and life plans develop afterward. Some begin vocational training, others pursue university studies, many are already working, and some have already started a family. A differentiated analysis is necessary to account for these differences and understand their impact on the overall values of the group.

The focus is on gender-specific differences among full-time employees, trainees, and students aged 18 to under 29. This includes 75 percent of young men and 64 percent of young women.

Table 38

Mobility indicators and socioeconomic characteristics – young workers, trainees and students by gender

only persons aged 18 to 29	young full-time Employed persons		young trainees, young		students		
	Men	Women	Men	Women	Men	Women	
Proportion of the total group of 18 to 29-year-olds	46	33	14	12	15	19	%
Mobility rate (outside the office on the reporting date)	85	89	85	87	82	85	%
average number of trips (per person per day)	3,7	3,5	3,0	3,4	2,6	2.8	Number
average distance	15,4	13,8	18,1	12,2	12,8	13,9	km
Daily distance,	56	48	54	41	33	39	km
average travel time	1:40	1:33	1:41	1:24	1:24	1:35	h:min
Percentage of households with a car	84	83	84	80	64	65	%
with public transport pass *	31	37	54	56	82	85	%
urban region	58	66	60	62	76	76	%
rural region	42	34	40	38	24	24	%
socioeconomic status							
very low low	3	3	11	17	23	22	%
	7	7	9	10	12	11	%
medium	30	32	31	29	20	21	%
high	47	44	38	36	34	36	%
very high	13	14	12	7	11	9	%

* Germany ticket, monthly subscription ticket, annual ticket, regional/local job ticket, company subscription, Semester ticket or comparable offers

A certain percentage of the young women were recorded and included in the analysis. The missing percentage, for example, is completing voluntary or military service, is still in school, or works – particularly women – part-time. Furthermore, women of this age are more likely than men to live in a family household. Since part-time employment and family life are significantly more common in older age groups, these aspects are initially excluded from the present analysis. Moreover, determining whether the young person in a family household is a parent or a child would be extremely difficult.

The largest group in this age group consists of full-time employees, comprising 46 percent of young men and 33 percent of young women. Young women are slightly more likely to be studying (19 percent) than young men (15 percent), while the proportion of those in vocational training is close, at 12 and 14 percent respectively. Differences are most pronounced between occupational groups. Students are more likely to live in households with lower economic status and in urban areas. Due to the urban environment and, in particular, the semester ticket, they are far more likely to have a public transport subscription, while their households are less likely to own a car.

Student mobility is characterized by shorter daily distances and a slightly lower mobility rate. If regular work-related commutes, which occur during working hours and are only relevant for trainees and employed individuals, are excluded, the differences are often not very significant.

However, employed persons and trainees generally exhibit a significantly higher level of mobility.

Regarding gender, the differences between occupational groups are greater for men than for women. For example, full-time employed men travel 56 kilometers per day, male trainees 54 kilometers, and male students only 33 kilometers. For women, the comparable figures are 48 kilometers (full-time) and 41 kilometers (trainees). The daily distance traveled by female students, at 39 kilometers, is only slightly lower than that of female trainees, but significantly higher, by 6 kilometers, than that of male students.

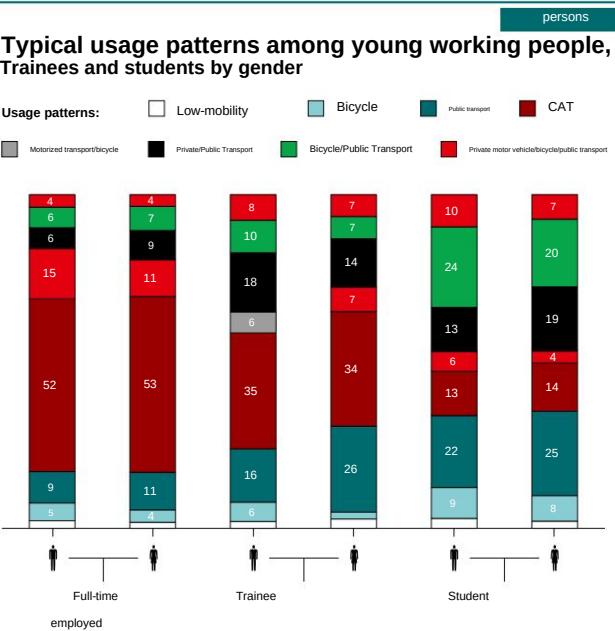
Overall, female students exhibit higher commuting distances, travel times, and daily distances, contradicting the general gender-specific trend in the data. However, when comparing all 18- to 29-year-olds, men exhibit more traditional male mobility characteristics than women. This is primarily due to the higher proportion of employed men, who achieve the highest level of mobility compared to all other groups.

Students take advantage of diverse mobility options, while young professionals rely on cars.

Significant differences are also evident in the mobility behavior of the three occupational groups. First, the modal groups are considered (Figure 77). These represent the set of transport options (car, bicycle, and public transport) typically used by an individual over the course of a week. While the differences between the occupational groups are pronounced, the pattern within the groups is very homogeneous between the sexes. For example, just over half of the employed, just over a third of the trainees, and only just over a tenth of the students belong to the group of people who exclusively use a car.

The values differ between men and women by only one percentage point.

Figure 77



Figures in percent, persons aged 18 to 29

For students, but also for trainees of both genders, the use of public transport – whether as the sole means of transport or in combination with others – plays an important role.

Despite similar modal group shares, the modal split of journeys differs between genders (Figure 78). For example, employed young men make more journeys by car than employed young women. For them, walking, which is not considered in modal group analysis, is more important. Furthermore, multimodal behavior – the use of more than one mode of transport over the course of a week – can be associated with very different journey modal shares.

Middle-aged women with larger Range of lifestyles as men

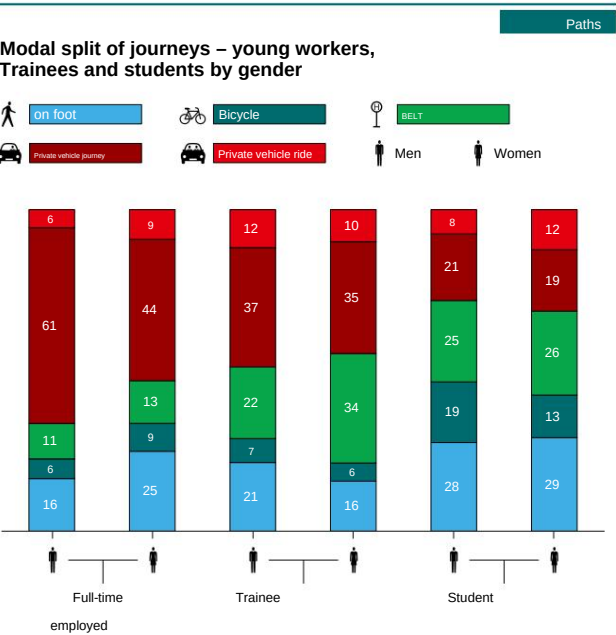
The following analysis examines gender-specific differences in the mobility behavior of middle-aged individuals – defined as those aged 30 to under 50. Here, too, distinctions are made based on employment status (full-time, part-time, and unemployed) as well as household type. The analysis includes individuals from households with children under 14 years of age (56 percent) and individuals from households with...

Ten percent without children (36 percent). This excludes the two small groups of people with children between 14 and 18 years old and single parents. The latter, with their specific mobility needs and behaviors, represent a special group that can only be adequately captured when considered separately. People with older children are not included, as children of this age already exhibit a high degree of autonomy and have less of an impact on their parents' mobility. Otherwise, average figures for family households would lose their significance.

However, overall 92 percent, and thus by far the largest part, of this age cohort will be included in the analysis.

Initially, employment status varies considerably between genders in this age group (Table 39). While 80% and 85% of men, respectively, are employed full-time, regardless of whether there are children in the household, women's lifestyles in this age group are more diverse. Sixty percent of women from households with children work part-time, while 20% are unemployed or work full-time. Even without children in the household, the proportion of part-time workers is significantly higher among women than among men. However, full-time employment is still the most common form of employment for women in this household constellation (64%).

Figure 78



Figures in percent, persons aged 18 to 29

While the mobility behavior of middle-aged men is primarily shaped by full-time employees, the average values for women reflect very different life situations.

Children shape women's mobility, Men remain unaffected.

Full-time employees exhibit a significantly higher level of mobility, particularly compared to those who are not employed. This manifests itself in longer commutes, greater daily distances, and longer travel times. Full-time employed men achieve higher levels of mobility than full-time employed women; children have no impact on their mobility. In contrast, the mobility level of full-time employed women is lower when children live in the household.

Men still rarely work part-time, although the trend is rising. It is striking that part-time working men include both those with and those without children.

Table 39

Mobility indicators and socioeconomic characteristics – middle-aged persons by household type, occupation and gender

HOUSEHOLD WITH CHILDREN UNDER 14 YEARS OLD							
only people from 30 to 49 years	Full-time employed		Part-time employed		not employed		
	Men	Women	Men	Women	Men	Women	
proportion of people	85	20	9	60	6	20 %	
Mobility rate (outside the office on the reporting date)	90	89	92	90	74	81 %	
average number of trips (per person per day)	3,6	3,4	3,7	3,7	2,1	2.6 <i>Number</i>	
average distance	15,8	11,0	11,8	9,4	9,5	9,0 <i>km</i>	
Daily distance,	57	37	43	35	20	24 <i>km</i>	
average travel time	1:40	1:25	1:37	1:26	1:07	1:13 <i>h:min</i>	
Percentage of households with a car	96	93	83	95	73	87 %	
with public transport pass *	15	21	29	15	26	16 %	
urban region	63	65	76	64	60	64 %	
rural region	37	35	24	36	40	36 %	
SOCIO-ECONOMIC STATUS							
very low low	2	1	8	2	33	12 %	
	4	3	9	4	21	11 %	
medium	25	16	24	23	19	31 %	
high	44	46	40	44	22	34 %	
very high	24	34	19	27	4	12 %	
HOUSEHOLD WITHOUT CHILDREN							
only people from 30 to 49 years	Full-time employed		Part-time employed		not employed		
	Men	Women	Men	Women	Men	Women	
proportion of people	80	64	8	22	12	13 %	
Mobility rate (outside the office on the reporting date)	88	87	85	86	75	76 %	
average number of trips (per person per day)	3,7	3,3	4,1	3,2	2,0	2.2 <i>Number</i>	
average distance	15,5	13,2	7,9	11,4	9,6	9,5 <i>km</i>	
Daily distance,	57	43	32	37	19	21 <i>km</i>	
average travel time	1:45	1:35	1:28	1:33	1:04	1:25 <i>h:min</i>	
Percentage of households with a car	81	80	58	72	53	49 %	
with public transport pass *	28	33	46	36	36	40 %	
urban region	67	70	78	67	69	71 %	
rural region	33	30	22	33	31	29 %	
SOCIO-ECONOMIC STATUS							
very low low	2	2	17	11	39	34 %	
	6	6	12	9	14	11 %	
medium	25	24	23	24	16	21 %	
high	47	47	39	43	27	30 %	
very high	20	20	9	13	4	3 %	

* Germany ticket, monthly subscription ticket, annual ticket, regional/local job ticket, company subscription, Semester ticket or comparable offers

They live disproportionately often in urban areas. It is possible that less traditional lifestyles are already more prevalent in urban contexts. Part-time working men and women with children travel more, but shorter distances, with overall shorter daily commutes than their full-time working counterparts. While the differences between full-time and part-time working parents are particularly large for men, they are smaller for women.

Non-working individuals in both household types have by far the fewest trips, as well as the shortest trip lengths and daily distances. The very low socioeconomic status of this group is striking. This applies to both sexes in childless households, but only to men in households with children. Non-working women with children will, in most cases, have a full-time employed man.

Households without children are generally more common in urban areas. The significantly lower overall car ownership rate and higher rate of public transport subscriptions among these groups can therefore be explained not only by the absence of children, but also partly by the more urban environment.

The data shows that the underlying factors are complex and affect mobility in different ways. Overall, having children leads to far less of a change in employment status and the associated mobility patterns for men. Women with children mostly work part-time and, even when employed full-time, restrict their mobility more than men.

Working women are less affected by Accompanying paths relieved as men

Figure 79 shows the distribution of route purposes. In childless households, full-time and part-time employed men have a significantly higher proportion of work-related travel and a lower proportion of leisure travel compared to women. No gender-specific differences exist among non-employed individuals in this type of household.

If children live in the household, women in all three employment groups (full-time, part-time, unemployed) travel more routes accompanying their children than men. In full-time employment,

For employed women, the proportion of trips involving accompanying persons is 16 percent, the same as for part-time working men. Part-time working women, in turn, undertake a quarter of their trips involving accompanying persons, a proportion matched by unemployed men. Thus, employed women are not relieved of accompanying persons to the same extent as men. The greatest contrast is between full-time employed men, for whom accompanying persons account for only 10 percent of trips, and unemployed women, for whom accompanying persons account for 30 percent.

Employment status and children in the household determine dependence on a car.

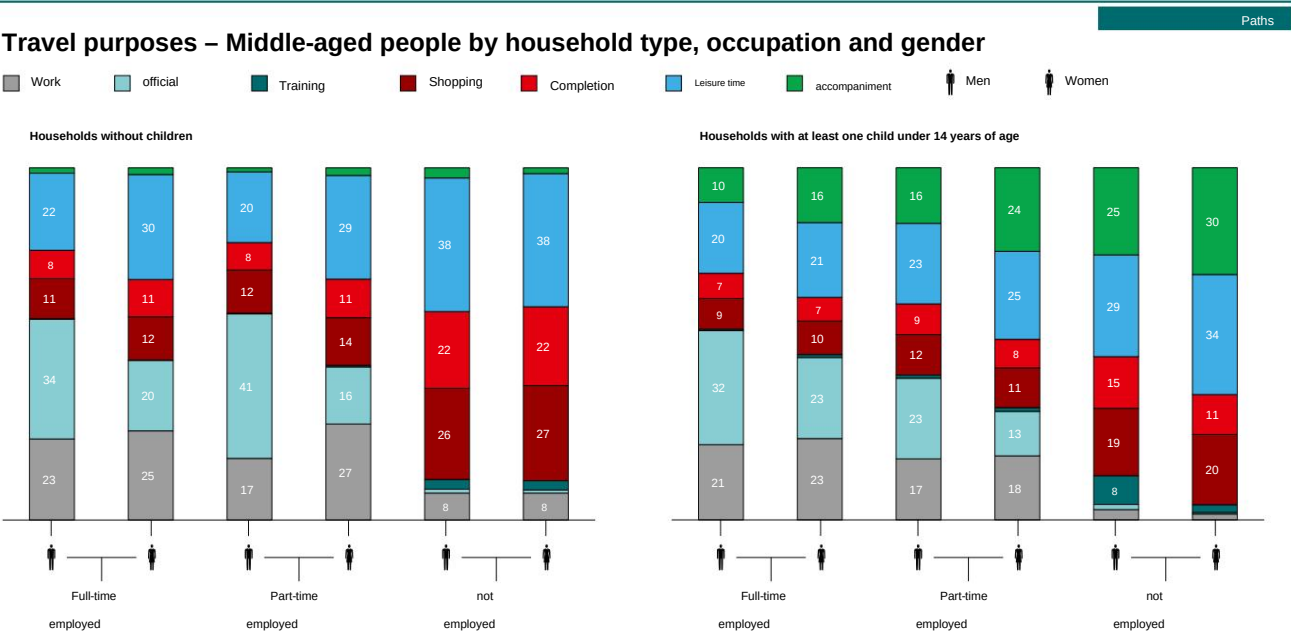
Middle-aged people are characterized by high car usage. However, this is strongly dependent on employment status and the presence of children in the household. One indicator of this is the proportion of monomodal car users – individuals who exclusively use their car during a typical week (Figure 80). In childless households, around half of full-time employed men and women belong to this group, while the proportion is significantly lower for part-time employees and those who are not employed. With children in the household, the proportion of monomodal car users increases sharply, particularly among part-time employees and those who are not employed. For women, employment status no longer plays a role in this case.

Approximately half of respondents, regardless of employment status, travel exclusively by car. Among part-time working men, this proportion is significantly lower, even among those with children. Part-time working mothers thus appear to have a greater reliance on their cars in everyday life than fathers in comparable situations. The group of part-time working fathers is still relatively small. Their lower car dependency is partly attributable to the higher proportion of them living in urban areas (Table 39).

Driving licence possession and self-employment are Key factors for mobility in old age

The greatest differences in mobility behavior between men and women emerge in later life, even though a key driver – employment – largely ceases during this phase. The following study examines how driver's license ownership, mobility limitations, and the

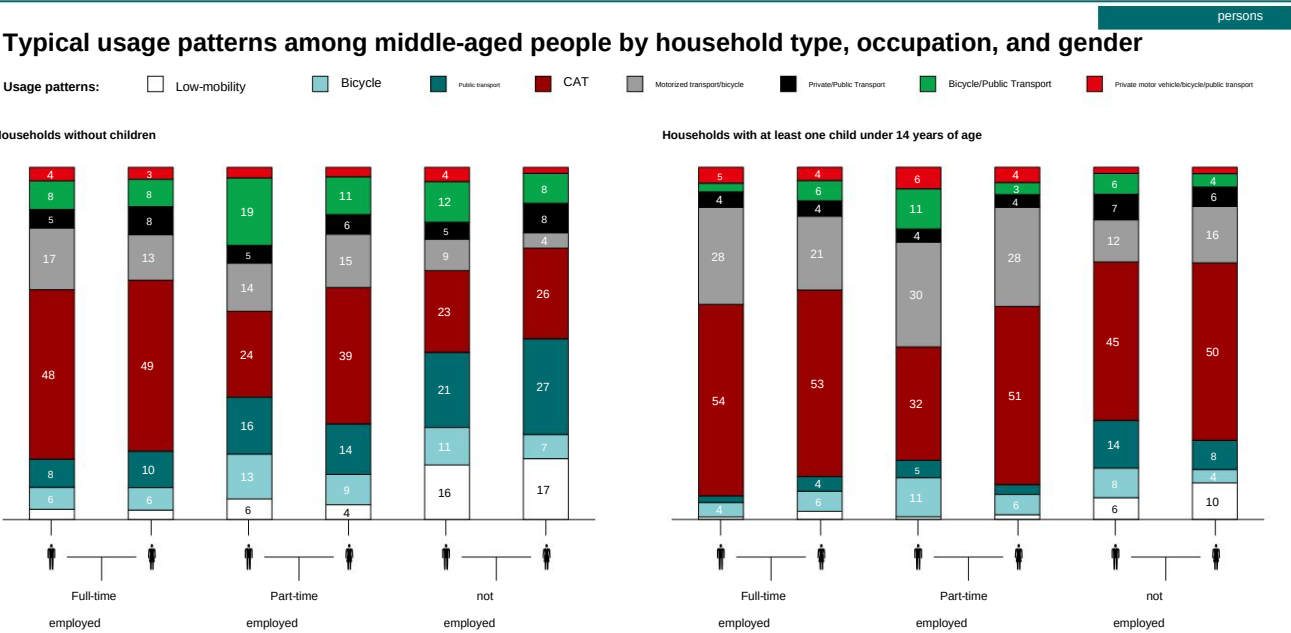
Figure 79



Figures in percent, persons aged 30 to 59

MiD 2023 | Results Report | Source: MiD 2023

Figure 80



Figures in percent, persons aged 30 to 59

MiD 2023 | Results Report | Source: MiD 2023

The need to be accompanied on journeys that affect the mobility of older women and men.

The analysis focuses on individuals aged 70 and over. 96 percent of this age group are no longer employed, thus the study examines mobility in retirement.

Older women are not only less likely to have a driver's license, but they are also more frequently dependent on being driven, picked up, or accompanied by others (Table 40). There is a close correlation between these aspects. Women without a driver's license report being dependent on others for their mobility more often than average.

Overall, a quarter of women aged 70 and over do not have a driver's license or are dependent on assistance from others. Among men of the same age group, this applies to only 6% and 15%, respectively.

8 percent.

Both lacking a driver's license and being dependent on others significantly reduce mobility for both men and women. The dependence on others has a particularly strong impact. Men and women are affected to a similar degree; for example, their daily commute is only about a third of that of people without this restriction. Men without a driver's license, however, are still somewhat more mobile than women without one, albeit at a very low level.

Mobility limitations due to health impairments are widespread in old age. 33 percent of men and 43 percent of women are affected. These limitations lead to a significant reduction in mobility levels for both sexes, although the differences between men and women are comparatively small.

Table 40

Mobility indicators of people aged 70 and over according to car driving licence possession, personal limitations and gender					
only people aged 70 and over	share of in total- group	Mobility quote	average number of trips	average daily distance	average travel time
WITH A CAR DRIVER'S LICENSE	%	%	Number	km	h:min
Men	94	76	2,2	24	1:12
Women	77	71	2,0	16	1:01
WITHOUT A CAR DRIVER'S LICENSE					
Men	6	62	1,5	10	0:56
Women	23	54	1,3	8	0:41
WITHOUT MOBILITY RESTRICTIONS					
Men	67	81	2,5	27	1:21
Women	57	77	2,3	19	1:10
WITH MOBILITY RESTRICTIONS					
Men	33	66	1,8	15	0:55
Women	43	66	1,4	10	0:42
NOT REQUIRED GUARDING					
Men	92	81	2,5	32	1:23
Women	75	74	2,2	18	1:06
REQUIRES ASSISTANCE					
Men	8	46	0,9	11	0:37
Women	25	50	1,0	6	0:26

Mobility limitations in old age lead to a growing importance of walking.

The consequences of reduced mobility levels in old age are particularly evident in the distribution of modal groups. In the general population, the group of people with low mobility—those who do not use any means of transport (car, bicycle, or public transport) over the course of a week and travel exclusively on foot—plays a negligible role. However, this group becomes significantly more important in old age when limitations are present.

This is most evident in the dependence on other people, followed by the lack of a driver's license and health limitations.

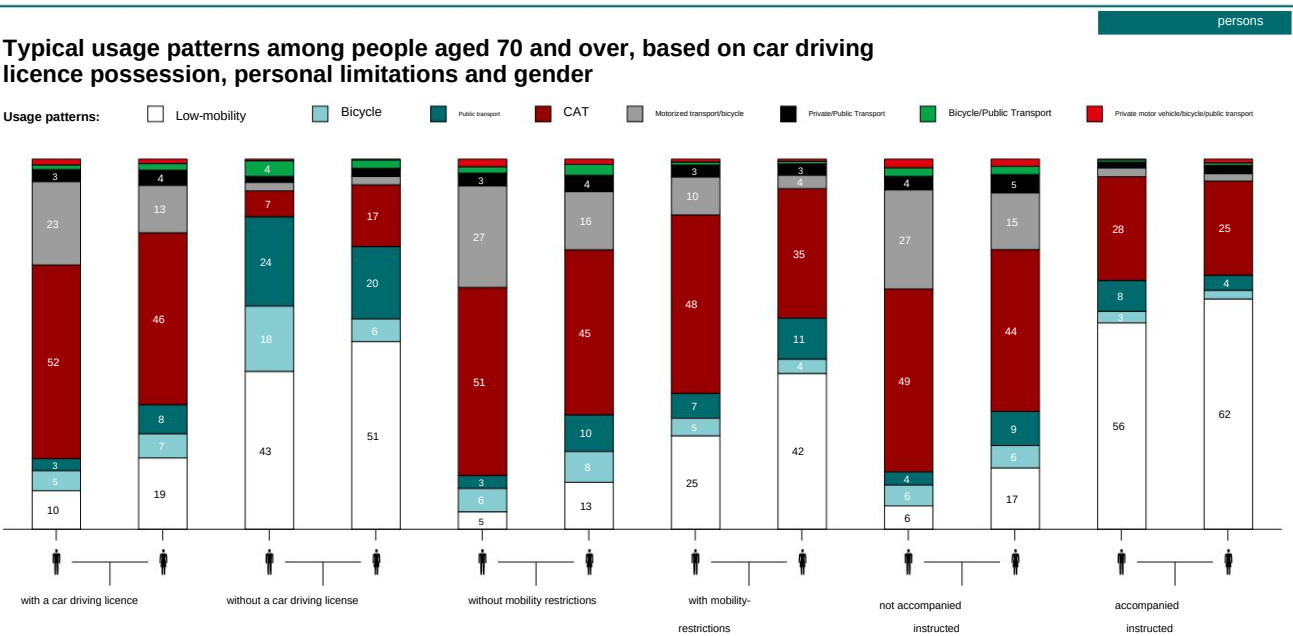
Women are more affected than men. For example, 62 percent of women who are dependent on others belong to this group, compared to 56 percent of men. If health limitations are present, 42 percent of women belong to this group, but only 25 percent of men (see also the specific chapter on mobility limitations). Without limitations, the group of people with limited mobility is significantly smaller, especially among men. For these individuals, the strong dominance of the car for everyday mobility is evident.

Good traffic planning requires differentiated

The analysis

Transport planning is often geared towards commutes and educational routes, thus primarily considering the needs of specific population groups. For example, public transport timetables and frequencies are primarily aligned with typical commuting and educational travel times. The diverse lifestyles and associated mobility requirements of society as a whole are inadequately reflected in this approach. This is particularly evident in complex journey chains – such as the combination of commutes, childcare, and errands – which have so far received little attention in transport planning. The mobility needs of women who adjust their employment and daily commutes to accommodate family caregiving responsibilities, as well as those of older women with varying mobility limitations, are also frequently neglected. A differentiated analysis of gender-specific mobility patterns can therefore provide an important foundation for more needs-based and inclusive transport planning.

Figure 81



Figures in percent, persons aged 70 and over

13. Migration background

– Mobility as diverse as the regions of origin

German society has long been shaped by immigration. The diversity of the country's population in terms of geographical origin and cultural background has been increasing for decades. This also raises important questions regarding mobility, for example, whether this migration trend influences mobility trends or whether it results in specific demands on the transport system. In the 2023 Migration Survey (MiD 2023), individuals aged 14 and older were asked about their migration background. This allows for analyses of the relationship between migration and everyday mobility in Germany that were previously not possible with this level of depth and reliability. In the MiD 2023, people with a migration background are defined as those who were either born abroad themselves or have at least one parent who is from abroad.

Together, they make up 1 in the realized sample. one fifth of all persons aged 14 and over (Figure 82).

Figure 83 shows the distribution of people with a migration background according to their place of birth. Nearly 80 percent of people with a migration background are in Europe (including Germany, the

Ukraine, Turkey and other European countries). Africa and the Middle East, Asia and Oceania, and North and Latin America follow.

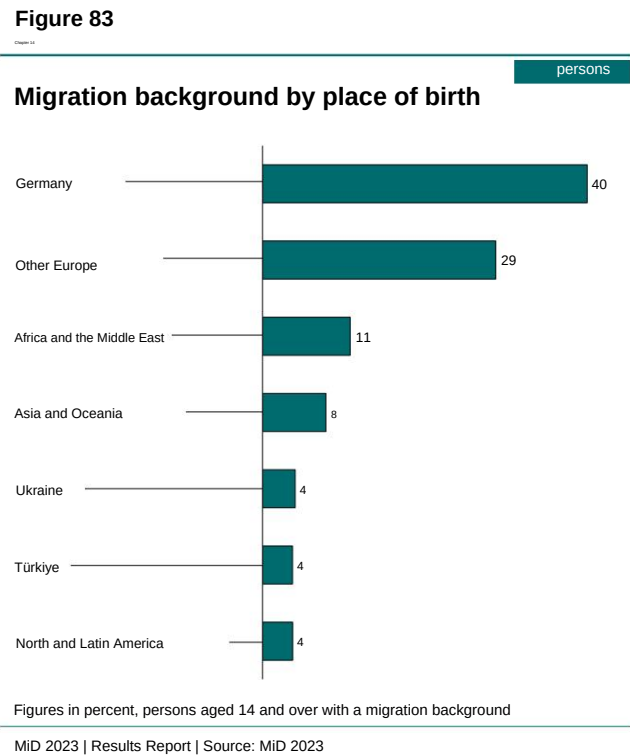
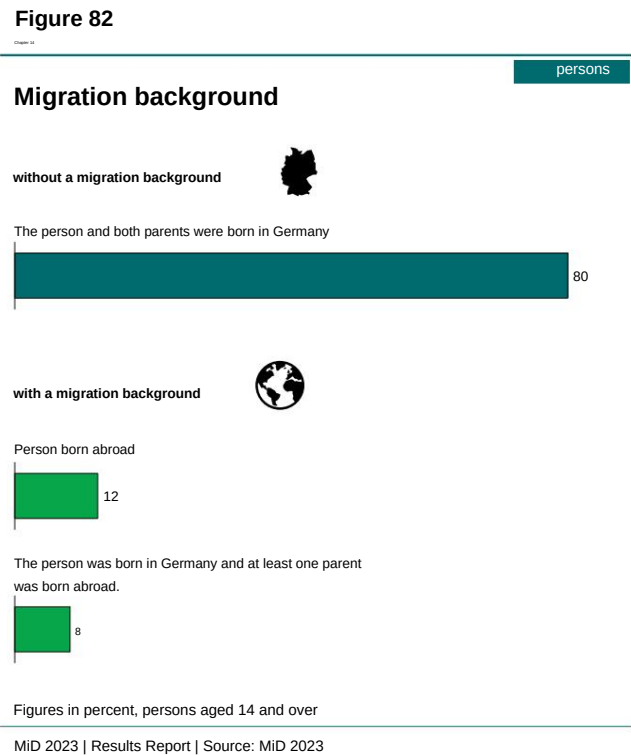
This chapter explores the question, relevant to an immigration society, of how the mobility of people from different regions of origin differs from the German average and from each other. The following results provide an impression of the great heterogeneity of life situations and the resulting mobility patterns of people with a migration background in Germany.

Significant socioeconomic differences depending on the country of origin.

The living situations of people with a migration background differ significantly from those of the population born in Germany, depending on their region of origin, but also among themselves.

This becomes apparent when looking at distributions by gender, age, economic status, and place of residence in Germany. For example, the proportions of men and women among people from Ukraine,

1 In this context, special methodological precautions were taken, such as the provision of documents in foreign languages and an additional onomastic qualification of the gross sample, see MiD Methods Report.



The population from Africa and the Middle East, as well as from Turkey, is significantly unbalanced: People living here from Ukraine are predominantly female (72 percent), while people from Africa and the Middle East (65 percent) and Turkey (59 percent) are predominantly male.

While gender imbalances do exist in some of the other regions of origin, they are significantly less pronounced.

People with a migration background – including those born in Germany – are on average younger than people without a migration background (Figure 84). In particular, the proportion of older people is low among migrants from outside Europe. The proportion of adolescents and young adults also varies significantly depending on the country of origin. For example, the proportions of these age groups are rather low among immigrants from other European countries and Turkey, while they are above average among people born in Germany with a migration background.

The economic situation of

Immigrants are made aware of the vast differences between the realities of life in the various countries.

Groups of people with a migration background:

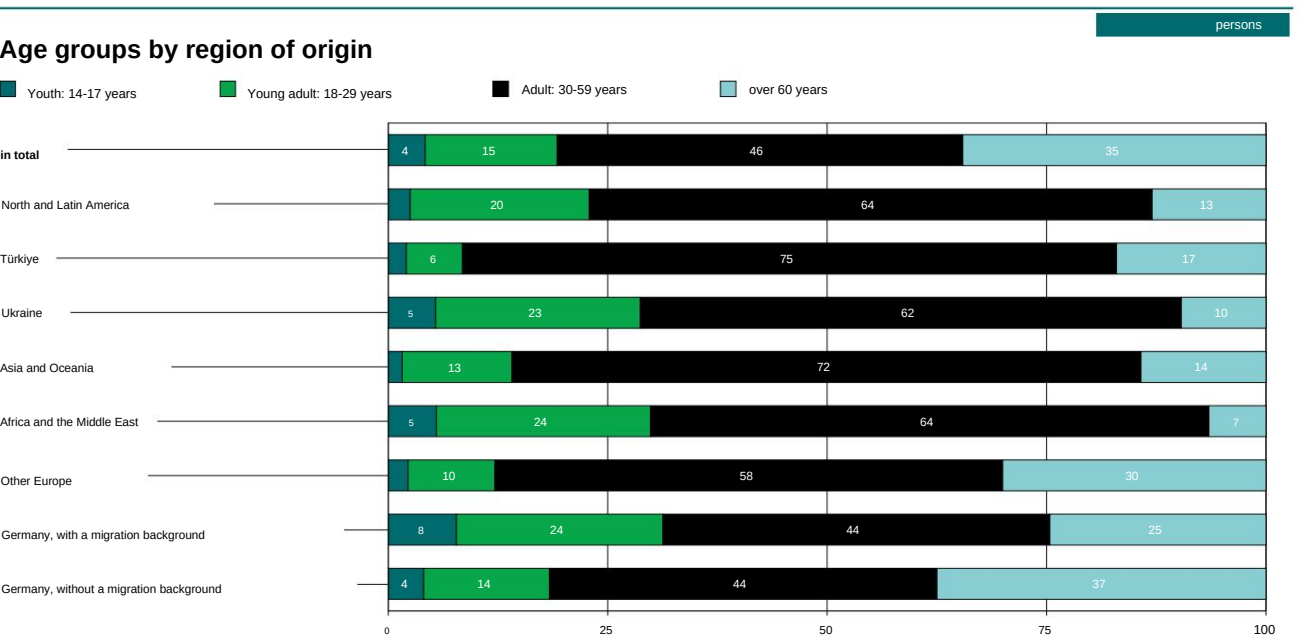
While only 8 percent of people without migraine-

While 45 percent of people from Ukraine live in households with a very low economic status, this is the case for those with a migration background (Figure 85). For the African and Middle Eastern groups, it is about one-third in each case, and one-fifth in the case of Turkey. In contrast, people from North and Latin America are generally very well off economically. For the other regions of origin, as well as for those born in Germany with at least one parent from abroad, the distribution according to economic status corresponds more or less to the German average.

Differences also exist regarding the places of residence of people with a migration background in Germany. Overall, a slightly larger proportion of them live in metropolitan areas and cities compared to the German average (Figure 86). However, here too, there are significant differences between the regions of origin.

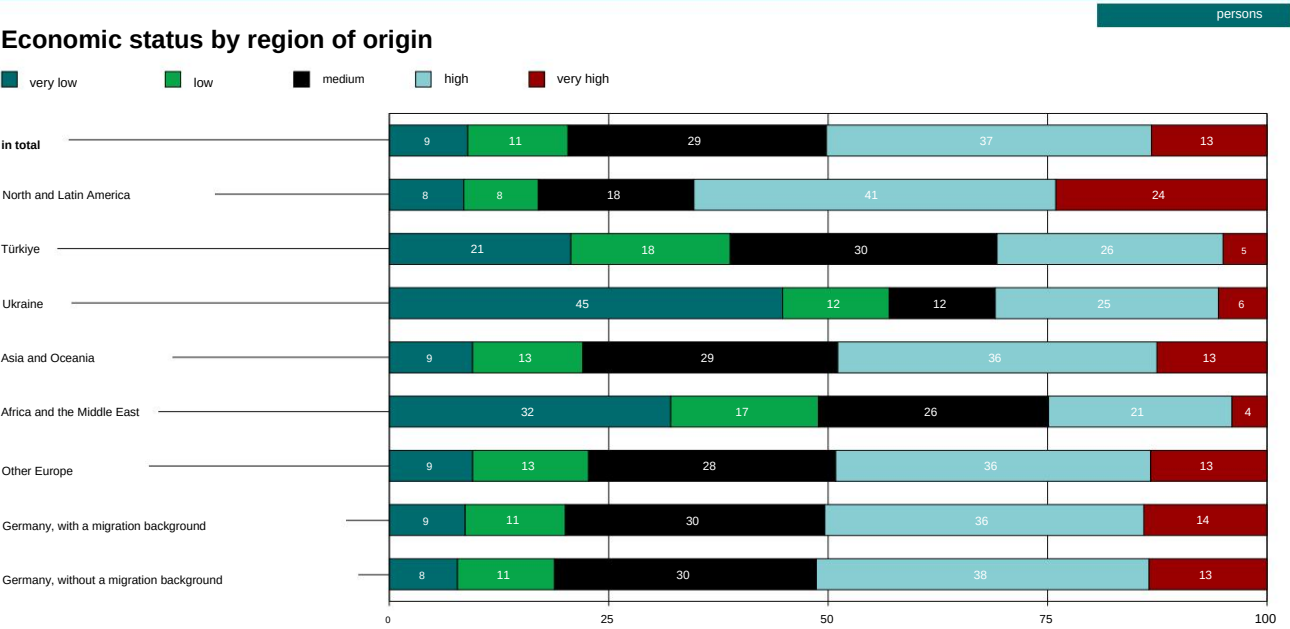
While the distribution of people from Ukraine across different geographical areas is almost in line with the German average, significantly smaller proportions of people from Turkey, Africa, and the Middle East are found in rural regions. Particularly high proportions of the population from North and Latin America live in cities and metropolitan areas.

Figure 84



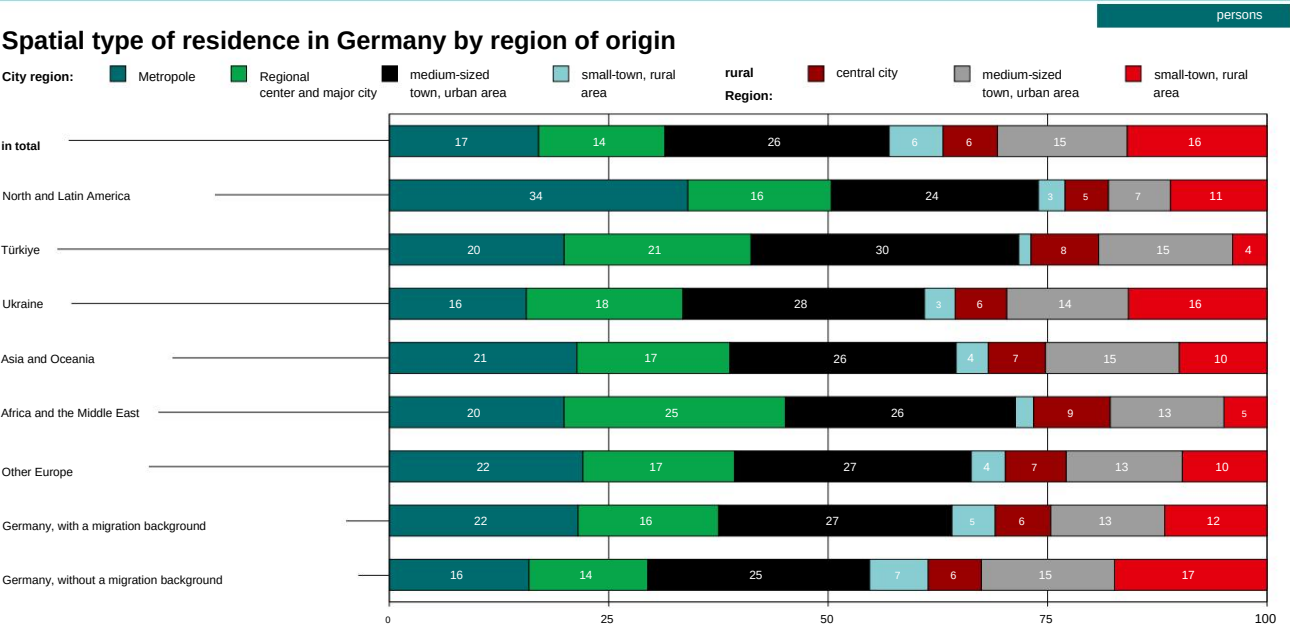
Figures in percent, persons aged 14 and over

Figure 85



MiD 2023 | Results Report | Source: MiD 2023

Figure 86



MiD 2023 | Results Report | Source: MiD 2023

People with a migration background are particularly likely to live in households without a car.

Significant differences in the mobility of people with a migration background are already evident in their choice of transportation. Figure 87 shows the proportion of people with cars per household, depending on their migration background and region of origin. People with a migration background are generally more likely to live in households without a car than average. The proportion of people without a car is particularly high in Ukraine (55 percent), Africa and the Middle East (39 percent), and North and Latin America (33 percent). If these households own a car, it is usually only one vehicle.

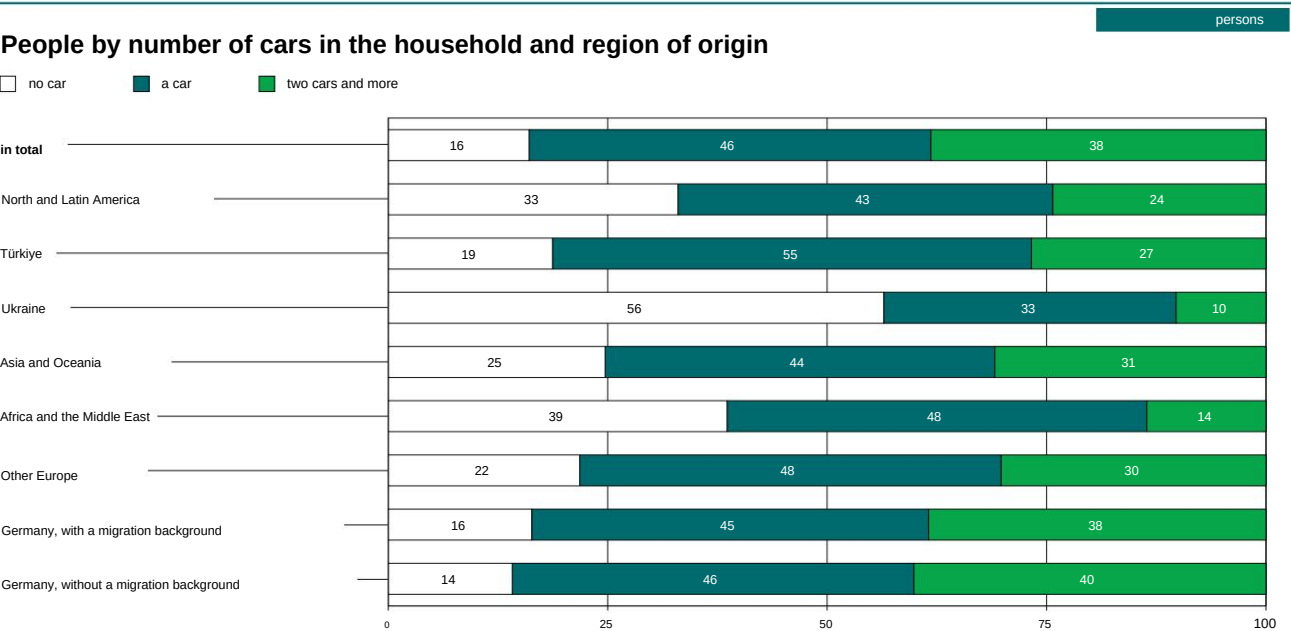
Car ownership is closely linked to economic status, which at least partially explains car ownership among immigrant households in most regions of origin. It is striking that a third of immigrants from America do not own a car, despite their comparatively high economic status. One possible explanation is that a large proportion of them live in a metropolis or large city (Figure 86) and therefore have little or no need for a car.

Low bicycle availability among people from Africa and the Middle East

Like car ownership, bicycle ownership is generally lower than average among people with a migration background (Figure 88). Among people from Africa and the Middle East, more than half do not own a bicycle, making them the only group where there are more people without bicycles than with them. In the case of people from Turkey, Ukraine, Asia, and Oceania, the proportion of people without bicycles is also significantly higher than average. However, bicycle availability among people with a migration background who were born in Germany or America is similar to that of people without a migration background.

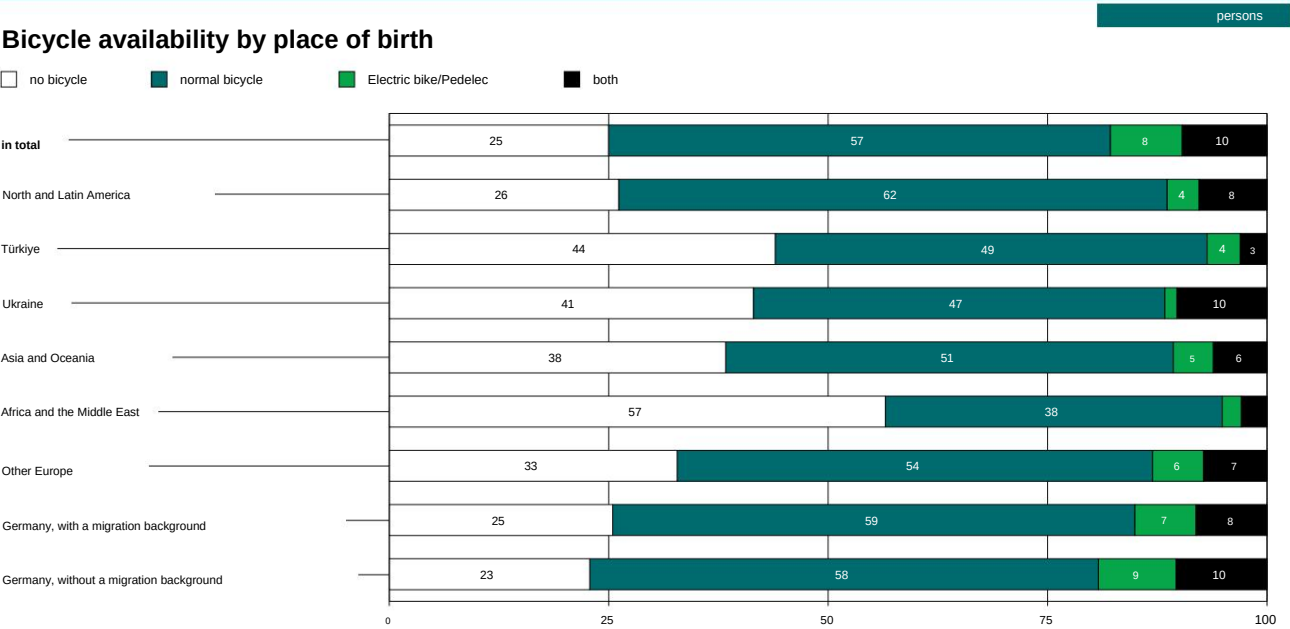
Depending on the region of origin, striking differences in bicycle ownership by gender are also evident (Figure 89). While bicycle ownership rates diverge between women and men in all population groups, with the proportion of men owning bicycles being on average 7 percent higher than that of women, the difference between the sexes is particularly pronounced for people from Ukraine (20 percent), Africa and the Middle East (18 percent), and Latin and North America (16 percent).

Figure 87



Figures in percent, persons aged 14 and over

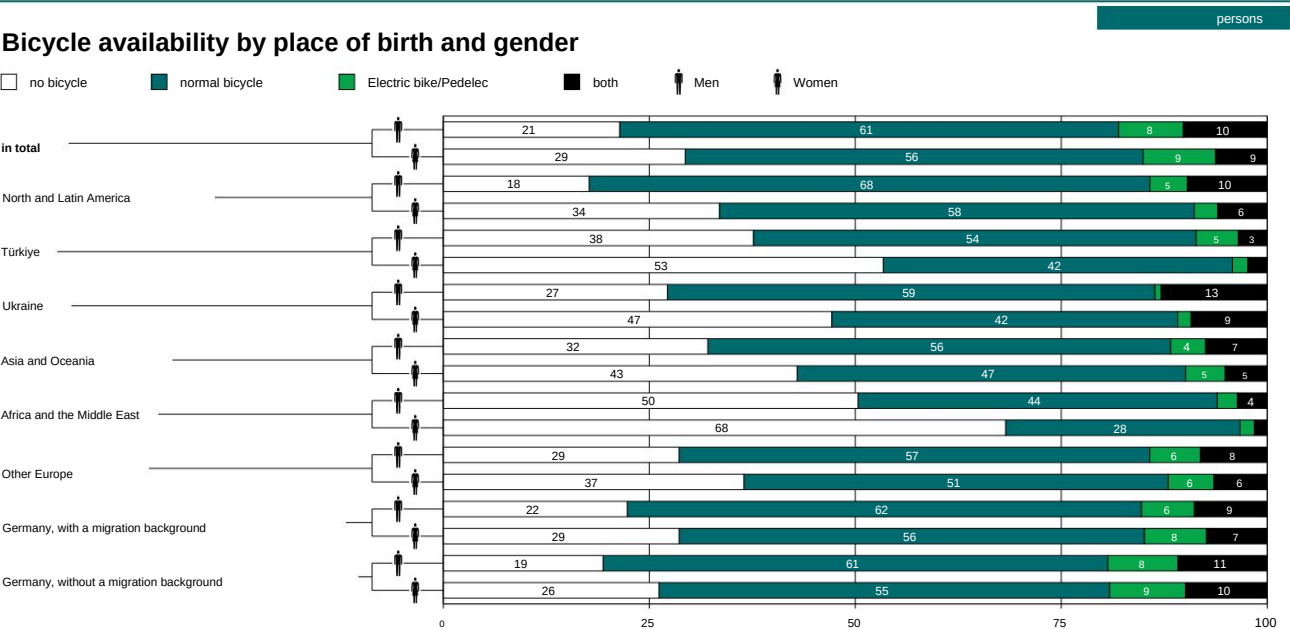
Figure 88



Figures in percent, persons aged 14 and over

MiD 2023 | Results Report | Source: MiD 2023

Figure 89



Figures in percent, persons aged 14 and over

MiD 2023 | Results Report | Source: MiD 2023

The smallest differences in bicycle ownership between men and women are found among people born in Germany or the rest of Europe.

Lower traffic volume and performance per capita for people from abroad

Table 41 shows key mobility indicators per capita depending on migration background or... Region of origin for persons aged 14 and over. On average, around four-fifths of the persons were out of the home on the reference date, which also applies to most regions of origin. Only for persons from Africa and the Middle East (72 percent) and Asia and Oceania (75 percent) are the mobility rates noticeably below average.

Regarding the number of trips per person per day, there are hardly any noticeable differences between the average person and those with a migration background who were born in Germany. However, people born abroad report slightly fewer trips per day than average. The above-average figure of 4.1 trips per mobile person per day for people from Africa is striking.

the Middle East. The traffic volume of this group therefore seems to be characterized by the fact that on a given day, a relatively large number stay at home, but those who do leave the house then make many journeys.

The average daily commute per person (aged 14 and over) is 39 kilometers, or 48 kilometers for those who are mobile. Here too, people not born in Germany are slightly below average – consistent with their lower mobility rate and fewer trips. Even immigrants who were mobile on the reference date traveled 5 kilometers or more below the average for all mobile individuals. Daily commute per person is particularly low for people from Turkey.

Fewer journeys by car and bicycle, more journeys on foot and by public transport

People born in Germany, the rest of Europe, or Asia and Oceania travel over half of their journeys by private motorized transport (Figure 90). For people from other countries,

Table 41

Mobility indicators by place of birth or migration background					
PLACE OF BIRTH	Mobility quote	Paths		Daily output	
		Pro Person and day	Pro mobile person and Tag	Pro Person and day	Pro mobile person and Tag
People with a migration background %		Number of routes	Number of routes km		km
IN TOTAL	81	3,1	3,9	39	48
North and Latin America	83	2,7	3,3	34	42
Türkiye	79	2,5	3,4	23	31
Ukraine	78	2,3	3,2	30	41
Asia and Oceania	75	2,5	3,5	30	42
Africa and the Middle East	72	2,7	4,1	27	42
other parts of Europe	79	2,7	3,5	33	43
Germany, with Migration background	83	3,1	3,8	37	45
Germany, without Migration background	82	3,2	3,9	40	49
Person aged 14 or older					

Figure 90

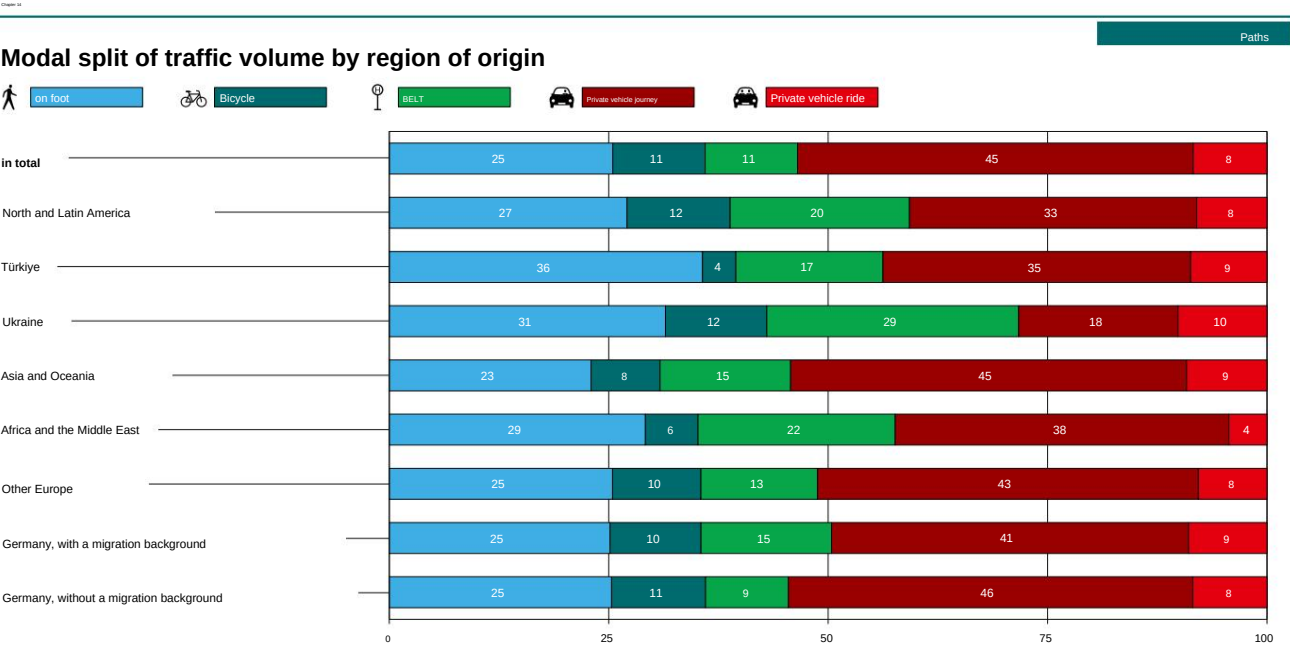
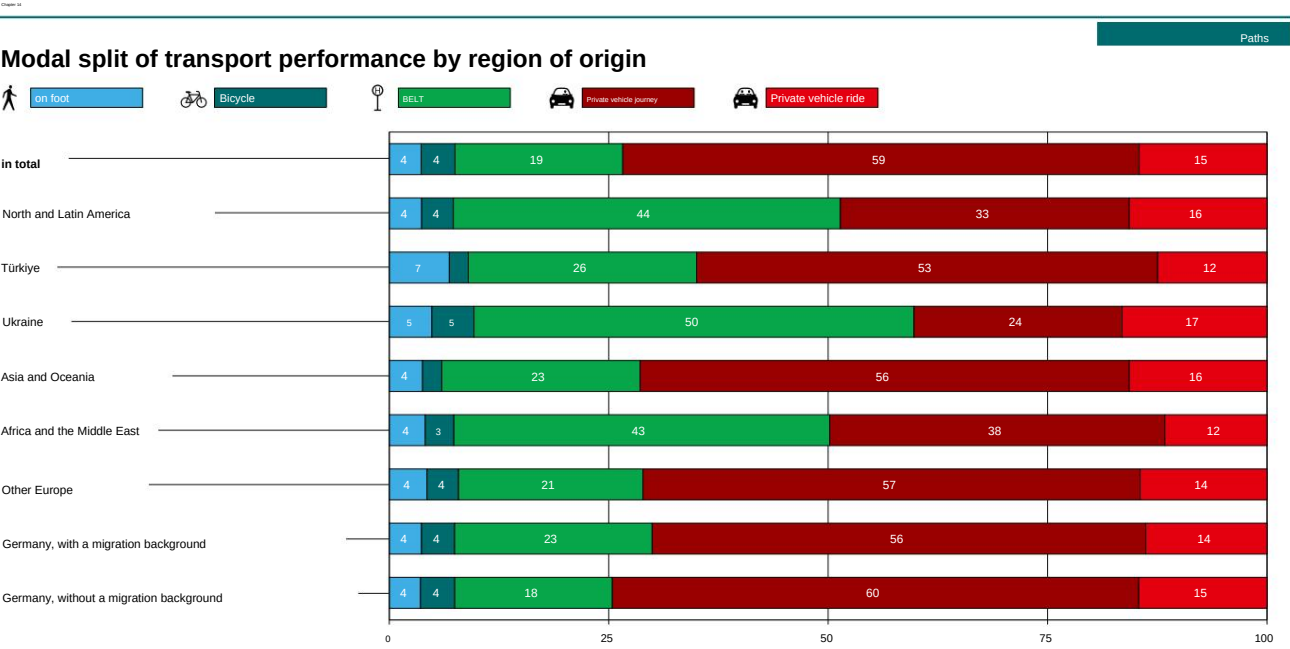


Figure 91



In developing regions, walking and public transport play a significantly larger role: The share of environmentally friendly modes of transport is considerably higher for people from Ukraine (72 percent) than for people without a migration background (45 percent). Public transport and walking also account for a significantly above-average share of journeys for immigrants from Africa and the Middle East, Turkey, and the Americas. In contrast, people from Turkey, Africa and the Middle East, as well as Asia and Oceania, use bicycles at a below-average rate.

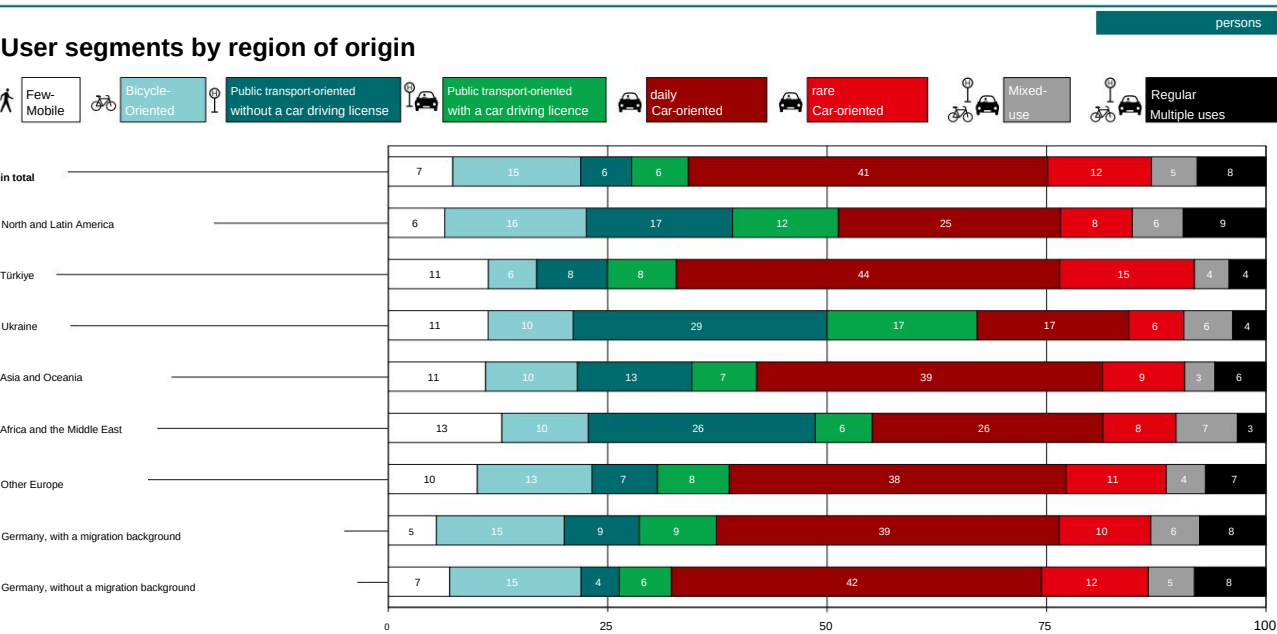
High public transport shares in transport performance for People from Ukraine, America, Africa and the Middle East

People born in Ukraine, North or Latin America, or Africa and the Middle East cover half or more of their daily distance (60 percent in Ukraine) using environmentally friendly modes of transport (Figure 91). This reflects the low car ownership rate among these groups. For all other individuals, private motorized transport clearly dominates, and the modal split per kilometer is not far from the German average.

Significant differences in the shares for car, public transport and multimodality according to region of origin.

The analyses of the data from specific reporting days indicate that people with a migration background sometimes use public transport for a significantly larger proportion of their mobility needs. This picture, based on individual reporting days, is supplemented below by segmentation according to the general frequency of use of different modes of transport. Consistent with previous findings, the analysis of typical mode usage also reveals a high proportion of car drivers among people born in Germany, Turkey, Asia and Oceania, or other parts of Europe (Figure 92). It is striking, however, that the proportion of car drivers is highest among those born in Turkey, even though their share of private motorized transport in the modal split is lower than average. In contrast, only 6 percent of people from Turkey are cyclists, a comparatively small proportion compared to 10 to 16 percent for the other regions of origin.

Figure 92



Figures in percent, persons aged 14 and over

The proportion of public transport users is particularly high among people from Ukraine, the Americas, Africa, and the Middle East. This corresponds to the low car availability and the high public transport usage rate on the reference date for these groups. When analyzing public transport usage in relation to car availability, the question arises as to what extent people choose public transport despite having alternatives, or whether they are dependent on it. People who rely on public transport due to a lack of car availability are also referred to as "public transport captives." The relative proportions of public transport captives among public transport users, broken down by migration background, are shown in Figure 93.

The analysis shows that the majority of public transport users are captive users. The proportion of captive users is highest among people from Ukraine (75 percent), which is more than twice the proportion of people without a migration background. A high proportion is also observed among people from Africa and the Middle East (59 percent) as well as North and Latin America (44 percent). The high proportion of public transport captive users underscores the importance of the public transport system, on which many people from diverse backgrounds depend.

How strongly do changes in migration shares influence the MiD results in the time series?

A closer look at the group of migrants reveals a picture that differs slightly from that of the general population. The most striking differences are their somewhat reduced basic mobility and, in some groups, their greater reliance on public transport. This finding is not new and is also evident in the MiD 2017 study, despite a different operationalization of migration background, as well as in other studies.

Furthermore, the cross-section of all persons with a migration background largely corresponds to that of the "average population" with regard to their orientation towards means of transport. Therefore, changes in the structure of traffic volume and performance due to a specifically increasing proportion of migrants, for example through the immigration of Ukrainians, cannot be completely dismissed, but they only explain a small part of the changes observed in this respect compared to 2017.

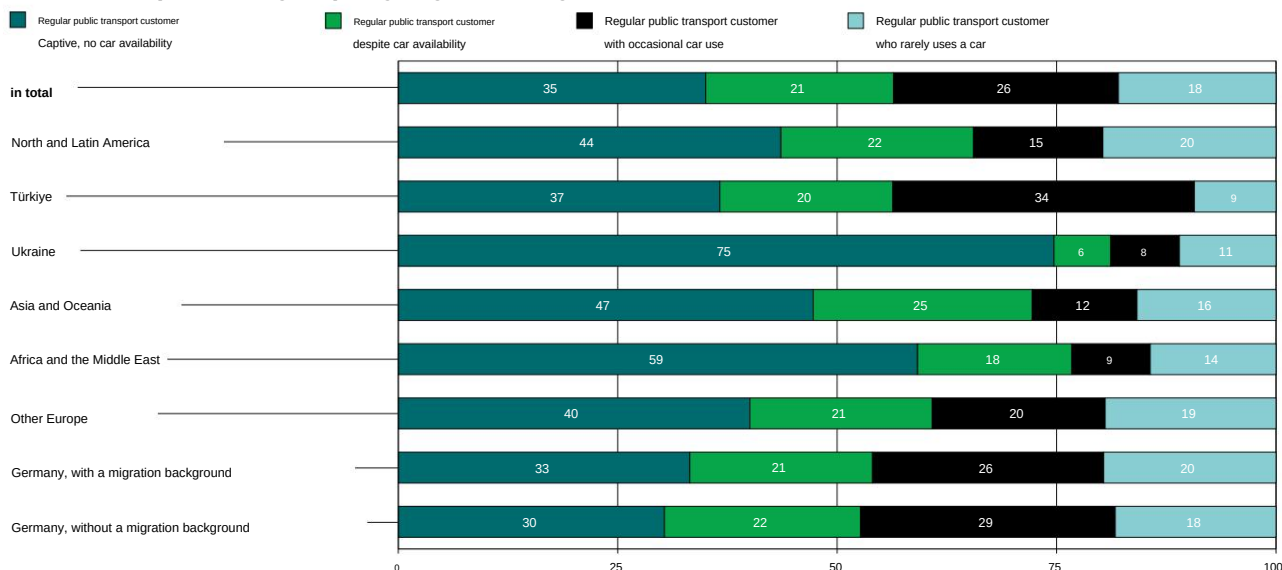
Here, it is more likely that a cluster of causes exists rather than simple causality – and there are stronger influences, which are also presented in this report, such as the ongoing home office trend or economic effects.

In 2017, a self-assessment of migration status was collected, thus using a different, not directly comparable, definition.

Figure 93

Chapter 14

Public transport user groups by region of origin



Figures in percent, persons aged 14 and over, public transport users only.

14. Children and seniors on the go: often the view through the windshield

The study design and large sample size of the MiD also allow for detailed analyses of the mobility of specific population groups. This chapter utilizes this option and focuses on the phases outside of middle age and employment: children on the one hand and senior citizens on the other. Within both groups, further subdivisions can be made along the age distribution. This allows children and young people to be tracked through the phases from kindergarten – or even before – to secondary school or the start of vocational training. They are interviewed either on behalf of their parents or guardians, or, from the age of 14, ideally directly. Among older citizens, different age phases can also be considered separately. Interviews are sometimes available for very elderly individuals and people with disabilities, provided they receive appropriate support at home or in a care facility. An overview of life-phase-related mode choices was already presented in [Chapter 4](#), which illustrated the modal split by individual birth cohorts.

Transportation equipment for children and young people: Are fewer and fewer people using bicycles?

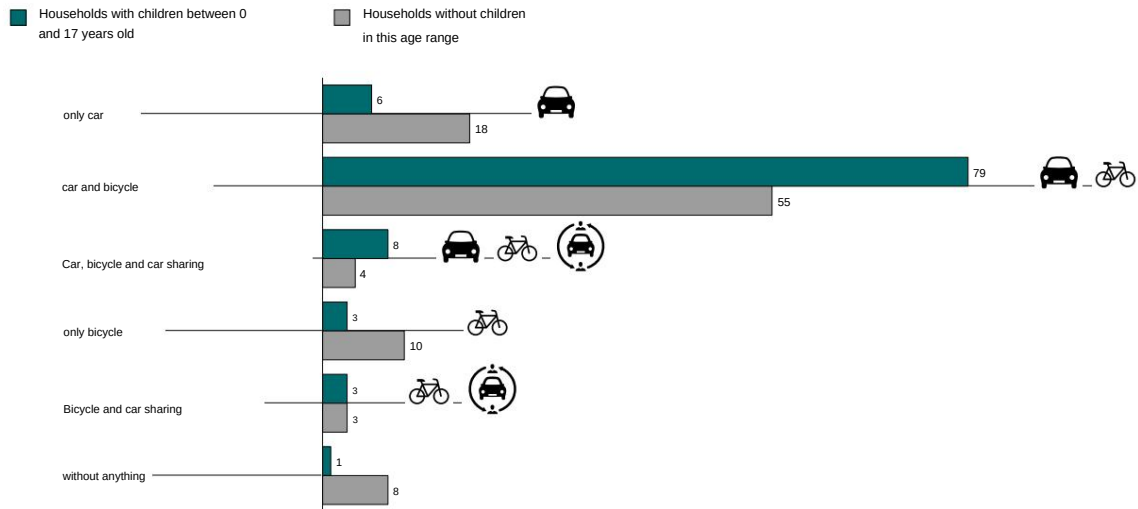
The everyday mobility of children and young people is a central aspect of their development, as it enables them to independently find their way around, reach places, and gain experience. This is not only a matter of physical mobility but also an important step towards autonomy and independence. Drop-off and pick-up routes, where parents or other adults accompany children to school and leisure activities, play a significant role in the formation of mobility routines. These routines shape preferences for specific mobility options and influence the daily activity patterns of children and young people. Despite the importance of independence, safety must always be paramount: the journey to school is often the first step towards independence and an important part of the daily routine of children and young people. The step towards independence requires safe infrastructure, early mobility education for children and parents, and diverse and flexible mobility options. Furthermore, mutual consideration in road traffic is essential to ensure the safety of children and young people as they develop their mobility.

Figure 94

Chapter 14, Page 12

Mobility equipment of households in which children and young people live

households



Figures in percent; households with at least one child aged 0 to 17 years not shown: pure carsharing shares and share of car with carsharing, each only one percent.

Households with children and teenagers equipped with a car and bicycle

Against this background, this section examines which means of transport are available to children and young people, how they use them, how and whether they are accompanied, and whether their independence differs between urban and rural areas. Equally important is the analysis of distances to places like daycare or school and whether these have changed in the MiD time series since 2002.

Households with children and young people up to the age of 17 are more diversely equipped compared to households without children in this age range. Figure 94 shows that 79 percent of households with children and young people own a car and a bicycle, compared to 55 percent of households without children and young people. When a car is needed, households with children are also more likely to use car sharing than households with only adults. Having only one mobility option available, such as a car or a bicycle, is less typical for households with children and young people. This already indicates that children's mobility often takes place in the back seat of a car.

However, having at least one bicycle in the household doesn't necessarily mean that even the youngest members can hop onto their own bicycle saddle. The MiD therefore also collects data for the

The availability of bicycles is a concern for the youngest children. However, it remains far from the fully equipped models one might expect. Availability increases from a respectable 45 percent among children up to 5 years old to 85 percent among 14- to 17-year-olds. And it depends on the parents' economic status: In low-status families, the aforementioned range spans from 35 to approximately 70 percent across all age groups. In high- or very high-status families, it reaches a range between roughly 50 and 90 percent. This is not only a question of economic means, but also of the parents' appreciation for cycling.

If the parents report owning their own bicycle, this usually also applies to the children.

Modal split and mobility indicators of children and adolescents: car-oriented, but less mobile than adults

Both the modal split by traffic volume and the modal split by transport performance show clear differences between the under-18 age groups, reflecting different life stages on the path to adulthood. While the share of the modal split as a passenger in private motorized transport decreases continuously with increasing age, the share of public transport increases, indicating growing independence in children's mobility. Cycling gains significant importance as a mobility alternative from the age of 10, and some 14- to 17-year-olds are already cycling.

Table 42

Key parameters for routes taken by children and young people

	average Number of trips (per person per day)	average Distance traveled (per person; in kilometers)	average Daily distance (per person; in kilometers)	average Travel time (per person per day; in h:min)	share of Because of Accompanying person
Persons under 18 years of age Number		km	km	h:min	%
IN TOTAL	2.6	9	22	1:03	80
ALTER					
0 to 5 years	2.5	8	22	0:59	99
6 to 9 years	2.7	7	19	0:57	87
10 to 13 years	2.6	8	22	1:05	69
14 to 17 years	2.4	11	27	1:11	50

already on the road as a private motor vehicle driver, be it by scooter or by car with a driver's license from the age of 17-
ren.

Most of the routes taken by children with
Accompanying persons

Children make an average of 2.6 trips per day, covering an average distance of 9 kilometers, a daily distance of 22 kilometers, and spending more than an hour traveling. It is noteworthy that 80 percent of these trips are made accompanied by an adult.

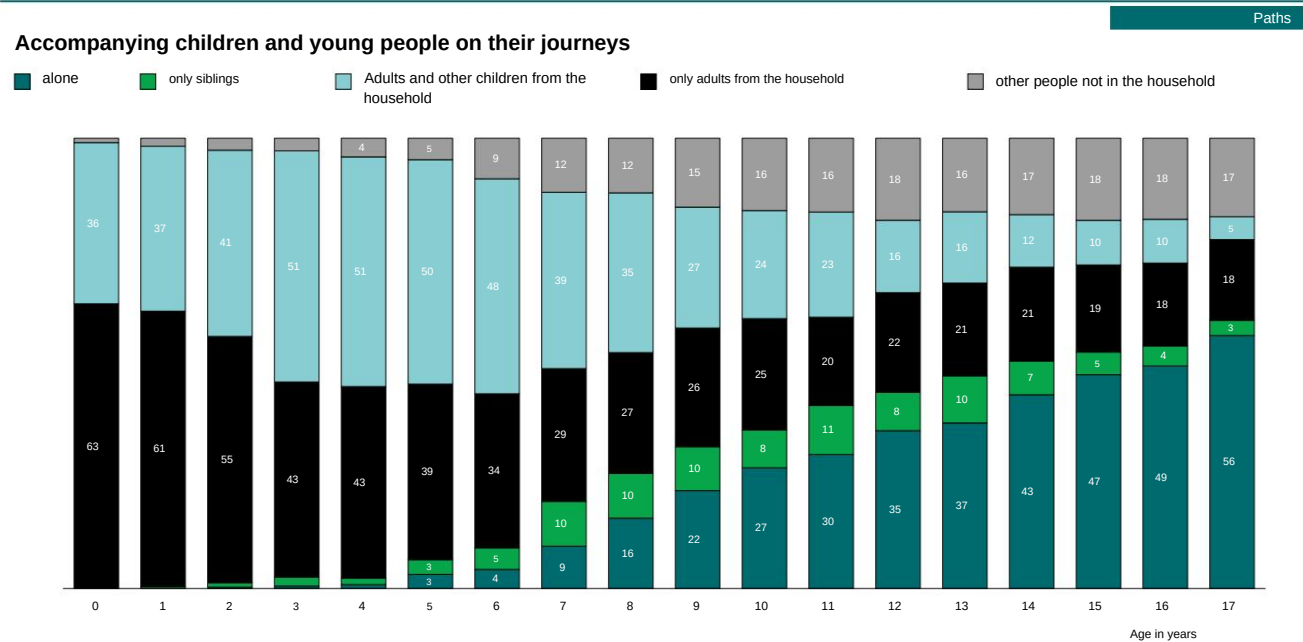
The youngest children (0 to 5 years) make 2.5 trips daily, averaging 8 kilometers in length and 22 kilometers in distance. The 6- to 9-year-old age group shows a slight increase in the number of trips to 2.7 per day. The trip length decreases to 7 kilometers. Nevertheless, 87 percent of these trips are accompanied by an adult (compared to 99 percent for 0- to 5-year-olds), indicating gradual steps towards independence. Children between 10 and 13 years old make 2.6 trips daily, averaging 8 kilometers. The proportion of accompanied trips drops noticeably to 69 percent, suggesting growing independence. Adolescents, aged 14 to 17, make fewer trips (2.4 per day), but cover greater distances (11 kilometers), thus...

The daily distance traveled increases to 27 kilometers. This is naturally related to the fact that new mobility alternatives such as public transport and private motorized transport are increasingly available to them, as previously mentioned.

The proportion of accompanied journeys decreases steadily with increasing age. The fact that even 50 percent of journeys made by 14- to 17-year-olds are still accompanied may initially seem surprising. However, it's important to note that the MiD survey initially only asked generally about accompanying persons. Only in subsequent questions did it specify whether the accompanying person was from the child's own household and whether they were an adult. A closer analysis of the accompanying persons reveals that with increasing age, fewer journeys are accompanied by adults from the child's own household, while more and more journeys involve people from outside the household. This also points to increasing independence, where parental accompaniment is gradually replaced by outings with friends.

Figure 95

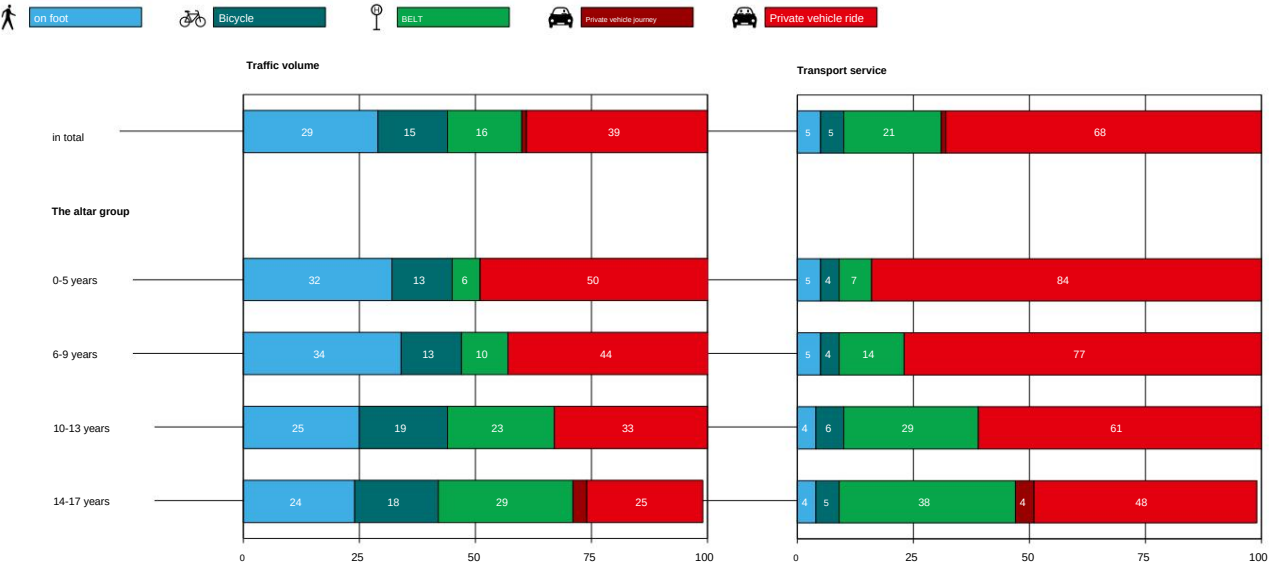
Chapter 14, Page 13



Figures in percent; routes taken by children and young people

Figure 96

Modal split of children in traffic volume and traffic performance by age group

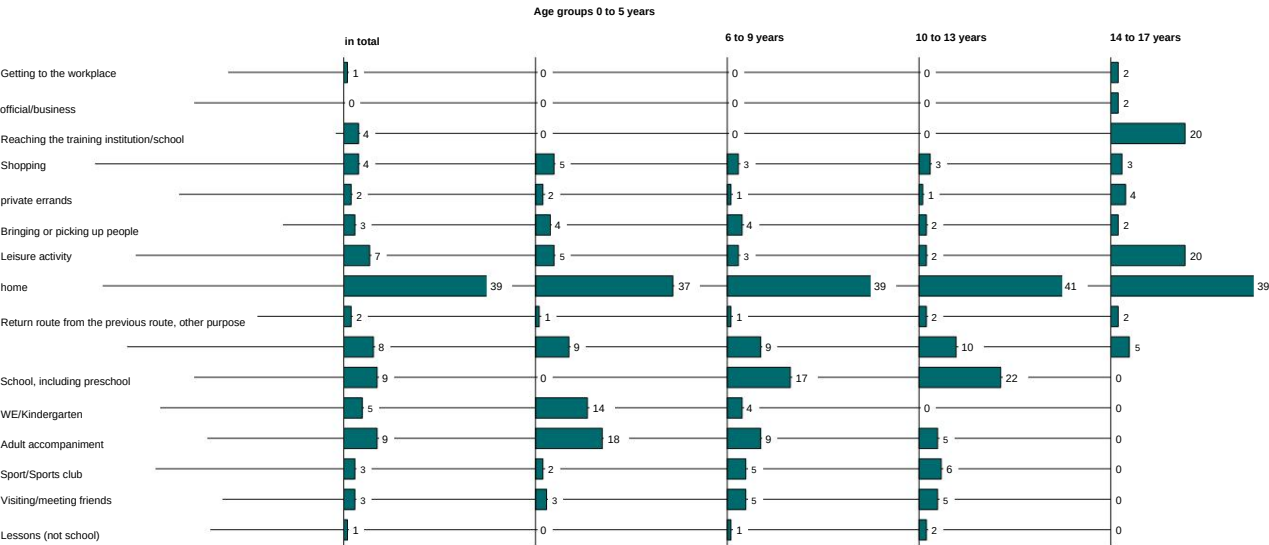


Figures in percent; persons under 18 years of age; all routes

MID 2023 | Results Report | Source: MID 2023

Figure 97

Route purposes for routes used by children and young people



Figures in percent; persons under 18 years of age; all routes

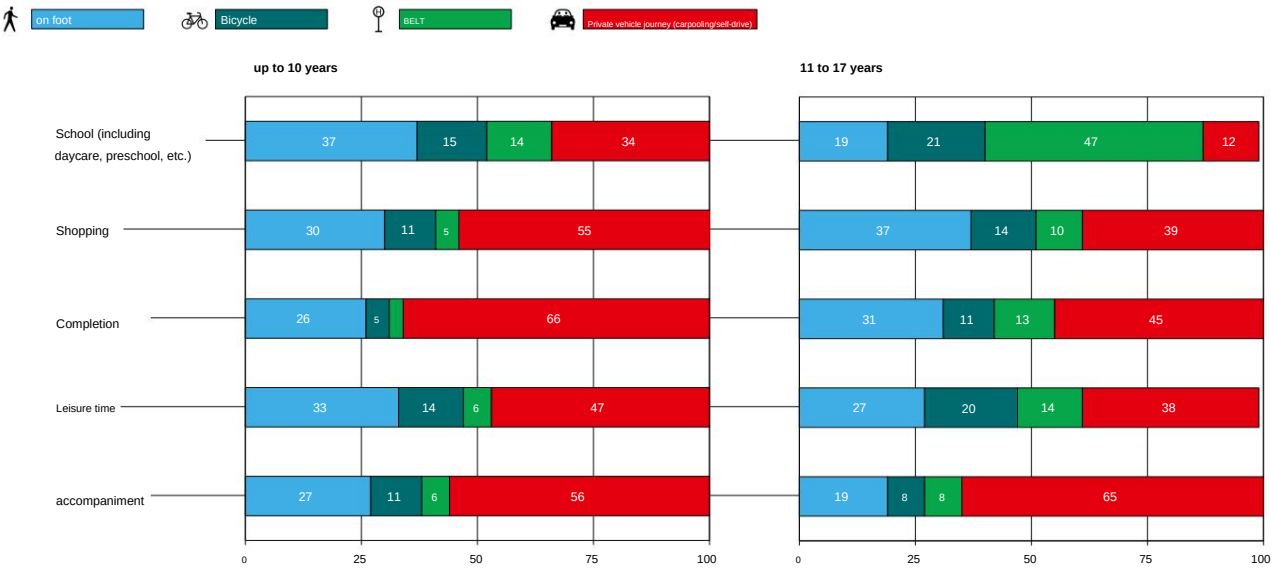
MID 2023 | Results Report | Source: MID 2023

Figure 98

Cap 14, A88

Use of transport for different journey purposes by children and young people

Paths



Data in percent, totaling 100%. Missing: don't know, no answer; routes taken by children and young people

MiD 2023 | Results Report | Source: MiD 2023

Journey destinations for children: from kindergarten to vocational training

The distribution of travel purposes also reflects the different life stages of children on their path to adulthood very well. For children aged 0-5, the journeys to daycare or kindergarten are replaced by journeys to preschool and school for those aged 6-9 and 10-13, and by journeys to vocational training or school for those aged 14-17. The children's travel purposes thus reflect their development through educational institutions. Furthermore, the fact that 20 percent of journeys are for leisure activities among 14- to 17-year-olds indicates an increasing degree of independent leisure activity.

Are there any changes over time?

The status quo in 2023 is one thing; another question concerns changes over time. For this, past MiD surveys, which also included interviews with children, can be consulted. A comparison of the mobility rate and the number of trips shows that since 2002, children and young people have been spending less time outdoors. According to MiD, in 2002, 91 percent of them were out and about.

The MiD 2023 shows a decrease of 9 percentage points. The number of trips has also been steadily declining since 2002, with almost constant daily distances traveled but shorter travel times. While the MiD 2023 also notes a decline in mobility for adults, it is not to this extent or with such consistent regularity. Therefore, there is strong evidence to suggest that the amount of out-of-home activity among children and adolescents has gradually decreased over the past two decades.

School commutes in rural areas are twice as long – But young people everywhere primarily use public transport.

The daily values shown are derived from the sum of journeys and the destinations to be reached along the way. Infrastructure and distance play a key role in this calculation. However, the time series from 2002 to 2023 shows hardly any changes. The average distance to kindergarten is around 3 kilometers at all survey points. The situation is similar for the slightly longer journeys to primary school. Only in the secondary school years do the average school journey lengths double to around 7 kilometers, but they also remain relatively stable over the years.

Leisure trips are often longer, but their average distances do not increase over time. For example, they remain in the range of 12 to 14 kilometers for young people.

As children grow older, the distances they travel and the means of transport they use for specific occasions change. The distance to school, for example, changes with age: while it's around 3 kilometers for children up to age 10 in kindergarten and primary school, as mentioned, teenagers between 11 and 17 years old have twice that distance to travel to school and vocational training. When transitioning to secondary school, it's usually no longer possible to reach the school on foot, and public transport now accounts for 47 percent of journeys. However, trips to accompany children are increasingly made by car from the age of 11 onwards.

For younger children, these routes are often accompanied by adults, and as they get older, they also include drop-off and pick-up routes. Cycling also becomes more important for the school commute from the age of 11 and is increasingly used for leisure trips, which children now manage independently. Leisure destinations are already around 6 kilometers away for younger children up to 10 years old, roughly the same distance as for teenagers. Walking and cycling routes remain comparatively shorter at up to 3 kilometers for children up to 10 years old and for teenagers up to 17 years old.

Journeys by public transport and car, on the other hand, are longer, covering up to 10 kilometers.

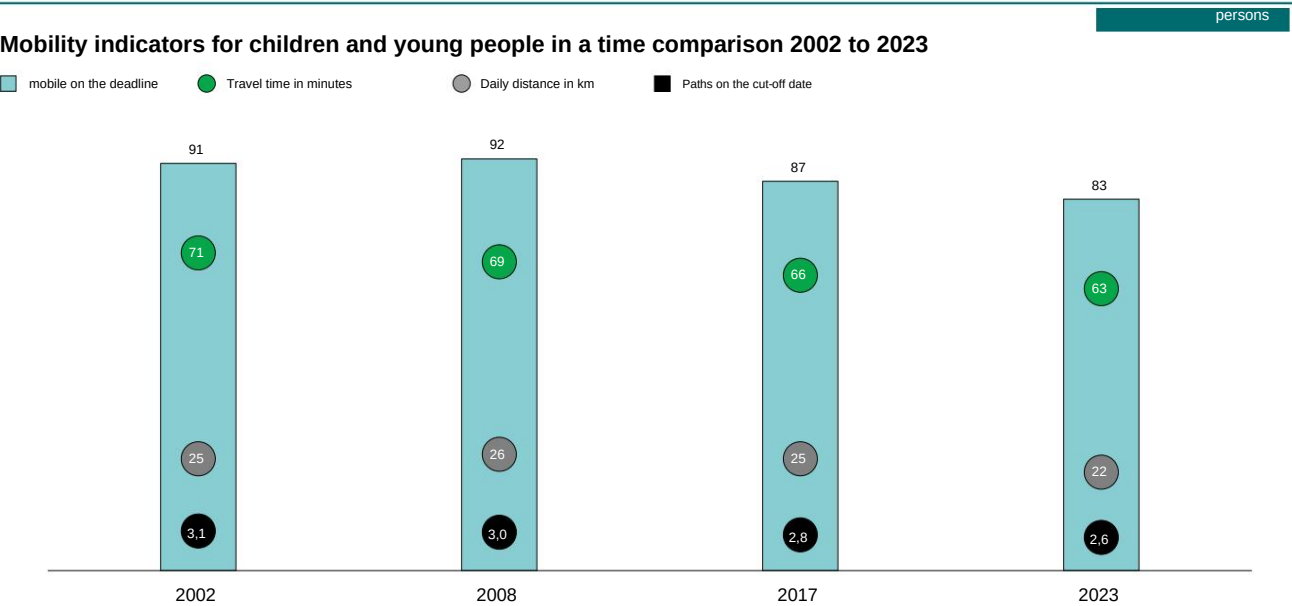
The length of school commutes varies greatly depending on the type of area: While children and teenagers in metropolitan areas and large cities travel a maximum of 3.8 kilometers to secondary school, this distance is already exceeded in rural regions for 7- to 10-year-olds during primary school. In villages, they have to cover up to 4.9 kilometers to get to school. To reach secondary school, the distance is then more than twice as long.

This means that the question of how children and young people use transportation remains dependent on the purpose of the journey: How are the different distances managed on different occasions?

Children are most often driven to daycare by car, except in metropolitan areas. This changes when they start school: except in rural areas where public transport plays a major role, most children now walk to school. When they move on to secondary school, public transport becomes more important, but cycling is also widely used in central cities and rural regions. Young people aged 14 and over continue to rely heavily on public transport. However, cycling gains popularity in metropolitan areas and...

Figure 99

Cap 14, Abb.



Data in percentages and analytical characteristics; persons between 0 and 17 years of age

Table 43

Kindergarten and school route lengths by room type and age group

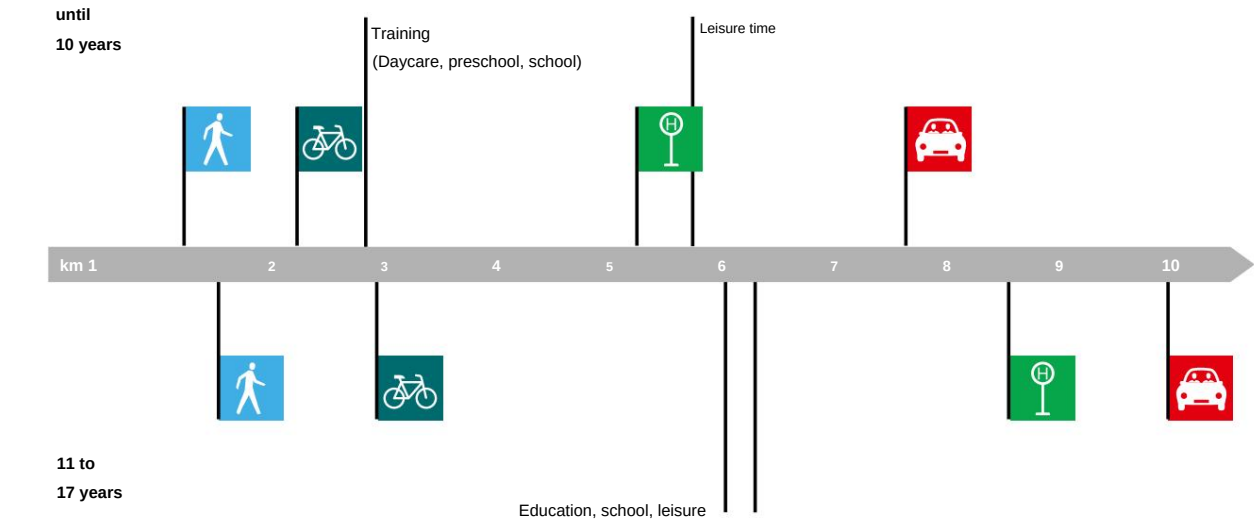
	0-6 years	7-10 years	11-13 years	14-17 years
Persons aged 0 to 17 years	km	km	km	km
ROOM TYPE				
urban region				
Metropole	2	2	4	4
Regional center and major city	2	2	4	4
medium-sized town, urban area	3	3	6	7
small-town, rural area	3	5	9	9
rural region				
central city	3	2	4	5
medium-sized town, urban area	3	4	6	7
small-town, rural area	3	5	9	10

MiD 2023 | Results Report | Source: MiD 2023

Figure 100

Cap 14_A08

How far are the distances in the immediate vicinity (up to 50 kilometers)?



Average distances in km; distances traveled by children and young people in the immediate vicinity up to 50 kilometers

MiD 2023 | Results Report | Source: MiD 2023

In large cities, it is becoming increasingly important, and in central cities in rural areas, the proportion of journeys made on foot is also increasing.

In summary, the patterns of journey purposes show that children's mobility decisions are strongly age-dependent and, with increasing age, are also characterized by growing autonomy and personal responsibility. The significant changes that children undergo from early childhood to adolescence are clearly visible through the diverse journey purposes and their development within the age groups.

Seniors on the go: at a reduced level, but still car-centric.

Following the insightful look at the youngest road users, we will now turn to the promised perspective on citizens at the other end of the age spectrum. The car also significantly shapes the mobility of this age group. Men almost always hold a car driving license, and older women are increasingly doing so, although they haven't quite caught up yet – as already shown in the chapter on car use.

When households with senior citizens are compared to other households, their relatively low mobility level is immediately apparent. The number of trips per person per day is closer to two than three. The average trip length also differs...

The difference is not significant, but the daily distance traveled is shorter than in younger households due to lower basic mobility.

When these figures are broken down into three age groups within the older population, a clear reduction with increasing age becomes apparent. The mobility rate, i.e., the proportion of people who are mobile outside their homes on the reference date, is only slightly below the average for the population at 78 percent for those aged 65 to 74.

However, this figure drops to 72 percent for those aged 75 to 84 and to just 56 percent for those aged 85 and over. This decline is accompanied by decreases in the number of trips made per person per day, the daily distance traveled, and the daily travel time. In the oldest group, these figures are even lower: only 1.3 trips, just under 40 minutes, and less than 10 kilometers per day.

Car ownership tells a different story: Three-quarters of households with senior citizens (all persons aged 65 or older) own at least one car, and 11 percent even own two or more. The differences between urban and rural areas, with higher car ownership rates outside of cities, are consistent with those in the general population. This is a comparatively good level of ownership relative to household size, but it shouldn't obscure the different circumstances of economically constrained households. If a household with senior citizens belongs to the lowest status group, almost half of them do not own a car. As is so often the case, the average masks both precarious and very comfortable living situations. This also applies to the category of the pre-

Table 44

Mobility indicators for journeys, travel time and daily distance for people aged 65 and over				
	Mobility quota (outside the home at Reporting day)	Paths	Average values per person per day	
			Travel time	Daily distance
	%	Number	min	km
65 to 74 years	78	2,5	81	27,1
75 to 84 years	72	2,0	62	17,5
85 years and older	56	1,3	39	9,0

Cars available: A comparatively large number of households with senior citizens own vans or SUVs, even exceeding the general trend in this regard.

The bicycle ownership situation is different again: In almost 60 percent of the households with senior citizens examined here, the residents own a bicycle, and of these, a rising trend shows that around 40 percent of bicycle owners also own a pedelec (electric bicycle). This is higher than the figure for electrically assisted bicycles compared to younger households and likely extends active individual cycling in many cases. However, this generally requires cycling socialization in younger years. The path from not cycling to using an e-bike – as other results show – is long and rare.

When considering the modal split for this household type, it is close to the population average, particularly with regard to private motorized transport, although with somewhat higher walking rates, reflecting a mobility pattern that is often limited to short distances. The share of bicycles, and especially e-bikes, is also comparatively high.

Public transport usage is comparatively low. When broken down by age group within the senior segment, the proportion of journeys made on foot increases with age. This also applies to the use of public transport.

It also increases in the older age groups, while the proportion of private motorized transport decreases accordingly.

However, this figure remains between 40 and 50 percent even in the oldest age groups. The car is therefore often no longer a daily companion, but still frequently perceived as indispensable.

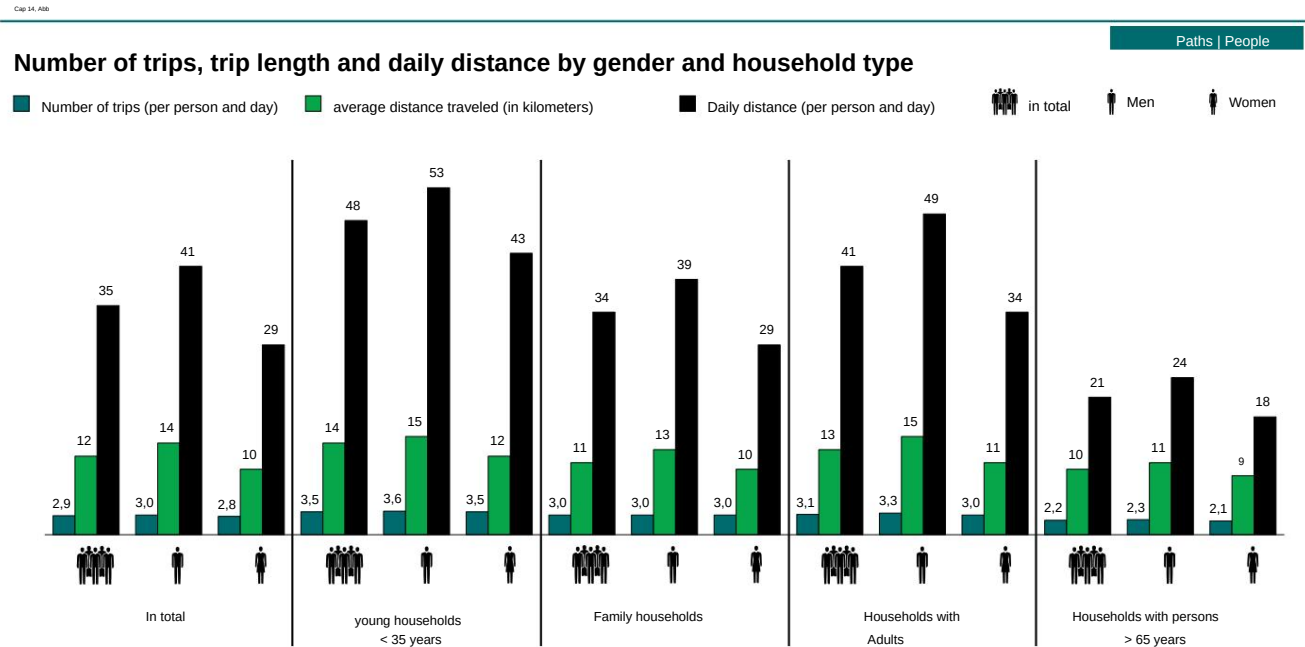
Numerous reasons for not traveling, but also specific mobility needs in old age

However, not every day counts as a day spent traveling for the group of senior citizens.

The mobility rate in these households reaches only 74 percent per person, which is lower than in the general population. The reasons for this are as multifaceted as they are age-specific and were collected as multiple responses. 55 percent of the older respondents stated that they had no reason to travel. 31 percent only did things at home. 11 percent – fewer than in younger groups! – reported being ill themselves, and 2 percent had to care for other household members.

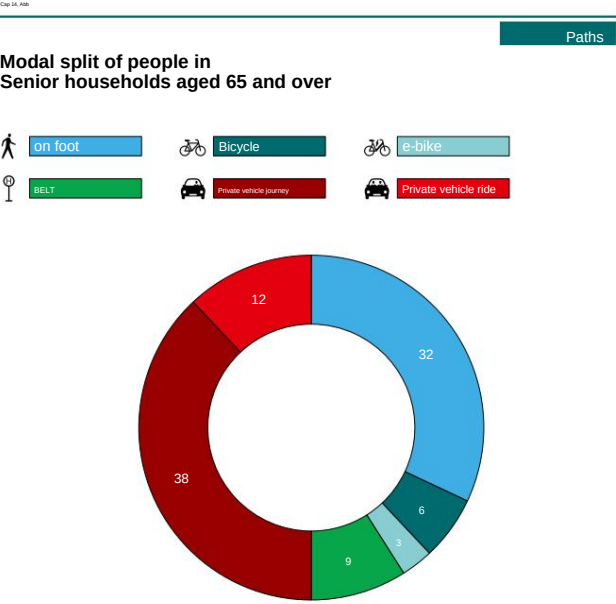
However, there is another side to consider.

Figure 101



Figures in percent; all persons, all routes
MiD 2023 | Results Report | Source: MiD 2023

Figure 102



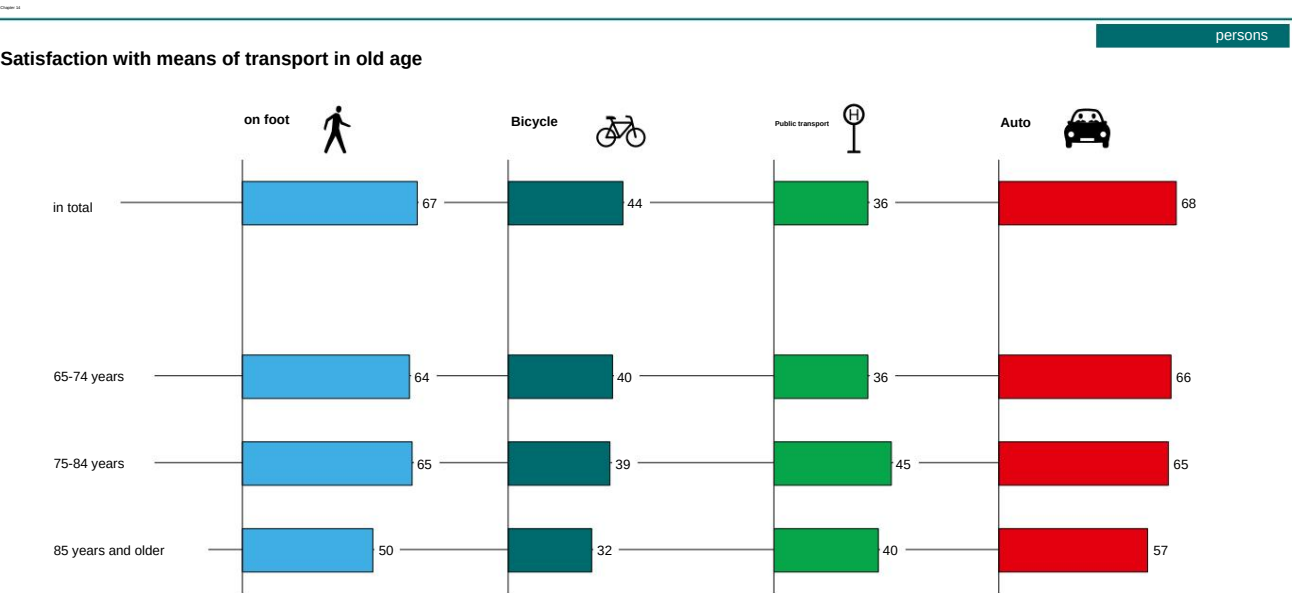
Figures in percent; routes
MiD 2023 | Results Report | Source: MiD 2023

Eight percent of senior citizens were traveling on the reference date, which is only slightly below the average for all age groups. However, these trips tend to be longer than average for seniors, with four out of ten trips including at least four overnight stays. The average for all trips is only 30 percent. It should also be noted that economic circumstances play a role. For example, the proportion of trips taken on the reference date in the highest-earning group (around 15 percent) is three times higher than in the lowest-earning group.

Satisfaction with transportation decreases with age.

Insightful for the everyday mobility of older people are not only the usual mobility indicators, but also the subjective satisfaction levels with the options for walking, cycling, driving, or using public transport in their local area. The perceived quality of these options significantly influences which activities are (still) pursued and with which means of transport. For example, poor pedestrian conditions or limited accessibility on buses and trains can make everyday activities more difficult not only for older citizens. This-

Figure 103



Figures in percent, TopBox satisfaction scores "very good" and "good", persons aged 65 and over, and overall score from age 14.
MiD 2023 | Results Report | Source: MiD 2023

The MiD results show that satisfaction with these options decreases with age.

This is especially true for the oldest group, those aged 85 and over. They rate all the aforementioned services lower than people under 65 and also less favorably than seniors aged 65 to 84. For example, only half of this segment gives the footpaths in their neighborhood a good rating. Regarding cycling conditions, it's only a third, for public transport only four out of ten respondents, and even for cars only slightly more than half. Some of these ratings may also be related to increasing personal uncertainties, but at least for walking and public transport, better conditions would be desirable for this group as well – quite apart from the fact that such improvements ultimately benefit all age groups, and that the overall population's rating of walking, cycling, and public transport also has room for improvement and demands better conditions.

Mobility limitations in old age: one third affected

Let's turn our attention to disabilities. Such mobility limitations don't only affect older citizens, although they do affect them to a greater extent. The 2023 Mobility in Germany (MiD) survey delves deeper into this topic than the editions from 2002 to 2017. While those earlier surveys already quantified the extent and nature of potential mobility limitations, at least a third of senior citizens reported experiencing such limitations. However, the 2023 MiD survey also included, for the first time, questions about dependence on assistive devices such as walkers or wheelchairs.

This is reported by four out of ten households in this age group who are generally affected by a mobility impairment. However, a specific chapter of this report examines these findings and the impact of this situation, considering all affected groups regardless of age.

15. Mobility and individual economic conditions: opportunities, participation and "mobility poverty"

The following chapter addresses mobility from the perspective of households' economic status and subjective assessments of their own mobility options. While considering the results based on an individual's or household's economic situation has been part of the Mobility in Germany (MiD) survey since 2002, the current edition adds questions about perceived personal mobility and activity opportunities. These opportunities can be more or less unlimited, but can also be influenced by individual financial or health limitations, living situation, poor infrastructure, and limited accessibility to everyday destinations.

These aspects are increasingly discussed under the heading of "mobility poverty," but they have always been an important dimension of participation and actually reflect more accurately "activity poverty" or even "opportunity poverty." Being mobile is not an end in itself, but a prerequisite for a fully integrated life. For an individual, a group of adults living together, or a family with children, their financial situation has a decisive influence on their ability to be mobile and participate in social and economic life.

Economic status as an important analytical perspective

The MiD distinguishes five different levels with regard to the economic status of a household and the persons living there. These are derived from the equivalent income per capita calculated using the weighted household size. More detailed information on the calculation of this characteristic is available in the MiD Data Usage Handbook.

Before this variable is used analytically in conjunction with mobility variables, a brief description of the resulting distributions will be given. This analytically derived, rather than directly queried, status differs according to sociodemographic characteristics. It changes primarily with age, educational attainment, employment status, and family situation. Young people between 18 and 29 years of age are similarly represented in the status groups. Since this life phase is very

Since the population is heterogeneous, the scope of employment and educational status provide further information. In contrast, middle-aged groups tend to be more heavily represented in the higher economic status groups. People of retirement age and older, on the other hand, are more strongly represented in the lower to middle groups, but presumably also have other forms of economic security than the monthly income used as the basis for this analysis.

Full-time employed persons and persons with University degrees are predominantly found in households with very high economic status.

Employment status is closely linked to the economic status of the household. More than half of the people in households with high and very high economic status are employed full-time, and about one-fifth work part-time. People with low and very low economic status, on the other hand, are primarily pensioners who have access to a lower income.

Besides employment status, the level of education also has a direct influence on a household's economic status. The higher the individual's educational attainment, the higher the household's economic status. 51 percent of people with a university degree live in households with a very high economic status. Only 8 to 19 percent of households with a secondary school leaving certificate (Realschulabschluss, Mittlere Reife, or Hauptschulabschluss) have a very high economic status. Household type is also related to the household's economic status. For example, 45 percent of households with a very high economic status are families with children, whereas only 9 percent of people live in households with young people under 35.

Economic status is thus shaped on the one hand by life stage, and on the other hand by educational attainment as well as the type and extent of employment. While these are not the sole determining factors, they are paramount. And as the further results show, clear correlations emerge with the type and extent of everyday mobility.

Table 45

Socio-demographic characteristics according to economic status					
	very low low		medium	high	very high
Basic households	%	%	%	%	%
CATEGORIES SHARE TOTAL	9	12	29	37	13
Column percentage, base persons	%	%	%	%	%
ALTER					
18 to 29 years	19	13	15	17	15
30 to 39 years	10	9	13	19	23
40 to 49 years	11	10	12	16	25
50 to 59 years	13	13	14	22	23
60 to 64 years	11	9	9	9	7
65 to 74 years	19	24	18	10	5
75 to 84 years	13	17	14	5	3
85 years and older	5	6	5	2	1
ACTIVITY OF THE PERSON					
Full-time employed (including trainees)	14	23	37	52	61
Part-time/marginally employed (11 to under 35 hours per week) employed with other scope or without specification of scope Pupil, including preschool	14	15	15	19	22
	2	2	1	2	1
	1	1	1	1	1
Student/in	7	3	2	3	2
Housewife/Househusband	3	3	3	2	1
Retired person, pensioner, including early retirement	37	46	36	16	7
Other (e.g. unemployed)	22	8	6	6	4
EDUCATION					
(still) without a secondary school	4	2	1	1	1
leaving certificate, 8th grade at a secondary school	38	40	32	18	8
Secondary school leaving certificate, intermediate school leaving certificate, 10th grade at the POS (Polytechnic Secondary School)	24	30	31	29	19
University entrance qualification, high school diploma	12	11	14	18	19
university degree (University of Applied Sciences or University)	13	13	18	30	51
other degree	9	4	4	4	2
HOUSEHOLD TYPE					
young households (under 35 years old)	14	9	10	11	9
Family households	16	17	25	31	45
Households with adults	35	31	33	46	41
Households with persons aged 65 and over	34	43	32	12	6

Status is a determining factor for car ownership, but less so for public transport subscriptions.

Regarding transportation options, a household's economic status primarily affects car ownership. While only 3 percent of respondents with very high economic status cannot own a car, either personally or as a passenger, 29 percent of those with very low status have no access to a car. A similar disparity is evident in bicycle ownership: only 59 percent of people in the lowest status group own a bicycle. This figure increases across the status groups, reaching a maximum of 88 percent in the highest status group.

The MiD 2023 also reveals the reasons for not owning a car. In urban areas, it is primarily a conscious decision to forgo car ownership. In rural regions, the inability to afford a car is more common, often due to age or health reasons. This is mainly due to the socio-demographic structure of rural areas compared to younger metropolitan areas.

In contrast, public transport season ticket ownership shows a significantly weaker correlation with the economic status of a household or the people living there.

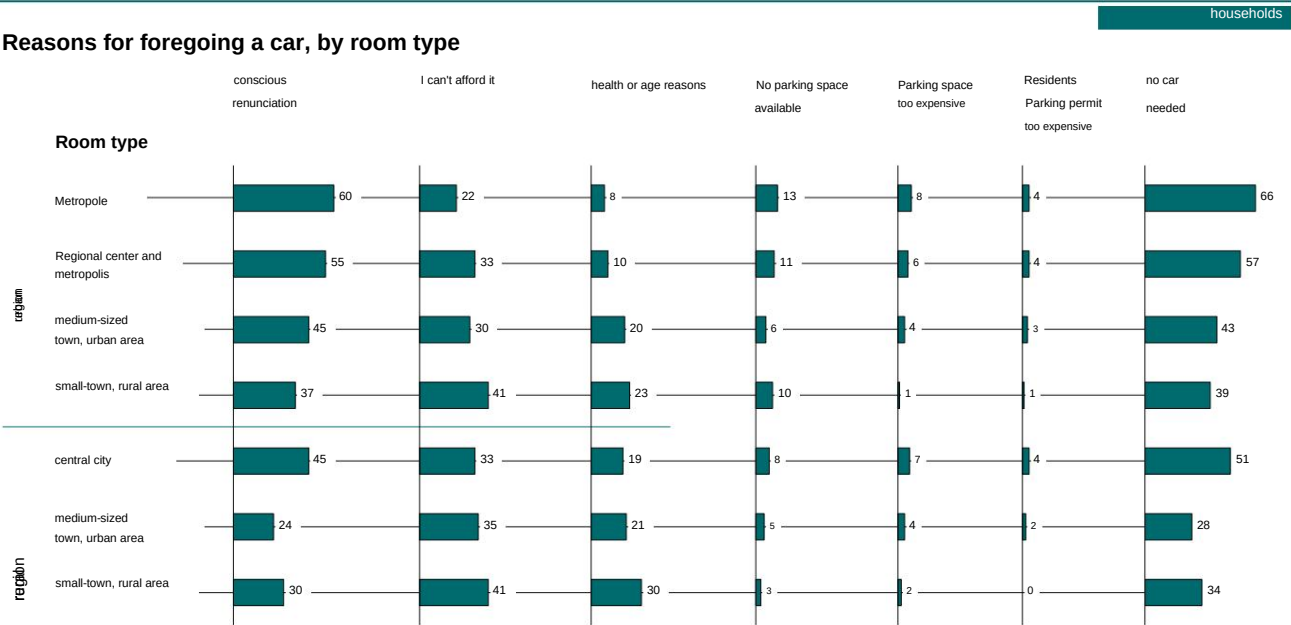
The figures show that between 14 and 19 percent of people with both low and high economic status own a Deutschlandticket (Germany-wide public transport ticket). While the ownership rate in the lowest status group is significantly higher than in the other status segments when additional season ticket options are offered, no significant differences emerge between these segments.

Households with low economic status are less mobile – similar to before

Differences in key mobility indicators relating to the reporting day are more obvious than with cars and tickets, i.e., "mobility equipment". The mobility rate, the average number of trips made by a person per day, the average daily trip length and distance traveled by a person, as well as the daily travel time, show a close correlation with the economic status of the household in which the person lives.

The higher the economic status of a household, the more frequently the people living in that household leave the home on the reporting day. People in households with a very high economic status spend an average of 12 minutes more outside the home each day.

Figure 104



Data in percent; multiple responses; households without car ownership; spatial type RegioStaR7 according to BMV

People traveling in the middle of the world complete 0.4 more journeys per day, travel an average of 6 kilometers longer distances and, with an average of 44 kilometers, have a daily distance more than twice as long as people from households with very low economic status.

Without a clear causal link, these differences are related to the type and extent of employment, the often time-saving availability of a car, and travel distances. A higher economic status often brings changes in all these aspects. Compared to the last MiD survey in 2017, as well as the measurements in 2002 and 2008, this structural difference has hardly changed.

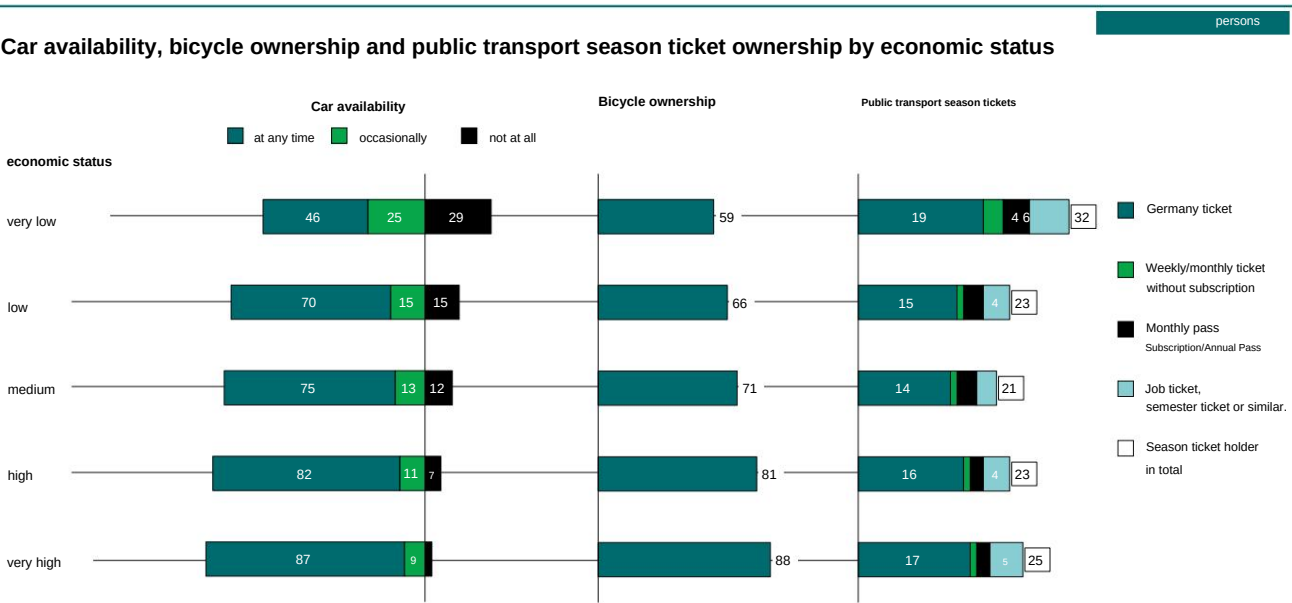
While mobility levels decreased by 5 to 10 percent in 2023 compared to 2017, and even more significantly in terms of daily distances traveled, this reduction, already shown in previous sections of the report, is more or less evident in all socioeconomic groups. It is somewhat more pronounced for each indicator in the lowest socioeconomic group, thus widening the gap between this group and the rest of the population, albeit not significantly.

In any case, the “mobility gap” between particularly well-off and particularly poorly-off people will remain in 2023 as well.

Growth increases with economic status
Route lengths and daily distances

An examination of the reasons for daily mobility reveals a close correlation between the purpose of the journey and the household's economic status. While individuals from households with very high economic status cover an average daily distance of over 100 kilometers for work-related travel, those with very low economic status travel an average of only 14 kilometers for this purpose. The length of individual commutes also increases with economic status. While these average 11 kilometers for individuals in households with low economic status, they rise to almost 19 kilometers for those with high economic status. Leisure journeys also become longer with increasing household economic status. Due to the greater availability of a car in households with higher economic status, longer journeys are easier to undertake and, in this segment, also more affordable. This is reflected in leisure activities taking place further afield, as well as in workplaces located further away.

Figure 105



Figures in percent; Car availability: driving or riding along, persons aged 14 and over; Bicycle ownership: persons aged 14 and over; Public transport season tickets: persons aged 14 and over

Economic status also has an impact
Choice of transport

Besides the number of journeys and their reasons, the choice of transport mode is related to a household's economic status. While households

While people with very low economic status travel in similar proportions on foot and by car, this ratio shifts in favor of the car as the household's status increases.

In higher status groups, over half of all journeys are made by car.

Table 46

Mobility indicators, travel time, routes, route lengths, daily distance according to the economic status of the household

	Mobility quote	average travel time per Person and day	routes per Person and day (Mean)	distance (Mean)	medium Daily distance per person
	%	minutes	Number	km	km
ECONOMIC STATUS					
very low	74	76	2.6	8	21
low	76	76	2.7	10	26
medium	80	81	2.9	11	33
high	84	86	3,0	13	39
very high	86	88	3,0	14	44

MiD 2023 | Results Report | Source: MiD 2023

Table 47

Daily journey length and daily distance per person according to journey purpose and economic status of the household

Average values across all routes (per category/person)	distance					Daily distance per person for this purpose				
	ECONOMIC STATUS very low	low	medium	high	very high	very low	low	medium	high	very high
	km	km	km	km	km	km	km	km	km	km
IN TOTAL	8	10	11	13	15	15	25	34	51	76
WAY PURPOSES										
Work	11	12	14	16	19	11	15	21	35	57
official	5	9	12	15	21	14	24	38	59	106
Training	7	7	7	6	5	8	11	9	11	11
Shopping	5	5	5	5	5	4	7	9	13	18
Completion	9	10	9	11	13	9	15	20	28	42
Leisure time	11	12	15	17	18	13	21	29	42	60
accompaniment	8	8	9	9	10	11	16	17	22	29

MiD 2023 | Results Report | Source: MiD 2023

Despite the previously described differences in bicycle ownership rates, cycling on the reporting day, and indeed in the modal split of all journeys, also varies according to economic status – with the exception of the lower income groups, whose use of bicycles is also somewhat above average. Public transport, on the other hand, is primarily the mode of transport for very low-income households, especially in terms of travel volume. When travel volume is considered in relation to the kilometers traveled, it becomes clear that for these individuals, almost a third (30 percent) of the kilometers are traveled using public transport. For journeys made by individuals from households with very high economic status, the corresponding share is only a fifth (21 percent). Instead, the share of private motorized transport increases, while walking and cycling do not benefit from this growth. From the middle income bracket upwards, the car thus dominates in terms of kilometers traveled even more than in terms of total journeys.

Here, the share of cars is consistently well above the 70 percent mark.

Economic status has little influence on satisfaction with means of transport.

One possible explanation for the differing modal split shown, depending on the segment of economic status, could be differing quality assessments. However, an examination of satisfaction with the local transport situation reveals no significant differences for economic status, although a few details do offer some insightful observations.

The car consistently scores best, closely matching the results for walking conditions. However, the rating for car travel is somewhat more favorable among higher-status groups. In contrast, there is broad agreement regarding cycling, with rather poor ratings. This is less pronounced when it comes to the quality of public transport: While satisfaction levels are generally in need of improvement, higher-status groups not only use public transport less frequently but also rate it somewhat lower. This suggests a different perception of service and overall experience quality.

Nevertheless, the previously shown differences in usage with regard to means of transport can hardly be explained by satisfaction with these transport services. The life situations already described

and, above all, the more pronounced availability of cars in the higher status groups are more likely to be decisive factors.

Abstaining from activity: most likely young and older people and those with mobility impairments Affected

The differences between the economic segments presented provide a striking picture of varying mobility patterns. These differences are rooted in several factors, and the most important influencing structural characteristics have been identified.

Individual preferences as well as objective life requirements come into play, whether related to the specific life situation or other factors such as place of residence. However, this does not answer the question of whether these differences are also perceived as restrictive or diminish participation.

In order to investigate this aspect as much as possible in the MiD, the 2023 survey program was extended to include some simple operationalizations in this regard, insofar as this was feasible within the MiD with regard to the survey length.

Newly included are various aspects explaining why people have to forgo mobility and are thus deprived of opportunities to participate in activities.

Across all status groups and individual reasons, one in four people reported having to forgo everyday activities more frequently. Four reasons were offered for foregoing journeys and activities: not being able to afford travel, not being able to easily reach destinations with available means of transport, not having nearby options for leisure activities, shopping or errands, or being dependent on being picked up, dropped off, or accompanied by other people.

The breakdown of the aggregated results by sociodemographic characteristics shows that, according to their own statements, women have to forgo mobility somewhat more often than men. With increasing age—possibly also due to better financial resources—the foregoing of mobility decreases, only to rise significantly again after the age of 80. In the youngest age groups, the "foregoing value" is...

Figure 106

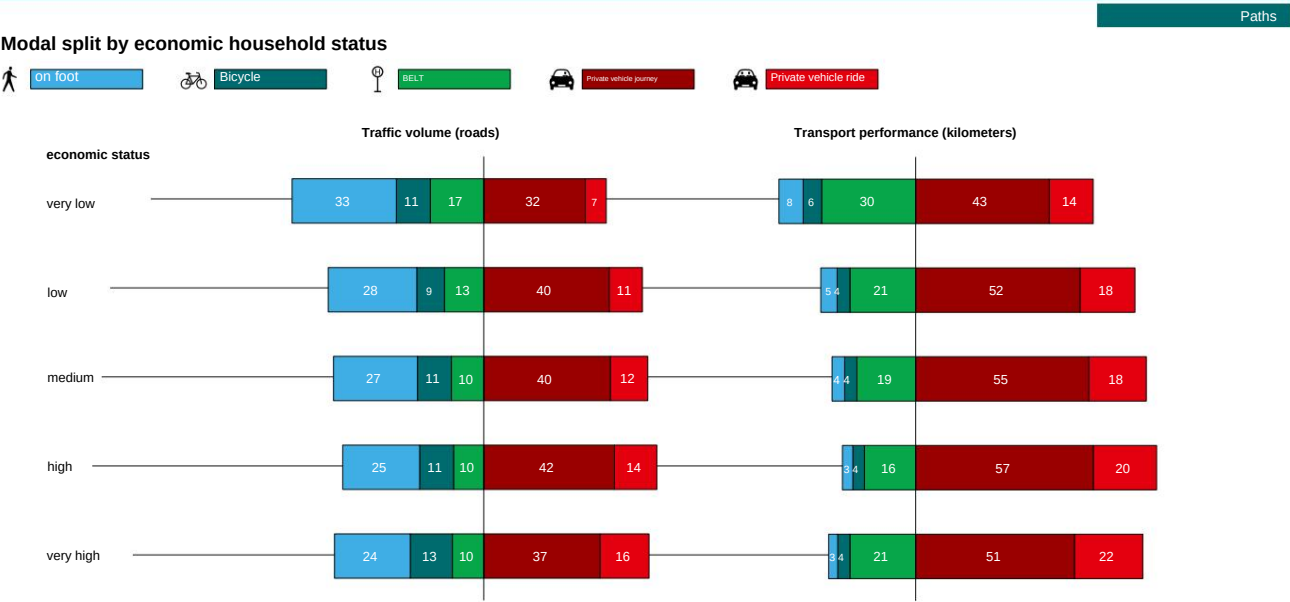


Figure 107

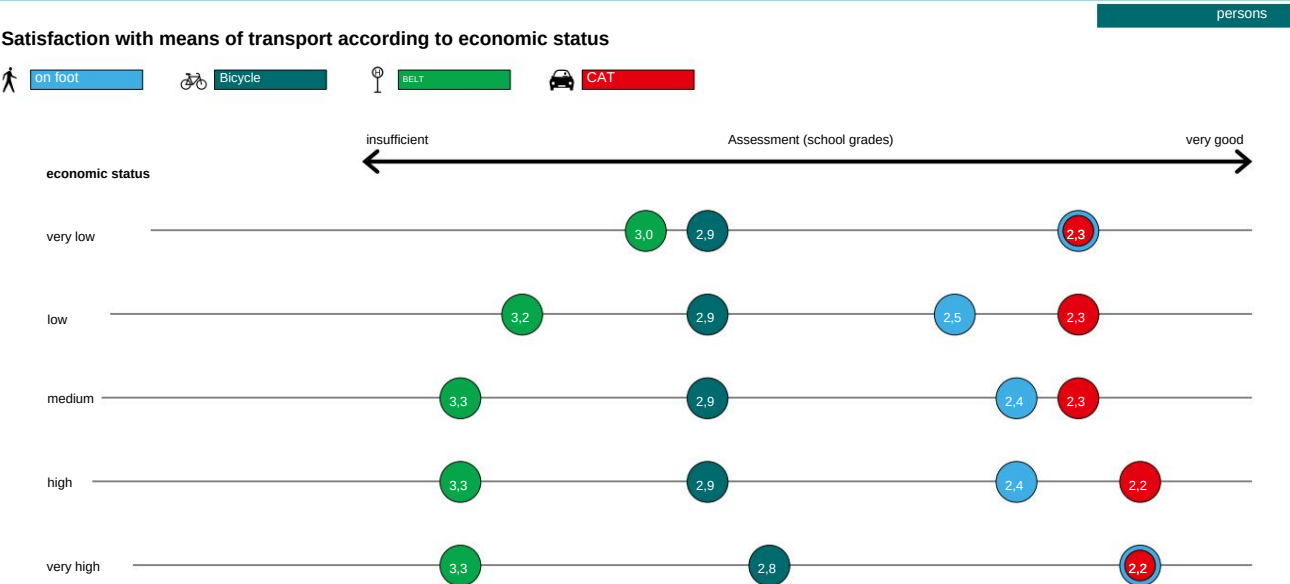
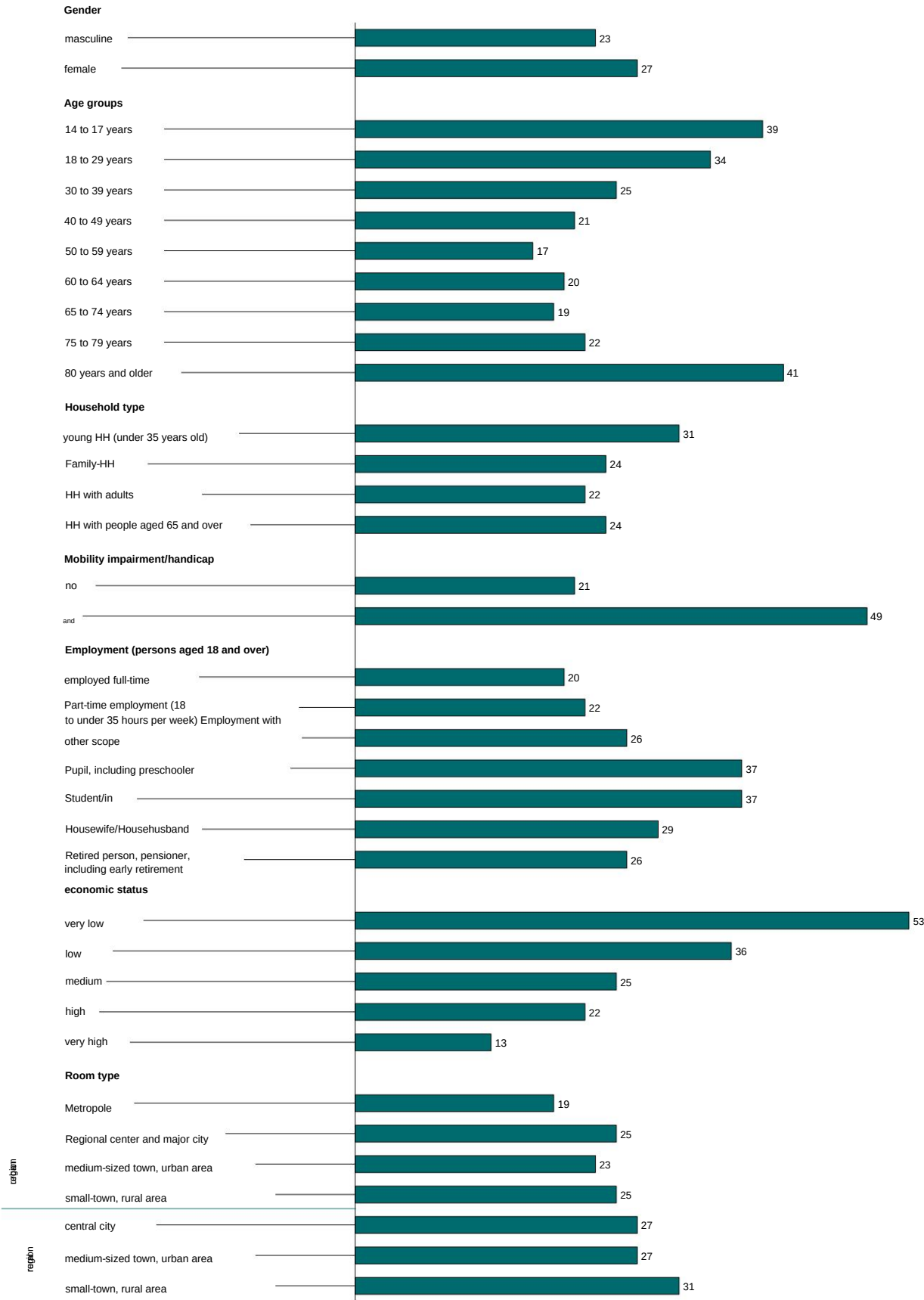


Figure 108

General abstinence from activity and mobility based on personal characteristics

persons



Figures in percent, proportions "yes"; persons aged 14 and over; Summary of individual reasons, at least one reason mentioned

30 to 40 percent, then constantly at around 20 percent, and only in old age reaching approximately 40 percent. Compared to other occupational groups, students have to forgo mobility the most, while full-time employees have to forgo it the least. Furthermore, a mobility restriction strongly influences the perception of a mobility restriction.

People who do not have to forgo mobility tend to live in metropolitan areas, are middle-aged, more likely to be male, and have no mobility restrictions.

Some of these differences suggest that there is a direct influence of income on the reported reduction in activity and mobility.

However, a look at the economic status of the household shows that both households with very low and very high economic status believe they must forgo mobility.

More decisive factors appear to be other life circumstances and individual opportunities or lack thereof.

Possibilities. Furthermore, the general question underlying these results asks for subjective self-assessment, not objective criteria. This subjective positioning can differ despite objectively similar circumstances. However, it remains to be noted that, albeit for very different reasons, approximately one-fifth of the respondents reported such a renunciation.

The main reason for abstaining from activity is unattainable goals, but also individual factors. Restrictions and lack of car access

As described, individual reasons for a perceived reluctance to engage in activities were also collected.

Before these are prioritized and presented in a segmentation at the end of the chapter, the individual results will be illustrated. The individual reasons for this reduction in mobility, which is also reflected in the travel indicators, include:

Table 48

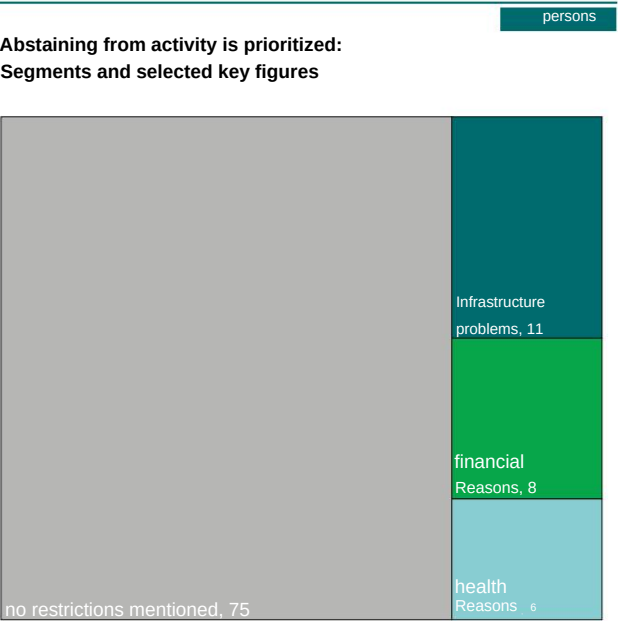
Individual reasons for foregoing mobility according to personal characteristics				
	Can the Cannot afford the trip	Destinations not reachable with available means of transport	Hardly any/no leisure activities offered	for the pickup/ The reason/ Instructed to be accompanied by other people
	%	%	%	%
ALL PERSONS 14 YEARS AND OLDER	8	13	7	8
Column percentage, affected persons only %				
GENDER				
masculine	51	48	51	35
female	49	52	49	65
AGE GROUPS				
14 to 17 years	3	6	5	14
18 to 29 years	25	28	26	23
30 to 39 years	23	20	22	13
40 to 49 years	18	12	15	8
50 to 59 years	13	13	13	7
60 to 64 years	8	7	6	4
65 to 74 years	7	7	8	8
75 to 79 years	2	2	2	4
80 years and older	1	5	4	19
Multiple responses, persons aged 14 and over indicating "rejection of mobility"				

Financial aspects, destinations that are difficult for the individuals concerned to reach, a lack of suitable means of transport, and dependence on companions are among the reasons cited. The most frequently mentioned reason is the inability to reach destinations using available means of transport, and thus the lack of suitable mobility options. Indications of limited infrastructure therefore predominate. However, taken together, over a tenth of the respondents also reported individual reasons such as financial constraints or dependence on others. Specifically, 8 percent each cited one of these two reasons.

When the reasons for avoiding activities are differentiated by gender, differences between women and men emerge only with regard to the reliance on being picked up, dropped off, and accompanied by another person. This is the main reason for avoiding mobility, especially for those in the 18-29 age group and the oldest age group. Here again, it is primarily women for whom this is a more significant factor than for men.

This reason also applies to people aged 75 and over. Here too, it is more often women for whom this is a

Figure 109



Figures in percent; persons aged 14 and over

MiD 2023 | Results Report | Source: MiD 2023

Table 49

Prioritizing reduced activity and mobility: selected key performance indicators

	all Persons aged 14 and over men	no restrictions	Infrastructure problems	financial Reasons	health- related Reasons	
average age	46	47	43	41	48	in years
Proportion of women	48	47	48	49	63	%
Possession of a car driving licence*	89	92	89	72	60	%
Percentage of households without a car	16	11	20	33	36	%
The proportion of economic status is very low.	8	5	8	31	16	%
Out-of-home quota	82	85	86	82	72	%
average number of trips per day, average	3	3,1	3,1	2,7	2.1	Number
daily distance, average travel	41	42	51	32	23	km
time per day	89	90	101	87	62	in Min.
Percentage of journeys on foot	25	24	26	30	33	%

*Persons aged 17 and over

The reason is to forgo mobility.

The unavailability of transportation can also lead to a lack of mobility. This is the most important reason for young people up to the age of 29 to forgo mobility – and is cited more often outside of cities than in urban areas. Men and women do not differ in this regard.

Subjectively perceived inactivity: the picture in key mobility indicators

In light of this finding, the question arises whether groups with or without self-reported activity abstinence actually differ along their objective mobility indicators?

Differences do indeed emerge. People who report having to forgo mobility have a slightly lower rate of time spent outside the home, a shorter average daily commute, travel more on foot and using public transport, and have less access to a car. Taken together, this shows that it is not necessarily the amount of time spent outside the home that is affected, but rather the nature and extent of mobility, reflected primarily in a different mix of transport modes and a smaller geographical area of activity.

Reasons for waiving prioritized: Segmentation based
The basis for the aforementioned individual aspects

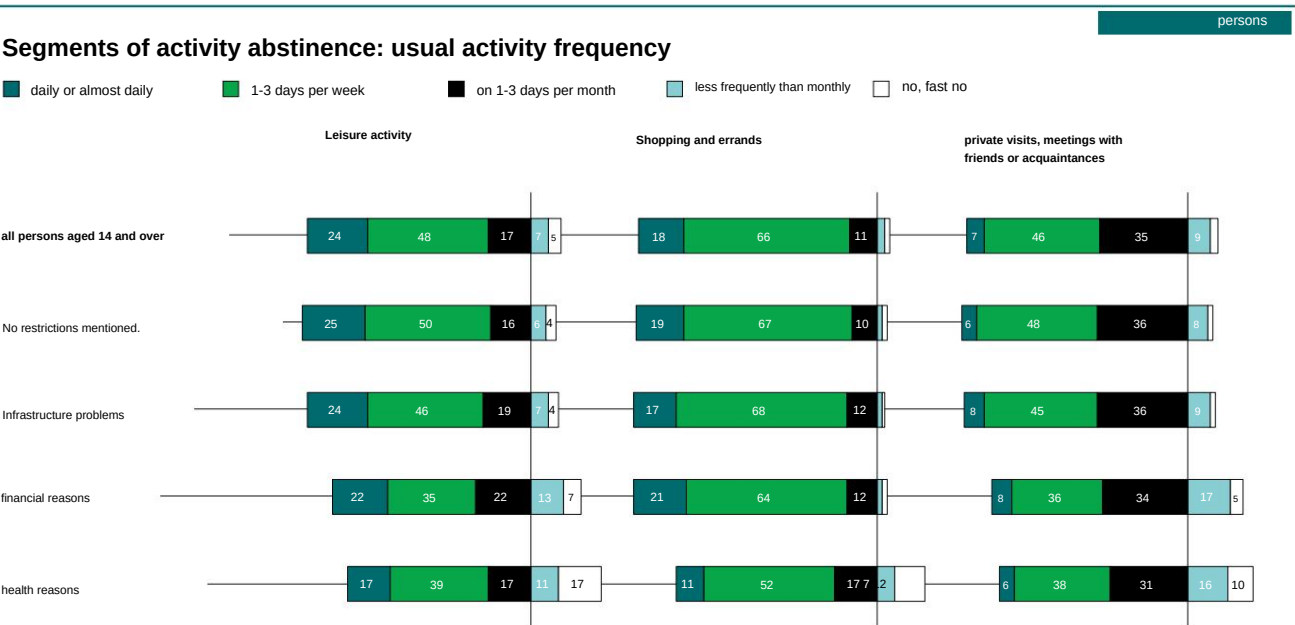
Finally, as announced, the individually mentioned reasons will be summarized in a segmentation. This classification prioritizes the stated causes. Financial reasons are listed first, followed by health-related reasons. If these were not mentioned, infrastructural reasons follow.

The proportion of respondents who did not cite any of the given reasons remains unchanged at 75 percent, meaning they presumably do not feel restricted. After prioritization, infrastructure problems rank second at 11 percent, followed by financial reasons at 8 percent, and health reasons at 6 percent.

If the four prioritized groups thus formed are considered in a similar way to the group of people with reported "activity poverty" as a whole with regard to their characteristics and mobility indicators relating to a specific date, and in a second step with regard to their usual activities in the leisure and errand sectors, further background information on the respective profiles emerges.

It is confirmed once again that people from households with high economic status are more mobile.

Figure 110



Figures in percent; persons aged 14 and over; CATI/CAWI only, random selection 50% for module

People travel longer distances and use their cars more frequently. Reduced mobility occurs at all economic levels, but is particularly prevalent among people with mobility impairments, as well as very young and very old people. Furthermore, reduced mobility is more common in rural areas due to limited access to cars and less extensive public transport infrastructure.

Participation deficits are also apparent.

Finally, looking at the usual frequency of activities provides clues to possible participation deficits.

Across all activities surveyed – leisure activities in general, shopping and errands, as well as private visits and meetings as key elements of participation – individuals with reported health deficits exhibit the lowest activity levels. This is particularly pronounced in general leisure activities and errands, and somewhat less pronounced, but still clearly visible, in the social sphere of visits and meetings. In this group, 28 percent engage in no or infrequent leisure activities, 19 percent do little shopping and errands, and 26 percent have infrequent or no visits or meetings. When financial or infrastructural limitations are more prominent, significantly fewer participation restrictions are observed – at least as far as the simplified operationalization used here, due to the aforementioned capacity constraints, can be applied. However, it provides strong indications that the topic of "limitations" deserves more attention – and is therefore explored in greater depth in the report chapter on health-related mobility limitations.

16. Health-related mobility limitations: Do we need more attention?

The recording of mobility impairments, such as walking or visual impairments, has been part of the standard MiD (Mobility in Germany) data collection since 2002. The focus is on capturing subjective assessments, not official disability ratings. This significantly facilitates access to the analysis. However, until 2017, information on the use of necessary assistive devices, such as walkers, wheelchairs, white canes, or other mobility aids, was lacking. This requirement was integrated into the data collection program in 2023. This allows for a comprehensive view of key mobility indicators in these life situations, as well as the extent of assistive device use. While this cannot fully address all aspects of this group's situation, the following findings can raise awareness of this aspect of everyday mobility.

More than one in ten people have limitations, and this number increases with age.

Physical conditions and abilities influence the possibilities for being mobile in everyday life and participating in social life. The majority of the population aged 18 and over does not have a permanent health-related restriction in their mobility. However, about one in seven people reports a permanent health impairment or

A disability that impairs their mobility. The majority of these individuals (9 percent) can leave their homes without assistance despite their mobility impairment. A smaller group (5 percent), however, occasionally or permanently requires support in their daily lives to be able to leave their homes.

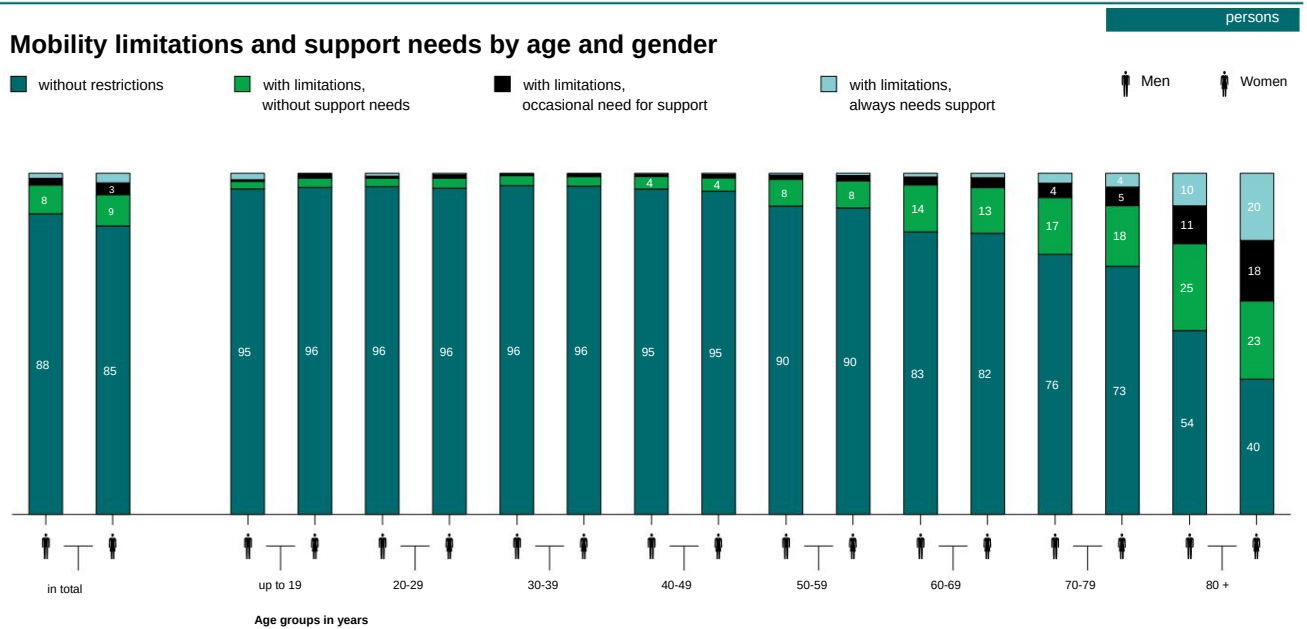
While permanent mobility limitations and the need for assistance when leaving the home are not limited to older people, they occur significantly more frequently in older age groups. From around the age of 40, the proportion of people with health-related mobility limitations exceeds five percent.

In the age group aged 60 and over, this proportion approaches 20 percent; among those over 80, more than half have limited mobility.

Walking difficulties are the most common – often requiring assistive devices.

Only a small number of people in the younger and middle age groups have permanent mobility limitations. Those affected in these age groups can manage their limitations, if necessary with the help of assistive devices, but without support from others.

Figure 111



Figures in percent; persons aged 18 and over
 MiD 2023 | Results Report | Source: MiD 2023

Figure 112



Figures in percent, multiple responses possible; persons aged 18 and over.

MiD 2023 | Results Report | Source: MiD 2023

Table 50

Type of assistive devices required for mobility impairments

	(usually) no aids	Wheelchair, Rollator, Electric car	Long or Taststock, Guide dog	other Aids	not except house despite Aids
	%	%	%	%	%
IN TOTAL	57	27	3	11	2
Men	64	20	3	12	2
Women	52	33	3	11	3
People aged 60 and over	50	34	3	12	3
People with disabilities, without support needs	76	11	2	9	1
People with disabilities, occasional need for support	34	46	6	17	2
People with disabilities always need support.	9	70	4	14	9

Persons aged 18 and over with mobility impairments, figures in percent, multiple responses possible

MiD 2023 | Results Report | Source: MiD 2023

The most common form of health-related mobility limitations is walking difficulties (9 percent). Somewhat less frequently reported are unspecified other health problems (6 percent) and visual impairments (2 percent). Compared to the general population aged 18 and over, the proportion of such limitations doubles in people aged 60 and over. Furthermore, the more support is needed to leave the home, the more frequently these limitations occur, or rather, the greater the need for support increases with the number of limitations reported.

Among the assistive devices used, walking aids such as rollators, wheelchairs, or mobility scooters are the most widespread (27 percent), followed by other unspecified aids (11 percent) and aids to compensate for visual impairments – such as white canes, walking sticks, or guide dogs (3 percent). Two percent of respondents aged 18 and over stated that they are unable to leave their homes even with assistive devices. This percentage increases fivefold once individuals are permanently dependent on assistance when leaving their homes. In this group, the use of walking aids such as rollators, wheelchairs, or mobility scooters also rises to 70 percent.

These figures demonstrate, firstly, the high demand for accessible living spaces and infrastructure. On the other hand, they illustrate that only a relatively small proportion of people with mobility impairments correspond to the classic image of a wheelchair user.

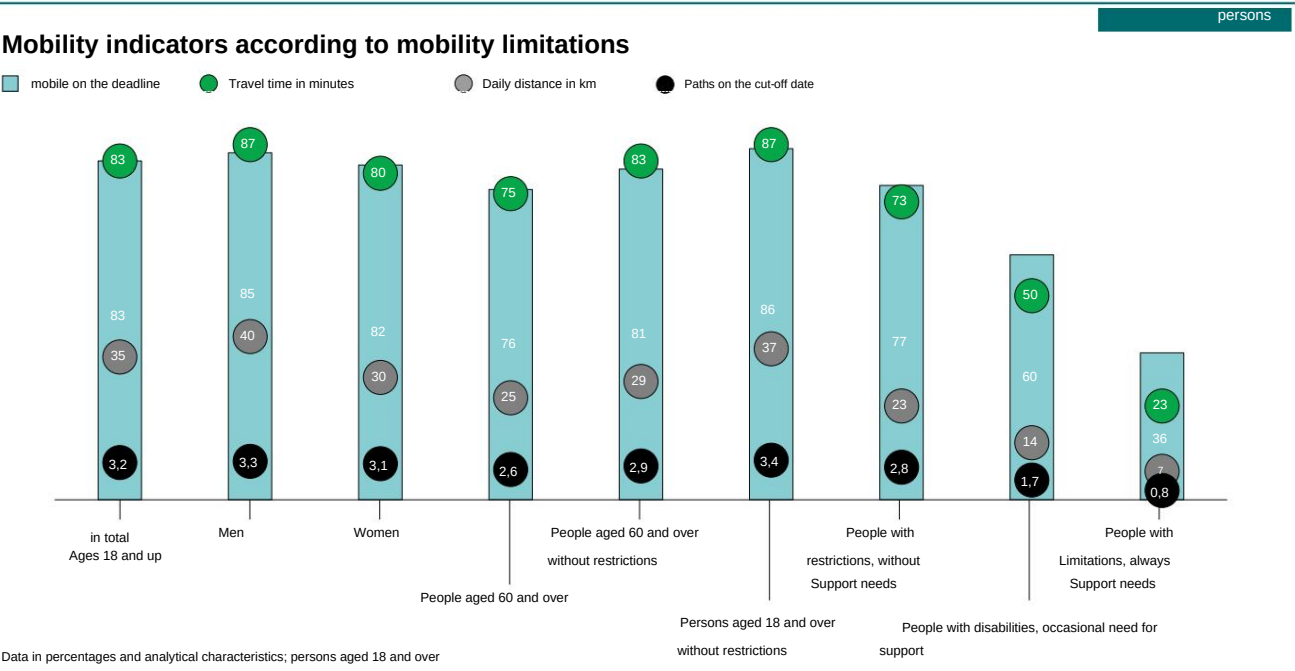
A large proportion of respondents indicate that they do not usually use assistive devices in their daily lives. Therefore, a broader focus is needed in the design of infrastructure and living spaces, one that goes beyond wheelchair-accessible entrances.

Reduced everyday mobility

The effects of health-related mobility limitations are evident in everyday mobility and activities. Since mobility limitations increase with age, individuals aged 60 and over without health-related mobility limitations are used as an additional comparison group to better understand life-phase-specific differences in mobility behavior.

People aged 60 and over without mobility limitations have a slightly lower level of mobility than the entire population aged 18 and over without limitations. The average distance traveled and the number of trips per day are significantly reduced. People with permanent mobility limitations

Figure 113



People with disabilities are significantly less likely to be out and about, regardless of age, especially if they require support.

As the need for support increases, the proportion of people who are out of the house on an average day decreases: from 86 percent in the group without permanent mobility limitations, to over 81 percent in the group aged 60 and over without limitations, down to just over a third for people who cannot leave their homes without assistance. Even in the group that leaves their homes without help despite permanent mobility limitations, the proportion of people who are out of the house on an average day is significantly lower, at 77 percent.

The lower mobility rate is also reflected in the average number of trips, the distance traveled, and the time required. In the group with the most severe mobility limitations, the number of daily trips is reduced to 25 percent (0.8 trips) compared to the general population aged 18 and over. People with less severe limitations, who can usually leave their homes without assistance, show an average reduction of about 10 percent in the number of trips (2.8 trips). People with mobility limitations

People with disabilities typically travel shorter distances. This is especially true for people who rely on support.

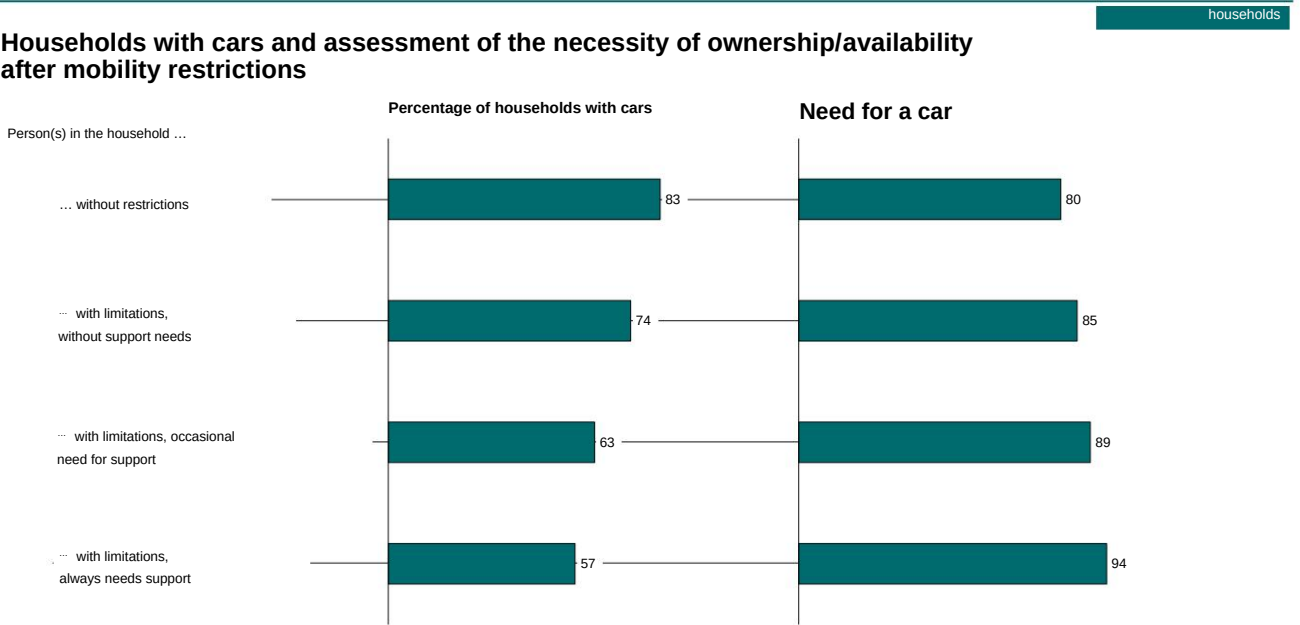
Increasingly rare car access

For people with mobility limitations, a car can provide practical support in everyday life. Walking distances, often necessary when using buses and trains, can be significantly reduced, and assistive devices can be transported more easily.

Households with at least one person with a mobility impairment particularly emphasize the importance of a car for everyday mobility. Nevertheless, car availability and driver's license ownership decrease significantly with increasing mobility impairment, and the proportion of people in households without a car rises with increasing support needs.

This trend is somewhat mitigated when, in addition to privately owned cars, vehicles that can be used by arrangement are also taken into account (extended car access, including car-sharing vehicles). The proportion of people with extended car access is over 90 percent in the total population aged 18 and over. Although the increase is more pronounced among people with mobility impairments, their extended car availability remains significantly lower.

Figure 114



Data in percent, households with information on members' mobility limitations and car ownership add-on module, necessity of car ownership: additionally only households with a car

The percentage of people with limited access to cars lags significantly behind the average and remains below the 80 percent mark in the most restricted group. The comparatively low rate of car driving licenses in this group suggests that, given the increased availability of cars, the primary focus is often on ride-sharing. Therefore, this group typically relies not only on arranging a car for their desired activities but also on having someone to drive.

A comparison with people aged 60 and over without mobility limitations shows that the difference in car availability is not primarily due to age. On average, this group even owns a car slightly more often than the general population aged 18 and over.

Information about the reasons for lower car access is provided by data from households without a car. In such households, health factors dominate the explanations as soon as at least one member has a mobility impairment and requires assistance leaving the home. Mobility impairments thus frequently occur together with other health conditions that make driving difficult.

In general, with increasing mobility-

Restrictions and increasing support needs are affecting the frequency of use of all means of transport.

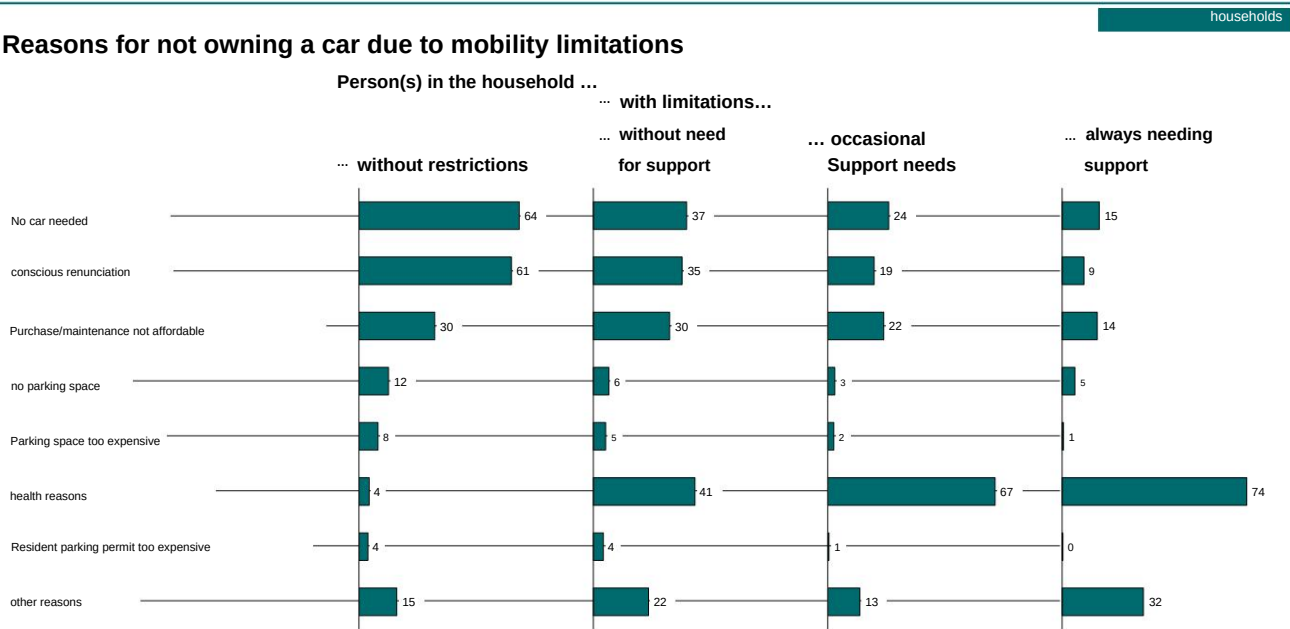
Less frequently and more slowly on the road

The limited access to cars and the overall lower use of means of transport mean that, with increasing mobility limitations and rising support needs, the time required for distances traveled does not decrease to the same extent as the daily distances.

The number of journeys made independently – by car, bicycle, or public transport – is declining particularly sharply. At the same time, the proportion of journeys made as passengers in cars is increasing.

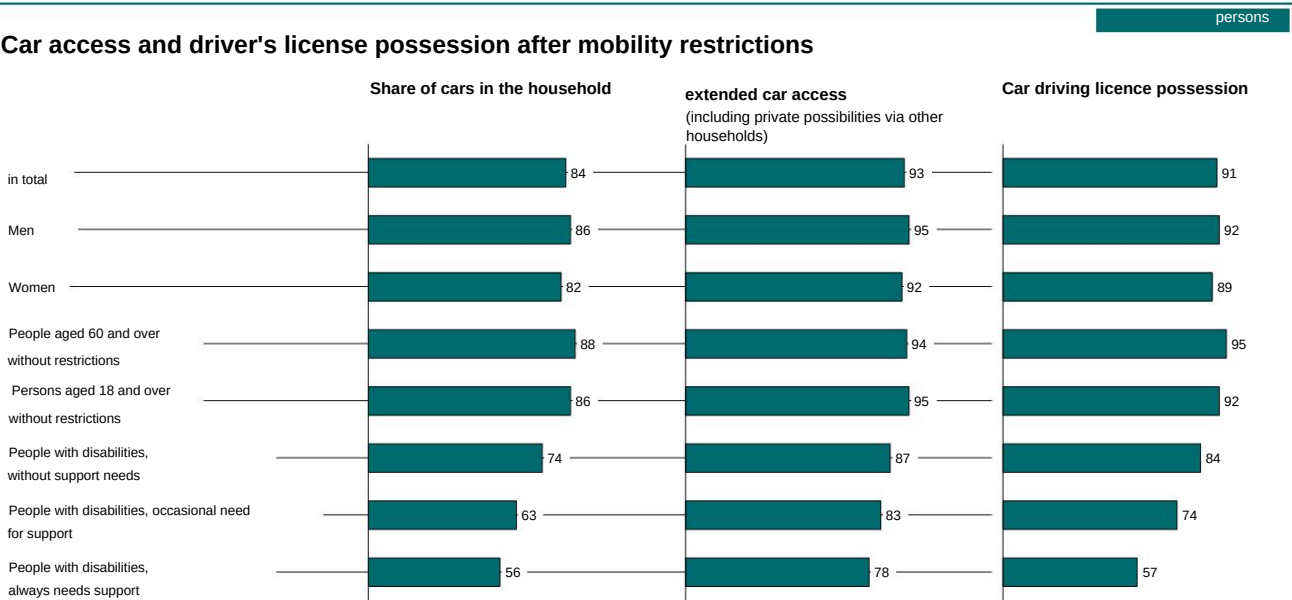
The number of journeys on foot initially remains largely constant and only decreases significantly when there is a high need for support. Consequently, the average speed at which destinations are reached decreases. These differences exist both in comparison to the general population aged 18 and over and to people aged 60 and over without mobility limitations. Age-related effects alone cannot explain them.

Figure 115



Data in percent, multiple responses possible, households with information on members' mobility limitations, car supplement module and without car

Figure 116



Figures in percent, persons aged 18 and over, extended car access: persons who live in households with a car or who occasionally/always have access to a car as a driver or passenger, including car sharing.

MiD 2023 | Results Report | Source: MiD 2023

Table 51

Average number of journeys per person, day and main mode of transport according to mobility limitations

	on foot	Bicycle	BELT	Private vehicle journey	Private vehicle ride
	Number	Number	Number	Number	Number
IN TOTAL	0,8	0,3	0,3	1,4	0,2
Men	0,8	0,3	0,3	1,7	0,1
Women	0,8	0,3	0,3	1,2	0,3
People aged 60 and over without restrictions	0,8	0,3	0,2	1,2	0,2
Persons aged 18 and over without restrictions	0,8	0,4	0,3	1,5	0,2
People with disabilities, without support needs	0,8	0,2	0,3	1,1	0,2
People with disabilities, occasional need for support	0,6	0,1	0,3	0,5	0,3
People with disabilities always need support.	0,2	0	0,2	0,1	0,3

Average number of trips per person per day, persons aged 18 and over

MiD 2023 | Results Report | Source: MiD 2023

Less suitable transport infrastructure?

Among the general population, satisfaction with the traffic situation is highest for driving and walking. Ratings for cycling and especially public transport are significantly lower. Satisfaction decreases noticeably with increasing mobility limitations and the need for support. This is particularly true for assessments of conditions for walking.

Given the central role that footpaths play for people with mobility impairments, this lower level of satisfaction is particularly concerning. Overall, the assessments indicate that infrastructure and living spaces meet the needs of this group even less than those of the general population.

Reduced everyday activity radii

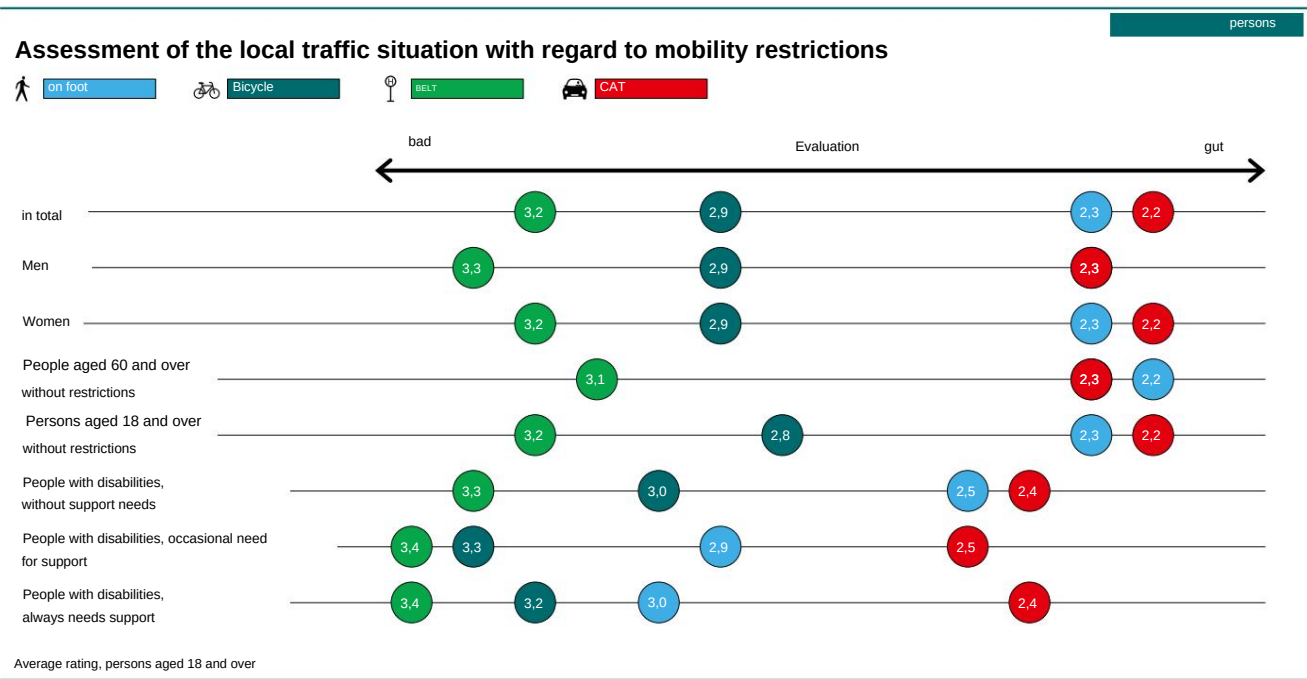
The previously observed lower average daily distance traveled by people with mobility impairments is evident across all situations. It is already significantly lower when the person can manage daily life without assistance from others.

can be out and about. Shopping and errands are an exception: People with mobility impairments who do not require assistance cover slightly longer distances on average than the general population. However, these distances are still significantly exceeded by people aged 60 and over without mobility impairments.

For all other occasions – such as business trips, journeys to work, accompanying people or leisure activities – the average distances traveled by people with reduced mobility are sometimes significantly below the level of the general population.

The shorter distances traveled for work-related reasons can primarily be explained by the lower proportion of employed individuals. However, there are no obvious age-related causes for the differences in trips undertaken for companionship and leisure activities. As soon as individuals require more frequent assistance when leaving their homes, the total daily distances traveled across all occasions decrease significantly once again.

Figure 117



Restriction and renunciation often go hand in hand.

The previous chapter has already shown that leisure activities, shopping and errands, as well as visits to or meetings with friends, are part of the weekly routine for the majority of the population aged 18 and over. Specifically, 87 percent go shopping or run an errand at least once a week, 73 percent participate in a leisure activity, and 53 percent meet acquaintances or friends.

For people with mobility impairments, these proportions are significantly lower – and they decrease further with increasing support needs. More than half of the people who always require assistance when leaving their homes do not undertake any of the three activities surveyed during a typical week.

Mobility restrictions are therefore accompanied by a noticeable reduction in everyday activities. These limitations can be attributed, at least in part, to external factors. For example, the proportions cited for all the reasons listed for occasionally foregoing activities outside one's own home are increasing: In the total population aged 18 and over, the proportion is around

a quarter, rises to a good third for people without support needs and reaches more than three quarters for people with permanent support needs.

As expected, a lack of accompaniment or Providing assistance with drop-off and pick-up for the most restricted group of people is the most frequent reason for having to forgo activities. 72 percent of these individuals report this, which pushes the influence of other obstacles into the background. For people with lower support needs in everyday life, the other factors become more important: difficulties in reaching desired destinations with available means of transport, financial hurdles, and a lack of regional services hinder their activities significantly more often than in the general population – even if they are fundamentally capable of traveling independently.

More user-friendly living spaces are needed.

Demographic trends are leading to an increasing number of older people and thus, presumably, to a higher proportion of people with permanent mobility impairments.

Table 52

Activity radii according to occasion and mobility limitations							
	Work	official	Out of- education	Shopping e	leddi- gung	Leisure time	Be- gliding
	km	km	km	km	km	km	km
IN TOTAL	8,2	8,1	0,5	2,3	4,0	14,0	1,6
Men	10,1	13,3	0,6	2,3	4,2	14,4	1,5
Women	6,3	3,2	0,5	2,3	3,8	13,6	1,7
People aged 60 and over without restrictions	3,7	4,3	0,0	2,9	4,8	13,8	1,3
Persons aged 18 and over without restrictions	9,0	9,0	0,6	2,3	4,1	15,0	1,7
People with disabilities, without support needs	4,2	3,4	0,1	2,5	4,1	9,5	1,1
People with disabilities, occasional need for support	1,4	0,6	0,1	1,7	3,2	6,5	0,8
People with disabilities always need support.	0,7	0,3	0,2	0,6	1,1	3,9	0,1
Persons aged 18 and over, average daily distance traveled per person according to occasion (main purpose of the trip).							

Table 53

Typical activities during the week following mobility restrictions

	at least weekly			no such Activities
	... Shopping activities	... Leisure time-activities	... visits/Meet	
	%	%	%	%
IN TOTAL	87	73	53	5
Men	84	73	50	6
Women	90	73	56	5
People aged 60 and over without restrictions	86	68	46	8
Persons aged 18 and over without restrictions	89	75	55	4
People with disabilities, without support needs	87	60	43	7
People with disabilities, occasional need for support	67	45	30	22
People with disabilities always need support.	34	21	17	56
Persons aged 18 and over				

MiD 2023 | Results Report | Source: MiD 2023

Table 54

Reasons for foregoing mobility after mobility restrictions

	Goals unattainable	can I Cannot afford the trip	depend on others	none/hardly any Offers for Activities	none of that
	%	%	%	%	%
IN TOTAL	13	8	7	7	76
Men	12	7	4	7	78
Women	14	8	9	7	74
People aged 60 and over	11	5	11	5	76
People aged 60 and over without restrictions	9	4	3	5	85
People without restrictions	12	7	4	7	79
People with disabilities, without support needs	20	15	12	9	62
People with disabilities, occasional need for support	28	16	52	9	32
People with disabilities always need support.	9	7	72	6	22

Multiple entries possible, persons aged 18 and over

MiD 2023 | Results Report | Source: MiD 2023

Support needs. The available data show that this group experiences limitations in its mobility and activity, and that these limitations increase with rising support needs.

Their opportunities for self-sufficiency and participation in social life are often limited. They frequently require support to compensate for health impairments, and their living environments are poorly adapted to their needs.

Only a relatively small proportion of this group of people regularly use walking aids such as walkers, wheelchairs, or mobility scooters. At the same time, less visible health limitations also significantly impair mobility. Improvements such as barrier-free access, easily accessible amenities, wide sidewalks, noise protection, greenery, and weather-protected seating could significantly increase the usability of the infrastructure.

A living environment oriented towards the needs of these individuals would also offer advantages to younger groups in their daily lives. Children, who have been excluded from previous considerations of mobility limitations due to their age, should be explicitly mentioned here. Given their age, most means of transport are also only available to them to a limited extent, which is why walking and cycling play a crucial role in their independent mobility. Living environments where important destinations can be reached safely and comfortably on foot or by bicycle enable children to engage in a high degree of independent activity.

Since working people – and thus decision-makers – rarely experience mobility limitations themselves, there is a risk that the needs of this growing population group will be insufficiently considered in the design of infrastructure and living spaces. It is therefore all the more important to integrate reliable data into planning processes and to design living spaces that are accessible to everyone. The present MiD results can contribute to this and increase awareness of these needs.

17. Home office and other online tasks: Drivers of changing mobility?

In recent years, a shift in work and consumption habits has become apparent, impacting daily mobility patterns. The increasing prevalence of working from home and the rise of online shopping are two key trends driving this change. While working from home affects how and when people commute to work or choose their place of residence and work, online shopping is transforming how goods are purchased and delivered, as well as other online activities. These developments have the potential to alter traditional traffic patterns and create new challenges and opportunities for spatial infrastructure and mobility. These changes necessitate an analysis and potentially an adaptation in the planning and management of transport networks to ensure efficient and sustainable mobility for the future.

with the option of working from home. The following sections present key findings on this group of people and their mobility on their commutes.

Between 2017 and 2023, the proportion of employed individuals with the option of working from home increased significantly across all groups. This growth is primarily attributable to two factors: Firstly, the COVID-19 pandemic acted as a catalyst, making working from home a short-term necessity in many sectors and thereby sustainably increasing its acceptance. Secondly, digitalization and technological developments have significantly improved the organizational and technical prerequisites for location-independent work.

Figure 118 shows how the possibility of working from home varies between different groups. The proportion of employed persons with the option of working from home increased from 13 percent in 2017 to 42 percent in 2023, highlighting the trend towards increasing flexibility in the working world.

From the street to the home office: Observations on mobility trends

The shift of professional activities into the private sphere has led to significant changes in mobility patterns in the professional world in recent years. Particularly due to the COVID-19 pandemic, working from home has become much more important in many professions. Even after the pandemic-related restrictions subsided, working from home has become an established component of hybrid work models in numerous companies and public administrations.

To better examine the societal changes associated with working from home and their related impact on mobility patterns, the 2023 MiD survey included a dedicated survey module on working from home and work-related mobility. The data allows for detailed analysis of which groups of people work from home, to what extent, how frequently this occurs, and the correlations with mobility patterns. Furthermore, the 2017 MiD survey already included initial questions on this topic. The "Working from Home and Work-Related Mobility" module surveyed approximately 210,000 employed individuals about working from home, including over 35,000 people in employment.

Definitions in the MiD regarding home office

For understanding the home office analysis, it is important to note that MiD deliberately chose a very broad definition compared to other studies on this topic: The surveyed practices include not only full-day or complete home office days, but also short periods or hybrid patterns, i.e., a mix of home office and workplace during the day. The usual amount of time spent working from home over the course of a normal week was also recorded.

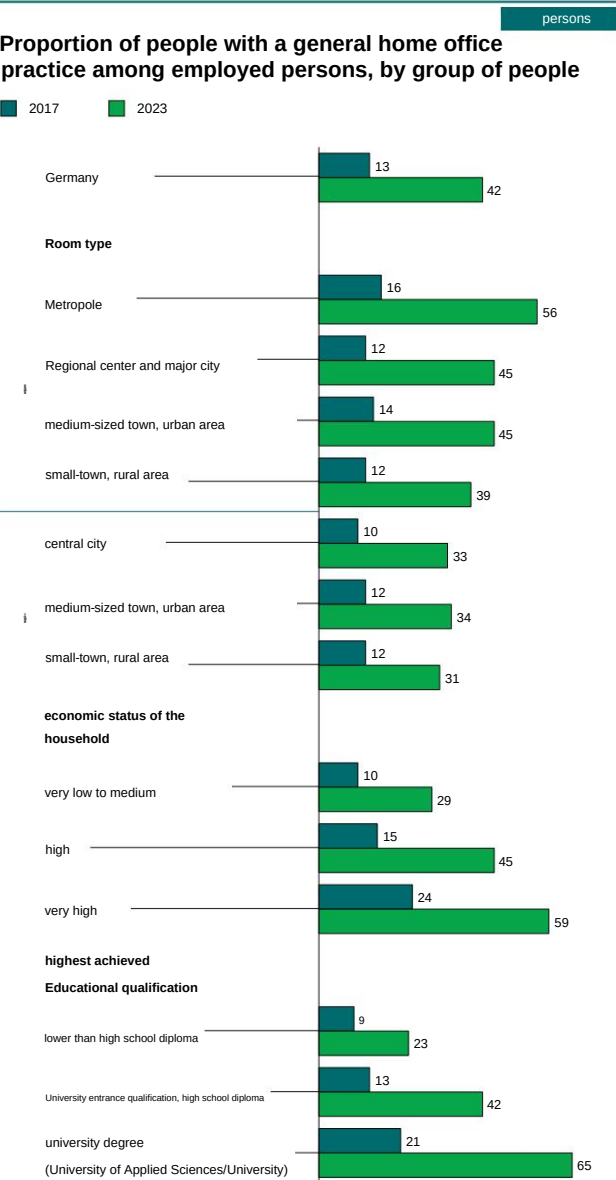
In addition to these general questions, a specific question was then asked, referring to the cut-off date, as to whether the option was exercised on that day.

In metropolitan areas, the overall percentage recorded is particularly high at 56 percent, as these large cities not only possess the necessary technical and organizational prerequisites for working from home, but also have a higher proportion of office jobs that are well-suited for working from home. This trend continues in other urban areas such as metropolitan regions, large cities, and

The trend continues in medium-sized cities, where around 40 percent of employees benefit from home office options. In contrast, only about a third of the workforce in smaller towns and villages has the opportunity to work from home. This, in turn, demonstrates that such jobs are less common in rural areas or less well-suited for working from home.

Interestingly, the breakdown by economic status reveals significant differences in the availability of working from home: While only 29 percent of people with a very low to middle economic status have the option of working from home, this percentage increases steadily with rising economic status. While just under half of the respondents with a high status have this option, it rises to nearly six out of ten for those with a very high status. This also shows in the MiD survey that higher-paid positions and the associated fields of activity are more likely to offer the technical and organizational conditions for working from home.

Figure 118



Figures in percent; employed persons aged 14 and over

A similar trend is reflected in educational attainment. People without a high school diploma rarely have the opportunity to work from home. For those with an academic degree, however, working from home is possible for around two out of three people. This suggests that higher educational qualifications often open access to professions that allow for flexible work arrangements. The higher the qualification, the greater the chance of working from home.

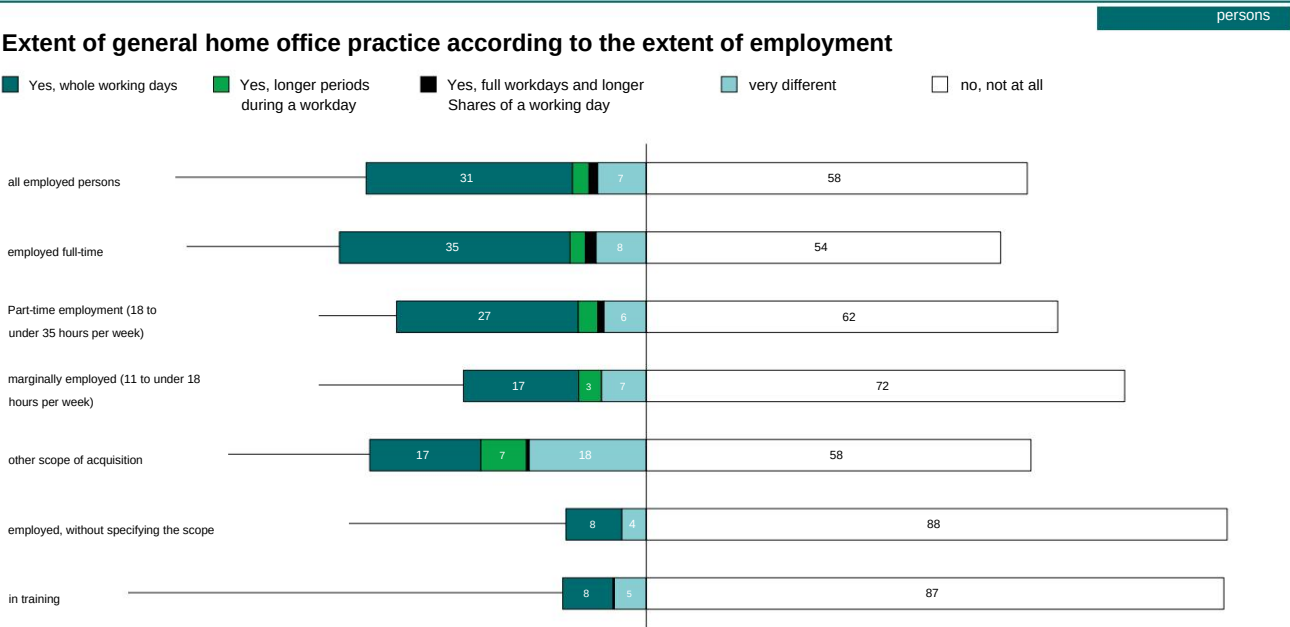
Working from home became widespread among the population during the survey period 2023/24 and remains prevalent, although it continues to be significantly more common among people in urban areas and in households with higher incomes and higher levels of education. This trend was already noticeable in 2017, but has intensified since then.

At the same time, working from home is widespread across all employment classes in 2023 (Figure 119). Of the approximately 45 million employed individuals in these groups, nearly 18 million work from home at least occasionally. This is particularly pronounced among full-time employees: more than a third regularly work entire days from home, with further proportions using home office part-time or flexibly. Nevertheless, the possibility of working from home remains completely unavailable to around half of full-time employees.

its.

Differences in home office usage are also evident between household types. Home office use is more frequent in family households and in households of young people under 35 than in...

Figure 119



Figures in percent; employed persons aged 14 and over
MiD 2023 | Results Report | Source: MiD 2023

Households consisting exclusively of adults or older persons (Figure 120). The proportion of people who do not work from home at all is highest in households with persons aged 65 and over. Furthermore, gender-specific differences are evident: In almost all household types, men more frequently report regularly spending entire workdays working from home. Women, on the other hand, are significantly more likely to be completely excluded from working from home. This pattern is particularly pronounced in family households and households of young people. This unequal distribution can possibly be attributed to gender-specific differences in career choices, the distribution of care work, and different working time models.

Working from home and commuting: a close connection

Table 55 shows the average distance to the workplace reported by respondents in the Home Office module. The analysis reveals a strong correlation between the possibility of working from home and the distance to the workplace. Employees who do not generally work from home have an average distance of 15 km to their workplace, while for those who do, it is [distance missing].

Table 55

Distance to the workplace after home office option and frequency of home office days

EMPLOYED *	share below employed	Distance nung to work-site
	%	km
no working from home	58	15
general use of home office, typical home	42	22
office days		
less than one day per week, one to less than	1	18
two days	7	20
two to under three days	8	24
three or more days,	13	26
very different *	13	20
employed persons aged 14 and over		

MiD 2023 | Results Report | Source: MiD 2023

For employed individuals who at least occasionally work from home, the average commute is 22 km. The results show that the at least occasional use of the option to work from home is also influenced by the geographical proximity of home and work.

Longer commutes are likely to be more readily accepted if working from home reduces the number of journeys. This suggests either self-selection or targeted offers from employers who encourage employees to live further away from their workplace or accept a position further away due to the home office conditions.

This means that working from home not only affects daily commuting habits, but can also lead to structural changes in the choice of residence and workplace.

As the number of typical home office days per week increases, so does the average distance to the workplace. While employees who work from home less than one day per week live an average of 18 kilometers from their workplace, the distance rises to approximately 26 kilometers for those working from home three or more days.

One added value of the MiD 2023 lies in the more differentiated data collection compared to previous surveys. In addition to the general home office-related house-

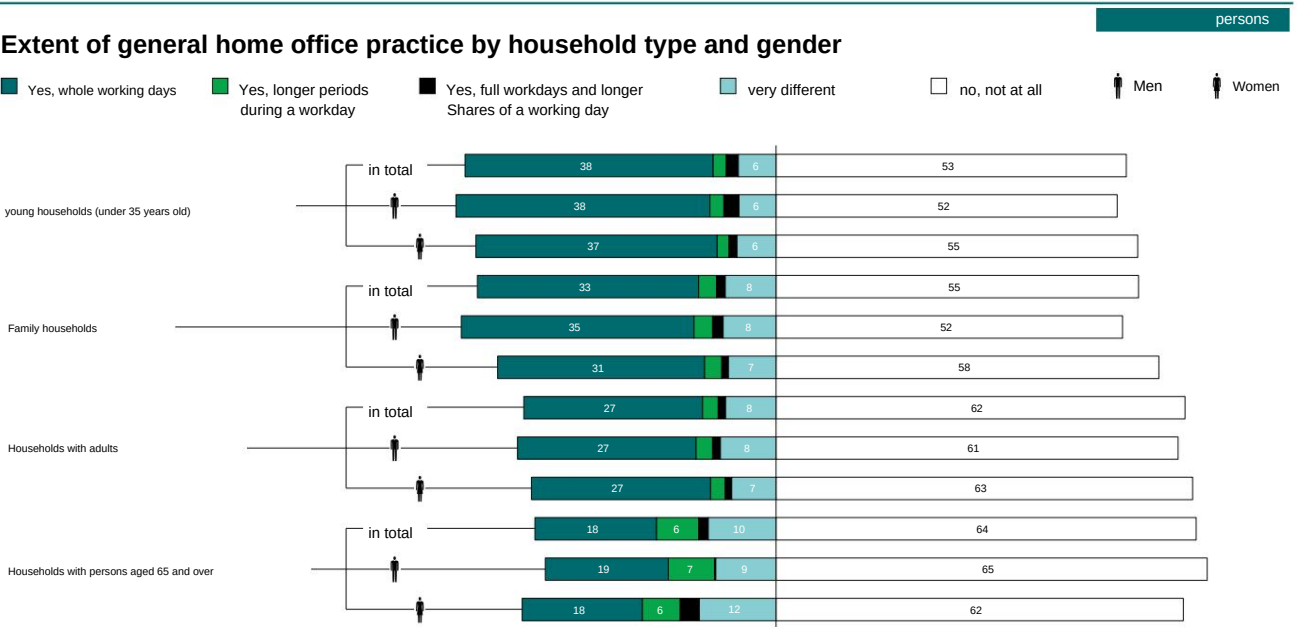
The survey also inquired about the usual intensity of home office use, i.e., whether entire workdays or only longer periods are spent at home. Figure 121 illustrates the correlations: Employees with higher frequencies of working from home are significantly more likely to work from home for full workdays and have longer commutes on average. In contrast, shorter distances are common with lower home office use or when working half-days. Particularly long commutes lead to flexible and irregular home office patterns.

Overall, this suggests that the geographical distance to the workplace is a key driver for the frequency and nature of working from home. Accordingly, working from home is primarily used to avoid particularly long commutes in full-time employment. Those who only work from home part-time tend to live closer.

Longer daily commutes with the option to work from home
Reporting day

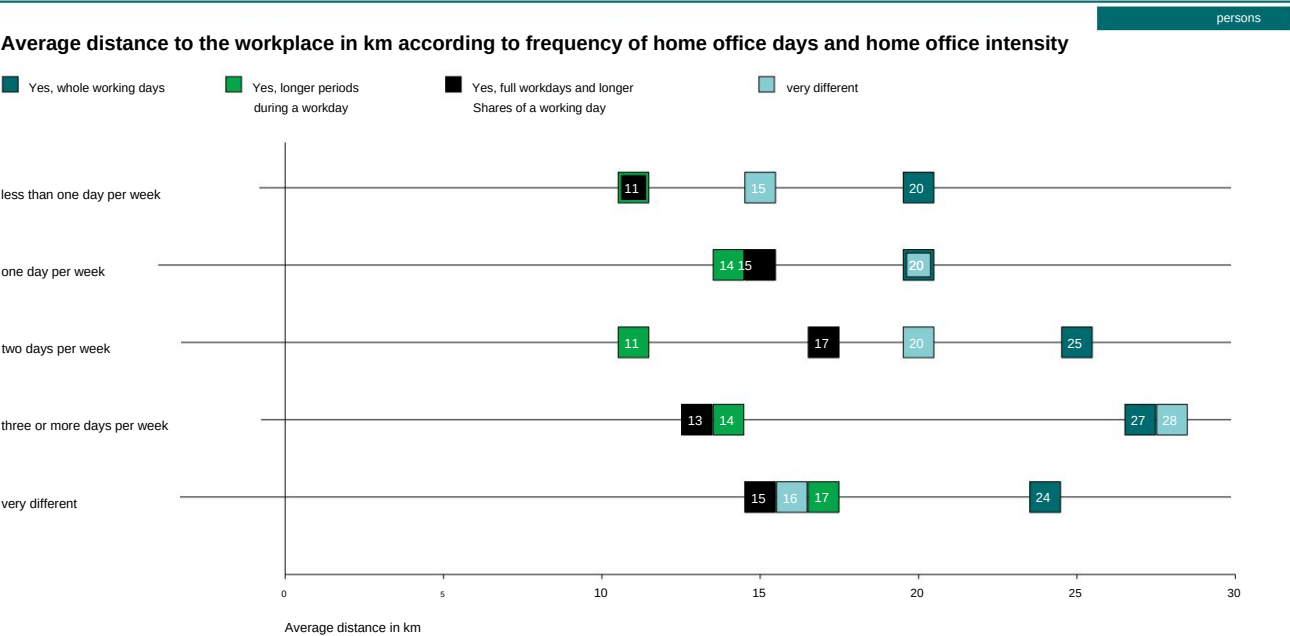
Figure 122 shows the average daily distance traveled by employees in the home office module, broken down by whether the person generally reports working from home and whether the person actually worked from home on the reporting day.

Figure 120



Figures in percent; employed persons aged 14 and over

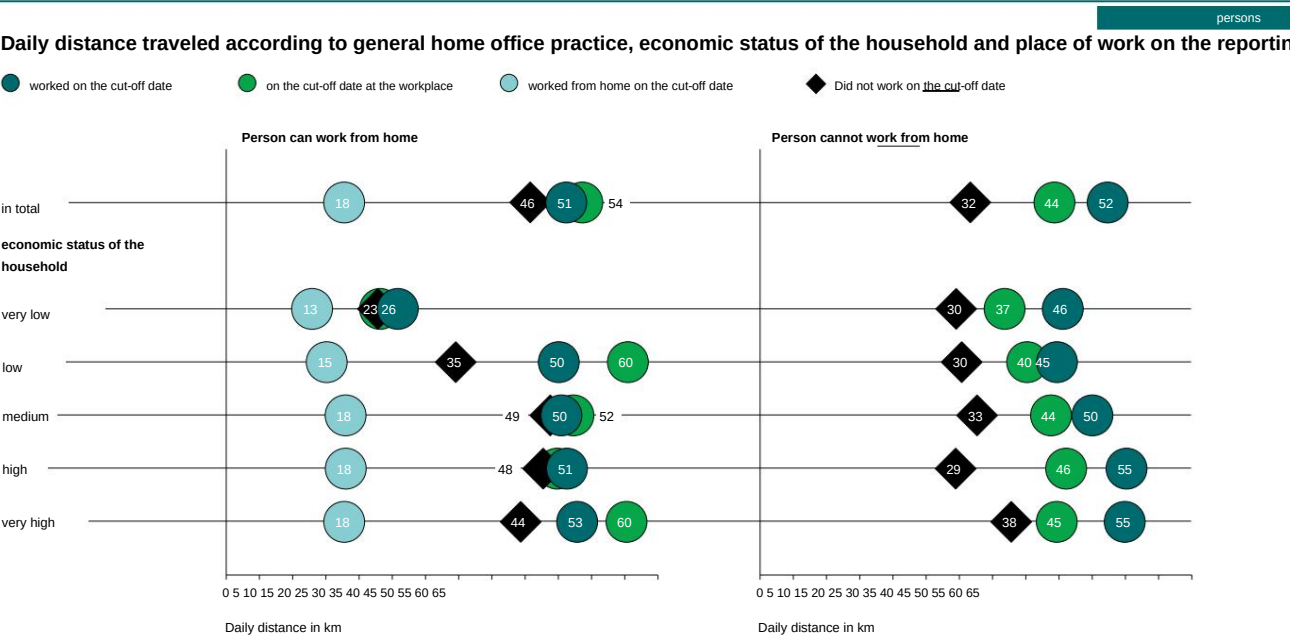
Figure 121



Figures in kilometers; employed persons aged 14 and over

MiD 2023 | Results Report | Source: MiD 2023

Figure 122



Data in kilometers; employed persons aged 14 and over; weekdays only (Mon-Fri); excluding public holidays

MiD 2023 | Results Report | Source: MiD 2023

The workplace was [location omitted]. Furthermore, the economic status of the household was considered in the evaluation.

Employees with the option to work from home travel an average of 50 kilometers per day, which is three kilometers more than those employed without this practice.

An analysis of the work location on the reporting day shows that employees with the option to work from home travel an average of 18 km on days they work from home, but significantly longer distances, averaging 54 km, on days when they commute to the workplace. In comparison, employees without the option to work from home travel an average of 44 km on days when they commute to the workplace. On days when they do not work, employees with the option to work from home travel even further.

When the economic status of the household is also considered, it becomes apparent that the daily commute increases with rising economic status for both employees with and without the option of working from home.

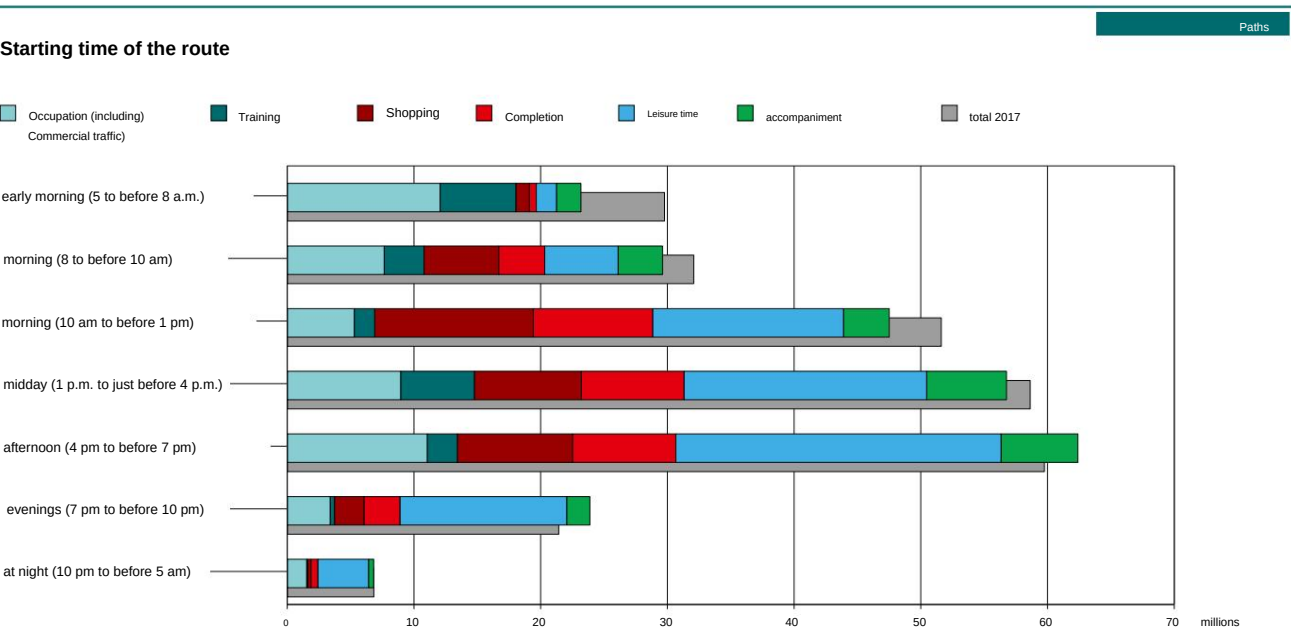
The daily traffic pattern has shifted somewhat towards the afternoon and evening.

The temporal distribution of traffic can be considered based on the start and arrival times of the reported routes collected in the MiD.

Figure 123 distributes the daily number of trips (excluding regular commercial traffic) across the different times of day and simultaneously differentiates them according to the presented trip purposes. For orientation, it also shows the total value for 2017 for each "time column" along the route. The familiar morning peak is still evident for commuter traffic. However, the highest overall traffic volumes occur in the afternoon and early evening, when work-related (return) trips coincide with other errands undertaken at that time.

This trend intensified further in 2023. Due in no small part to working from home, the morning and mid-morning figures decreased, while the afternoon and evening figures each increased slightly – despite the overall figure decreasing slightly from 257 to 250 million journeys per day.

Figure 123



Figures in millions of kilometers; (excluding regular work-related journeys, as they were collected without time data)

Work-related (non-)mobility on the reporting date: approximate savings

Table 56 categorizes employed individuals on the reporting date according to their commute and whether they worked from home that day. The analysis allows for a rough estimate of how many commutes and kilometers could be saved in the listed sectors through working from home. Of the employed individuals on the reporting date, 15 percent worked from home. The majority of employed individuals remain mobile. Only 2 percent were not mobile on the reporting date. Employees working from home have an average commute of 23 km. This again illustrates that working from home is primarily used where longer commutes exist. If the approximately 15 percent of those working from home on the reporting date had not been working from home, a conservative estimate suggests that around 10 percent of commutes and 15 percent of kilometers could be saved in this sector, as a complete elimination of commuting cannot be assumed. This equates to up to 8 million roads and 170 million passenger kilometers per day.

The proportion of business trips is higher among lower status groups, indicating activities in the transport and delivery sector.

More active ways to get around on home office days

In addition to distances and daily values, the choice of transport mode on the relevant days is significant. Figure 125 shows all journeys made by the home office subsample of employed individuals with a clearly defined workplace. The modal split differentiates the distribution of transport modes for all journeys made by individuals working from home on the reporting day compared to those who traveled to their workplace on the reporting day. On home office days, respondents use active transport modes such as bicycles and walking significantly more often than on traditional office days, although motorized transport still accounts for more than half of all journeys. This is likely due to greater temporal flexibility and the often shorter distances involved in everyday life.

Shorter shopping and errand trips, but longer leisure trips on home office days.

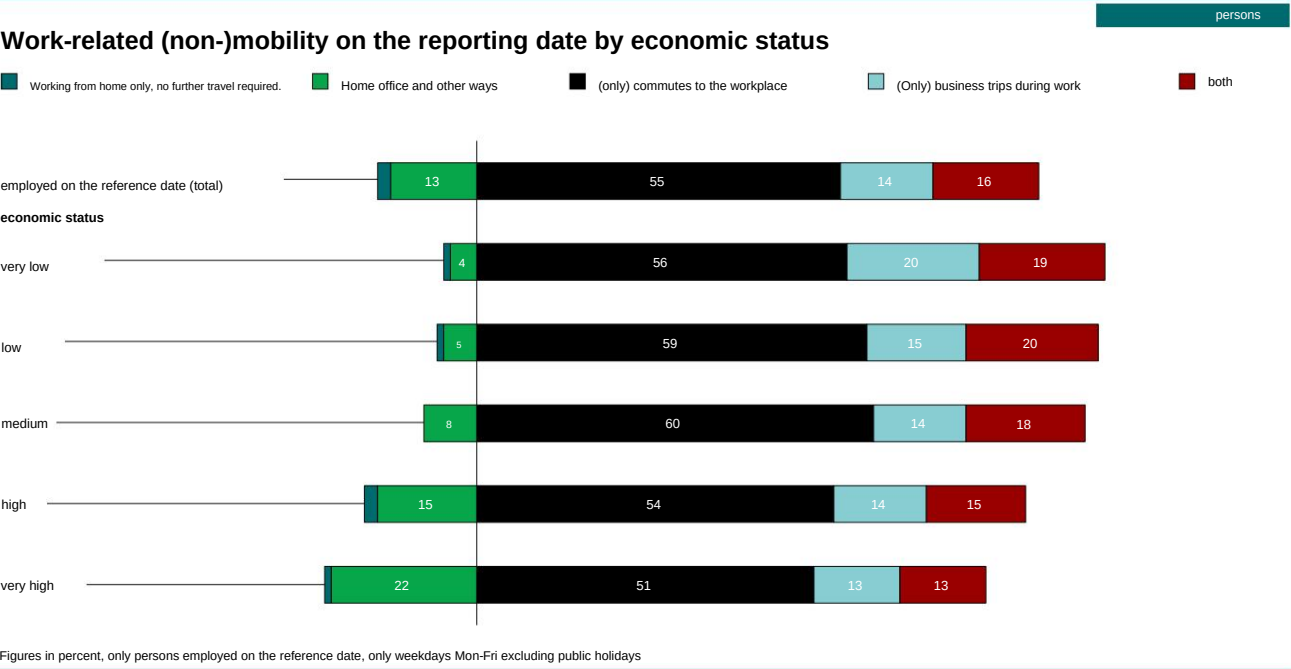
If the groups of employed persons are divided according to economic status on the reporting day (Figure 124), higher proportions of people working from home are evident with increasing rates of working from home is changing the structure of the

Table 56

Key figures and conservative estimate of the home office effect					
	Portion	Values for the reporting day (all activities) per person			specified Distance to the usual workplace
	%	Number of routes	km	Min.	km
Working from home only, without any travel on the deadline date	2	0	0	0	23
Home office and other routes (only)	13	2,8	19	63	
Commuting to the workplace (only) Business trips	55	3,4	40	87	16
during work	14	5,2	90	137	17
both	16	9,0	98	195	16
EMPLOYED PEOPLE	100	4,4	54	107	17
TOTAL AS OF THE CERTIFICATE DATE					

* Calculation capped at max. 150 km
Only persons employed on the cut-off date, only weekdays Mon-Fri excluding public holidays.

Figure 124



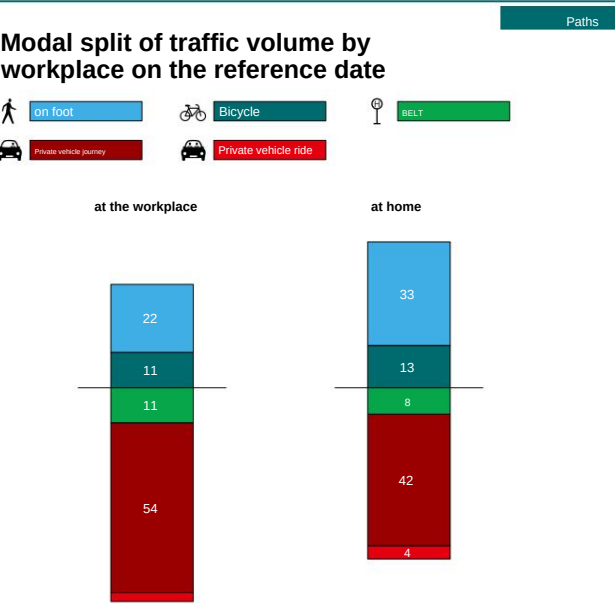
MiD 2023 | Results Report | Source: MiD 2023

Routes have changed (Figure 126). Shopping and errand trips are shorter on home office days than for employees at the workplace. This suggests that people working from home are more likely to use services closer to home. Leisure trips are, on average, longer when working from home than when in the office. This indicates that the time saved by eliminating commutes is used for leisure activities further afield. While work-related and training-related trips show a decrease in average trip lengths on home office days, trips related to work are longer in the office context, presumably because they are more frequently combined with work-related travel. An analysis of modes of transport shows that trips using active transport (walking and cycling) remain stable in length but occur significantly more often on home office days.

Limitations of MiD analysis

Unlike these point-in-time results, the MiD survey concept does not allow for the analysis of individual patterns over time, such as a week or a month. However, working from home can significantly alter everyday routines. For example, non-work-related travel may shift to days when working from home – perhaps resulting in a chain of trips that ultimately lead to work.

Figure 125



Figures in percent; employed persons aged 14 and over; weekdays only (Mon-Fri); excluding public holidays

MiD 2023 | Results Report | Source: MiD 2023

The reverse is also conceivable – and is indicated in the results of the reference date: The time saved on home office days due to unnecessary commutes can create options for other activities and associated journeys.

Such rebound effects can combine or even mutually limit each other. Therefore, the impact of working from home doesn't necessarily have to be reflected in reduced commuting and fewer kilometers traveled. Other mechanisms are also conceivable.

However, they can only be reliably detected with a longitudinal study design rather than the statistically oriented MiD cross-sectional study.

Another mobility changer?
Online shopping and mobility

Another digitally driven phenomenon that, alongside the increasing prevalence of home office work models, shapes current mobility is online shopping. The use of digital shopping platforms has increased dramatically in recent years, changing both consumer behavior and urban logistics. This growing reliance on the internet for purchasing goods and services is accompanied by a decline in the physical presence of brick-and-mortar retail stores.

This coincides with increased strain on supply chains and logistics processes. As a result, traffic flows are shifting: delivery traffic is increasing, and the demands on urban infrastructure are changing. Online shopping is therefore part of a broader digitalization process that is having a lasting impact on mobility patterns.

Therefore, MiD 2017 already addressed this topic with several questions and continues to do so in the current issue.

More and more people are shopping online.

The analyses illustrated in Figure 127 show that the use of online shopping increased significantly between 2017 and 2023. What was once a fringe phenomenon is now firmly established in the everyday lives of many people. Today, around 3 percent of those aged 14 and over order online almost daily. In 2017, it was barely more than one percent.

63 percent shop online at least once a month. In comparison, this figure was 42 percent in 2017. The increase is particularly strong among intensive online shoppers. In 2023, 20 percent of those aged 14 and over made an online purchase one or more times a week. In 2017, this figure was only 9 percent.

At the same time, the proportion of people who never or almost never shop online has decreased, and also

Figure 126



The proportion without internet access has decreased. Overall, there is a clear shift towards broader and more frequent use of online shopping.

Young adults and people with high economic status regularly use online shopping.

Figure 128 shows a more detailed analysis of the frequency of online shopping. The results illustrate that online shopping is now widespread across all population groups, but the intensity of use varies considerably.

While regional differences based on area type are only slight, meaning that usage is not higher in rural regions with poorer infrastructure, significant differences emerge based on age and economic status. Young adults and people with high economic status, in particular, shop online regularly, sometimes weekly or more often, while older and lower-income groups shop online considerably less frequently and often not at all – partly due to a lack of internet access. Overall, it is clear that while online shopping has become widespread, it remains unevenly distributed socially and demographically.

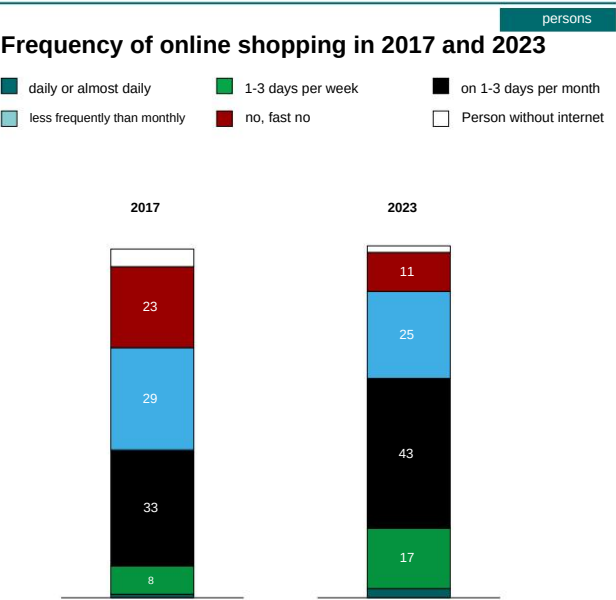
Frequent online shoppers are more mobile.

The question remains as to the impact on the recorded mobility. To gain an impression of this, Figures 129 and 130 are shown.

Daily distances traveled were analyzed according to age, purpose of travel, and frequency of online shopping. The results show that in almost all age groups, individuals who shop online at least once a month travel greater distances than those who shop online less frequently than once a month or almost never. The difference is most pronounced in middle-aged groups, while it is less significant for adolescents and older adults. This suggests that online shopping does not necessarily replace travel, but rather is used during periods of high mobility alongside longer daily commutes. It is therefore less a substitution and more an expression of "active lifestyle" that encompasses both online activities and physical mobility. Instead of replacing one with the other, potential time savings may be used for new activity patterns.

At the same time, it must be taken into account that a disproportionately large number of infrequent users are older people. These individuals exhibit different mobility behavior than younger people: they travel fewer journeys overall, and these journeys are also shorter. It should be noted that the comparison of the mobility-

Figure 127



Figures in percent

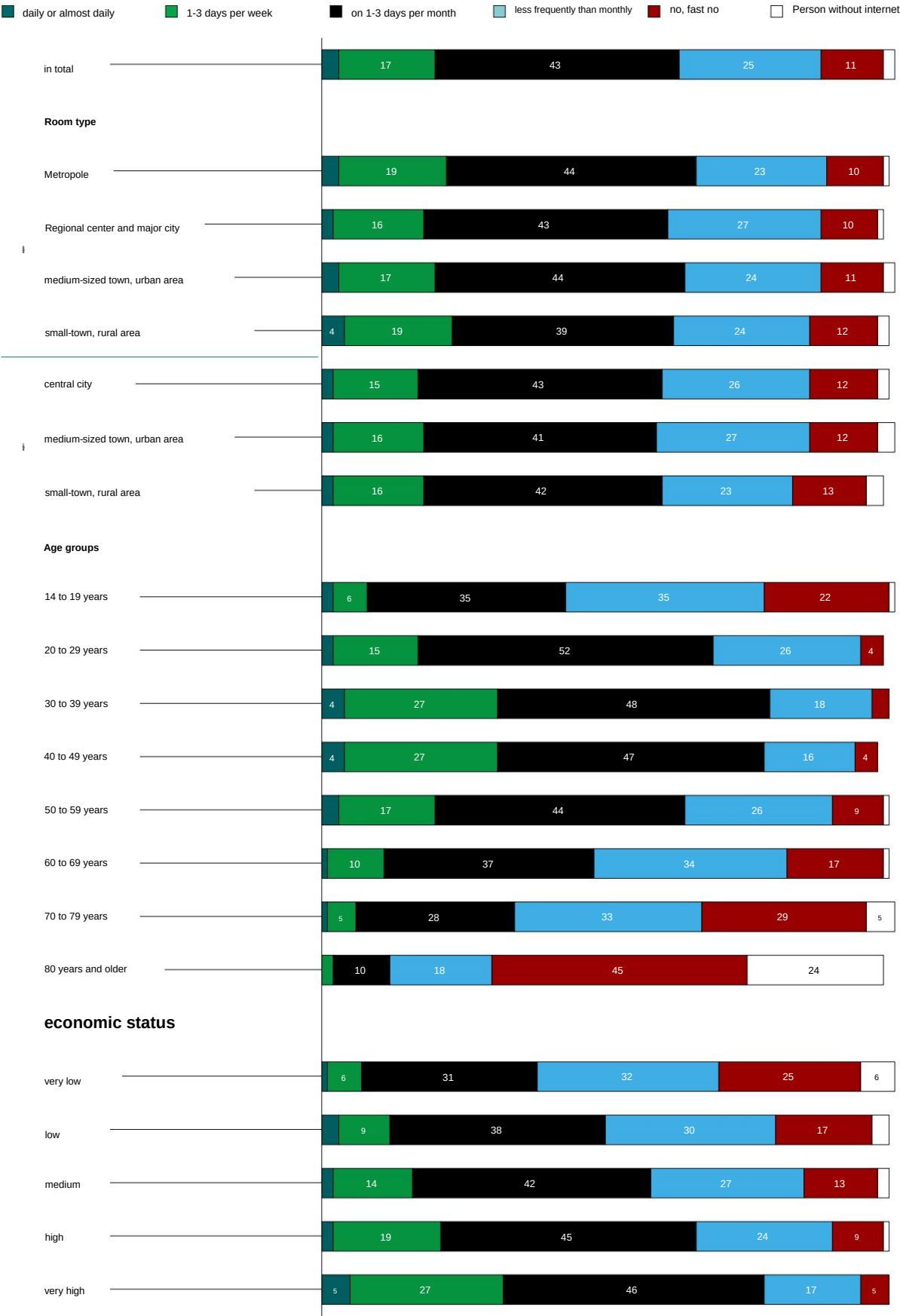
The behavior of frequent and infrequent online shoppers is thus strongly influenced by age-specific differences in mobility, similar to differences in economic status.

The analysis of daily distances traveled by purpose shows that the purpose of "shopping" accounts for only a small share of daily travel in both the 2017 and 2023 MiD (Midday Travel Survey). Across all age groups, the values in the 2023 MiD are mostly between 1 and 3 kilometers per day. No systematic differences in the proportion of trips made for shopping are apparent between frequent and infrequent online shoppers. Even when comparing 2017 and 2023, the decline in shopping trips is only slight. This suggests that while online shopping potentially replaces shopping trips, this is not clearly reflected in the average daily distance traveled. The purpose of "shopping" thus remains constant, but overall,

Figure 128

Frequency of online shopping by room type, age and economic status

persons



Figures in percent, persons aged 14 and over

Figure 129

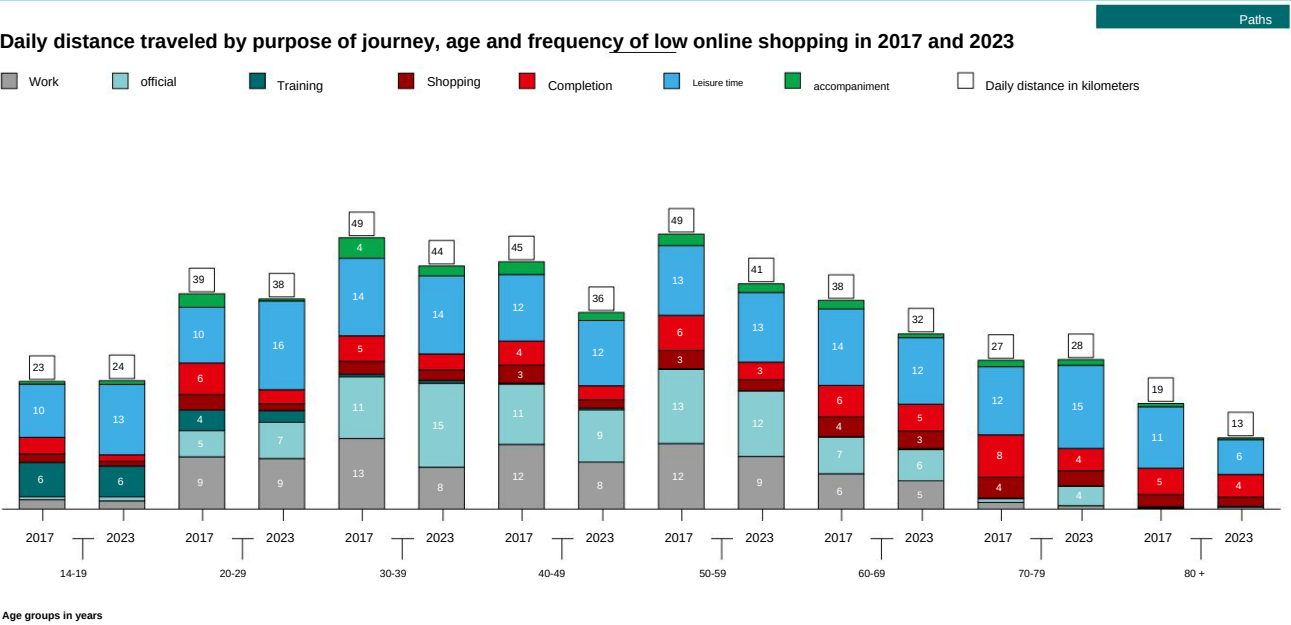
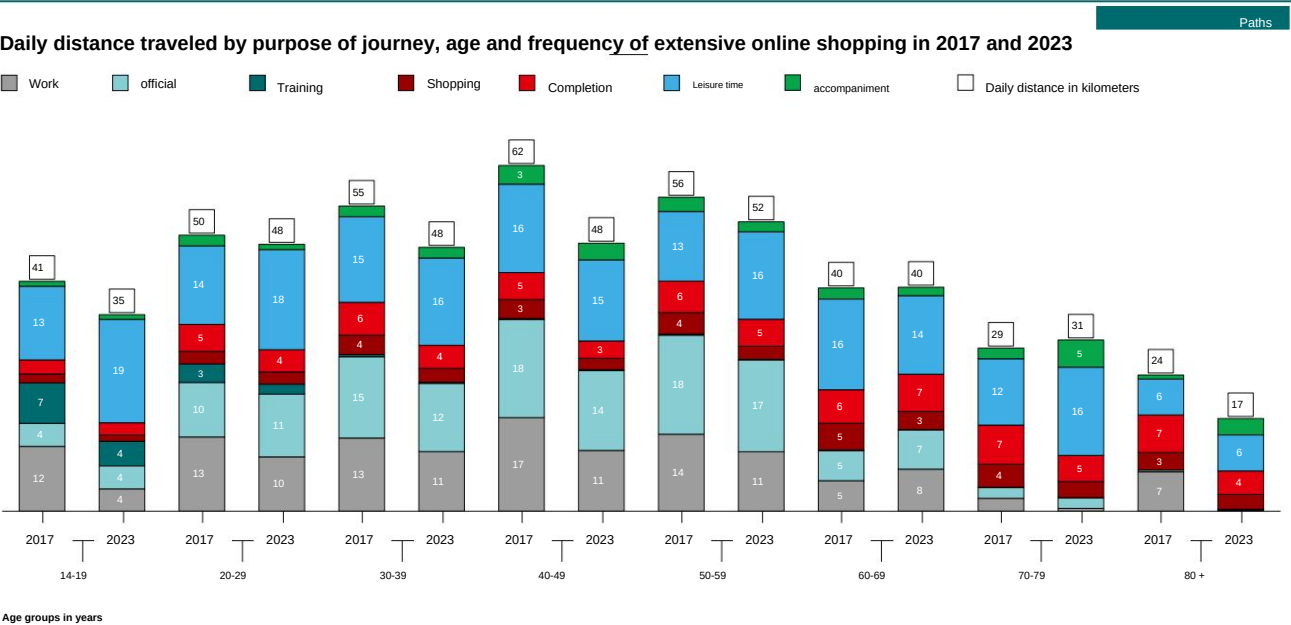


Figure 130



subordinate factor in daily mobility behavior.

Not just shopping

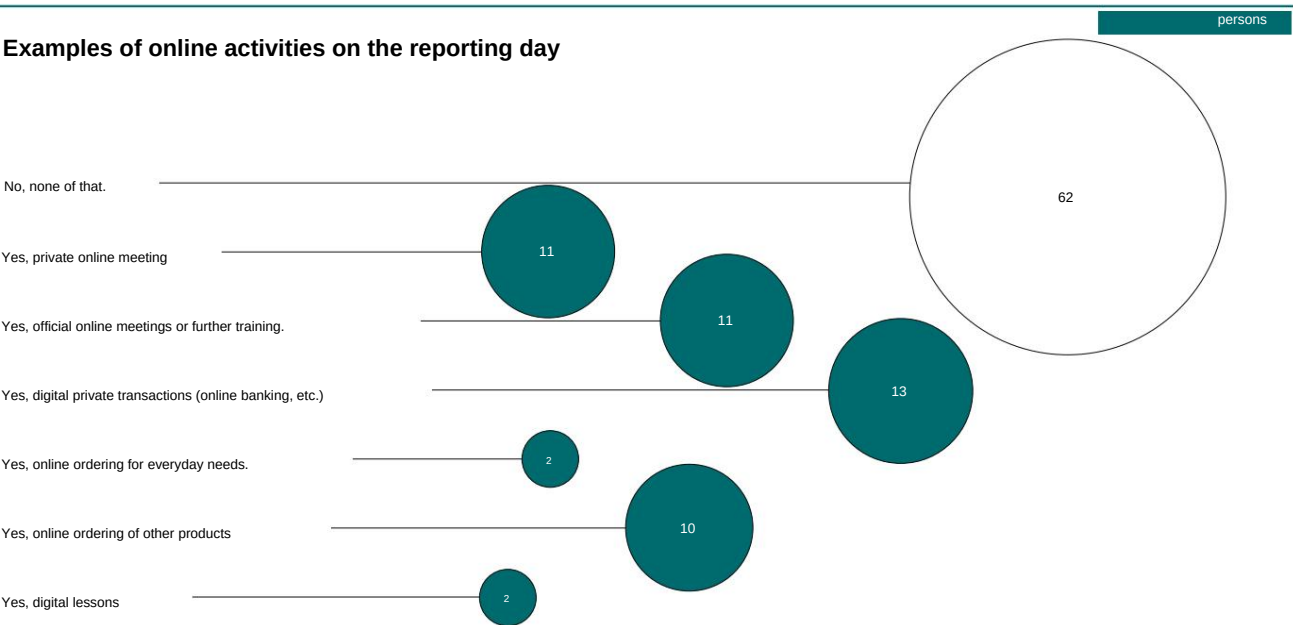
However, online shopping is much more than just shopping. Many activities can now be completed online without physically leaving the home. The MiD 2023 survey also addresses this issue, collecting data on selected online activities on the reference date (Figure 131). The results show that well over a third of the population aged 14 and over already reported engaging in at least one of the activities included. The focus is less on shopping, as previously discussed, and more on other errands. These include primarily professional and private social contacts – and errands beyond the typical shopping trip. However, the correlation is not straightforward: individuals who report online activities on the reporting date do not necessarily experience less travel time or fewer kilometers covered. As with shopping, this depends more on sociodemographic factors and different life stages.

A brief interim report on "digital mobility"

The analyses presented in this chapter have shown that the usage patterns of digital options are changing – and with them, actual spatial mobility. While working from home has indeed led to some relief, new options are also emerging, and new routines and opportunities for other reasons for mobility may be developing. Furthermore, the causal relationships are not straightforward, and interaction effects cannot be ruled out.

The MiD snapshot illuminates some of these effects, but certainly not all, not least due to the survey design being only partially suitable and its focus on the statistical aggregate of the reference date instead of a longitudinal study over a longer individual period. However, even the inventory of the current structure is helpful and contributes to a better understanding of emerging or consolidating trends. This also applies to the online purchases considered and other online activities. But here, too, the MiD results show that a direct conclusion about a reduction or relief of everyday mobility is not permissible. In some cases, the opposite effect is even more likely. This needs to be observed further and more closely in the future.

Figure 131



Data in kilometers; persons aged 14 and over, multiple selections allowed.

18. Not just everyday mobility: on trips, excursions and long commutes in and outside of Germany.

Mobility encompasses not only daily journeys in the familiar surroundings of one's home, but also travel, which is divided into day trips and overnight trips. Day trips start and end on the same day at one's place of residence and do not require an overnight stay, while overnight trips include at least one night away from home and thus trigger additional journeys and activities at the destination. Especially during vacation and holiday periods, overnight trips can account for a significant portion of regional traffic volume.

To ensure a sound basis for evaluating travel data, overnight trips have been recorded since the 2002 MiD (Migration in Germany) survey, specifically for the three months prior to the interview – and thus not just on the reporting date. However, until 2017, this procedure was not followed for day trips (i.e., trips without overnight stays). The 2023 MiD survey was expanded to include this travel category. A random subsample is asked about day trips taken in the last four weeks.

This provides travel data records for both overnight and day trips. Based on this data, travel figures can be extrapolated and the reported trips can be described in terms of their structure.

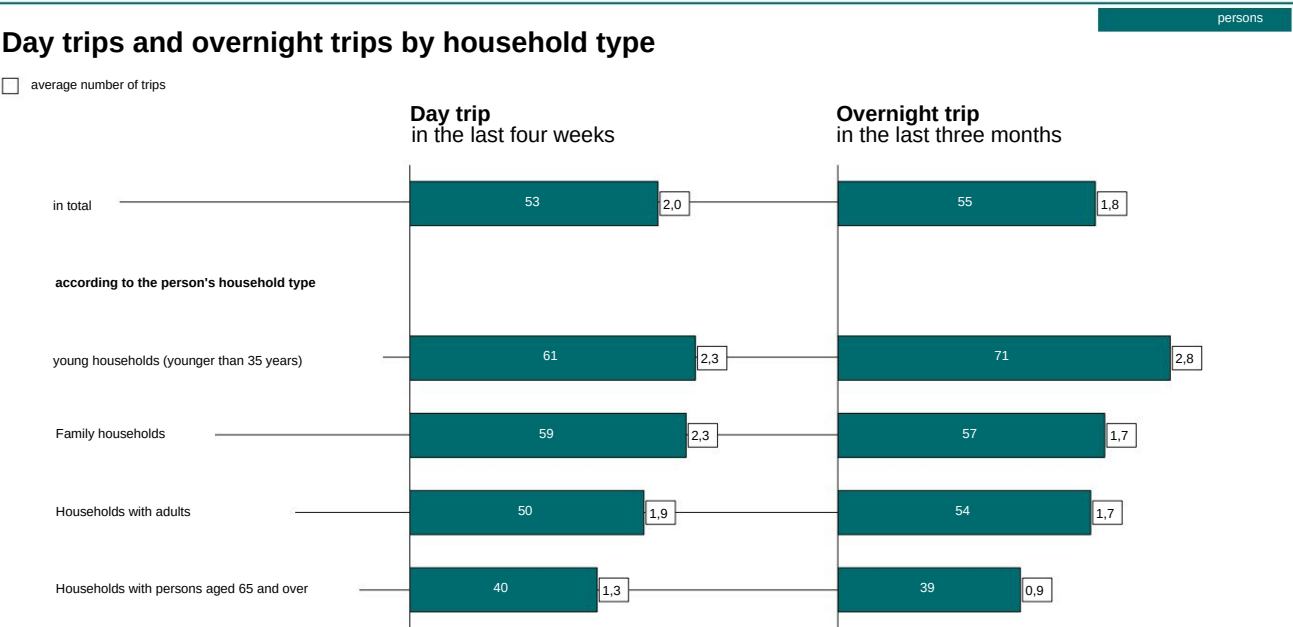
The MiD 2023 also includes expanded data on Day trips are available – and overnight trips as before.

The MiD (Mobility in Germany) surveys everyday mobility on a randomly selected reporting day. This provides a sufficient and reliable data basis for typical routes. Travel activities that took place on this day within Germany or on the way to or from abroad are also recorded. However, they represent only a small number of cases within the daily reports and, in the case of longer journeys, cannot always be reliably identified as such in the daily log.

Results on travel frequencies broken down by Day trips and overnight trips

The results show that, on average, the German resident population aged 14 and over takes two day trips per person per month. Extrapolated to a twelve-month period, this equates to approximately 1.7 billion day trips, or 24 such trips per person. For trips involving overnight stays, the figure is 1.8 trips per person, based on the three-month period used as the basis for the survey.

Figure 132

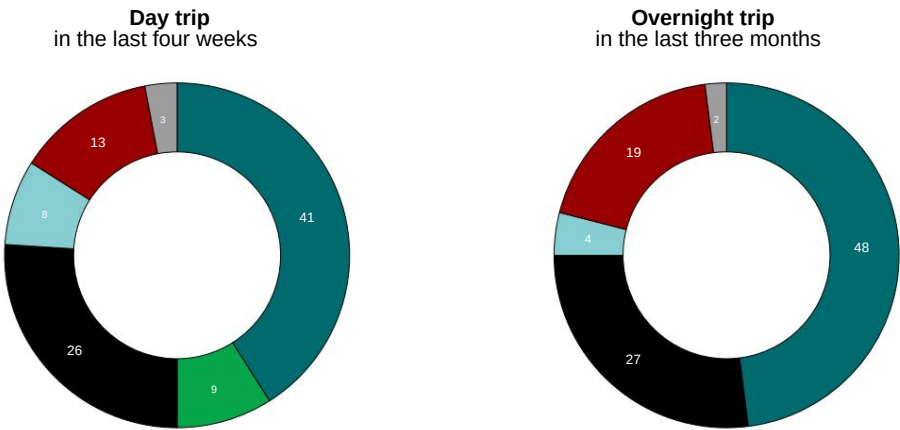


Figures in percent

Figure 133

Purpose of day trips and overnight trips

Day trip, vacation, short break, etc. Shopping/Supplies Visits other private trip business trip other



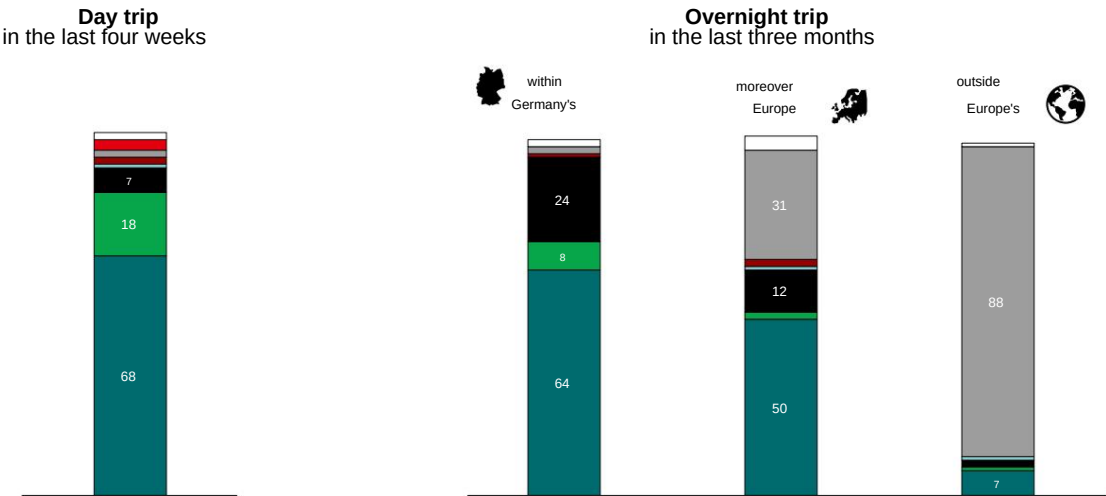
Figures in percent, travel

MiD 2023 | Results Report | Source: MiD 2023

Figure 134

Main means of transport for day trips and overnight trips

Car/Motorhome Bus/train in local transport Long-distance rail travel Long-distance bus in scheduled service coach Airplane Bicycle miscellaneous



Figures in percent, travel

MiD 2023 | Results Report | Source: MiD 2023

Extrapolating to a year, this equates to approximately 0.5 billion trips with at least one overnight stay – or roughly seven trips per person per year in the population aged 14 and over. Understanding these figures hinges on the MiD definition of a day trip. This definition is not tied to a distance threshold but is based on relevant surveys from Eurostat. It therefore encompasses all activities outside of routine routines. This includes excursions, visits, and short business trips.

These travel frequencies can be differentiated using the available data, both structurally and across various population groups. This will be done here using a simple household typology based on life stages and household composition. Regarding day trips, an average of 53 percent of citizens aged 14 and over report taking a day trip each month. This percentage is slightly higher for younger households, with or without children and families, than for older adult households.

Fifty-five percent of respondents reported overnight trips in the last three months. Here too, younger households and families proved to be more prone to such trips.

more eager to travel. However, if the economic household status is used as a characteristic, the lower groups show lower proportions of travelers by more than ten percentage points.

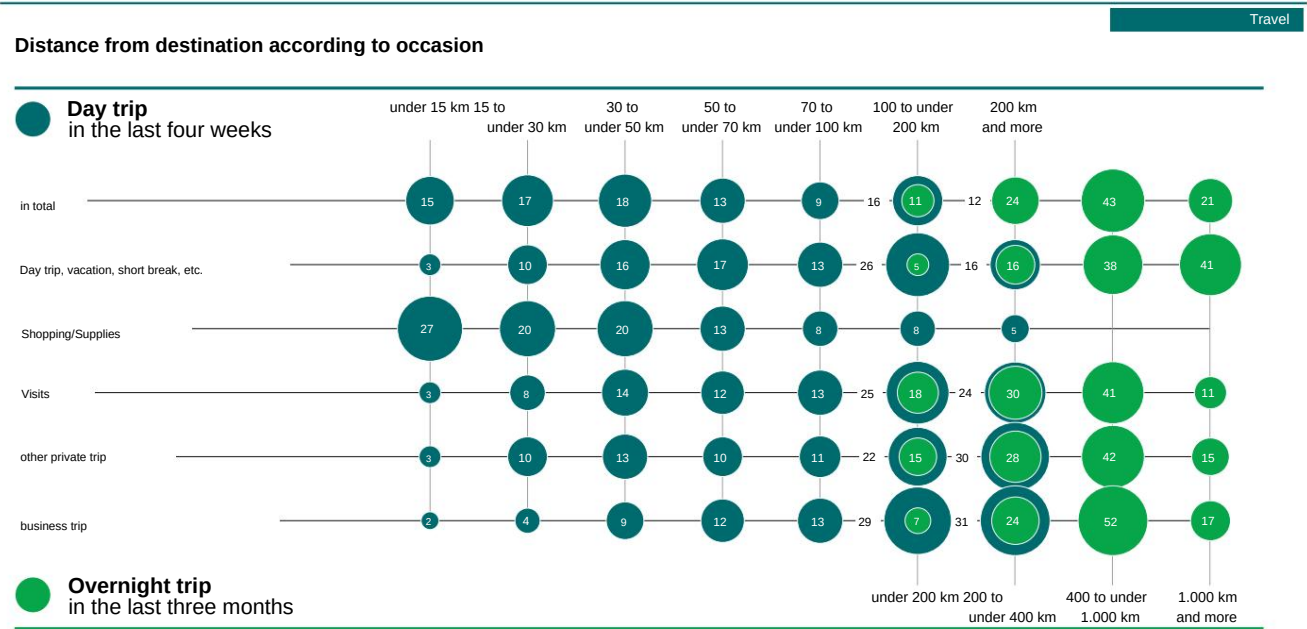
Numerous parameters are recorded for the trips, such as distance, reasons for travel, destinations, duration for multi-day trips, and the means of transport used for arrival or departure.

Private occasions are the most common reason for day trips. Day trips account for the largest share of these journeys at 41 percent, followed by visits to family or friends at 26 percent. Business or official purposes make up only 13 percent of day trips. The proportion of reasons for these trips is not significantly different for overnight trips. Here, vacations account for half of the trips, and business trips for about one-fifth.

Travel distances: Kilometers don't scare anyone.

The distances can be considered separately for both day trips and overnight trips, depending on the purpose. Day trips abroad are also undertaken without an overnight stay: 8 percent of all day trips lead beyond the German border, and for these trips, at least 30 percent of people also take on additional costs for private and business travel.

Figure 135



Figures in percent, line percentages, out of 100%. Missing: no information.

Day trips can cover distances exceeding 200 kilometers (Figure 135). However, the distances of day trips cannot be easily categorized into predefined groups; trips of less than 15 kilometers are undertaken with equal frequency, as are trips of over 200 kilometers. The median distance for all day trips undertaken is 50 kilometers.

Naturally, the distances for overnight trips vary. One in four trips is to a destination 200 to 400 kilometers away, and almost half are to a destination up to 1,000 kilometers away. The median distance for all overnight trips is 350 kilometers, significantly less than the 1,000 kilometers and more that are possible here. Only for vacation trips are far-flung destinations chosen, with one in four vacation trips leading to a destination at least 2,000 kilometers away. These destinations likely include worldwide and European locations such as Turkey, Spain, Portugal, the Canary Islands, and the Balearic Islands. To make the vacation worthwhile, 61 percent of trips typically last one to two weeks.

Duration of trips with overnight stays: one third with four overnight stays or more

But a surprisingly large amount of time is also spent on business trips. 30 percent of these trips are week-long with 4 to 5 overnight stays. Trips to visit family, relatives, friends, or girlfriends, as well as...

In contrast, visits with acquaintances tend to be short-term and primarily take the form of weekend visits, usually with two overnight stays (27 percent).

Provided the distance is correct, it also dominates in Travel by car

The differences are more pronounced in the chosen means of transport, which is mainly due to the greater distances involved in overnight trips. Day trips are predominantly within the country.

This means that cars, or private motorized transport, dominate, accounting for 68 percent of journeys. Public transport accounts for 18 percent, and long-distance trains for 7 percent. Cycling, with a 3 percent share of day trips, is also a significant mode of transport.

For overnight trips, it's advisable to differentiate based on a rough classification of destinations. Within Germany, the car is the preferred mode of transport for about two-thirds of trips, and for European trips, it's still the choice for every second trip. Public transport accounts for about one-third of trips within Germany. Within Europe, it's only chosen in about one out of ten cases.

Air travel, on the other hand, plays hardly any role within Germany. Within Europe, it is already chosen for every third trip, and for destinations outside Europe, it accounts for almost 90 percent.

Table 57

Duration of overnight trips by travel purpose							
	Overnight stays						
	one	two	three	4 to 5	6 to 7	8 to 14	15 and more
	%	%	%	%	%	%	%
IN TOTAL	21	25	14	15	10	10	4
Day trip, vacation, short break, etc.	2	9	11	17	21	29	11
Visits	11	27	18	19	11	10	4
other private trip	14	26	13	17	7	14	10
business trip	13	23	18	30	5	7	4
Travel, 100% absences: no information							

Not least, it's also part of travel: long-distance commuting.

Long commutes to work have been part of everyday life for some employees long before the advent of Corona and home office. A special case is so-called "long-distance commuting"—not a true private trip, not a genuine business trip, but often over long distances and with considerable time expenditure. Such journeys are also included in the MiD (Mission Involvement) data collection on the reporting date, but are not always identifiable as such, because the day before or after is missing for classification, and the stated route or reason does not necessarily allow for the reliable identification of such a journey. A first approximate indication of the scale can be obtained by looking at Sundays in the daily logs and the work-related journeys or commutes over particularly long distances recorded there. If the 100-kilometer threshold is used here, the proportions for commutes are just over one percent and for journeys classified as work-related, around three percent. The distinction between these two categories is probably not clear enough with regard to long-distance commuting, therefore both values are given here.

Regardless of the economic situation of the household, the distance traveled for long-distance commutes is often significantly greater than the minimum threshold used here. For 45 percent of cases, it ranges between 100 and 250 kilometers, for around 35 percent between 250 and 500 kilometers, and for 20 percent it is even greater. The affected population group therefore frequently has to, or wants to, accept long travel times and thus, as a rule, considerable costs associated with long-distance commuting.

However, in order to quantify the group of affected individuals more precisely, a survey was conducted outside the scope of the report regarding secondary residences and the reasons for them. Seven percent of adults stated that they have a secondary residence, and of these, 31 percent cited the need to be able to reach their workplace. If the 100-kilometer threshold is also applied here for long-distance commuting, two-thirds of this segment remains. This results in a proportion of between one and two percent of adults who belong to the group of (occasional)

Long-distance commuters traveling over the aforementioned distance are included. They belong to the economically high-status group to a somewhat larger extent than the general population average, but long-distance commuting is by no means limited to this segment of the population.

19. Summary and outlook – also regarding other sources

Between the MiD 2017 and the MiD 2023, some framework conditions have changed. Some of these factors also influence everyday mobility.

This includes indirect consequences of the Covid-19 pandemic, such as the establishment of home office options. At the same time, there is a growing population, as well as a continuing trend of urbanization and increased settlement density.

Added to this is the progressive aging of the population. Since 2017, the proportion of citizens over 60 in Germany has exceeded 30 percent, increasing by two percentage points. At the same time, there were easily measurable increases in the number of employed people. What was not easily measurable was whether, or to what extent, the discussed shifts between rich and poor occurred. However, at least in the years 2020-2023, there was no real wage growth for the first time in a long time.

Numerous influencing factors must be taken into account.

These more structural shifts are complemented by direct mobility characteristics. These include, first and foremost, a still-growing car fleet with increasing e-mobility and the introduction of the Germany Ticket, but also new sharing services and discussions about the quality of rail transport. Last but not least, the discussion about the effects of climate change has intensified, and expectations for more environmentally friendly mobility are more in focus in 2023 than in 2017.

Each of these developments, except for the climate aspects, is not serious in itself, progresses rather continuously and does not occur suddenly.

Each factor, taken individually, influences everyday mobility, both in patterns and habits as well as in extrapolated absolute values. The directions of these potential effects differ, sometimes canceling each other out or reinforcing one another.

This makes the analysis challenging. Changes are readily identifiable descriptively using the powerful and carefully developed MiD sample.

However, reliably identifying causal relationships is more difficult given the overlapping nature of changes. Furthermore, the effects of slow structural shifts are not easily separable from changes in attitudes and potential new behavioral routines.

Car use is declining in importance, but still dominates

Until now, mobility surveys have been characterized by the growing importance of car traffic. This also applies to the MiD time series from 2002 to 2017. The 2023 survey marks a slight change for the first time. While car ownership and availability continue to rise, and a driver's license remains a basic requirement for most citizens, the level of daily car use is lower for the first time in a current MiD survey than in the previous survey, both proportionally and in absolute terms of traffic volume and performance. This decrease does not represent a trend reversal, and it remains to be seen what the coming years will reveal before this term can be used with any certainty. The slowly growing electric vehicle (E) sector of car mobility will also be able to make a small contribution towards climate-neutral road transport, as long as the resulting CO₂ savings are offset by the size of the combustion engine fleet and vehicles.

No real winners can be identified among the other modes of transport.

No clear beneficiaries have yet emerged among the other modes of transport. Cycling is growing more strongly in some cities, but only slowly nationwide. Moreover, this growth is primarily driven by technological developments such as e-bikes and not across all population groups. Public transport is recovering from its temporary slump caused by the pandemic, presumably with the help of the newly introduced Germany-wide ticket. However, neither cycling nor public transport has experienced a real breakthrough – and both must face significantly more critical assessments of their infrastructure than before. They may be missing out on potential and could demonstrate greater growth if their infrastructure were in better condition.

Pedestrian traffic, on the other hand, has seen a considerable increase. However, it remains to be seen what further developments will bring. Higher pedestrian share figures than in 2023 were already observed in the 2008 MiD (Municipal Impact Assessment). It is possible that 2017 was an outlier rather than 2023 a new milestone. Moreover, clear reasons for this are difficult to pinpoint.

and the satisfaction of being able to cover distances well on foot has also decreased significantly.

It is certain that digital options such as working from home and online shopping are changing mobility demand and may also be largely responsible for its slightly declining level. However, we must also caution against overly simplistic conclusions. A decrease in commuter and private shopping and supply traffic is offset by an increase in leisure and commercial traffic. Mobility needs are thus met, and resources freed up in one area are at least partially deployed elsewhere, merely shifted – for example, from private shopping to delivery services. Likewise, there are trends where the negative impacts of traffic continue, not least due to high land consumption and increasing urban sprawl.

Quality is expensive – but necessary if you want to... traffic is to be changed

We should also be wary of overly short-sighted hopes. The MiD time series primarily shows that everyday mobility changes only slowly and depends, not least, on truly compelling new offerings that can counter the perceived advantages of the car with new qualities. While sharing services and other new options are finding customers, so far only at a low level and not to the extent sometimes hoped for. Moreover, they are economically demanding. Certainly, there is still potential for economies of scale, so that profitability will increase with wider adoption. However, according to current findings, they will require financial support in the long term, much like traditional public transport services, if they are to become a truly significant player.

This also applies, on a different level and to a much greater extent, to traditional public transport. The image of public transport as the backbone of less individualized mobility is often overused, but still accurate. Only comfortable and reliable mass transit systems have the potential for truly sustainable transport on a large scale. This also includes walking, because the journey to the stop-

Getting to the starting point or covering the "last mile" to the goal is often only possible this way.

Background factors often shape mobility more than optimizations of the Transport services themselves

Last but not least, the results underscore that the nature of everyday mobility is generally a consequence of its surrounding circumstances. Settlement structure decisively determines mobility demand, as do the housing and labor market situations. The economic environment also influences mobility demand. This is also demonstrated by the MiD time series. If more environmentally friendly everyday mobility is the goal, solutions must therefore be found not only in the transport services themselves, but especially regarding the underlying determinants. These solutions are often only achievable in the medium and long term, but they make a significant and lasting contribution to achieving the goal in the area of mobility.

Future measurement of mobility

The Mobility Data Database (MiD) is a key source for measuring everyday traffic patterns over time. With the additional modules added in 2023, its dataset extends significantly beyond statistical benchmarks for modal distribution at the level of traffic volume and performance. This includes data on the perception of transport services, attitudes, and provided geodata, as well as information on travel by residents of Germany. It offers the most comprehensive of the survey-based data sets. It is followed by the equally established system of representative transport surveys (SrV), although this only uses samples for individual cities or sub-regions and is not representative nationwide. For a long time, the now-discontinued Mobility Panel (MOP) held third place in this series. The current editions of the MiD and the SrV show very similar results and both indicate the changes described, which are summarized schematically in [Figure 136](#) – always bearing in mind that this represents a snapshot in the shadow of previous or still-ongoing crises.

The survey approaches underlying these assessments also need further development. For example, new digital technologies are available that simplify route recording and expand its scope. However, sample quality and participant participation remain crucial and often underestimated. The Federal Highway Research Institute (BAST), in coordination with the Federal Ministry of Transport (BMV), has assumed responsibility for continuing the MiD time series. This team is currently preparing a new wave of data collection, likely for 2028. New, supplementary data collection techniques are currently being tested for this purpose.

At the same time, interim surveys with a shortened data collection period are already taking place. These are intended to serve as a transition to an ongoing survey that will measure mobility throughout the year and be able to track current changes more flexibly.

Further data sources on everyday mobility

However, these survey-based measurement approaches are gaining increasing support. More and more sources are becoming available that can also be used to measure everyday mobility. These are less comprehensive and, unlike the MiD, often only cover parts of the overall picture.

or data relating to individual means of transport. Nevertheless, the final – and looking ahead – question arises as to what results these sources provide for current developments, how they should be classified, and whether they confirm, supplement, or call into question MiD results.

A comprehensive inventory cannot and should not be provided here, but a brief overview is warranted. This reveals that all currently available nationwide results show essentially similar findings to the current MiD: a slight reduction in car traffic, a slow recovery of public transport, especially rail, following the pandemic-related losses from 2020 to 2022, and only local or regional growth rates in cycling. The situation is more difficult with regard to pedestrian traffic alone, for which there are hardly any quantitative sources besides the MiD.

A brief attempt to list and roughly categorize the most important data sources yields the following list, without claiming to be exhaustive (for sources see the literature references below):

- For car traffic, count results are available for federal roads (motorways and federal highways),

Figure 136
Cap 18, ABB

What does all this mean – and what might happen next?



primarily under the responsibility of the Federal Highway Research Institute (BAST) – however, not nationwide and requiring careful processing.

- The same applies to road count results in some federal states with regard to state roads.

Similarly, some municipalities maintain data series on local traffic counts, including both motorized and bicycle traffic. However, these data must also be thoroughly processed and verified if they are to be reliably linked to or related to other sources.

- Also available for private motorized transport (PMT) are so-called “floating car data”. These primarily use navigation and Bluetooth data from the newer, equipped vehicles in the fleet and extrapolate this data to the entire fleet as far as possible, which is also a complex process.

- **Mobile network data** is increasingly being used across all modes of transport. During the COVID-19 pandemic, such data was available as experimental data from Destatis (the Federal Statistical Office) based on data from only one provider, meaning it needs to be calibrated for the entire population, insofar as this is even possible. Since this data is based on cell tower data and is not always sufficiently detailed for short distances, its analysis is often limited to journeys of 30 kilometers or more, representing about 10 percent of everyday journeys outside of freight transport.

This results in declines in car traffic and a slight increase in public transport.

- For public transport, **reports from transport companies on passenger numbers** and passenger-kilometers are available, increasingly based on automatic passenger counting systems (AFZS), which are published in an established time series by Destatis. These data currently show the effects of the Germany Ticket and – taking differences in definition into account – align with the MiD findings of a public transport recovery that is gradually reaching pre-COVID levels. However, the results are difficult to interpret regionally, as the reports to Destatis refer to the company headquarters and not the location where the service is provided.

- Analyses of **work-related commuting data** are available from Destatis, the Federal Employment Agency, and the Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR). When comparing these analyses to the MiD (Minimum Daily Inventory), the definition of commuting must be taken into account, as well as the distinction between employees, the self-employed, and civil servants. If this is done, similar results are obtained for commuting compared to the MiD.

- A key and valuable compilation of such and numerous other sources is provided by the BMV's "**Transport in Figures**" in very long time series. Passenger transport represents only a small segment of a comprehensive data collection and tabular presentation. The system regularly uses the MiD (Minimum Data on Transport) as the basis for its models and is therefore already compatible with it through this link.

- A new possibility, not yet established nationally in Germany, is mobility data within Google's "**Environmental Insights Explorer (EIE)**". This data is made available globally for public purposes in the form of regional aggregates.

Based on Google Maps data and using traffic recognition, they include a differentiated count of stages and kilometers. However, scientific transparency and further verification are necessary here, for example, regarding the representativeness and the underlying definition of routes. This is similar to the frequently used source of mobile phone data.

Brief synopsis and assessment

Each of the aforementioned sources is valuable in its own right and can contribute data and results to the assessment of the overall situation. However, the precise delimitation and quality assessment are crucial in each case. This is particularly true given that almost none of these sources provides a comprehensive overview, but rather, as described, only illuminate specific aspects. Furthermore, they must manage limited sample quality and be aware of their data collection conditions.

This can be illustrated using the example of traffic count data for roads and public transport. A report compiled a synopsis of data from 2019 to 2023 and concluded that private motorized transport (PMT) remained significantly slowed even after the post-COVID-19 normalization. Only with such a comprehensive analysis can the various data points be properly interpreted, taking into account their respective limitations. However, it remains to be seen whether the assessment and the forecast formulated therein—that no further recovery in PMT is to be expected after the now-compensated COVID-19-related decline—will prove accurate.

It remains to be seen where everyday mobility actually stands after this snapshot from the 2023 Mobility in Daylight (MiD) survey, and whether the widely observed decline in car traffic will disappear again. Some indicators suggest that the peak of car use has been passed or that it has reached a plateau. But other modes of transport must be able to fully utilize this decline to their advantage, instead of merely preventing a further increase in car use. There are also some indications that this is the case. However – and the 2023 MiD results also showed this – it doesn't yet appear that everything is sufficiently in place in terms of quality, comfort, and reliability, nor in terms of citizens' preferences.

List of regional participants

federal states

- Baden-Württemberg
- Brandenburg
- Free and Hanseatic City of Hamburg
- Free State of Bavaria
- Free State of Thuringia
- Hesse
- Mecklenburg-Vorpommern
- Lower Saxony
- North Rhine-Westphalia
- Rheinland-Pfalz
- Saarland

Transport associations, transport companies and Special-purpose associations

- Hamburger Verkehrsverbund GmbH (HVV)
- Hessen Mobil –
- ivm GmbH (Integrated Transport and Mobility Management
Region Frankfurt Rhine-Main)
- Kassel Transport Company AG
- Munich Transport Company (MVG)
- Munich Transport and Tariff Association (MVV)
- Paderborn/Höxter Public Transport Association (nph)
- North Hessian Transport Association (NVV)
- Rhine-Main Transport Association GmbH (RMV) –
- traffiQ Local Public Transport Company
Frankfurt am Main mbH
- Bremen Transport Association/
Lower Saxony GmbH (VBN)
- Transport Association of East Westphalia-Lippe (VVOWL)
- Rhine-Neckar Transport Association (VRN)
- Rhine-Ruhr Transport Association (VRR)
- Rhein-Sieg Transport Association (VRS)
- Zweckverband Mobilität Münsterland (ZVM)
- Zweckverband Mobilität Ruhr-Lippe (ZRL)
- Zweckverband Nahverkehr Rheinland GmbH (NVR)/
go.Rheinland –
- Zweckverband Nahverkehr Westfalen-Lippe (NWL)
- Public Transport Association
Westphalia-South (ZWS)

Supra-regional associations and metropolitan areas

- Region Stuttgart
- FrankfurtRheinMain Regional Association
- Regional Association Greater Braunschweig Area
- Neckar-Alb Regional Association
- Ruhr Regional Association

districts

- Hohenlohe district
- Cuxhaven district
- Fürth district
- Trier-Saarburg district
- Ostalbkreis
- Region Hannover
- Rhein-Erft district
- Rhein-Neckar district
- Rhein-Sieg district
- Rheinisch-Bergischer-Kreis
- Aachen City Region

Cities and municipalities

- City of Aachen
 - City of Bonn
 - City of Brühl
 - City of Darmstadt
 - City of Freiburg
 - Stadt Fürstenfeldbruck
 - City of Fürth
 - City of Fulda
 - City of Gütersloh
 - City of Ingolstadt
 - City of Koblenz
 - City of Leverkusen
 - City of Lindau
 - Stadt Ludwigsburg
 - Stadt Offenbach
 - City of Stein
 - City of Stuttgart
 - City of Wuppertal
(in cooperation with WSW mobil GmbH)
-

Explanation of important terms

The following section introduces the key terms from the results report. The focus is particularly on its importance in passenger transport.

Stage	A stage is a section of a route, either by means of transport or on foot.
Corona-Phase	Describes the measures taken to limit the spread of SARS-CoV-2 at the beginning of the Corona pandemic from 2020 to 2022, which also affected the mobility of citizens. These include contact and travel restrictions, within which home office regulations were introduced and expanded, schools were temporarily closed, curfews were imposed, and masks were mandatory in public spaces (such as public transport).
Main means of transport	If multiple modes of transport are used on a journey, they are grouped into the main mode of transport according to a hierarchy, with the highest-ranking mode being selected. The ranking is as follows: public transport, private motorized transport, bicycle, walking.
Intermodality	A special form of multimodality in which different means of transport are used on one journey.
CAT	Stands for "motorized individual transport", which includes cars, motorized two-wheelers (but not electric bicycles), trucks and other motorized vehicles.
Migration background	Different definitions are possible; in the MID, a person is someone who, or of whom at least one parent, was not born in Germany.
Mobility behavior	Changes in location of a person with their spatial, temporal and modal characteristics.
Modal Split	It depicts the distribution of the main means of transport by route or by passenger-kilometers in percentage shares or also in absolute figures.
Multimodality	Use of different means of transport when completing a person's journeys within a specific observation period (often one week).
Public transport (PT)	Includes all public transport, even on longer journeys (local buses, all trains, long-distance and coach buses, airplanes and taxis).
On-demand transport	A commercial, on-demand transport service where journeys can be booked flexibly. The distinction between this and similarly on-demand, but time- and location-bound, call-a-taxi services (AST) is not always clear-cut.
Passenger kilometers	It is a unit of measurement for transport performance and includes the distance traveled by one or all person(s) on a route or in a unit of time. ten Kilometers.
Reporting date/reporting date	Each respondent is assigned a reference date determined using a statistical random procedure. All routes taken on this date are to be provided. The reference dates are spread over 12 months and cover all days from Monday to Sunday.

Usual use of transport	Unlike the calculation of modal split, which reflects mode choice at the trip level, habitual mode use expresses an individual's average behavior over a longer period. While modal split is a statistical measure used to describe traffic volume, the question of habitual mode use allows us to describe individual mobility patterns.
Traffic	Aggregated changes in location of people within a spatial unit.
Traffic volume	Represents the absolute number or percentage distribution of all journeys made by the population in a specific period (e.g., per day or year).
Transport service	Represents the absolute number or percentage distribution of all person-kilometers traveled by the population in a specific period (e.g., per day or year).
Road	In the MiD survey, respondents are asked to list all journeys they made on a given day. A journey is defined as the route from the starting point to the destination for a specific purpose. If there are transfers or changes of transport, it remains one journey. If there are interruptions or longer stops, it becomes two journeys (for example, from work to home with a shopping stop).

